AP Calculus BC

Saint Ignatius College Preparatory

Textbook Companion

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"Ad Majorem Dei Gloriam"

For the Greater Glory of God.

Created with care by the Mathematics Department
Saint Ignatius College Preparatory, San Francisco, CA

How To Use This Textbook?

This textbook has been heavily edited in regards to the original, KQuattrin.com version. However, the structure that this textbook follows has not changed. There are 11 chapters within this textbook, and each chapter has varying numbers of sections. Within each section, one can expect to find the following:

- Explanation of material
- Example problems and solutions
- Practice problems
- Practice exam

Answers to practice problems can be found at the end of every chapter.

Information within these boxes is important information.

Information within these boxes is the objectives for the section.

Information within these boxes is interesting sidenotes and extra relevant information.

Information within these boxes is definitions for important terms.

Information within these boxes is example problems.

Information within these boxes is solutions to example problems.

`These symbols` depict buttons on your TI-84.

Chapter 1:

Review of Derivatives

Chapter 1 Overview: Review of Derivatives

The purpose of this chapter is to review the ”how” of differentiation. We will review all the derivative rules learned last year in Precalculus. In the next two chapters, we will review the ”why.” As a quick reference, here are those rules:

The Power Rule: $\dfrac{d}{dx}\left[u^n\right] = nu^{n-1}\dfrac{du}{dx}$

The Product Rule: $\dfrac{d}{dx}\left[u \cdot v\right] = u \cdot \dfrac{dv}{dx} + v \cdot \dfrac{du}{dx}$

The Quotient Rule: $\dfrac{d}{dx}\left[\dfrac{u(x)}{v(x)}\right] = \dfrac{v \cdot \dfrac{du}{dx} - u \cdot \dfrac{dv}{dx}}{v^2}$

The Chain Rule: $\dfrac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x)$

$$\frac{d}{dx}[\sin(u)] = (\cos(u))\frac{du}{dx} \qquad \frac{d}{dx}[\csc(u)] = (-\csc(u)\cot(u))\frac{du}{dx}$$

$$\frac{d}{dx}[\cos(u)] = (-\sin(u))\frac{du}{dx} \qquad \frac{d}{dx}[\sec(u)] = (\sec(u)\tan(u))\frac{du}{dx}$$

$$\frac{d}{dx}[\tan u] = \left(\sec^2(u)\right)\frac{du}{dx} \qquad \frac{d}{dx}[\cot(u)] = \left(-\csc^2(u)\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[e^u\right] = \left(e^u\right)\frac{du}{dx} \qquad \frac{d}{dx}[\ln u] = \left(\frac{1}{u}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[a^u\right] = \left(a^u \cdot \ln u\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\log_a u\right] = \left(\frac{1}{u \cdot \ln a}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[\sin^{-1}(u)\right] = \left(\frac{1}{\sqrt{1-u^2}}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\csc^{-1}(u)\right] = \left(\frac{-1}{|u|\sqrt{u^2-1}}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[\cos^{-1}(u)\right] = \left(\frac{-1}{\sqrt{1-u^2}}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\sec^{-1}(u)\right] = \left(\frac{1}{|u|\sqrt{u^2-1}}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[\tan^{-1}(u)\right] = \left(\frac{1}{u^2+1}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\cot^{-1}(u)\right] = \left(\frac{-1}{u^2+1}\right)\frac{du}{dx}$$

Here is a quick review from last year:

Identities: While all will eventually be used somewhere in Calculus, the ones that occur most often early are the Reciprocals and Quotients, the Pythagoreans, and the Double Angle Identities.

$$\tan(x) = \frac{\sin(x)}{\cos(x)}; \qquad \cot(x) = \frac{\cos(x)}{\sin(x)}; \qquad \sec(x) = \frac{1}{\cos(x)} = \frac{1}{\sin(x)}$$

$$\sin^2(x) + \cos^2(x) = 1; \qquad \tan^2(x) + 1 = \sec^2(x); \qquad \cot^2(x) + 1 = \csc^2(x)$$

$$\sin(2x) = 2\sin(x)\cos(x); \qquad \cos(2x) = \cos^2(x) - \sin^2(x)$$

Inverses: Because of the quadrants, taking an inverse yields two answers, only one of which your calculator can show. How the second answer is found depends on the kind of inverse:

$$\cos^{-1}(x) = \left\{ \begin{array}{l} \textit{calculator} \ \pm 2\pi n \\ -\textit{calculator} \ \pm 2\pi n \end{array} \right\} \qquad \sin^{-1}(x) = \left\{ \begin{array}{l} \textit{calculator} \ \pm 2\pi n \\ \pi - \textit{calculator} \ \pm 2\pi n \end{array} \right\}$$

$$\tan^{-1}(x) = \left\{ \begin{array}{l} \textit{calculator} \ \pm 2\pi n \\ \pi + \textit{calculator} \ \pm 2\pi n \end{array} \right\} = \textit{calculator} \ \pm \pi n$$

Logarithm Rules: Here are some logarithm rules which you should recall:

$$\log_a x + \log_a y = \log_a xy$$

$$\log_a x - \log_a y = \log_a \frac{x}{y}$$

$$\log_a x^n = n \log_a x$$

1.1: The Power and Exponential Rules with the Chain Rule

In Precalculus we developed the idea of the derivative geometrically. That is, the derivative initially arose from our need to find the slope of the tangent line. In Chapter 2 and 3, that meaning, its link to limits, and other conceptualizations of the derivative will be explored. In this chapter, we are primarily interested in how to find the derivative and what it is used for.

Derivative → Definition: $f(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

→ Means: The function that yields the slope of the tangent line.

Numerical Derivative → Definition: $f'(a) = \lim_{x \to a} \dfrac{f(x) - f(a)}{x - a}$

→ Means: The numerical value of the slope of the tangent line at $x = a$

Symbols for the Derivative

$\dfrac{dy}{dx}$ = "d - y - d - x" $f'(x)$ = "f prime of x" y' = "y prime"

$\dfrac{d}{dx}$ = "d - d - x" D_x = "d sub x"

OBJECTIVES

Use the Power Rule and Exponential Rules to Find Derivatives.
Find the Derivative of Composite Functions.

Key Idea from Precalculus: The derivative yields the slope of the tangent line. (But there is more to it than that).

The first and most basic derivative rule is the Power Rule. Among the last rules we learned in Precalculus were the Exponential Rules. They look similar to one another, therefore it would be a good idea to view them together.

The Power Rule:

$$\frac{d}{dx}\left[x^n\right] = nx^{n-1}$$

The Exponential Rules:

$$\frac{d}{dx}\left[e^x\right] = e^x$$

$$\frac{d}{dx}\left[a^x\right] = a^x \cdot \ln a$$

The difference between these is where the variable is. The Power Rule applies when the variable is in the *base*, while the Exponential Rules apply when the variable is in the *exponent*. The difference between the two Exponential rules is what the base is. $e = 2.718281828459\ldots$, while a is any positive number other than 1.

Ex 1.1.1: Find a) $\dfrac{d}{dx}\left[x^5\right]$ and b) $\dfrac{d}{dx}\left[5^x\right]$.

Sol 1.1.1: The first is a case of the Power Rule while the second is a case of the second Exponential Rule. Therefore,

a) $\dfrac{d}{dx}\left[x^5\right] = \boxed{5x^4}$ b) $\dfrac{d}{dx}\left[5^x\right] = \boxed{5^x \ln 5}$

There are a few other basic rules that we need to remember.

$$\frac{d}{dx}[\text{constant}] = 0$$

$$\frac{d}{dx}\left[cx^n\right] = (cn)x^{n-1}$$

$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$$

These rules allow us to easily differentiate a polynomial term by term.

Ex 1.1.2: $y = 3x^2 + 5x + 1$; find $\dfrac{dy}{dx}$.

Sol 1.1.2:

$$\frac{dy}{dx} = \frac{d}{dx}\left[3x^2 + 5x + 1\right]$$
$$= (3 \cdot 2)x^{2-1} + (5 \cdot 1)x^{1-1} + 0$$
$$= \boxed{6x + 5}$$

Ex 1.1.3: $f(x) = x^2 + 4x - 3 + e^x$; find $f'(x)$.

Sol 1.1.3:

$$f'(x) = \frac{d}{dx}\left[x^2 + 4x - 3 + e^x\right]$$
$$= (1 \cdot 2)x^{2-1} + (4 \cdot 1)x^{1-1} - 0 + e^x$$
$$= \boxed{2x + 4 + e^x}$$

Ex 1.1.4: $y = \sqrt{x^3} + \dfrac{4}{\sqrt{x}} - \sqrt[4]{x^3} + e^4$; find $\dfrac{dy}{dx}$.

Sol 1.1.4:

$$y = \sqrt{x^3} + \frac{4}{\sqrt{x}} - \sqrt[4]{x^3} + e^4$$
$$= x^{\frac{3}{2}} + 4x^{-\frac{1}{2}} - x^{\frac{3}{4}} + e^4$$
$$\frac{dy}{dx} = \frac{d}{dx}\left[x^{\frac{3}{2}} + 4x^{-\frac{1}{2}} - x^{\frac{3}{4}} + e^4\right]$$
$$= \left(1 \cdot \frac{3}{2}\right)x^{\frac{3}{2}-1} + \left(4 \cdot -\frac{1}{2}\right)x^{-\frac{1}{2}-1} - \left(1 \cdot \frac{3}{4}\right)x^{\frac{3}{4}-1} + 0$$
$$= \boxed{\frac{3}{2}x^{\frac{1}{2}} - 2x^{-\frac{3}{2}} - \frac{3}{4}x^{-\frac{1}{4}}}$$

Note in Ex 1.1.4 that e^4 is a constant. Therefore, its derivative is 0.

As we have seen, when the variable was in the exponent, we use the Exponential Rules. When the variable was in the base, we used the Power Rule. But what if the variable is in both places, such as $\frac{d}{dx}\left[(2x-1)^{x^2}\right]$? It is definitely an exponential problem, but the base is not a constant as the rules above have. The Change of Base Rule allows us to clarify the problem:

$$\frac{d}{dx}\left[(2x-1)^{x^2}\right] = \frac{d}{dx}\left[e^{x^2\ln(2x-1)}\right]$$

but we will need the Product Rule for this derivative. Therefore, we will save this for later.

Ex 1.1.5: If $y = (x^2+1)(x^3-4x)$, find $\frac{dy}{dx}$.

Sol 1.1.5:

$$\begin{aligned} y &= (x^2+1)(x^3-4x) \\ &= x^5 - 4x^3 + x^3 - 4x \\ &= x^5 - 3x^3 - 4x \\ \frac{dy}{dx} &= \frac{d}{dx}\left[x^5 - 3x^3 - 4x\right] \\ &= \boxed{5x^4 - 9x^2 - 4} \end{aligned}$$

Ex 1.1.6: If $y = \frac{x^2-4x+6}{\sqrt[3]{x}}$, find $\frac{dy}{dx}$.

Sol 1.1.6:

$$\begin{aligned} y &= \frac{x^2-4x+6}{\sqrt[3]{x}} \\ &= \frac{x^2-4x+6}{x^{\frac{1}{3}}} \\ &= x^{\frac{5}{3}} - 4x^{\frac{2}{3}} + 6x^{-\frac{1}{3}} \end{aligned}$$

$$\frac{dy}{dx} = \frac{d}{dx}\left[x^{\frac{5}{3}} - 4x^{\frac{2}{3}} + 6x^{-\frac{1}{3}}\right]$$

$$= \boxed{\frac{5}{3}x^{\frac{2}{3}} - \frac{8}{3}x^{-\frac{1}{3}} - 2x^{-\frac{4}{3}}}$$

The Chain Rule

Composite Function → Definition: A function made of two other functions, one within the other.

→ For example, $y = \sqrt{16x - x^3}$, $y = \sin\left(x^3\right)$. $y = \cos^3(x)$, and $y = \left(x^2 + 2x - 5\right)^3$. The general symbol is $f(g(x))$.

Ex 1.1.7: Given $f(x) = \cos^{-1}(x)$, $g(x) = x^2 - 1$, and $h(x) = \sqrt{1 + x^2}$, find a) $f\left(g\left(\sqrt{2}\right)\right)$, b) $h(g(1))$, and c) $f(h(g(1)))$.

Sol 1.1.7:

(a) $g\left(\sqrt{2}\right) = \left(\sqrt{2}\right)^2 - 1 = 1,$ so $f\left(g\left(\sqrt{2}\right)\right) = f(1) = \cos^{-1}(1) = \boxed{0}$.

(b) $g(1) = 0,$ so $h(g(1)) = h(0) = \sqrt{1 + 0^2} = \boxed{1}$

(c) $g(1) = 0$ and $h(g(1)) = h(0) = \sqrt{1 + 0^2} = 1,$ so $f(h(g(1))) = \cos^{-1}(1) = \boxed{0}$

So. How do we take the derivative of a composite function? There are two (or more) functions that must be differentiated, but, since one is inside the other, the derivatives cannot be taken at the same time. Just as a radical cannot be distributed over addition, a derivative cannot be distributed concentrically. The composite function is like a matryoshka (Russian doll) that has a doll inside a doll. The derivative is akin to opening them. They cannot both be opened at the same time and, when one is opened, there is an unopened one within. The result is two open dolls adjacent to each other.

The Chain Rule: $\dfrac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x)$

If you think of the inside function (the $g(x)$) as equaling u, we could write the Chain Rule like this:

$$\frac{d}{dx}[f(u)] = \frac{df}{du} \cdot \frac{du}{dx}$$

This is the way that most derivatives are written with the Chain Rule.

The Chain Rule is one of the cornerstones of Calculus. It can be embedded within each of the other rules, as seen in the introduction to this chapter. So the Power Rule and Exponential Rules in the last section really should have been stated as:

The Power Rule:

$$\frac{d}{dx}\left[u^n\right] = nu^{n-1} \cdot \frac{du}{dx}$$

The Exponential Rules:

$$\frac{d}{dx}\left[e^u\right] = e^u \cdot \frac{du}{dx}$$

$$\frac{d}{dx}\left[a^u\right] = \left(a^u \cdot \ln a\right) \cdot \frac{du}{dx}$$

(where u is a function of x)

Ex 1.1.8: $\dfrac{d}{dx}\left[\left(4x^2 - 2x - 1\right)^{10}\right]$

Sol 1.1.8:

$$u = 4x^2 - 2x - 1 \text{ and } f(u) = u^{10}$$

$$\begin{aligned}
\frac{d}{dx}\left[f(u)\right] &= f'(u) \cdot \frac{du}{dx} \\
&= 10u^9 \cdot (8x - 2) \\
&= \boxed{10\left(4x^2 - 2x - 1\right)^9 (8x - 2)}
\end{aligned}$$

Ex 1.1.9: $\dfrac{d}{dx}\left[e^{4x^2}\right]$

Sol 1.1.9:

$$u = 4x^2$$

$$\begin{aligned}\frac{d}{dx}\left[e^u\right] &= e^u \cdot \frac{du}{dx} \\ &= e^{4x^2} \cdot 8x \\ &= \boxed{8xe^{4x^2}}\end{aligned}$$

Ex 1.1.10: If $y = \sqrt{16 - x^3}$, find $\dfrac{dy}{dx}$.

Sol 1.1.10:

$$u = 16 - x^3 \text{ and } f(u) = \sqrt{u}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left[f(u)\right] = f'(u) \cdot \frac{du}{dx} \\ &= \frac{1}{2\sqrt{u}} \cdot -3x^2 \\ &= \boxed{\frac{-3x^2}{2\sqrt{16 - x^3}}}\end{aligned}$$

Ex 1.1.11: $\dfrac{d}{dx}\left[\sqrt{\left(x^2+1\right)^5 + 7}\right]$

Sol 1.1.11:

$$u = x^2 + 1,\ g(u) = u^5 + 7,\ \text{and } f(g(u)) = \sqrt{g(u)}$$

$$
\begin{aligned}
\frac{d}{dx}\left[f(g(u))\right] &= f'(g(u)) \cdot g'(u) \cdot \frac{du}{dx} \\
&= \frac{1}{2\sqrt{g\left(u\right)}} \cdot 5u^4 \cdot 2x \\
&= \frac{5\left(x^2+1\right)\left(2x\right)}{2\sqrt{\left(x^2+1\right)^5+7}} = \boxed{\frac{5x\left(x^2+1\right)}{\sqrt{\left(x^2+1\right)^5+7}}}
\end{aligned}
$$

1.1 Free Response Homework

Find the derivatives of the given functions. Simplify where possible.

1. $f(x) = x^2 + 3x - 4$
2. $f(t) = \frac{1}{4}\left(t^4 + 8\right)$
3. $y = x^{-\frac{2}{3}}$
4. $y = 5e^x + 3$
5. $v(r) = \frac{4}{3}\pi r^3$
6. $g(x) = x^2 + \frac{1}{x^2}$
7. $y = \frac{x^2 + 4x + 3}{\sqrt{x}}$
8. $u = \sqrt[3]{t^2} + 2\sqrt{t^3}$
9. $z = \frac{A}{y^{10}} + Be^y$
10. $y = e^{x+1} + 1$

Complete the following.

11. $\frac{d}{dx}\left[x^7 - 4\sqrt[8]{x^7} + 7^x - \frac{1}{\sqrt[7]{x^4}} + \frac{1}{5x}\right]$
12. $\frac{d}{dx}\left[x^6 - 3\sqrt[6]{x^7} + 5^x - \frac{1}{\sqrt[3]{x^5}} + \frac{1}{8x}\right]$
13. $\frac{d}{dx}\left[x^4 - 14\sqrt[7]{x^9} + 8^x - \frac{1}{\sqrt[3]{x^7}} + \frac{1}{8x}\right]$
14. $\frac{d}{dx}\left[(x-1)\sqrt{x}\right]$
15. $\frac{d}{dz}\left[\left(z^2 - 4\right)\sqrt{z^3}\right]$
16. $\frac{d}{dx}\left[\left(x^2 - 4x + 3\right)\sqrt{x^5}\right]$
17. $\frac{d}{dt}\left[\left(4t^2 + 1\right)\left(3t^3 + 7\right)\right]$
18. $\frac{d}{dx}\left[\left(x^3 + 4x - \pi\right)^{-7}\right]$
19. $\frac{d}{dx}\left[\sqrt{3x^2 - 4x + 9}\right]$
20. $\frac{d}{dx}\left[\sqrt[7]{x^3 - 2x}\right]$
21. $\frac{d}{dy}\left[\frac{4y^3 - 2y^2 - 5y}{\sqrt{y}}\right]$
22. $\frac{d}{dv}\left[\frac{v^2 - 4v + 7}{2\sqrt{v}}\right]$
23. $\frac{d}{dw}\left[\frac{7w^2 - 4w + 1}{5w^3}\right]$
24. $\frac{d}{dw}\left[\frac{5w^2 - 3w - 4}{7w^2}\right]$
25. $f(x) = \sqrt[4]{1 + 2x + x^3}$, find $f'(x)$
26. $f(x) = \sqrt[5]{\left(\frac{1}{x} + 2x + e^x\right)^3}$, find $f'(x)$
27. $f(x) = \left(x^3 + 2x\right)^{37}$, find $f'(x)$
28. $f(x) = 3x^5 - 5x^3 + 3$, find $f'(x)$
29. $g(2) = 3,\ g'(2) = -4,\ f(x) = e^{g(x)}$, find $f'(2)$
30. $y = e^{\sqrt{x}}$, find $\frac{dy}{dx}$

31. $f(x) = \sqrt{4 - \frac{4}{9}x^2}$, find $f'\left(\sqrt{5}\right)$

32. $f(x) = e^{\sqrt{9-x^2}}$, find $f'(x)$

33. $v(t) = \sqrt{\left(\frac{E(t)}{3} + 3t\right)^{\frac{3}{7}} - 4}$, find $v'(t)$

34. $v(t) = \sqrt[3]{\left(\frac{C(t)}{7} + 4t^2\right)^{\frac{5}{7}} - 1}$, find $v'(t)$

1.1 Multiple Choice Homework

1. If $f(x) = x^{\frac{3}{2}}$, then $f'(4) =$

a) -6 b) -3 c) 3 d) 6 e) 8

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2. The derivative of $\sqrt{x} - \frac{1}{x\sqrt[3]{x}}$

a) $\frac{1}{2}x^{-\frac{1}{2}} - x^{-\frac{4}{3}}$ b) $\frac{1}{2}x^{-\frac{1}{2}} + \frac{4}{3}x^{-\frac{7}{3}}$ c) $\frac{1}{2}x^{-\frac{1}{2}} - \frac{4}{3}x^{-\frac{1}{3}}$

d) $-\frac{1}{2}x^{-\frac{1}{2}} + \frac{4}{3}x^{-\frac{7}{3}}$ e) $-\frac{1}{2}x^{-\frac{1}{2}} - \frac{4}{3}x^{-\frac{1}{3}}$

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3. Given $f(x) = \frac{1}{2x} + \frac{1}{x^2}$, find $f'(x)$

a) $-\frac{1}{2x^2} - \frac{2}{x^3}$ b) $-\frac{2}{x^2} - \frac{2}{x^3}$ c) $\frac{2}{x^2} - \frac{2}{x^3}$

d) $-\frac{1}{2x^2} + \frac{2}{x^3}$ e) $\frac{1}{2x^2} - \frac{2}{x^3}$

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4. If $f(x) = e^{5x^2} + x^4$, then $f'(1) =$

a) $e^5 + 1$ b) $5e^4 + 4$ c) $5e^5 + 1$ d) $10e + 4$ e) $10e^5 + 4$

---

5. If h is the function defined by $h(x) = e^{5x} + x + 3$, then $h'(0)$ is

a) 2 b) 4 c) 5 d) 6 e) 8

---

6. If $y = \left(x^4 + 4\right)^2$, then $\dfrac{dy}{dx} =$

a) $2\left(x^4 + 4\right)$ b) $\left(4x^3\right)^2$ c) $2\left(4x^3 + 4\right)$

d) $4x^3\left(x^4 + 4\right)$ e) $8x^3\left(x^4 + 4\right)$

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7. If $h(x) = [f(x)]^2\, g(x)$ and $g(x) = 3$, then $h'(x) =$

a) $2f'(x)g'(x)$ b) $6f'(x)f(x)$ c) $g'(x)\,[f(x)]^2 + 2f(x)f'(x)g(x)$

d) $2f'(x)g(x) + g'(x)\,[f(x)]^2$ e) 0

---

8. Which of the following statements must be true?

I. $\dfrac{d}{dx}\left[\sqrt{e^x + 3}\right] = \dfrac{e^x}{2\sqrt{e^x + 3}}$

II. $\dfrac{d}{dx}\left[5^{3x^2}\right] = 6x\ln(5)\left(5^{3x^2}\right)$

III. $\dfrac{d}{dx}\left[6x^3 - \pi + \sqrt[3]{x^8} - \dfrac{2}{x^3}\right] = 18x^2 + \dfrac{8}{3}\sqrt[3]{x^5} + \dfrac{6}{x^4}$

a) I only b) II only c) I and III only

d) I and III only e) I, II, and III

1.2: Trig, Trig Inverse, and Log Rules

Trigonometric → Definition: A function (sin, cos, tan, sec, csc, or cot) whose independent variable represents an angle measure.

→ Means: An equation with sine, cosine, tangent, secant, cosecant, or cotangent in it.

Logarithmic → Definition: The inverse of an exponential function.

→ Means: An equation with log or ln in it.

Trig Derivative Rules

$$\frac{d}{dx}[\sin u] = (\cos u)\frac{du}{dx} \qquad \frac{d}{dx}[\csc u] = (-\csc u \cot u)\frac{du}{dx}$$

$$\frac{d}{dx}[\cos u] = (-\sin u)\frac{du}{dx} \qquad \frac{d}{dx}[\sec u] = (\sec u \tan u)\frac{du}{dx}$$

$$\frac{d}{dx}[\tan u] = \left(\sec^2 u\right)\frac{du}{dx} \qquad \frac{d}{dx}[\cot u] = \left(-\csc^2 u\right)\frac{du}{dx}$$

Log Derivative Rules

$$\frac{d}{dx}[\ln u] = \left(\frac{1}{u}\right)\frac{du}{dx} \qquad \frac{d}{dx}[\log_a u] = \left(\frac{1}{u \cdot \ln a}\right)\frac{du}{dx}$$

Note that all these rules are expressed in terms of the Chain Rule.

OBJECTIVES

Find Derivatives Involving Trig, Trig Inverse, and Logarithmic Functions.

Ex 1.2.1: $\frac{d}{dx}\left[\sin^3(x)\right]$

Sol 1.2.1:

$$\frac{d}{dx}\left[\sin^3(x)\right] = \boxed{3\sin^2(x)\cos(x)}$$

Ex 1.2.2: $\dfrac{d}{dx}\left[\sin\left(x^3\right)\right]$

Sol 1.2.2:

$$\frac{d}{dx}\left[\sin\left(x^3\right)\right] = \boxed{3x^2\cos\left(x^3\right)}$$

Ex 1.2.3: $\dfrac{d}{dx}\left[\ln\left(4x^3\right)\right]$

Sol 1.2.3:

$$\begin{aligned}\frac{d}{dx}\left[\ln\left(4x^3\right)\right] &= \frac{1}{4x^3}\cdot 12x^2\\ &= \boxed{\frac{3}{x}}\end{aligned}$$

We could have also simplified algebraically before taking the derivative:

$$\begin{aligned}\ln\left(4x^3\right) &= \ln 4 + \ln x^3\\ &= \ln 4 + 3\ln x\\ \frac{d}{dx}\left[\ln 4 + 3\ln x\right] &= 0 + 3\cdot\frac{1}{x}\\ &= \boxed{\frac{3}{x}}\end{aligned}$$

Of course, composites can involve more than two functions. The Chain Rule has as many derivatives in the chain as there are functions.

Ex 1.2.4: $\dfrac{d}{dx}\left[\sec^5\left(3x^4\right)\right]$

Sol 1.2.4:

$$\begin{aligned}\frac{d}{dx}\left[\sec^5\left(3x^4\right)\right] &= 5\sec^4\left(3x^4\right)\cdot\sec\left(3x^4\right)\tan\left(3x^4\right)\cdot\left(12x^3\right)\\ &= \boxed{60x^3\sec^5\left(3x^4\right)\tan\left(3x^4\right)}\end{aligned}$$

Ex 1.2.5: $\dfrac{d}{dx}\ln\left(\cos\left(\sqrt{x}\right)\right)$

Sol 1.2.5:

$$\begin{aligned}\frac{d}{dx}\ln\left(\cos\left(\sqrt{x}\right)\right) &= \frac{1}{\cos\left(\sqrt{x}\right)}\cdot\left(-\sin\left(\sqrt{x}\right)\right)\cdot\frac{1}{2\left(\sqrt{x}\right)}\\ &= -\tan\left(\sqrt{x}\right)\cdot\frac{1}{2\left(\sqrt{x}\right)}\\ &= \boxed{\frac{-\tan\left(\sqrt{x}\right)}{2\sqrt{x}}}\end{aligned}$$

General inverses are not all that interesting. We are more interested in particular *transcendental* inverse functions, like the natural log. Another particular kind of inverse function that bears more study is the trig inverse function. Interestingly, as with the log functions, the derivatives of these transcendental functions become algebraic functions.

Inverse Trig Derivative Rules

$$\frac{d}{dx}\left[\sin^{-1}u\right] = \left(\frac{1}{\sqrt{1-u^2}}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\csc^{-1}u\right] = \left(\frac{-1}{|u|\sqrt{u^2-1}}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[\cos^{-1}u\right] = \left(\frac{-1}{\sqrt{1-u^2}}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\sec^{-1}u\right] = \left(\frac{1}{|u|\sqrt{u^2-1}}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[\tan^{-1}u\right] = \left(\frac{1}{u^2+1}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\cot^{-1}u\right] = \left(\frac{-1}{u^2+1}\right)\frac{du}{dx}$$

Ex 1.2.6: $\dfrac{d}{dx}\left[\tan^{-1}\left(3x^4\right)\right]$

Sol 1.2.6:

$$\frac{d}{dx}\left[\tan^{-1}\left(3x^4\right)\right] = \frac{1}{(3x^4)^2+1}\cdot\left(12x^3\right)$$

$$= \boxed{\frac{12x^3}{9x^8+1}}$$

Ex 1.2.7: $\dfrac{d}{dx}\left[\sec^{-1}\left(x^2\right)\right]$

Sol 1.2.7:

$$\frac{d}{dx}\left[\sec^{-1}\left(x^2\right)\right] = \frac{1}{|x^2|\sqrt{(x^2)^2-1}}\cdot 2x$$

$$= \frac{2x}{(x^2)\sqrt{(x^2)^2-1}}$$

$$= \boxed{\frac{2}{x\sqrt{x^4-1}}}$$

General Inverse Derivative

$$\frac{d}{dx}\left[f^{-1}(x)\right] - \frac{1}{f'\left[f^{-1}(x)\right]}$$

Ex 1.2.8: If $f(x) = x^2+2x+3,\ g(x) = f^{-1}(x),$ and $g(1) = 2;$ find $g'(1)$.

Sol 1.2.8:

$$f'(x) = 2x+2 \ \therefore\ f'\left(g(x)\right) = 2\left(g(x)\right)+2$$

$$\frac{d}{dx}\left[f^{-1}(x)\right] = \frac{1}{f'\left[f^{-1}(x)\right]} = \frac{1}{f'\left(g(x)\right)}$$

$$g'(1) = \frac{1}{f'\left(g(1)\right)} = 2\left(g(1)\right) + 2 = \boxed{6}$$

1.2 Free Response Homework Set A

Find the derivatives of the given functions. Simplify where possible.

1. $y = \sin(4x)$
2. $y = 4\sec\left(x^5\right)$
3. $f(t) = \sqrt[3]{1 + \tan t}$
4. $f(\theta) = \ln\left(\cos(\theta)\right)$

5.* $y = a^3 + \cos^3(x)$

6.* $y = \cos\left(a^3 + x^3\right)$

7. $f(x) = \cos\left(\ln x\right)$
8. $f(x) = \sqrt[5]{\ln x}$
9. $f(x) = \log_{10}\left(2 + \sin(x)\right)$
10. $f(x) = \log_2(1 - 3x)$
11. $y = \sin^{-1}\left(e^x\right)$
12. $y = \tan^{-1}\left(\sqrt{x}\right)$

*Note that a is a constant.

Complete the following.

13. $\dfrac{d}{dx}\left[\sin^{-1}\left(e^{3x}\right)\right]$
14. $\dfrac{d}{dx}\left[\cot^{-1}\left(e^{2x}\right)\right]$
15. $\dfrac{d}{dx}\left[\tan^{-1}\left(x^2\right)\right]$
16. $\dfrac{d}{dx}\left[\cot^{-1}\left(\dfrac{1}{x}\right) - \tan^{-1}(x)\right]$
17. $\dfrac{d}{dx}\left[3e^{x^2+2x}\right]$
18. $\dfrac{d}{dx}\left[3\cos\left(x^2 + 2x\right)\right]$
19. $\dfrac{d}{dx}\left[\sqrt[3]{16 + x^3}\right]$
20. $\dfrac{d}{dx}\left[\sec^{-1}\left(2x^2\right)\right]$
21. $\dfrac{d}{dx}\left[5e^{\tan(7x)}\right]$
22. $\dfrac{d}{dx}\left[\sqrt{\cos\left(1 - x^2\right)}\right]$
23. $\dfrac{d}{dx}\left[\ln^3\left(x^2 + 1\right)\right]$
24. $\dfrac{d}{dx}\left[\ln\left(\sin\left(x^3\right)\right)\right]$
25. $\dfrac{d}{dx}\left[\ln\left(\sec(x)\right)\right]$
26. $\dfrac{d}{dx}\left[\cos\left(x^2\right)\right]$
27. $f(x) = \ln\left(x^2 + 3\right)$, find $f'(x)$
28. $g(x) = \ln\left(x^2 - 4x + 4\right)$, find $g'(x)$
29. $h(x) = \sqrt{x^2 + 5}$, find $h'(x)$
30. $F(x) = \sqrt[3]{3x^2 - 6x + 1}$, find $F'(x)$
31. $y = \sin^{-1}\left(\cos(x)\right)$, find y'
32. $y = \sin\left(\cos^{-1}(x)\right)$, find y'
33. $y = \tan^2\left(3\theta\right)$, find y'
34. $y = \cot^7\left(\sin(\theta)\right)$, find y'

35. $y = \sin^{-1}\left(x\sqrt{2}\right)$, find y'

36. $y = \sin^{-1}(2x+1)$, find y'

1.2 Free Response Homework Set B

Find the derivatives of the given functions. Simplify where possible.

1. $y = \cos^{-1}\left(e^{3z}\right)$

2. $y = \tan^{-1}\left(\sqrt{x^2-1}\right)$

3. $y = \sec^{-1}(4x) + \csc^{-1}(4x)$

4. $f(x) = \ln\left(\tan^{-1}(5x)\right)$

5. $g(w) = \sin^{-1}(5w) + \cos^{-1}(5w)$

6. $f(t) = \sec^{-1}\left(\sqrt{9+t^2}\right)$

Complete the following.

7. $\dfrac{d}{d\theta}\left[e^{\csc(\theta)} + \ln\left(\cot\left(\theta^2\right)\right) - \sec(\theta)\right]$

8. $\dfrac{d}{dx}\left[\ln\left(\sec^3\left(x^3 + 5\ln x + 7\right)\right)\right]$

9. $\dfrac{d}{dx}\left[\ln\left(\tan^3\left(x^2 + 5e^x + 7\right)\right)\right]$

10. $\dfrac{d}{dx}\left[\dfrac{\cos\left(\ln\left(5x^2\right)\right)}{\sin\left(\ln\left(5x^2\right)\right)}\right]$

11. $\dfrac{d}{dx}\left[\ln\left(\sqrt{x^2+4x-5}\right)\right]$

12. $\dfrac{d}{dt}\left[\sin^5\left(\ln(7t+3)\right)\right]$

13. $\dfrac{d}{dx}\left[\csc\left(\ln\left(7x^2+x\right)\right)\right]$

14. $\dfrac{d}{dx}\left[\ln\left(\sqrt{e^{4t^2+6}}\right)\right]$

15. $\dfrac{d}{dx}\left[\sqrt{9x - 27x^2 + \dfrac{5}{x^3}}\right]$

16. $\dfrac{d}{dx}\left[\sec(5x) + \cot\left(e^x\right) - 10\ln x\right]$

17. $z = \ln(\cos(t)) + \sec\left(e^t\right) + 7\pi^2$, find $\dfrac{dz}{dt}$

18. $z = \ln(\tan(t)) + \sin\left(e^t\right) + 7\pi^2$, find $\dfrac{dz}{dt}$

19. $z = \ln(\cot(\theta)) + \sec(\ln\theta) + 7\pi^2$, find $\dfrac{dz}{d\theta}$

20. $z = \ln(\cos(\theta)) + \sin(\ln\theta) + 7\pi^2$, find $\dfrac{dz}{d\theta}$

21. If $g(3) = \dfrac{\pi}{2}$, $g'(3) = \dfrac{\pi}{4}$, and $f(x) = x^3 g(x) + g\left(-3\cos\left(\dfrac{\pi}{3}x\right)\right) - e^{\sin(g(x))}$, find $f'(3)$

1.2 Multiple Choice Homework

1. If $y = \sin^{-1}\left(e^{3\theta}\right)$, then $\dfrac{dy}{d\theta} =$

a) $\dfrac{1}{\sqrt{1-e^{3\theta}}}$ b) $\dfrac{1}{\sqrt{1-e^{6\theta}}}$ c) $\dfrac{1}{\sqrt{1-e^{9\theta^2}}}$

d) $-3e^{3\theta}\cos^{-1}\left(e^{3\theta}\right)$ e) $\dfrac{3e^{3\theta}}{\sqrt{1-e^{6\theta}}}$

---

2. If $f(x) = \tan^{-1}(\cos x)$, then $f'(x) =$

a) $-\csc(x)\sec^{-2}(\cos(x))$ b) $-\sin(x)\sec^{-2}(\cos(x))$ c) $-\cos(x)\csc^{-2}(\cos(x))$

d) $\dfrac{-\cos(x)}{1-\sin^2(x)}$ e) $\dfrac{-\sin(x)}{\cos^2(x)+1}$

---

3. If $h(x) = \ln\left(x^2\right)\tan^{-1}(x)$, then $h'(1) =$

a) $\dfrac{\pi}{4}$ b) $\dfrac{\pi}{4}+1$ c) $\dfrac{\pi}{2}$ d) $\dfrac{\pi}{2}+1$ e) $\dfrac{\pi}{2}+2$

---

4. If $f(t) = t\sqrt{1-t^2}+\cos^{-1}(t)$, then $f'(t) =$

a) $\dfrac{t-2}{2\sqrt{t^2-1}}$ b) $\dfrac{-2t^2}{\sqrt{1-t^2}}$ c) $\dfrac{-2t^2+2}{\sqrt{1-t^2}}$ d) $\dfrac{-1-t^2}{\sqrt{1-t^2}}$ e) $\dfrac{1-t^2}{\sqrt{1-t^2}}$

---

5. If h is the function defined by $h(x) = e^{5x}+x+3$, then $h'(0) =$

a) 2 b) 4 c) 5 d) 6 e) 8

---

6. Given that $f(x) = 8\sin^2(5x)$, find $f'\left(\dfrac{\pi}{30}\right)$

a) $20\sqrt{3}$ b) $20\sqrt{2}$ c) 20 d) 100 e) 0

---

7. If $g(x) = \cos^2(2x)$, then $g'(x)$ is

a) $2\cos(2x)\sin(2x)$ b) $-4\cos(2x)\sin(2x)$ c) $2\cos(2x)$

d) $-2\cos(2x)$ e) $4\cos(2x)$

---

8. If $f(x) = \sin^2(3-x)$, then $f'(0) =$

a) $-2\cos(3)$ b) $-2\sin(3)\cos(3)$ c) $6\cos(3)$

d) $2\sin(3)\cos(3)$ e) $6\sin(3)\cos(3)$

---

9. If $f(x) = \cos^2(3-x)$, then $f'(0) =$

a) $-2\cos(3)$ b) $-2\sin(3)\cos(3)$ c) $6\cos(3)$

d) $2\sin(3)\cos(3)$ e) $6\sin(3)\cos(3)$

---

10. The function $f(x) = \tan(3^x)$ has one zero in the interval $[0, 1.4]$. The derivative at this point is

a) 0.411 b) 1.042 c) 3.451 d) 3.763 e) undefined

1.3: Trig, Trig Inverse, and Log Rules

Remember:

$$\text{The Product Rule: } f'(x) = U \cdot \frac{dV}{dx} + V \cdot \frac{dU}{dx}$$

$$\text{The Quotient Rule: } f'(x) = \frac{V \cdot \frac{dU}{dx} - U \cdot \frac{dV}{dx}}{V^2}$$

OBJECTIVES

Find the Derivative of a Product or Quotient of Two Functions.

The Product Rule

Ex 1.3.1: $\frac{d}{dx}\left[x^2 \sin(x)\right]$

Sol 1.3.1:

$$\frac{d}{dx}\left[x^2 \sin(x)\right] = x^2 \cdot \cos(x) + \sin(x) \cdot (2x)$$

$$= \boxed{x^2 \cos(x) + 2(x)\sin(x)}$$

Ex 1.3.2: $\dfrac{d}{dx}\left[5^x \cos(x)\right]$

Sol 1.3.2:

$$\begin{aligned}\frac{d}{dx}\left[5^x \cos(x)\right] &= 5^x \cdot (-\sin(x)) + \cos(x) \cdot (5^x \ln 5) \\ &= \boxed{5^x\left(\ln(5)\cos(x) + \sin(x)\right)}\end{aligned}$$

The product rule is pretty straightforward. The tricky part is simplifying the algebra.

Ex 1.3.3: If $f(x) = x^2 e^{-\frac{x}{2}}$, find $f'(x)$

Sol 1.3.3:

$$U = x^2,\ \frac{dU}{dx} = 2x$$

$$V = e^{-\frac{x}{2}},\ \frac{dV}{dx} = e^{-\frac{x}{2}} \cdot \left(-\frac{1}{2}\right) = -\frac{1}{2}e^{-\frac{x}{2}}$$

$$\begin{aligned}f'(x) &= x^2 \cdot \left(-\frac{1}{2}e^{-\frac{x}{2}}\right) + e^{-\frac{x}{2}} \cdot 2x \\ &= \boxed{xe^{-\frac{x}{2}}\left(-\frac{1}{2}x + 2\right)}\end{aligned}$$

Ex 1.3.4: $\dfrac{d}{dx}\left[x\sqrt{1-x^2}\right]$

Sol 1.3.4:

$$U = x,\ \frac{dU}{dx} = 1$$

$$V = \sqrt{1-x^2} = \left(1-x^2\right)^{\frac{1}{2}},\ \frac{dV}{dx} = \frac{1}{2}\left(1-x^2\right)^{-\frac{1}{2}} \cdot (-2x) = -\frac{x}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\left[x\sqrt{1-x^2}\right] = x\cdot\left(-\frac{x}{\sqrt{1-x^2}}\right)+\sqrt{1-x^2}\cdot 1$$

$$= \frac{-x^2+(1-x^2)}{\sqrt{1-x^2}}$$

$$= \boxed{\frac{1-2x^2}{\sqrt{1-x^2}}}$$

Ex 1.3.5: $\frac{d}{dx}\left[(2x-3)^8\left(3x^2-1\right)^7\right]$

Sol 1.3.5:

$$U=(2x-3)^8,\ \frac{dU}{dx}=8(2x-3)^7\cdot 2=16(2x-3)^7$$

$$V=\left(3x^2-1\right)^7,\ \frac{dV}{dx}=7\left(3x^2-1\right)^6\cdot 6x=42x\left(3x^2-1\right)^6$$

$$\frac{d}{dx}\left[(2x-3)^8\left(3x^2-1\right)^7\right]=(2x-3)^8\cdot 42x\left(3x^2-1\right)^6+\left(3x^2-1\right)^7\cdot 16(2x-3)^7$$

This, then, is factorable.

$$\frac{d}{dx}\left[(2x-3)^8\left(3x^2-1\right)^7\right] = 42x(2x-3)^8\left(3x^2-1\right)^6+16\left(3x^2-1\right)^7 16(2x-3)^7$$

$$= 2(2x-3)^7\left(3x^2-1\right)^6\left(21x(2x-3)+8\left(3x^2-1\right)\right)$$

$$= 2(2x-3)^7\left(3x^2-1\right)^6\left(42x^2-63x+24x^2-8\right)$$

$$= \boxed{2(2x-3)^7\left(3x^2-1\right)^6\left(66x^2-63x-8\right)}$$

Remember that in Section 1.1 we said that we would need the Product Rule to deal with the derivative of a function where the variable is in both the base and the exponent. We can now address that situation.

Ex 1.3.6: $\frac{d}{dx}\left[(\cos(x))^{x^2}\right]$

Sol 1.3.6:

$$\frac{d}{dx}\left[(\cos(x))^{x^2}\right] = \frac{d}{dx}\left[e^{x^2\ln(\cos(x))}\right]$$

$$= e^{x^2\ln(\cos(x))} \cdot \left(x^2 \cdot \frac{1}{\cos(x)} \cdot -(\sin(x)) + \ln(\cos(x)) \cdot 2x\right)$$

$$= \boxed{(\cos(x))^{x^2}\left(2x\ln(\cos(x)) - x^2\tan(x)\right)}$$

The Quotient Rule

Ex 1.3.7: $\frac{d}{dx}\left[\frac{6x}{x^2+4}\right]$

Sol 1.3.7:

$$U = 6x,\ \frac{dU}{dx} = 6$$

$$V = x^2+4,\ \frac{dV}{dx} = 2x$$

$$\frac{d}{dx}\left[\frac{6x}{x^2+4}\right] = \frac{\left(x^2+4\right)\cdot 6 - 6x \cdot 2x}{\left(x^2+4\right)^2}$$

$$= \frac{6x^2+24-12x^2}{(x^2+4)}$$

$$= \boxed{\frac{24-6x^2}{(x^2+4)}}$$

Ex 1.3.8: $\frac{d}{dx}\left[\frac{x^2+2x-3}{x-4}\right]$

Sol 1.3.8:

$$U = x^2+2x-3,\ \frac{dU}{dx} = 2x+2$$

$$V = x-4,\ \frac{dV}{dx} = 1$$

$$\begin{aligned}
\frac{d}{dx}\left[\frac{x^2+2x-3}{x-4}\right] &= \frac{(x-4)\cdot(2x+2)-(x^2+2x-3)\cdot 1}{(x-4)^2} \\
&= \frac{2x^2-6x-8-x^2-2x+3}{(x-4)^2} \\
&= \boxed{\frac{x^2-8x-5}{(x-4)^2}}
\end{aligned}$$

Ex 1.3.9: $\frac{d}{dx}\left[\frac{x^2-4x+3}{2x^2-5x-3}\right]$

Sol 1.3.9: Notice that this problem becomes much easier if we simplify before applying the Quotient Rule.

$$\begin{aligned}
\frac{d}{dx}\left[\frac{x^2-4x+3}{2x^2-5x-3}\right] &= \frac{d}{dx}\left[\frac{(x-1)(x-3)}{(2x+1)(x-3)}\right] \\
&= \frac{d}{dx}\left[\frac{x-1}{2x+1}\right]
\end{aligned}$$

$$U = x-1,\ \frac{dU}{dx} = 1$$

$$V = 2x+1,\ \frac{dV}{dx} = 2$$

$$\frac{d}{dx}\left[\frac{x-1}{2x+1}\right] = \frac{(2x+1)\cdot 1-(x-1)\cdot 2}{(2x+1)^2}$$

$$= \boxed{\frac{3}{(2x+1)^2}}$$

Ex 1.3.10: $\dfrac{d}{dx}\left[\dfrac{\cot(3x)}{x^2+1}\right]$

Sol 1.3.10:

$$U = \cot(3x),\ \frac{dU}{dx} = -\csc^2(3x)\cdot 3 = -3\csc^2(3x)$$

$$V = x^2+1,\ \frac{dV}{dx} = 2x$$

$$\frac{d}{dx}\left[\frac{\cot(3x)}{x^2+1}\right] = \frac{(x^2+1)\cdot(-3\csc^2(3x)) - \cot(3x)\cdot 2x}{\left(x^2+1\right)^2}$$

$$= \frac{-3x^2\csc^2(3x) - 3\csc^2(3x) - 2x\cot(3x)}{\left(x^2+1\right)^2}$$

$$= \boxed{-\frac{\csc^2(3x)\left(3x^2+3\right) + 2x\cot(3x)}{\left(x^2+1\right)^2}}$$

As with the Product Rule, the difficulty with the Quotient Rule arises from the algebra needed to simplify our answer.

Ex 1.3.11: If $y = \dfrac{4x}{\sqrt{x^2+4}}$, find $\dfrac{dy}{dx}$

Sol 1.3.11:

$$U = 4x,\ \frac{dU}{dx} = 4$$

$$V = \sqrt{x^2+4} = \left(x^2+4\right)^{\frac{1}{2}},\ \frac{dV}{dx} = \frac{1}{2}\left(x^2+4\right)^{-\frac{1}{2}}\cdot 2x = \frac{2x}{2\sqrt{x^2+4}}$$

$$\frac{dy}{dx} = \frac{\sqrt{x^2+4}\cdot 4 - 4x\cdot\dfrac{2x}{2\sqrt{x^2+4}}}{x^2+4}$$

$$= \frac{\dfrac{4\left(x^2+4\right)}{\sqrt{x^2+4}} - \dfrac{4x^2}{\sqrt{x^2+4}}}{x^2+4}$$

$$= \frac{4x^2+16-4x^2}{\left(x^2+4\right)^{\frac{3}{2}}}$$

$$= \boxed{\frac{16}{\left(x^2+4\right)^{\frac{3}{2}}}}$$

Ex 1.3.12: Find the equation of the tangent line to $f(x) = \dfrac{x}{\sqrt{x^2+9}}$ at $x = -\sqrt{7}$.

Sol 1.3.12: As we recall, for the equation of a line, we need a point and a slope.

$$\text{The point: } f\left(-\sqrt{7}\right) = \frac{-\sqrt{7}}{\sqrt{\left(-\sqrt{7}\right)^2+9}}$$

$$= -\frac{\sqrt{7}}{4} \to \left(-\sqrt{7}, -\frac{\sqrt{7}}{4}\right)$$

The slope is the derivative at the given x-value:

$$U = x,\ \frac{dU}{dx} = 1$$

$$V = \sqrt{x^2+9} = \left(x^2+9\right)^{\frac{1}{2}},\ \frac{dV}{dx} = \frac{1}{2}\left(x^2+9\right)^{-\frac{1}{2}}\cdot 2x = \frac{2x}{2\sqrt{x^2+9}}$$

$$\frac{dy}{dx} = \frac{\sqrt{x^2+9}\cdot 1 - x\cdot\dfrac{2x}{2\sqrt{x^2+9}}}{x^2+9}$$

Rather than simplify the algebra, we can find the slope by substituting $x = -\sqrt{7}$:

$$\left.\frac{dy}{dx}\right|_{x=-\sqrt{7}} = \frac{\sqrt{\left(-\sqrt{7}\right)^2+9}\cdot 1 - \left(-\sqrt{7}\right)\cdot\dfrac{2\left(-\sqrt{7}\right)}{2\sqrt{\left(-\sqrt{7}\right)^2+9}}}{\left(-\sqrt{7}\right)^2+9}$$

$$= \frac{4 - \frac{7}{4}}{16}$$

$$= \frac{9}{64}$$

The tangent line equation is therefore:

$$\boxed{y + \frac{\sqrt{7}}{4} = \frac{9}{64}\left(x + \sqrt{7}\right)}$$

1.3 Free Response Homework Set A

Find the derivatives of the given functions.

1. $y = t^3 \cos(t)$

2. $y = (2x-5)^4 \left(8x^2-5\right)^{-3}$

3. $y = \dfrac{\tan(x)-1}{\sec(x)}$

4. $y = \dfrac{\sin(x)}{x^2}$

5. $y = xe^{-x^2}$

6. $y = \dfrac{r}{\sqrt{r^2+1}}$

7. $y = e^{x\cos(x)}$

8. $y = e^{-5x}\cos(3x)$

9. $y = x\sin\left(\dfrac{1}{x}\right)$

10. $y = \ln\left(e^{-x} + xe^{-x}\right)$

11. $y = \dfrac{\sec^{-1}(x)}{x}$

12. $y = \dfrac{\sec(x)}{x^3}$

13. $y = \left(1+x^2\right)\tan^{-1}(x)$

14. $y = \ln\left(x^2+4\right) - x\tan^{-1}\left(\dfrac{x}{2}\right)$

15. $f(x) = x\sqrt{\ln x}$

16. $g(x) = (1+4x)^5\left(3+x-x^2\right)^8$

17. $f(x) = x\cos^{-1}(x) - \sqrt{1-x^2}$

18. $g(x) = \cos^{-1}(x) + x\sqrt{1-x^2}$

Complete the following.

19. $\dfrac{d}{dx}\left[\dfrac{3x^2+4x-3}{x^2-9}\right]$

20. $\dfrac{d}{dx}\left[\dfrac{x^3-2x^2-5x+6}{x+2}\right]$

21. $\dfrac{d}{dx}\left[\dfrac{x^5-12x^3-19x}{3x^3}\right]$

22. $\dfrac{d}{dx}\left[\dfrac{3x+3}{x^3+1}\right]$

23. $\dfrac{d}{dx}\left[\dfrac{x-4}{x^2-9x+20}\right]$

24. $\dfrac{d}{dx}\left[\dfrac{\tan(x)+5}{\sin(x)}\right]$

25. $\dfrac{d}{dx}\left[\dfrac{\sin(x)}{1-\cos(x)}\right]$

26. $\dfrac{d}{dx}\left[\dfrac{x^2}{\cos(x)}\right]$

27. $y = \dfrac{x^2-3}{x^2-4}$, find $\dfrac{dy}{dx}$

28. $f(x) = \dfrac{x^2+2x-8}{x^2-x-3}$, find $f'(x)$

29. $y = \dfrac{x^2+2x-3}{x-4}$, find y'

30. $f(x) = \dfrac{x}{\ln x}$, find $f'(x)$

31. $h(t) = \left(\frac{1+t^2}{1-t^2}\right)^{17}$, find $h'(t)$

32. $y = \frac{\tan(x)}{\cos(x)-3}$, find $\frac{dy}{dx}$

33. $f(x) = \left(x\sin(2x) + \tan^4\left(x^7\right)\right)^5$, find $f'(x)$

34. $f(x) = e^x - x^2 \arctan x$, find $f'(x)$

35. $f(x) = \frac{\tan(x)}{\tan(x)+1}$, find $f'\left(\frac{\pi}{4}\right)$

36. $y = x^2\sqrt{5-x^2}$, find $y'(1)$

1.3 Free Response Homework Set B

Complete the following.

1. Find the first derivative for the following function: $x(t) = e^{t^2}\sin\left(t^2 - 5t^4\right)$

2. Find the first derivative for the following function: $x(t) = e^{5t}\tan\left(3t^4\right)$

3. Find the first derivative for the following function: $y = \frac{x^2+2x-15}{x-3}$

4. Find the first derivative for the following function: $x(t) = e^t\left(t^2 - 5t^4\right)$

5. $\frac{d}{dx}\left[\frac{e^x + 7x^2 + 5}{\sin\left(x^3\right)}\right]$

6. $\frac{d}{dx}\left[e^{\sin(x)}\ln\left(\cot\left(e^x\right)\right)\right]$

7. $\frac{d}{dx}\left[x^2\sin\left(x^2\right) + \frac{x+1}{\ln x}\right]$

8. $\frac{d}{dx}\left[x^2\cos\left(x^2\right) + \frac{e^x}{x}\right]$

9. $\frac{d}{dx}\left[x^5\ln(5x+4) + \frac{x}{\ln x}\right]$

10. $\frac{d}{dx}\left[\frac{\cos\left(x^2-3\right)}{e^{-5x}}\right]$

11. $\frac{d}{dx}\left[e^{x^2}\cos(x)\right]$

12. $\frac{d}{dx}\left[\frac{1+\tan(x)}{\ln(4x)}\right]$

13. $\frac{d}{dx}\left[\sin(t)\tan(t)\right]$

14. $\frac{d}{dx}\left[\frac{1+\ln x}{\csc(x)}\right]$

15. $\frac{d}{dx}\left[e^{5x^4}\ln\left(\sin(x)\right)\right]$

16. $\frac{d}{dx}\left[5x\sin(x) + e^{2x} - \ln\left(3x^2+1\right) + \frac{x}{x^2+1}\right]$

17. $\frac{d}{dx}\tan\left(e^x\right)\left(x^4 - 5x^3 + x\right)$

18. $\frac{d}{dx}\left[\frac{5x+2}{\ln(3x+7)}\right]$

19. $\dfrac{d}{dx}\left[\dfrac{x^5 - 12x^3 - 19x}{3x^3}\right]$

20. $\dfrac{d}{dx}\left[8\cos(4x+2)\sin(4x+2)\right]$

21. $g(z) = \left(\dfrac{e^{5z}}{1+\ln z}\right)^{118}$, find $g'(z)$

22. $g(t) = \left(\dfrac{t^2-4}{1-t^2}\right)^{15}$, find $g'(t)$

23. $y = \tan^{-1}\left(\dfrac{2e^x}{1-e^{2x}}\right)$, find y'

24. $f(x) = x^2 \arccos(x)$, find $f'(x)$

25. $y = \ln\left(u^2+1\right) - u\cot^{-1}(u)$, find $\dfrac{dy}{du}$

26. $y = \cos^{-1}\left(\dfrac{x-1}{x+1}\right)$

27. $f(t) = c\sin^{-1}\left(\dfrac{t}{c}\right) - \sqrt{c^2-t^2}$, find $f'(t)$

28. $y = 4\sin^{-1}\left(\dfrac{1}{2}x\right) + x\sqrt{4-x^2}$

29. If $h(1) = 5$ and $h'(1) = 3$, find $f'(1)$ if $f(x) = (h(x))^4 + x\ln(h(x))$

1.3 Multiple Choice Homework

1. If $y = x^2\cos(2x)$, then $\dfrac{dy}{dx} =$

   a) $-2x\sin(2x)$ b) $-4x\sin(2x)$ c) $2x\left(\cos(2x) - \sin(2x)\right)$

   d) $2x\left(\cos(2x) - x\sin(2x)\right)$ e) $2x\left(\cos(2x) + \sin(2x)\right)$

---

2. If $x(t) = 2t\cos\left(t^2\right)$, find $x'(t)$.

   a) $x'(t) = \sin\left(t^2\right) + 3$ b) $x'(t) = -\sin\left(t^2\right) + 4$ c) $x'(t) = \sin\left(t^2\right) + 2$

   d) $x'(t) = -4t^2\sin\left(t^2\right)$ e) $x'(t) = -4t^2\sin\left(t^2\right) + 2\cos\left(t^2\right)$

---

3. If $f(x) = x\tan(x)$, then $f'\left(\dfrac{\pi}{4}\right) =$

   a) $1 - \dfrac{\pi}{2}$ b) $1 + \dfrac{\pi}{2}$ c) $1 + \dfrac{\pi}{4}$ d) $1 - \dfrac{\pi}{4}$ e) $\dfrac{\pi}{2} - 1$

---

4. If f is a function that is differentiable throughout its domain and is defined by $f(x) =$

$\dfrac{1+e^x}{\sin(x^2)}$, then the value of $f'(0) =$

a) -1 b) 0 c) 1 d) e e) nonexistent

---

5. If $y = \dfrac{5x-4}{4x-5}$, then $\dfrac{dy}{dx} =$

a) $-\dfrac{9}{(4x-5)^2}$ b) $\dfrac{9}{(4x-5)^2}$ c) $\dfrac{40x-41}{(4x-5)^2}$ d) $\dfrac{40x+41}{(4x-5)^2}$ e) $\dfrac{5}{4}$

---

6. If $y = \dfrac{3-2x}{3x+2}$, then $\dfrac{dy}{dx} =$

a) $\dfrac{12x+2}{(3x+2)^2}$ b) $\dfrac{12x-2}{(3x+2)^2}$ c) $\dfrac{13}{(3x+2)^2}$ d) $-\dfrac{13}{(3x+2)^2}$ e) $-\dfrac{2}{3}$

---

7. If $y = \dfrac{3}{4+x^2}$, then $\dfrac{dy}{dx} =$

a) $-\dfrac{6x}{(4+x^2)^2}$ b) $\dfrac{3x}{(4+x^2)^2}$ c) $\dfrac{6x}{(4+x^2)^2}$ d) $-\dfrac{3x}{(4+x^2)^2}$ e) $\dfrac{3}{2x}$

---

8. Which of the following statements must be true?

I. $\dfrac{d}{dx}\left[x\tan(x)\right] = x\tan(x) + x\sec^2(x)$

II. $\dfrac{d}{dx}\left[\dfrac{3}{4+x^2}\right] = \dfrac{-6x}{(4+x^2)^2}$

III. $\dfrac{d}{dx}\left[\sqrt{1-x}\right] = \dfrac{1}{2\sqrt{1-x}}$

a) I only b) II only c) III only

d) I and II only e) I, II, and III

1.4: Higher Order Derivatives

What we've been calling the derivative is actually the first derivative. There can be successive uses of the derivative rules, and they have meanings other than the slope of the tangent line. In this section, we will explore the process of finding the higher-order derivatives.

Second Derivative $\rightarrow$ Definition: The derivative of the derivative.

Just as with the first derivative, there are several symbols for the second derivative.

Higher Order Derivative Symbols

$$\frac{d^2y}{dx^2} = \text{“d squared y, d - x squared”} \rightarrow \frac{d^3y}{dx^3}, \frac{d^4y}{dx^4} \dots \frac{d^ny}{dx^n}$$

$$\frac{d^2}{dx^2} = \text{“d squared, d - x squared”} \rightarrow \frac{d^3}{dx^3}, \frac{d^4}{dx^4} \dots \frac{d^n}{dx^n}$$

$$f''(x) = \text{“f double prime of x”} \rightarrow f'''(x),\ f^{IV}(x) \dots f^n(x)$$

$$y'' = \text{“y double prime”}$$

OBJECTIVES

Find Higher Order Derivatives.

Ex 1.4.1: $\dfrac{d^2}{dx^2}\left[x^4 - 7x^3 - 3x^2 + 2x - 5\right]$

Sol 1.4.1:

$$\begin{aligned}\frac{d^2}{dx^2}\left[x^4 - 7x^3 - 3x^2 + 2x - 5\right] &= \frac{d}{dx}\left[\frac{d}{dx}\left[x^4 - 7x^3 - 3x^2 + 2x - 5\right]\right] \\ &= \frac{d}{dx}\left[4x^3 - 21x^2 - 6x + 2\right] \\ &= \boxed{12x^2 - 42x - 6}\end{aligned}$$

Ex 1.4.2: Find $\dfrac{d^3y}{dx^3}$ if $y = \sin(3x)$.

Sol 1.4.2:

$$y = \sin(3x)$$

$$\frac{dy}{dx} = \cos(3x) \cdot 3 = 3\cos(3x)$$

$$\frac{d^2y}{dx^2} = 3(-\sin(3x)) \cdot 3 = -9\sin(3x)$$

$$\frac{d^3y}{dx^3} = -9\cos(3x) \cdot 3 = \boxed{-27\cos(3x)}$$

More complicated functions, in particular composite functions, have a more complicated process. When the Chain Rule is applied, the result often includes a product or a quotient. Therefore, the second derivative will require the Product or Quotient Rules, as well as possibly the Chain Rule.

Ex 1.4.3: $y = e^{3x^2}$, find $\frac{d^y}{dx^2}$.

Sol 1.4.3:

$$\frac{dy}{dx} = e^{3x^2} \cdot 6x = 6xe^{3x^2}$$

$$\frac{d^2y}{dx^2} = 6x\left(e^{3x^2} \cdot 6x\right) + e^{3x^2} \cdot 6$$

$$= 36x^2e^{3x^2} + 6e^{3x^2}$$

$$= \boxed{6e^{3x^2}\left(6x^2+1\right)}$$

Ex 1.4.4: $y = \sin^3(x)$, find y''.

Sol 1.4.4:

$$y' = 3\sin^2(x) \cdot \cos(x)$$

$$y'' = 3\sin^2(x)(-\sin(x)) + \cos(x)(6\sin(x)\cdot\cos(x))$$

$$= \boxed{3\sin(x)\left(2\cos^2(x) - \sin^2(x)\right)}$$

Ex 1.4.5: $f(x) = \ln\left(x^2 + 3x - 1\right)$, find $f''(x)$.

Sol 1.4.5:

$$f'(x) = \frac{1}{x^2+3x-1}(2x+3) = \frac{2x+3}{x^2+3x-1}$$

$$f''(x) = \frac{(x^2+3x-1)(2) - (2x+3)(2x+3)}{(x^2+3x-1)^2}$$

$$= \frac{(2x^2+6x-2) - (4x^2+12x+9)}{(x^2+3x-1)^2}$$

$$= \boxed{-\frac{2x^2-6x-11}{(x^2+3x-1)^2}}$$

Ex 1.4.6: $g(x) = \sqrt{4x^2+1}$, find $g''(x)$

Sol 1.4.6:

$$g'(x) = \frac{1}{2}\left(4x^2+1\right)^{-\frac{1}{2}}(8x) = \frac{4x}{(4x^2+1)^{\frac{1}{2}}}$$

$$
\begin{aligned}
g''(x) &= \frac{\left(4x^2+1\right)^{\frac{1}{2}}(4) - (4x)\left[\frac{1}{2}\left(4x^2+1\right)^{-\frac{1}{2}}(8x)\right]}{\left[\left(4x^2+1\right)^{\frac{1}{2}}\right]^2} \\
&= \frac{\left(4x^2+1\right)^{\frac{1}{2}}(4) - \dfrac{16x^2}{\left(4x^2+1\right)^{\frac{1}{2}}}}{\left(4x^2+1\right)} \\
&= \frac{\left(4x^2+1\right)(4) - 16x^2}{\left(4x^2+1\right)^{\frac{3}{2}}} \\
&= \boxed{\frac{4}{\left(4x^2+1\right)^{\frac{3}{2}}}}
\end{aligned}
$$

1.4 Free Response Homework

Find the second derivatives of the given functions. Simplify where possible.

1. $f(x) = x^5 + 6x^2 - 7x$
2. $h(x) = 5x^4 + 9x^3 - 4x^2 + x - 8$
3. $y = \left(x^3 + 1\right)^{\frac{2}{3}}$
4. $H(t) = \tan(3t)$
5. $g(t) = t^3 e^{5t}$
6. $y = e^{3x^2}$
7. $y = \sin^4(x)$
8. $f(t) = t\cos(t)$
9. $y = -\dfrac{4x}{x^2 + 4}$
10. $y = \dfrac{x^2 - 1}{x^2 - 4}$
11. $f(x) = x\sqrt{8 - x^2}$
12. $y = \dfrac{1}{2}x + \sin(x)$
13. $g(t) = te^{-t}$
14. $y = e^{-x^2}$
15. $y = \dfrac{x}{x^2 - 9}$
16. $B(x) = 2x - x^{\frac{2}{3}}$
17. $y = x^3 + x^2 - 7x + 15$
19. $y = 3x^4 - 20x^3 + 42x^2 - 36x + 16$

Complete the following:

21. $y = \cos\left(x^2\right)$, find y''
22. $y = \tan^2(x)$, find y''
23. $y = \sec(3x)$, find $\dfrac{d^2y}{dx^2}$
24. $y = xe^{2x}$, find $\dfrac{d^2y}{dx^2}$
25. $f(x) = \ln\left(x^2 + 3\right)$, find $f''(x)$
25. $g(x) = \ln\left(x^2 - 4x + 4\right)$, find $g''(x)$
27. $h(x) = \sqrt{x^2 + 5}$, find $h''(x)$
28. $F(x) = \sqrt{3x^2 - 2x + 1}$, find $F''(x)$
29. $y = \dfrac{x^2 - 3}{x^2 - 10}$, find $\dfrac{d^2y}{dx^2}$
30. $y = \dfrac{3x + 3}{x^3 + 1}$, find $\dfrac{d^2y}{dx^2}$

1.4 Multiple Choice Homework

1. If f and g are twice differentiable and if $h(x) = g(f(x))$, then $h''(x) =$

   a) $g''(f(x))$ b) $g''(f(x))\,f''(x)$ c) $g''(f(x))\,[f'(x)]^2$

d) $g'\left(f(x)\right)\left[f'(x)\right]^2 + f'(x)\left(f''(x)\right)$ e) $g'\left(f(x)\right)f''(x) + \left[f'(x)\right]^2 g''\left(f(x)\right)$

---

2. Find $\dfrac{d^2y}{dx^2}$ if $y = \dfrac{x+2}{x-3}$

a) $-\dfrac{2}{(x-3)^2}$ b) 0 c) $\dfrac{10}{(x-3)^3}$ d) $\dfrac{2}{(x-3)^2}$ e) None of these

---

3. If $y = \ln(\cos(x))$ and $0 \le x \le \pi$, then $\dfrac{d^2y}{dx^2}$ is

a) $-\tan(x)$ b) $-\sec^2(x)$ c) $\tan(x)$ d) $\sec^2(x)$ e) $\sec(x)\tan(x)$

---

4. If $y = \ln\left(x^2+4\right)$, then $\dfrac{d^2y}{dx^2}$ is

a) $\dfrac{1}{x^2+4}$ b) $\dfrac{2x}{x^2+4}$ c) $\dfrac{-2x^2+8}{x^2+4}$ d) $\dfrac{2x}{\left(x^2+4\right)^2}$ e) $\dfrac{-2x^2+8}{\left(x^2+4\right)^2}$

---

5. If $y = e^{x^2}$, then $\dfrac{d^2y}{dx^2} =$

a) e^{x^2} b) $2e^{x^2}\left(2x^2+1\right)$ c) $2xe^{x^2}$ d) $4x^2e^{x^2}$ e) $2e^{x^2}\left(2x^2-1\right)$

---

6. If $h(t) = \ln\left(t^2+1\right)$, then $h''(-1) =$

a) $\ln 2$ b) 0 c) -1 d) -2 e) DNE

---

7. If $y = \sin\left(e^x\right)$, then $\dfrac{d^2y}{dx^2} =$

a) $\cos\left(e^x\right)$ b) $e^x\cos\left(e^x\right)$ c) $e^x\sin\left(e^x\right) + e^x\cos\left(e^x\right)$

d) $-e^x\sin\left(e^x\right) + e^x\cos\left(e^x\right)$ e) $-e^x\left(e^x\sin\left(e^x\right) - \cos\left(e^x\right)\right)$

1.5: Implicit Differentiation and Second Derivative Applications

Implicit differentiation is a technique that is considered here because of its direct impact on related rates in section 8 of this chapter. Implicit differentiation is an application of the Chain Rule where the y-function is not easily defined explicitly.

One of the most useful aspects of the Chain Rule is that we can take derivatives of more complicated equations that would be difficult to take the derivative of otherwise. One of the key elements to remember is that we already know the derivative of y with respect to x — that is, $\dfrac{dy}{dx}$. This can be a powerful tool as it allows us to take the derivative of relations as well as functions while bypassing a lot of tedious algebra. When y cannot easily be isolated, we can treat y like we treat $g(x)$. In other words:

$$\frac{d}{dx}\left[f\left(g(x)\right)\right] = f'\left(g(x)\right) \cdot \left[g'(x)\right] \text{ is the same as } \frac{d}{dx}\left[f(y)\right] = f'(y) \cdot \left(\frac{dy}{dx}\right)$$

OBJECTIVES

Take Derivatives of Relations Implicitly.
Use Implicit Differentiation to Find Higher Order Derivatives.
Use the Second Derivative Test to Determine Whether a Point is at a Maximum, Minimum, or Neither.

Ex 1.5.1: Find $\dfrac{dy}{dx}$ if $x^2 + y^2 = 25$

Sol 1.5.1:

$$\frac{d}{dx}\left[x^2 + y^2 = 25\right]$$

$$\frac{d}{dx}\left[x^2\right] + \frac{d}{dx}\left[y^2\right] = \frac{d}{dx}[25]$$

$$\rightarrow 2x + 2y\frac{dy}{dx} = 0$$

We can now isolate $\dfrac{dy}{dx}$

$$2y\frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = \boxed{-\frac{x}{y}}$$

With this function, notice that y could have been isolated and $\frac{dy}{dx}$ could've been found **explicitly**.

$$x^2 + y^2 = 25$$

$$y^2 = 25 - x^2$$

$$y = \sqrt{25 - x^2}$$

$$\frac{dy}{dx} = -\frac{x}{\sqrt{25 - x^2}}$$

Notice that this is the same answer as we found with implicit differentiation. You could substitute y for $\sqrt{25 - x^2}$ in the denominator and come up with the same derivative, $\frac{dy}{dx} = -\frac{x}{y}$.

Ex 1.5.2: Find the derivative of $x^2 - 3y^2 + 4x - 12y - 2 = 0$ implicitly.

Sol 1.5.2:

$$\frac{d}{dx}\left[x^2 - 3y^2 + 4x - 12y - 2 = 0\right]$$

$$2x - 6y\frac{dy}{dx} + 4 - 12\frac{dy}{dx} = 0$$

$$(-6y - 12)\frac{dy}{dx} = -2x - 4$$

$$\frac{dy}{dx} = \boxed{\frac{-2x - 4}{-6y - 12}}$$

When considering functions, implicit differentiation may not seem to be a particularly powerful tool, because it is often simple to isolate y. But consider a non-function, like this circle, ellipse, or hyperbola, where y is not so easily isolated.

Ex 1.5.3: Find $\frac{dy}{dx}$ for the hyperbola $x^2 - 3xy + 3y^2 = 2$

Sol 1.5.3: It would be very difficult to solve for y here, so implicit differentiation is really our only option.

$$\frac{d}{dx}\left[x^2 - 3xy + 3y^2 = 2\right]$$

Note that $-3xy$ is a product. It will require the Product Rule.

$$2x - 3x\frac{dy}{dx} - 3y + 6y\frac{dy}{dx} = 0$$

$$(-3x + 6y)\frac{dy}{dx} = -2x + 3y$$

$$\frac{dy}{dx} = \boxed{\frac{-2x + 3y}{-3x + 6y}}$$

Ex 1.5.4: Find the equation of the line tangent to $x^3 - y^2 + 6y = -3$ at $y = 1$

a) $3x^2 - 2y = 6$

b) $3x - y = -7$

c) $3x + y = -5$

d) $x + 3y = 1$

e) $x - 3y = -5$

Sol 1.5.4: First, let's find the point of tangency.

$$x^3 - y^2 + 6y = -3 \to x^3 - (1)^2 + 6(1) = -3$$

$$x^3 = -8 \therefore x = -2$$

$$\frac{d}{dx}\left[x^3 - y^2 + 6y = -3\right]$$

$$3x^2 - 2y\frac{dy}{dx} + 6\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-3x^2}{-2y + 6}$$

Now, let's plug in our point that we found, $(-2, 1)$

$$\frac{dy}{dx} = \frac{-3(-2)^2}{-2(1) + 6}$$

$$\frac{dy}{dx} = -3$$

$$y - 1 = -3(x + 2)$$

$$y - 1 = -3x - 6$$

$$\rightarrow \boxed{\text{c) } 3x + y = -5}$$

Of course, if we want to find a second derivative, we can use implicit differentiation a second time.

Ex 1.5.5: Given $\dfrac{dy}{dx} = \dfrac{x+2}{3y+6}$, find $\dfrac{d^2y}{dx^2}$

Sol 1.5.5:

$$\frac{d}{dx}\left[\frac{dy}{dx} = \frac{x+2}{3y+6}\right]$$

$$\frac{d^2y}{dx^2} = \frac{(3y+6) - (x+2)\left(3\dfrac{dy}{dx}\right)}{(3y+6)^2}$$

Since we already know $\dfrac{dy}{dx}$, we can substitute

$$\begin{aligned}
\frac{d^2y}{dx^2} &= \frac{(3y+6) - (x+2)(3)\left(\dfrac{x+2}{3y+6}\right)}{(3y+6)^2} \\
&= \frac{(3y+6) - (x+2)(3)\left(\dfrac{x+2}{3y+6}\right)}{(3y+6)^2} \cdot \frac{3y+6}{3y+6} \\
&= \boxed{\frac{(3y+6)^2 - 3(x+2)^2}{(3y+6)^3}}
\end{aligned}$$

Ex 1.5.6: Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ for $\sin(y) = 2\cos(3x)$.

Sol 1.5.6:

$$\frac{d}{dx}\left[\sin(y) = 2\cos(3x)\right]$$

$$\cos(y)\frac{dy}{dx} = -6\sin(3x)$$

$$\frac{dy}{dx} = \boxed{-\frac{6\sin(3x)}{\cos(y)}}$$

$$\frac{d^2y}{dx^2} = \frac{-18\cos(y)\cos(3x) - 6\sin(3x)\sin(y)\dfrac{dy}{dx}}{\cos^2(y)}$$

$$= \frac{-18\cos(y)\cos(3x) - 6\sin(3x)\sin(y)\left(-\dfrac{6\sin(3x)}{\cos(y)}\right)}{\cos^2(y)}$$

$$= \boxed{\frac{-18\cos(y)\cos(3x) + 36\sin^2(3x)\sin(y)}{\cos^3(y)}}$$

AP-Style Implicit Differentiation Problems

Common Sub-topics:

- Demonstrating implicit differentiation
- Finding the equation of a tangent line
- Finding points where the tangent line is horizontal and/or vertical
- Finding points on a curve with a particular slope
- Finding the particular solution (this will have to wait for the next chapter)

OBJECTIVES

Take Derivatives of Relations Implicitly.
Use Implicit Differentiation to Find Higher Order Derivatives.
Use Separation of Variables to Find the Particular Solution to a Differential Equation.

Ex 1.5.7: Consider the curve given by $x^2 + 4xy + y^2 = -12$.

(a) Show that $\dfrac{dy}{dx} = -\dfrac{x+2y}{2x+y}$

(b) Find the point(s) where the equation of the tangent line(s) is/are horizontal.

(c) Find the value(s) of $\dfrac{d^2y}{dx^2}$ at the point(s) found in part (b). Does the curve have a local maximum, a local minimum, or neither at those points? Justify your answer.

Sol 1.5.7:
a) Because we have a $4xy$ term, we need to use the product rule

$$\frac{d}{dx}\left[x^2 + 4xy + y^2 = -12\right]$$

$$2x + 4x\frac{dy}{dx} + 4y(1) + 2y\frac{dy}{dx} = 0$$

$$4x\frac{dy}{dx} + 2y\frac{dy}{dx} = -2x - 4y$$

$$(4x+2y)\frac{dy}{dx} = -2x - 4y$$

$$\frac{dy}{dx} = \frac{-2x-4y}{4x+2y} = \boxed{-\frac{x+2y}{2x+y}}$$

b) Horizontal lines have a slope of 0, so we need to find when $\dfrac{dy}{dx} = 0$.

$$\frac{dy}{dx} = 0$$

$$-\frac{x+2y}{2x+y} = 0$$

$$x + 2y = 0$$

$$x = -2y$$

To be on the curve, $x = -2y$ must satisfy the original equation.

$$(-2y)^2 + 4(-2y)y + y^2 = -12$$

$$4y^2 - 8y^2 + y^2 = -12$$

$$-3y^2 = -12$$

$$y^2 = 4 \therefore y = \pm 2$$

$$x = -2y \rightarrow \boxed{(-4, 2) \text{ and } (4, -2)}$$

c) This part requires a technique which we will revisit in Chapter 4.

Be Careful!! There is a lot of algebraic simplification that happens in these problems, and it is easy to make mistakes. Take your time with the simplifications so that you don't make careless mistakes.

1.5 Free Response Homework

Find $\dfrac{dy}{dx}$ for each of these equations, first by implicit differentiation, then by solving for y and differentiating. Show that $\dfrac{dy}{dx}$ is the same in both cases.

1. $x^2 + y^2 = 1$
2. $x^3 + 4y^2 = 16$
3. $\dfrac{1}{x} + \dfrac{1}{y} = 1$
4. $\sqrt{x} + \sqrt{y} = 4$

Find $\dfrac{dy}{dx}$ for each of these equations by implicit differentiation.

5. $x^2 + xy = 10$
6. $x^3 + 10x^2y + 7y^2 = 60$
7. $x^2 + xy - 4y - 1 = 0$
8. $xy + 2x + 3x^2 = 4$
9. $x^2 + 4xy - 5y^2 = 4$
10. $3x^2 + xy - 4y^2 = 5$
11. $x^2 = \dfrac{x-y}{x+y}$
12. $x^2 + y^2 = \dfrac{x}{y}$
13. $y^2 = \dfrac{x-y}{x+y}$
14. $y^2 = \dfrac{x^2-1}{x+2}$
15. $x^2y^2 + x\sin(y) = 4$
16. $4\cos(x)\sin(y) = 1$
17. $e^{x^2y} = x + y$
18. $\tan(x-y) = \dfrac{y}{1+x^2}$
19. Find the equation of the line tangent to $x^2 - y^2 - 6y - 3 = 0$ at $\left(\sqrt{3}, 0\right)$.
20. Find the equation of the line tangent to $9x^2 + 4y^2 + 36x - 8y - 32 = 0$ at $(0, 2)$.
21. Find the equation of the line tangent to $12x^2 - 4y^2 + 72x + 16y + 44 = 0$ at $(-1, -3)$.
22. Find the equation of the line tangent to $x^3 + \dfrac{y}{x} + y^2 = 7$ at $(1, 2)$.
23. Find the equation of the lines tangent and normal to $y - \dfrac{4}{\pi^2}x^2 = 2e^{y\sin(x)} + y^3 - 3$ through the point $\left(\dfrac{\pi}{2}, 0\right)$.

24. Find the equation of the lines tangent and normal to $x^2 + 3xy + y^2 = 11$ through the point $(1, 2)$.

25. Find $\dfrac{d^2y}{dx^2}$ if $xy + y^2 = 1$.

26. Find $\dfrac{d^2y}{dx^2}$ if $4x^2 + 9y^2 = 36$.

27. Find $\dfrac{d^2y}{dx^2}$ if $x^2 + y^2 = 1$

28. Find $\dfrac{d^2y}{dx^2}$ if $x^3 + 4y^2 = 16$.

29. Consider the curve given by $3x^2 - 4xy + 5y^2 = 25$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{3x - 2y}{2x - 5y}$.

(b) Determine point(s) P on the curve for which the x-coordinate is equal to 2.

(c) Find the equation(s) of the line(s) tangent to $3x^2 - 4xy + 5y^2 = 25$ at the point(s) P found in part (b).

(d) Find the point(s) on $3x^2 - 4xy + 5y^2 = 25$ where the tangent line is horizontal.

30. Consider the curve given by $x^2 - xy + y^2 = 4$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{y - 2x}{2y - x}$.

(b) Determine point(s) P on the curve for which the x-coordinate is equal to 2.

(c) Find the equation(s) of the line(s) tangent to $x^2 - xy + y^2 = 4$ at the point(s) P found in part (b).

(d) Find the point(s) on $x^2 - xy + y^2 = 4$ where the tangent line is vertical.

31. Consider the curve given by $2x^2 - xy + y^2 = 44$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{4x - y}{x - 2y}$.

(b) Determine point(s) P on the curve for which the x-coordinate is equal to 5.

(c) Find the equation(s) of the line(s) tangent to $2x^2 - xy + y^2 = 44$ at the point(s) P found in part (b).

(d) Find the point(s) on $x^2 - xy + y^2 = 4$ where the tangent line is vertical.

32. Consider the curve given by $x^2 + xy + y^2 = 12$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{-y-2x}{2y+x}$.

(b) Find the point(s) P on $x^2 + xy + y^2 = 12$ where the tangent line is horizontal.

33. Consider the curve given by $xy + y^3 = 4x$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{4-y}{3y^2 = x}$.

(b) Show that there are no points on the curve where the tangent line is horizontal.

(c) Find the point(s) on $xy + y^3 = 4x$ where the tangent line is vertical.

34. Consider the curve given by $x^2 + xy + y^2 = 4$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{-2x-y}{x+2y}$.

(b) Find the point(s) on $x^2 + xy + y^2 = 4$ where the tangent line is horizontal.

(c) Find the y-coordinates of the point(s) where the tangent line is vertical.

1.5 Multiple Choice Homework

1. Use implicit differentiation to find the points on $x^3 - y^2 + x^2 = 0$ where the tangent line is vertical.

a) $(0, 0)$ only b) $(-1, 0)$ only c) $\left(1, \sqrt{2}\right)$ only

d) $(-1, 0)$ and $(0, 0)$ e) No such points exist

---

2. If $x^2 + xy = 10$, then when $x = 2$, $\dfrac{dy}{dx} =$

a) $-\dfrac{7}{2}$ b) -2 c) $\dfrac{2}{7}$ d) $\dfrac{3}{2}$ e) $\dfrac{7}{2}$

---

3. What is the slope of the line tangent to the curve $y^2 + x = -2xy - 5$ at the point $(2, 1)$.

a) $-\dfrac{4}{3}$ b) $-\dfrac{3}{4}$ c) $-\dfrac{1}{2}$ d) $-\dfrac{1}{4}$ e) 0

4. Given $3x^3 - 4xy - 4y^2 = 1$, determine the change in y with respect to x.

a) $\dfrac{6x-4y}{4x+4}$ b) $\dfrac{9x^2-4}{4x+8y}$ c) $\dfrac{9x^2-4}{4+8y}$ d) $\dfrac{9x^2-4y}{4x+8y}$ e) $\dfrac{9x^2-4y}{4+8y}$

---

5. Given $x + xy + 2y^2 = 6$, then $\left.\dfrac{dy}{dx}\right|_{(2,1)} =$

a) $\dfrac{2}{3}$ b) $\dfrac{1}{3}$ c) $-\dfrac{1}{3}$ d) $-\dfrac{1}{5}$ e) $-\dfrac{3}{4}$

---

6. Consider the closed curve in the xy-plane given by $2x^2 + 5x + y^2 + y = 8$. Which of the following is correct?

a) $\dfrac{dy}{dx} = -\dfrac{4x+5}{8x+2y+1}$ b) $\dfrac{dy}{dx} = \dfrac{4x+5}{2y+1}$ c) $\dfrac{dy}{dx} = -\dfrac{4x+5}{8x+2y}$

d) $\dfrac{dy}{dx} = \dfrac{4x+5}{8x+2y}$ e) $\dfrac{dy}{dx} = \dfrac{4x+5}{2y+1}$

---

7. The slope of the line tangent to $xy - y^3 + 6 = 0$ at $(1, 2)$ is

a) 0 b) $-\dfrac{1}{12}$ c) $\dfrac{2}{11}$ d) $\dfrac{1}{6}$ e) $\dfrac{1}{4}$

---

8. Find the equation of the line tangent to the curve $\sec\left(x^2\right) + xy^3 = 2 - y$ at $x = 0$.

a) $y = -x$ b) $y - 1 = -x$ c) $y - 2 = -x$ d) $y - 1 = x$ e) $y - 2 = x$

---

9. If $\sin^{-1}(x) = \ln y$, then $\dfrac{dy}{dx} =$

a) $\dfrac{y}{\sqrt{1-x^2}}$ b) $\dfrac{xy}{\sqrt{1-x^2}}$ c) $\dfrac{y}{1+x^2}$ d) $e^{\sin^{-1}(x)}$ e) $\dfrac{e^{\sin^{-1}(x)}}{1+x^2}$

---

10. If $x^2y + yx^2 = 6$, then at $(1, 3)$, $\dfrac{d^2y}{dx^2} =$

a) -18 b) -6 c) 6 d) 12 e) 18

---

11. If $y = x + \sin(xy)$, then $\dfrac{dy}{dx} =$

a) $1 + \cos(xy)$ b) $1 + y\cos(xy)$ c) $\dfrac{1}{1 - \cos(xy)}$

d) $\dfrac{1}{1 - x\cos(xy)}$ e) $\dfrac{1 + y\cos(xy)}{1 - \cos(xy)}$

---

12. If $\sin(xy) = x^2$, then $\dfrac{dy}{dx} =$

a) $2x\sec(xy)$ b) $\dfrac{\sec(xy)}{x^2}$ c) $2x\sec(xy) - y$

d) $\dfrac{2x\sec(xy)}{y}$ e) $\dfrac{2x\sec(xy) - y}{x}$

---

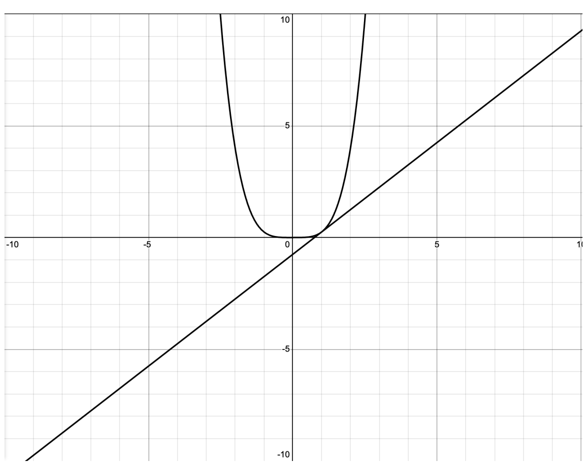
13. Given $y = \ln\left(x^2 + y^2\right)$, find $\dfrac{dy}{dx}$ at the point $(1, 0)$.

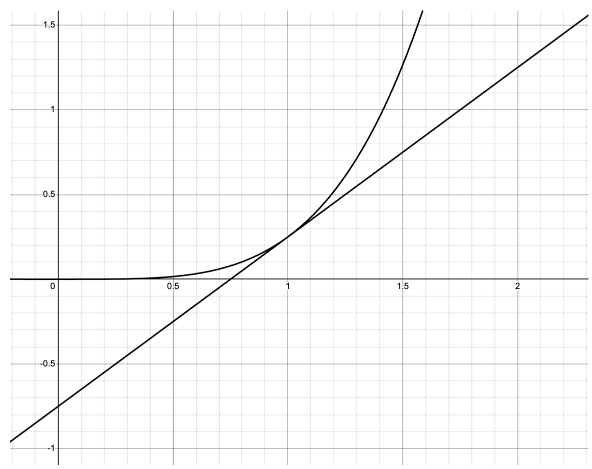
a) 0 b) 0.5 c) 1 d) 2 e) undefined

1.6: Local Linearity, Euler's Method, and Approximations

Before calculators, one of the most valuable uses of the derivative was to find approximate function values from a tangent line. Since the tangent line only shares one point on the function, y values on the line are very close to y values on the function. This idea is called local linearity—near the point of tangency, the function curve appears to be a line. This can be easily demonstrated with the graphing calculator by zooming in on the point of tangency.

Consider the graphs of $y = \dfrac{1}{4}x^4$ and its tangent line at $x = 1$, given by the equation $y = x - \dfrac{3}{4}$:





The closer you zoom in, the more the line and the curve become one. The y values on the line are good approximations of the y values on the curve. For a good animation of this concept, see the following:

tangent line approximation animation

Since it's easier to find the y value of a line arithmetically than for other functions — especially transcendental functions — the tangent line approximation is useful if you have no calculator.

OBJECTIVES

Use the Equation of a Tangent Line to Approximate Function Values.

Ex 1.6.1: Find the equations of the line tangent to $f(x) = x^4 - x^3 - 2x^2 + 1$ at $x = -1$.

Sol 1.6.1: The slope of the tangent line will be $f'(-1)$

$$f'(x) = 4x^3 - 3x^2 - 4x$$

$$f'(-1) = 4(-1)^3 - 3(-1)^2 - 4(-1) = -3$$

(Note that we could've gotten this more easily with the nDeriv function on our calculator.)

$$f(-1) = 1, \therefore \boxed{y - 1 = -3(x + 1)} \text{ or } \boxed{y = -3x - 2}$$

One of the many uses of the tangent line is based on the idea of local linearity. This means that in small areas, algebraic curves act like lines — namely their tangent lines. Therefore, one can get an approximate y value for points near the point of tangency by plugging x values into the equation of the tangent line.

Ex 1.6.2: Use the tangent line equation found in **Ex 1.6.1** to get an approximate value of $f(-0.9)$.

Sol 1.6.2: While we can find the exact value of $f(-0.9)$ with a calculator, we can get a quick approximation from the tangent line.

If $x = -0.9$ on the tangent line, then:

$$f(-0.9) \approx y(-0.9) = -3(-0.9) - 2 = \boxed{0.7}$$

This last example was somewhat trite in that we could've just plugged -0.9 into $f(x) = x^4 - x^3 - 2x^2 + 1$ and figured out the exact value even without a calculator. It would have been a pain, but it is doable. Consider the next example, though.

Ex 1.6.3: Find the tangent line equation to $f(x) = e^{2x}$ at $x = 0$ and use it to approximate the value of $e^{0.2}$.

Sol 1.6.3: Without a calculator, we could not find the exact value of $e^{0.2}$. In fact, even the calculator gives us an approximate value.

$$f'(x) = 2e^{2x} \text{ and } f'(0) = 2e^{2(0)} = 2$$

$$f(0) = e^0 = 1$$

So the tangent line equation is $\boxed{y - 1 = 2(x - 0)}$ or $y = \boxed{2x + 1}$

$$e^{0.2} \approx 2(0.2) + 1 = \boxed{1.2}$$

Note that the value you get from a calculator of $e^{0.2}$ is $1.221403\ldots$. Our approximation of 1.2 seems very reasonable.

Though not as useful as practically useful (in 2 dimensions) as the tangent line, another context for the derivative is finding the equation of the normal line.

Normal Line $\rightarrow$ Definition: The line perpendicular to a curve.

Ex 1.6.4: Find the equation of the line normal to $f(x) = x^4 - x^3 - 2x^2 + 1$ at $x = -1$.

Sol 1.6.4: In **Ex 1.6.1**, we saw that the slope of the tangent line was $f'(-1) = -3$. The normal line is perpendicular to the tangent line and, therefore, has the negative reciprocal slope of $\frac{1}{3}$. This gives us

$$\boxed{y - 1 = \frac{1}{3}(x + 1)} \text{ or } \boxed{y = \frac{1}{3}x - \frac{4}{3}}$$

for the equation of the normal line.

Euler's Method

OBJECTIVES

Use Euler's Method to Approximate a Numerical Solution to a Differential Equation at a Given Point.

In the previous section, we learned a little regarding approximations with tangent lines. Euler's Method is just a better approximation method. It uses more than one tangent line to get the job done.

The process is similar to approximating with tangent lines. We use $\frac{dy}{dx}$ to find a tangent line, then use that tangent line to find an approximate value for y. We then use that y value and another x value to create another "tangent line." Of course, it isn't actually a tangent line because our y value wasn't actually on the curve. We then repeat the process until we get to the value we want to approximate.

Steps to Euler's Method

1. Identify your starting point and step size.
2. Use $\frac{dy}{dx}$ to find the slope and make it a tangent line.
3. Find an approximate y value by plugging in $x + (\textit{1 step size})$ to the tangent line.
4. Use the approximate y value and the next x step over to make a new tangent line.
5. Repeat steps 3 and 4 until you reach your final x value – the one you actually want an approximation for.

Ex 1.6.5: Use Euler's Method with a step size of $\frac{1}{2}$ to estimate $f(3)$, where $f(x) = \ln(x)$.

Sol 1.6.5:

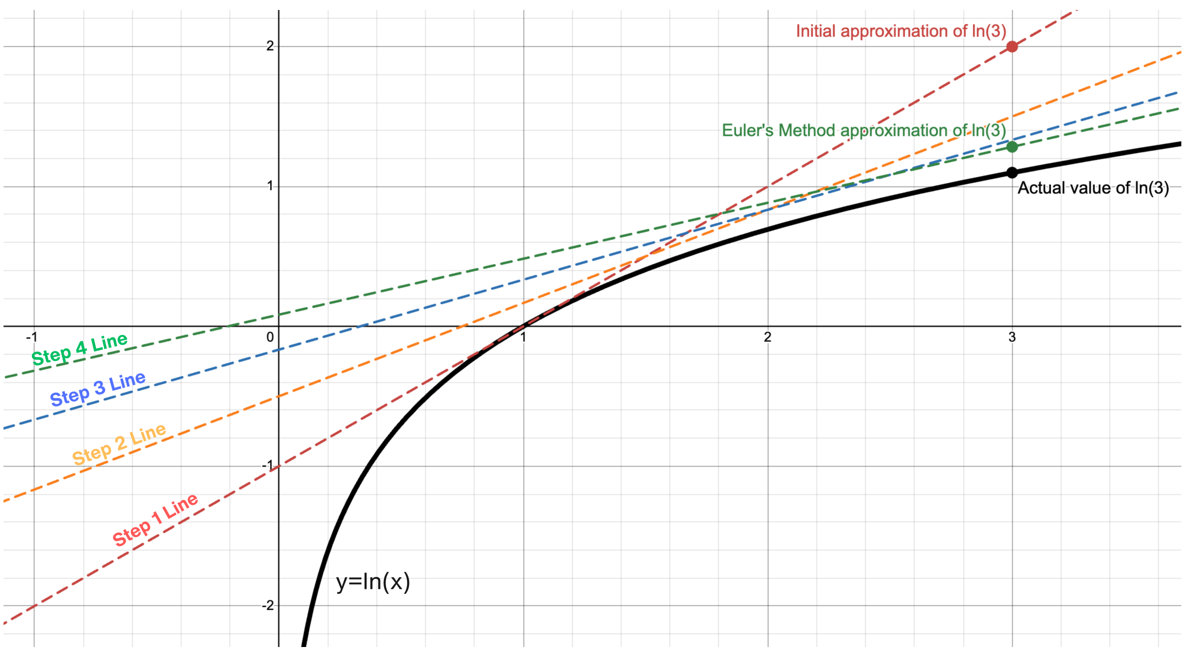
$f(1) = \ln 1 = 0$	We start with 1 because we know $\ln 1 = 0$.
$f'(x) = \frac{1}{x}$	We start by taking the derivative.

Note that in our chart below, we are getting our "New y" from our tangent line (step 3 above). Our "New x" comes from the "$x + (\textit{1 step size})$" step. Our new slope comes from plugging in the "New x" into $f'(x)$. For instance, for our first step below, the "New x" is equal to 1 plus the step size of $\frac{1}{2}$ given in the problem, our "New y" comes from plugging in $\frac{3}{2}$ for x and solving for y in $y - 0 = 1(x - 1)$, and our slope (for the next step) is a result of $f'\left(\frac{3}{2}\right) = \frac{1}{\frac{3}{2}} = \frac{2}{3}$.

Step	Point	$f'(x)$ (Slope)	Tangent Line Equation	New x	New y
1	$(1, 0)$	1	$y - 0 = 1(x - 1)$	$\frac{3}{2}$	$\frac{1}{2}$
2	$\left(\frac{3}{2}, \frac{1}{2}\right)$	$\frac{2}{3}$	$y - \frac{1}{2} = \frac{2}{3}\left(x - \frac{3}{2}\right)$	2	$\frac{5}{6}$
3	$\left(2, \frac{5}{6}\right)$	$\frac{1}{2}$	$y - \frac{5}{6} = \frac{1}{2}(x - 2)$	$\frac{5}{2}$	$\frac{13}{12}$
4	$\left(\frac{5}{2}, \frac{13}{12}\right)$	$\frac{2}{5}$	$y - \frac{13}{12} = \frac{2}{5}\left(x - \frac{5}{2}\right)$	3	$\frac{77}{60}$

So, $f(3) \approx y(3) = \boxed{\frac{77}{60}}$ or $\boxed{1.28\overline{3}}$. By way of comparison, $\ln 3 \approx 1.099$.

If we had just used the initial tangent line (the tangent line from step one) to get an approximation, we would've gotten $f(3) \approx 2$. Euler's method got us a much closer approximation.



We could also use this process to approximate a value for a curve when we only know its derivative and an initial value on the curve.

Ex 1.6.6: Use Euler's method with a step size of $\frac{1}{2}$ to estimate $f\left(\frac{5}{2}\right)$ for the function whose derivative is given by $\frac{dy}{dx} = 2x + y$ with an initial value of $f(1) = 4$.

Sol 1.6.6: For this problem, to find the slope for each step, we simply we need to plug in our point into the given differential equation.

Step	**Point**	$\frac{dy}{dx}$ **(Slope)**	**Tangent Line Equation**	**New** x	**New** y
1	$(1, 4)$	6	$y - 4 = 6(x - 1)$	$\frac{3}{2}$	7
2	$\left(\frac{3}{2}, 7\right)$	10	$y - 7 = 10\left(x - \frac{3}{2}\right)$	2	12
3	$(2, 12)$	16	$y - 12 = 16(x - 2)$	$\frac{5}{2}$	20

$f\left(\frac{5}{2}\right) \approx y\left(\frac{5}{2}\right) = \boxed{20}$. We cannot get an exact value for this function, because we have not learned techniques regarding how to solve this differential equation yet.

Ex 1.6.7: Is the approximation in **Ex 1.6.6** an overestimate or underestimate? Why?

Sol 1.6.7: To determine this, we need to look at the concavity of the curve – this requires the second derivative.

$$\frac{dy}{dx} = 2x + y$$

$$\frac{d^2y}{dx^2} = 2 + \frac{dy}{dx} \qquad \text{Don't forget implicit differentiation!}$$

$$\frac{d^2y}{dx^2} = 2 + (2x + y)$$

$$\left.\frac{d^2y}{dx^2}\right|_{(1,4)} = 2 + 2(1) + 4 = 8 \qquad \text{Plug in our initial value}$$

Since the second derivative is positive, our curve is concave up at this point, which means our tangent line lies under the curve. This is indicative of an $\boxed{\text{underestimate}}$.

Generally:

→ Your approximation will be an **overestimate** if the curve is **concave down** (since your "tangent lines" will be above the curve).

→ Your approximation will be an **underestimate** if the curve is **concave up** (since your "tangent lines" will be below the curve).

Ex 1.6.8: Use Euler's method with four equal step sizes to approximate $f(2)$ for $\frac{dy}{dx} = 3y - x$, given $f(0) = 1$. Is this an overestimate or an underestimate?

Sol 1.6.8: First, let's figure out our step sizes. We know that we start with $x = 0$, and we need to end up at $x = 2$. Therefore, four equal step sizes means that each step must be $+\frac{1}{2}$.

Next, let's make our table:

Step	**Point**	$\frac{dy}{dx}$ **(Slope)**	**Tangent Line Equation**	**New** x	**New** y
1	$(0, 1)$	3	$y - 1 = 3(x - 0)$	$\frac{1}{2}$	$\frac{5}{2}$
2	$\left(\frac{1}{2}, \frac{5}{2}\right)$	7	$y - \frac{5}{2} = 7\left(x - \frac{1}{2}\right)$	1	6
3	$(1, 6)$	17	$y - 6 = 17(x - 1)$	$\frac{3}{2}$	$\frac{29}{2}$
4	$\left(\frac{3}{2}, \frac{29}{2}\right)$	42	$y - \frac{29}{2} = 42\left(x - \frac{3}{2}\right)$	2	$\frac{71}{2}$

We can see that $f(2) \approx y(2) = \boxed{\frac{71}{2}}$. Now, let's solve for the second derivative to find out if this is an overestimate or underestimate.

$$\frac{d^2y}{dx^2} = 3\frac{dy}{dx} - 1 = 9y - 3x - 1$$

$$\left.\frac{d^2y}{dx^2}\right|_{(0,1)} = 9(1) - 3(0) - 1 = 8$$

Because the positive second derivative indicates that y is concave up at $(0, 1)$, our Euler's Method result is going to be an $\boxed{\text{underestimate}}$.

1.6 Free Response Homework

Complete the following:

1. Find the equation of the line tangent to $y = x^4 + 2e^x$ at the point $(0, 2)$.

2. Find the equation of the line tangent to $y = x + \cos(x)$ at the point $(0, 1)$.

3. Find the equation of the line tangent to $y = \sec(x) - 2\cos(x)$ at the point $\left(\frac{\pi}{3}, 0\right)$.

4. Find the equation of the line tangent to $y = x^2 e^{-x}$ at the point $\left(1, \frac{1}{e}\right)$.

5. Find the equation of the line tangent to $y = \frac{2}{\pi}x + \cos(4x)$ when $x = \frac{\pi}{2}$.

6. Find the equation of the line tangent to $y = \frac{x^2 - 3}{x^2 - 4}$ when $x = -1$.

7. Find the equation of the line tangent to $f(x) = x\sqrt[4]{7 + x^2}$ when $x = 3$.

8. Find the equation of the line tangent to $y = e^{x\sin(4x)} + 2$ when $x = 0$.

9. Find the equation of the line tangent to $f(x) = x^5 - 5x + 1$ when $x = -2$ and use it to get an approximate value of $f(-1.9)$.

10. Find the equation of the line tangent to $f(x) = x\sqrt[3]{1 - x^2}$ when $x = 3$ and use it to get an approximate value of $f(3.1)$.

11. Find all points on the graph of $y = 2\sin(x) + \sin^2(x)$ where the tangent line is horizontal.

12. Find all points on the graph of $y = 2x^3 + 3x^2 - 12x + 1$ where the tangent line is horizontal.

13. Find the equation of the lines tangent and normal to $y = -\frac{2x}{x^2 + 16}$ at $x = -1$.

14. Find the equation of the lines tangent and normal to $y = -\frac{3x}{x^2 + 1}$ at $x = 1$.

15. Find the equation of the lines tangent and normal to $y = \dfrac{x^2 - 4x + 3}{2x^2 - 5x - 3}$ at $x = 2$.

16. Find the equation of the lines tangent and normal to $y = x \sin\left(\dfrac{\pi}{2} \ln x\right)$ when $x = e$.

17. Find the equation of the lines tangent and normal to $y = x \sin\left(\dfrac{1}{x}\right)$ when $x = \dfrac{4}{\pi}$.

18. Use Euler's Method with 2 equal step sizes to find an approximation for $f(0)$, given that $f(-1) = 2$ and $\dfrac{dy}{dx} = 6x^2 - x^2 y$.

19. Use Euler's Method with 4 equal step sizes to find an approximation for $f(1.4)$, given that $f(1) = 0$ and $f(x) = \ln(2x - 1)$.

20. Use Euler's Method with 3 equal step sizes to find an approximation for $f(2.6)$, given that $f(2) = -2$ and $\dfrac{dy}{dx} = 2x + y$.

1.6 Multiple Choice Homework

1. Let f be the function given by $f(x) = 2e^{4x^2}$. For what value of x is the slope of the line tangent to the graph of f at $(x, f(x))$ equal to 3?

a) 0.168 b) 0.274 c) 0.318 d) 0.342 e) 0.551

---

2. Which of the following is an equation of the line tangent to the graph of $f(x) = x^6 + x^5 + x^2$ at the point where $f'(x) = -1$?

a) $-3x - 2$ b) $-3x + 4$ c) $-x + 0.905$

d) $-x + 0.271$ e) $-x - 0.271$

---

3. At what point on the graph of $y = \dfrac{1}{2}x^2$ is the tangent line parallel to the line $2x - 4y = 3$?

a) $\left(\dfrac{1}{2}, -\dfrac{1}{2}\right)$ b) $\left(\dfrac{1}{2}, -\dfrac{1}{8}\right)$ c) $\left(\dfrac{1}{2}, -\dfrac{1}{4}\right)$ d) $\left(1, -\dfrac{1}{2}\right)$ e) $(2, 2)$

---

4. A normal line to the graph of a function f at the point $(x, f(x))$ is defined to be the line perpendicular to the tangent line at that point. An equation of the normal line to the curve $y = \sqrt[3]{x^2 - 1}$ at the point where $x = 3$ is

a) $y + 12x = 38$ b) $y - 4x = 10$ c) $y + 2x = 4$ d) $y + 2x = 8$ e) $y - 2x = -4$

---

5. Let $y = f(x)$ be the solution to the differential equation $\frac{dy}{dx} = \frac{4x}{y}$ with the initial condition $f(0) = 1$. What is the best approximation for $f(1)$ using Euler's Method, starting at $x = 0$ with a step size of 0.5?

a) 1 b) 2 c) $\sqrt{5}$ d) 2.5 e) 3

---

6. Let $y = f(x)$ be the solution to the differential equation $\frac{dy}{dx} = x - y^2$ with the initial condition $f(0) = 1$. What is the best approximation for $f(2)$ using Euler's Method, starting at $x = 0$ with a step size of 1?

a) -1 b) 0 c) 1 d) 2 e) 3

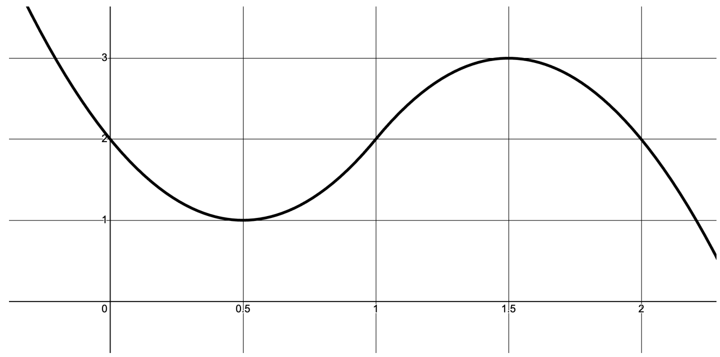
---

7. Let $y = f(x)$ be the solution to the differential equation $\frac{dy}{dx} = y - x$ with the initial condition $f(1) = 2$. What is the best approximation for $f(2)$ using Euler's Method, starting at $x = 1$ with a step size of 0.5?

a) 1 b) 2 c) 3 d) 4.5 e) 6

---

8. The graph of $y = f'(x)$ is given below. Use this information and the fact that $f(0) = 3$ to find an approximate value of $f(1)$ using Euler's method with 2 equal step sizes.



a) 2.5 b) 3.5 c) 4 d) 4.5 e) 5

---

9. The table below gives selected values for the derivative of a function g on the interval $-1 \le x \le 2$. If $g(-1) = -2$ and Euler's Method with a step size of 1.5 is used to approximate $g(2)$, what is the resulting approximation?

x	-1.0	-0.5	0	0.5	1.0	1.5	2.0
$f'(x)$	2	4	3	1	0	-3	-6

a) -6.5 b) -1.5 c) 1.5 d) 2.5 e) 3

---

10. The equation of the line **normal** to the graph of $y = \dfrac{3x+4}{4x-3}$ at $(1, 7)$ is

a) $25x + y = 32$ b) $25x - y = 18$ c) $7x - y = 0$

d) $x - 25y = -174$ e) $x + 25y = 176$

---

11. The equation of the line **normal** to the graph of $y = 3x\sqrt{x^2+6} - 3$ at $(0, -3)$ is

a) $3\sqrt{6}x + y = -3$ b) $3\sqrt{6}x - y = -3$ c) $x + 3\sqrt{6}y = -3$

d) $x - 3\sqrt{6}y = 9\sqrt{6}$ e) $x + 3\sqrt{6}y = -9\sqrt{6}$

1.7: Intro to AP: Basic Derivatives Numerically and Graphically

Traditionally, calculus was an algebraically heavy subject. One of the philosophical changes that the CollegeBoard made in the 1990s was to emphasize that calculus should be understood in a variety of modes. As they state in their enduring understanding:

> *"Students should be able to work with functions represented in a variety of ways: graphical, numerical, analytical or verbal. They should understand the connections among these representations."*

Later, they added that students should be able to verbalize their understanding and be able to communicate that understanding through proper writing. We will consider this later as we consider more context-oriented problems.

OBJECTIVES

Determine Derivative Values from Numerical or Graphical Data.

Ex 1.7.1: Assume $h(x) = f(x)g(x)$. Given the table of values below, find $h'(2)$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	2	4	6
2	1	8	5	7
3	7	2	7	9

Sol 1.7.1:

$$h(x) = f(x)g(x)$$

$$h'(x) = f(x)g'(x) + g(x)f'(x)$$

$$h'(2) = f(2)g'(2) + g(2)f'(2)$$

$$= 1 \cdot 7 + 8 \cdot 5$$

$$= \boxed{47}$$

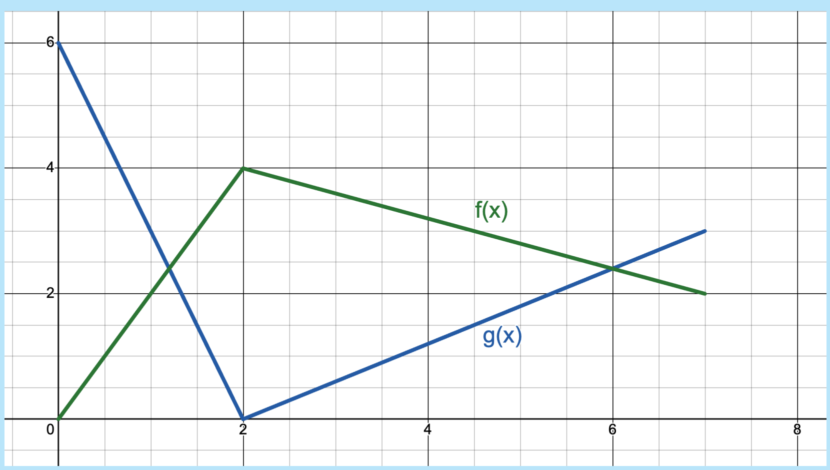
Ex 1.7.2: Using the table of values in **Ex 1.7.1:**, find $\dfrac{d}{dx}[f(g(x))]$ and $\dfrac{d}{dx}[g(f(x))]$ at $x = 1$.

Sol 1.7.2: These are two different but similar problems, so let's consider them individually.

$$\begin{aligned}\frac{d}{dx}\left[f(g(x))\right]\Big|_{x=1} &= f'(g(1)) \cdot g'(1) \\ &= f'(2) \cdot 6 \\ &= 5 \cdot 6 \\ &= \boxed{30}\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}\left[g(f(x))\right]\Big|_{x=1} &= g'(f(1)) \cdot f'(1) \\ &= g'(3) \cdot 4 \\ &= 9 \cdot 4 \\ &= \boxed{36}\end{aligned}$$

Ex 1.7.3: Given the graph below, find (a) $w'(1)$ if $w = \dfrac{g(x)}{f(x)}$ and (b) $v'(1)$ if $v = g(f(x))$.



Sol 1.7.3:

(a)

$$w = \frac{g(x)}{f(x)} \therefore w' = \frac{f(x)g'(x) - g(x)f'(x)}{[f(x)]^2}$$

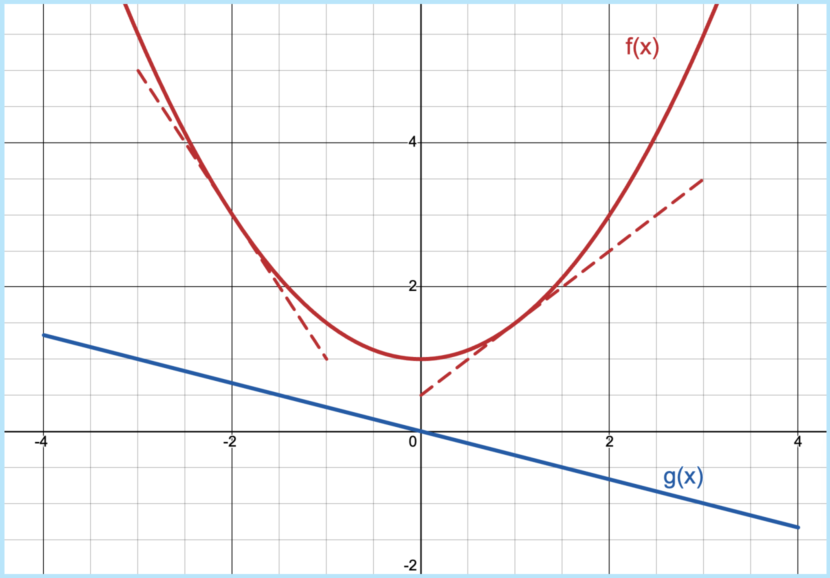
$$\begin{aligned} w'(1) &= \frac{f(1)g'(1) - g(1)f'(1)}{[f(1)]^2} \\ &= \frac{2 \cdot (-3) - 3 \cdot 2}{2^2} \\ &= \boxed{-3} \end{aligned}$$

(b)

$$\begin{aligned} v(x) &= g(f(x)) \\ v'(x) &= g'(f(x)) \cdot f'(x) \\ v'(1) &= g'(f(1)) \cdot f'(1) \\ &= g'(2) \cdot 2 \\ &= \boxed{DNE} \end{aligned}$$

(Note that $g'(2)$ does not exist. The slope cannot be determined at $x = 1$ because the slopes to the left and right of $x = 1$ are different. The is called a corner point, or a cusp point, and will be explored further in a later chapter.)

Ex 1.7.4: The figure below shows the graph of the functions f and g. The graphs of the lines tangent to the graph of f at $x = -2$ and $x = 1$ are also shown. If $B(x) = f(x) \cdot g(x)$, what is $B'(1)$?

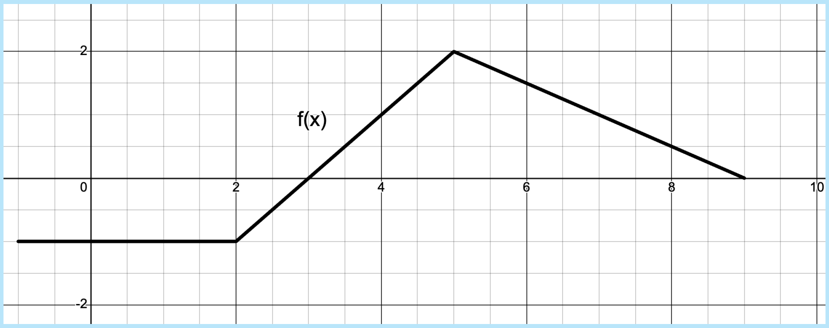


a) $-\frac{5}{6}$ b) $-\frac{1}{2}$ c) $\frac{1}{6}$ d) $\frac{1}{3}$ e) $\frac{1}{2}$

Sol 1.7.4:

$$
\begin{aligned}
B(x) &= f(x) \cdot g(x) \\
B'(x) &= f(x)g'(x) + g(x)f'(x) \\
B'(1) &= f(1)g'(1) + g(1)f'(1) \\
&= \frac{3}{2} \cdot \left(-\frac{1}{3}\right) + \left(-\frac{2}{3}\right) \cdot 1 \\
&= \boxed{\text{a) } -\frac{5}{6}}
\end{aligned}
$$

Ex 1.7.5: Let $f(x)$ be the function whose graph is given below and let $g(x)$ be a differentiable function with selected values for $g(x)$ and $g'(x)$ given in the table below. Furthermore, let h be the function defined by $h(x) = \ln\left(x^2 + 4\right)$.



x	$g(x)$	$g'(x)$
0	-1	1
2	1	3
4	3	6
6	6	12
8	4	8

(a) Find the equation of the line tangent to $f(x)$ at $x = 4$.

(b) Let K be the function defined by $K(x) = h(f(x))$. Find $K'(3)$.

(c) Let M be the function defined by $M(x) = g(x) \cdot f(x)$. Find $M'(6)$.

(d) Let J be the function defined by $J(x) = \dfrac{g(x)}{h\left(\frac{1}{2}x\right)}$. Find $J'(8)$.

Sol 1.7.5:

(a) $f(4) = 1$ and $f'(4) = 1$. Therefore, the tangent line equation is

$$\boxed{y - 1 = 1(x - 4)}$$

(b)

$$h(x) = \ln\left(x^2 + 4\right)$$

$$h'(x) = \frac{2x}{x^2 + 4}$$

$$K(x) = h(f(x)) \therefore K = h'(f(x)) \cdot f'(x)$$

$$\begin{aligned} K'(3) &= h'(f(3)) \cdot f'(3) \\ &= h'(0) \cdot 1 \\ &= 0 \cdot 1 \\ &= \boxed{0} \end{aligned}$$

(c)

$$M(x) = g(x) \cdot f(x)$$

$$M'(x) = g(x) \cdot f'(x) + f(x) \cdot g'(x)$$

$$\begin{aligned} M'(6) &= g(6) \cdot f'(6) + f(6) \cdot g'(6) \\ &= 6 \cdot \frac{1}{2} + 12 \cdot \frac{3}{2} \\ &= \boxed{15} \end{aligned}$$

(d)

$$J(x) = \frac{g(x)}{h\left(\frac{1}{2}x\right)}$$

$$J'(x) = \frac{h\left(\frac{1}{2}x\right) \cdot g'(x) - g(x) \cdot h'\left(\frac{1}{2}x\right) \cdot \frac{1}{2}}{\left[h\left(\frac{1}{2}x\right)\right]^2}$$

$$\begin{aligned} J'(8) &= \frac{h(4) \cdot g'(8) - g(8) \cdot h'(4) \cdot \frac{1}{2}}{[h(4)]^2} \\ &= \boxed{\frac{8\ln 8 - \frac{4}{5}}{\ln^2 8}} \end{aligned}$$

1.7 Free Response Homework

1. Given the following table of values, find the indicated derivatives.

x	$f(x)$	$f'(x)$
2	1	7
8	5	-3

a) $g'(2)$, where $g(x) = [f(x)]^3$

b) $h'2$, where $h(x) = f(x^3)$

2. The following table shows some values of $g(x)$, $g'(x)$, and $h(x)$, where $h(x) = g^{-1}(x)$.

x	$g(x)$	$h(x)$	$g'(x)$	$h'(x)$
1	2	3	$\frac{1}{2}$	$\frac{1}{3}$
3	1	2	-2	$\frac{1}{2}$

a) Find $g'(1)$

b) Find $h'(1)$

For problems 3 – 14, refer to the values in the table below.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	4	-2	8	1
4	2	8	4	3
8	8	-12	2	4

3. If $h(x) = f(g(x))$, find $h'(8)$

4. If $h(x) = f(g(x))$, find $h'(2)$

5. If $k(x) = g(f(x))$, find $k'(2)$

6. If $m(x) = f(f(x))$, find $m'(4)$

7. If $P_1(x) = f(x)g(x)$, find $P_1'(2)$

8. If $P_1(x) = f(x)g(x)$, find $P_1'(8)$

9. If $P_2(x) = f(2x)g(x)$, find $P_2'(2)$

10. If $P_3(x) = f(x)g\left(\frac{1}{2}x\right)$, find $P_3'(4)$

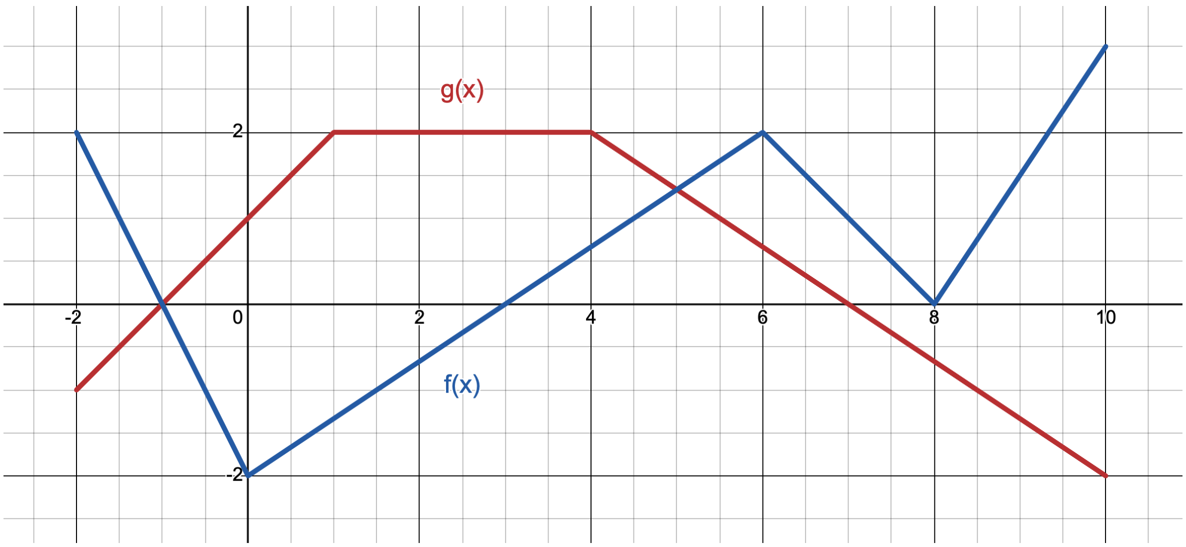
11. If $Q_1(x) = \dfrac{f(x)}{g(x)}$, find $Q_1'(2)$

12. If $Q_2(x) = \dfrac{g(x)}{f(x)}$, find $Q_2'(8)$

13. If $Q_3(x) = \dfrac{f(2x)}{g(x)}$, find $Q_3'(4)$

14. If $Q_4(x) = \dfrac{g\left(\frac{1}{2}x\right)}{f(2x)}$, find $Q_4'(4)$

For problems 15 – 26, the graphs of $f(x)$ and $g(x)$ are given below.



15. If $u = f(g(x))$, find $u'(2)$

16. $v = g(f(x))$, find $v'(4)$

17. If $w = g(g(x))$, find $w'(6)$

18. If $t = f(f(x))$, find $t'(8)$

19. If $P_1(x) = f(x)g(x)$, find $P_1'(2)$

20. If $P_1(x) = f(x)g(x)$, find $P_1'(8)$

21. If $P_2(x) = f(2x)g(x)$, find $P_2'(2)$

10. If $P_3(x) = f(x)g\left(\frac{1}{2}x\right)$, find $P_3'(2)$

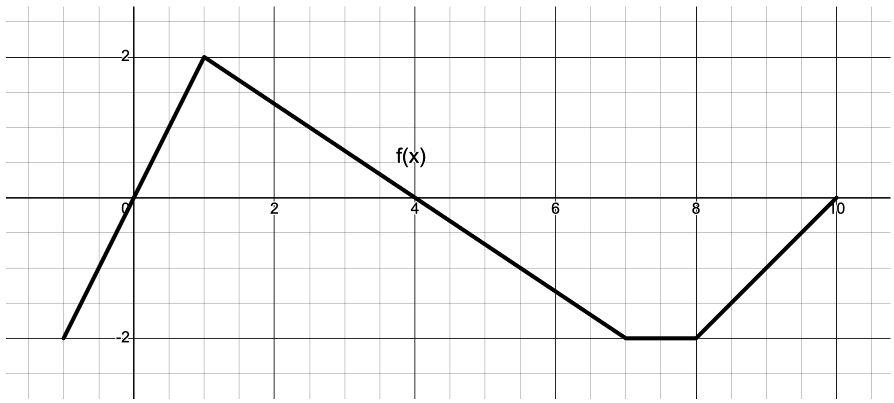
23. If $Q_1(x) = \dfrac{f(x)}{g(x)}$, find $Q_1'(2)$

24. If $Q_2(x) = \dfrac{g(x)}{f(x)}$, find $Q_2'(8)$

25. If $Q_3(x) = \dfrac{f(2x)}{g(x)}$, find $Q_3'(4)$

26. If $Q_4(x) = \dfrac{g\left(\frac{1}{2}x\right)}{f(2x)}$, find $Q_4'(4)$

27. Let $f(x)$ be the function whose graph is given below, and let $g(x)$ be a differentiable function with selected values for $g(x)$ and $g'(x)$ given in the table below.



x	$g(x)$	$g'(x)$
0	-1	1
2	1	3
4	3	6
6	6	12

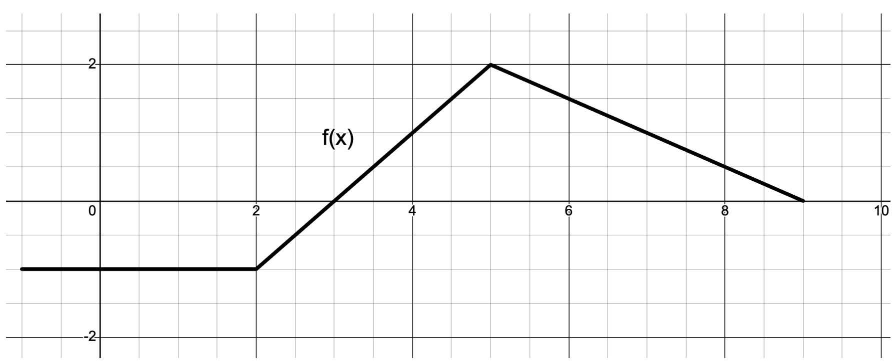
(a) Find the equation of the line tangent to $f(x)$ at $x = 4$.

(b) Let K be the function defined by $K(x) = g(f(x))$. Find $K'(1)$.

(c) Let M be the function defined by $M(x) = g(x) \cdot f(x)$. Find $M'(4)$.

(d) Let J be the function defined by $J(x) = \dfrac{f(2x)}{g(x)}$. Find $J'(2)$.

28. Let $f(x)$ be the function whose graph is given below, and let $g(x)$ be a differentiable function with selected values for $g(x)$ and $g'(x)$ given in the table below.



x	$g(x)$	$g'(x)$
0	-1	1
2	1	3
4	3	6
6	6	12
8	4	8

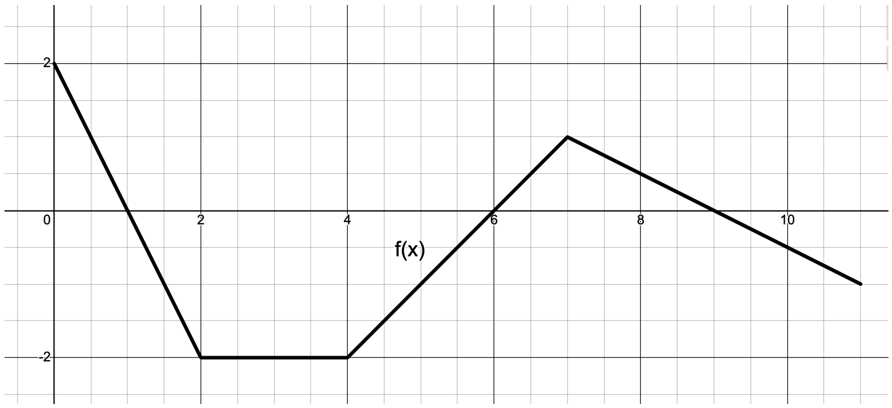
(a) Find the equation of the line tangent to $g(x)$ at $x = 4$.

(b) Let K be the function defined by $K(x) = g(g(x))$. Find $K'(8)$.

(c) Let M be the function defined by $M(x) = g(x) \cdot f(x)$. Find $M'(4)$.

(d) Let J be the function defined by $J(x) = \dfrac{g(2x)}{f(x)}$. Find $J'(1)$.

29. Let $f(x)$ be the function defined by $f(x) = 4x - x^3$, let $h(x)$ be the function whose graph is given below, and let $g(x)$ be a differentiable function with selected values of $g(x)$ and $g'(x)$ given in the table below.



x	$g(x)$	$g'(x)$
0	-1	1
2	1	3
4	3	6
6	6	12
8	4	8

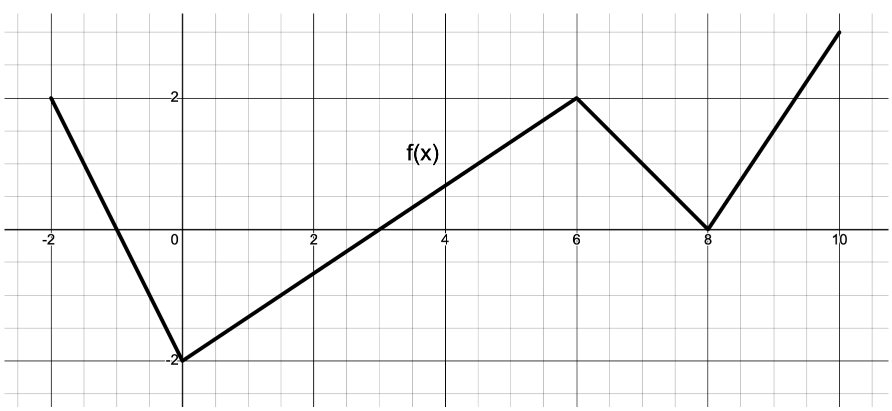
(a) Find the equation of the line tangent to $g(x)$ at $x = 4$.

(b) Let K be the function defined by $K(x) = h(f(x))$. Find $K'(1)$.

(c) Let M be the function defined by $M(x) = g(x) \cdot f(x)$. Find $M'(6)$.

(d) Let J be the function defined by $J(x) = \dfrac{g(x)}{f(x)}$. Find $J'(4)$.

30. Let h be the function defined by $h(x) = \sin(x) + e^{\cos(3x)}$, let f(x) be the function whose graph is given below, and let g(x) be a differentiable function with selected values of $g(x)$ and $g'(x)$ given in the table below.



x	$g(x)$	$g'(x)$
-4	3	2
-2	5	-1
0	7	0
2	5	-1
4	3	2

(a) Find the equation of the line tangent to $h(x)$ at $x = \dfrac{\pi}{2}$.

(b) Let K be the function defined by $K(x) = f(h(x))$. Find $K'\left(\dfrac{\pi}{2}\right)$.

(c) Let M be the function defined by $M(x) = f(x) \cdot g(x)$. Find $M'(0)$.

(d) Let J be the function defined by $J(x) = g(2x) \cdot f(x)$. Find $J'(2)$.

31. 2017 AP Calculus AB #6

1.7 Multiple Choice Homework

1. Let the function f be differentiable on the interval $[0, 2.5]$ and g be defined by $g(x) = f(f(x))$. Use the table to find $g'(1.5)$.

x	0	0.5	1	1.5	2	2.5
$f(x)$	0.5	1.5	2	2.5	1	0
$f'(x)$	0.1	0.3	0.6	1.1	2	2.2

a) 0 b) 1.24 c) 1.65 d) 2.08 e) 2.42

---

2. Given the functions $f(x)$ and $g(x)$ that are both continuous and differentiable, and that have values given in the table below, find $h'(2)$, where $h(x) = g(x) \cdot f(x)$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	4	-2	8	1
4	10	8	4	3
8	6	-12	2	4

a) -12 b) -2 c) 0 d) 30 e) 64

---

3. Let $f(x)$ and $g(x)$ be differentiable functions. The table below gives the values of $f(x)$ and $g(x)$, and their derivatives, at several values of x.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	2	4	-6
2	1	8	-5	7
3	7	-2	7	9

If $h(x) = \dfrac{f(x)}{g(x)}$, what is the value of $h'(2)$?

a) -4 b) -63 c) 51 d) $-\dfrac{47}{64}$ e) $-\dfrac{33}{64}$

---

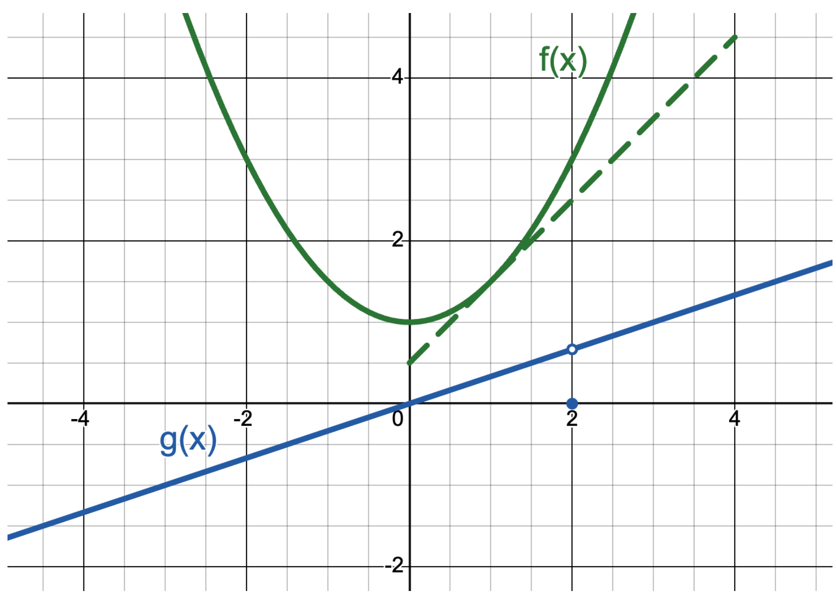
4. Let $h(x) = g(x) \cdot f\left(x^3\right)$. According to the table below, what is the value of $h'(2)$?

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-3	0	9	10
2	4	6	-4	1
4	9	2	3	3
8	-1	1	2	5

a) -6 b) 2 c) 11 d) 24 e) 143

---

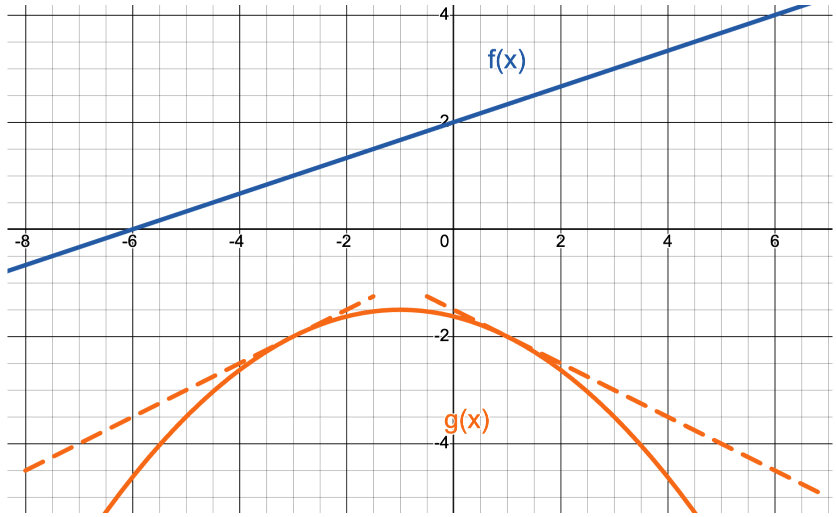
5. The figure below shows the graph of the functions f and g. The graph of the line tangent to the graph of f at $x = 1$ is also shown. If $B(x) = f(x) \cdot g(x)$, what is $B'(1)$?



a) $\dfrac{5}{6}$ b) $-\dfrac{1}{2}$ c) $-\dfrac{1}{6}$ d) $\dfrac{1}{3}$ e) $\dfrac{7}{6}$

---

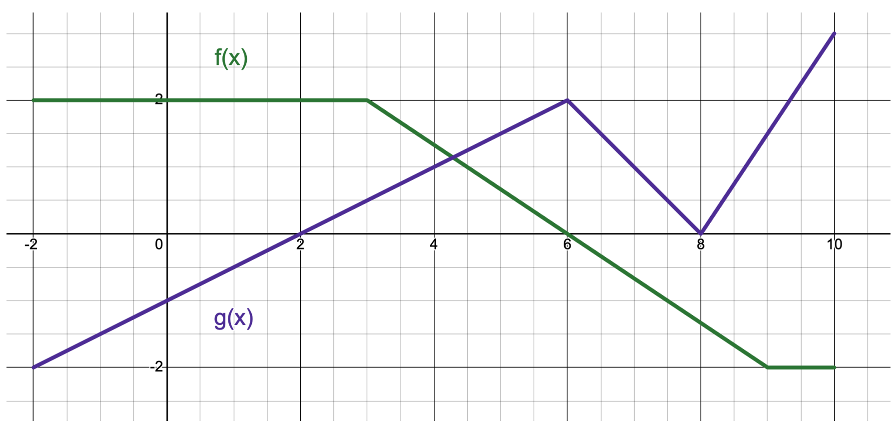
6. The figure below shows the graph of the functions f and g. The graphs of the lines tangent to the graph of g at $x = -3$ and $x = 1$ are also shown. If $B(x) = f(g(x))$, what is $B'(1)$?



a) $-\dfrac{1}{2}$ b) $-\dfrac{1}{6}$ c) $\dfrac{1}{6}$ d) $\dfrac{1}{3}$ e) $\dfrac{1}{2}$

---

7. Given the graphs of the two functions below and the fact that $B(x) = f(g(x))$, $B'(4) =$



a) 0 b) 1 c) $\dfrac{1}{2}$ d) $-\dfrac{1}{3}$ e) DNE

1.8: Related Rates

In this course, derivatives have primarily been interpreted as the slope of the tangent line. But, as with rectilinear motion, there are other contexts for the derivative. One overarching concept is that the derivative is a **rate of change**. The tendency is to think of rates as distance per time unit, like miles per hour or meters per second, but even slope is a rate of change—it is just that the rise and run are both measured as distances.

The idea behind related rates is two-fold. First, change is occurring in two or more measurements that are related to each other by the geometry (or algebra) of the situation. Second, an implicit Chain Rule situation exists in that the x and y-values are functions of time, which may or may not be a variable in the problem. Therefore, when taking the derivative of an x or y, an **implicit rate term** $\left(\dfrac{dx}{dx} \text{ or } \dfrac{dy}{dt}\right)$ often occurs.

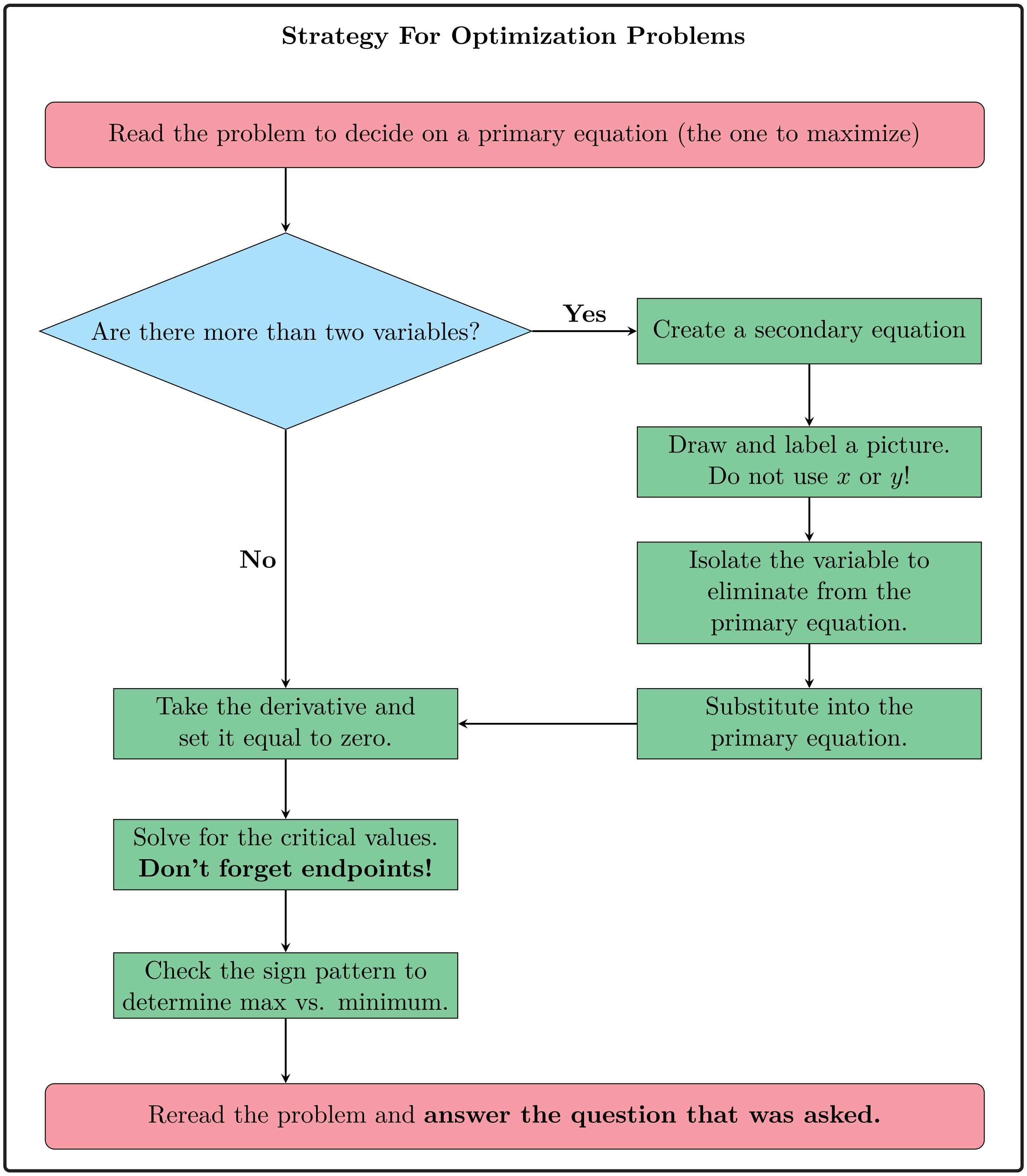
OBJECTIVES

Solve Related Rates Problems.

At first glance, related rates problems might seem like optimization problems that we've seen last year. Consider the following example:

Ex 1.8.1: The volume of a cylindrical cola can is 32π in^3. What is the minimum surface area for such a can?

The word "minimum" tells us that we have an optimization problem. Recall our workflow for tackling optimization problems:



So, let's tackle our example.

Sol 1.8.1: The problem asks to minimize surface area, which is determined by:

$$S = 2\pi r^2 + 2\pi rh$$

As there are more than two variables in this equation, either r or h needs to be eliminated in this formula before differentiating. The volume is $V = \pi r^2 h = 32\pi$, so $h = \dfrac{32}{r^2}$ and

$$\begin{aligned}
S &= 2\pi r^2 + 2\pi r\left(\frac{32}{r^2}\right)\\
&= 2\pi r^2 + \frac{64\pi}{r}\\
S' &= 4\pi r - \frac{64\pi}{r^2} = 0\\
\therefore 4\pi r &= \frac{64\pi}{r^2}\\
r^3 &= 16 \to r = 2.5198
\end{aligned}$$

Now, let's make a sign pattern to determine if this critical value is a minimum.

$$\begin{array}{cccc}
S' & - & 0 & + \\
 & \longleftarrow & \longrightarrow & \\
r & & 2.520 &
\end{array}$$

Because the derivative changes from negative to positive, we know that 2.5198 is a minimum. We can plug it back into our surface area equation to find our minimum surface area.

$$S(2.5198) = 2\pi(2.5198)^2 + \frac{64\pi}{2.5198}$$

$$S(2.5198) = 119.687 \text{ in}^2$$

Therefore, the minimum surface area of the cola can is $\boxed{119.687 \text{ in}^2}$.

A related rates problem is characterized by various measurements that are changing **in relation to each other**. The variables are still related to each other through a geometric or physical relationship. The key difference from other differentiation problems is that we differentiate implicitly with respect to time, rather than a variable like x. In other words,

Optimization Problems:	Apply $\dfrac{d}{dx}$
Related Rates Problems:	Apply $\dfrac{d}{dt}$

Now, let's take a look at this different (related rates) cola problem.

Ex 1.8.2: The volume of a cylindrical cola can is 32π in^3. The height of the can is changing at $\frac{1}{4}$ in/sec. If the radius changes at the same time so as to maintain the volume, how fast is the radius shrinking when the can is 4 inches tall?

Sol 1.8.2:

$$V = \pi r^2 h = 32\pi$$

$$h = \frac{32}{r^2}$$

$$\frac{d}{dt}\left[h = \frac{32}{r^2}\right]$$

$$\frac{dh}{dt} = -\frac{64}{r^2}\left(\frac{dr}{dt}\right)$$

Now that we have a method to find what we are looking for, $\frac{dr}{dt}$, let's substitute in the values that we are given in the problem.

$$h = 4 \to \frac{32}{r^2} = 4 \to r^2 = 8$$

$$\frac{1}{4} = -\frac{64}{8}\left(\frac{dr}{dt}\right)$$

$$\frac{dr}{dt} = \boxed{-\frac{1}{32}\text{ in/sec}}$$

As we can see, many of the steps that we take in the related rates cola problem is similar to those of the optimization cola problem.

Process For Related Rates Problems

1. Draw a visual for the problem.
2. Label the visual with what is given and what is being asked.
    a. Use variables for any quantities that are **changing**.
    b. Pay particular attention to the units.
3. Determine the equation(s) that relate the variables to each other.

a. Decide which equation will be differentiated.

b. If there is a product of two variables, eliminate the product by either multiplying the equation out or substituting a secondary equation.

4. **Differentiate in terms of time!** This is the key step.

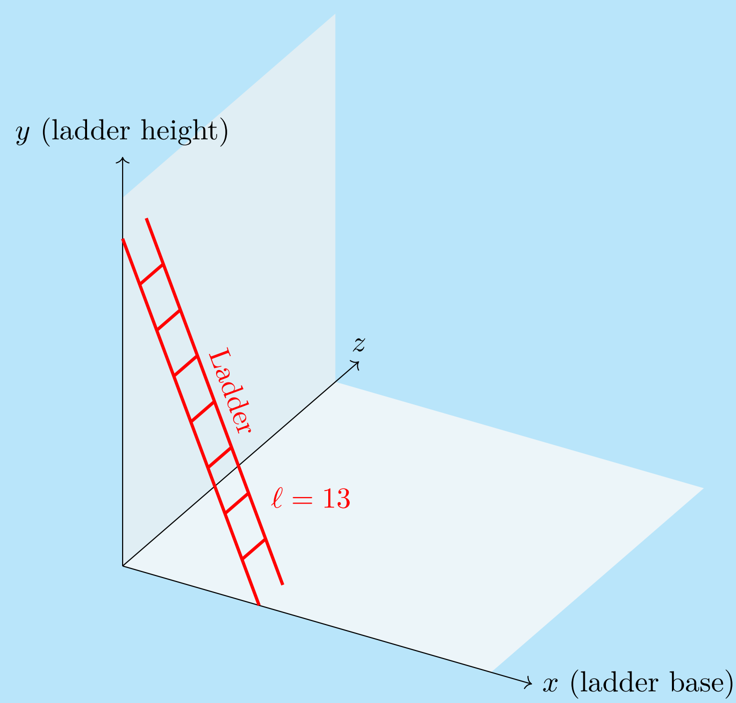
   a. Do not forget implicit fractions.

5. Substitute the given information and solve for the missing variable.

6. Reread the problem and make sure to answer the question that was asked.

A classic related rates problem is the falling ladder.

Ex 1.8.3: A 13-foot ladder is leaning against a wall. The bottom of the ladder slides away from the wall at 4 ft/sec. How fast is the top of the ladder moving down the wall when the ladder is 5 feet from the wall?



Sol 1.8.3: As can be seen in the visual, the height of the top of the ladder and the distance the bottom of the ladder is from the wall is related through the Pythagorean theorem. Both are variables, because the ladder is moving.

$$x^2 + y^2 = 13^2$$

We are given the information that the bottom of the ladder slides away from the wall at 4 ft/sec. Therefore, we can say that our $\frac{dx}{dt} = 4$. To find $\frac{dy}{dt}$, we differentiate $x^2 + y^2 = 13^2$ to get

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

Although this may seem like a complex four equation variable, we already know x and $\frac{dx}{dt}$. We also can determine that $y = 12$ by the Pythagorean theorem. So,

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$2(5)(4) + 2(12)\frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = \boxed{-\frac{5}{3}\text{ ft/sec}}$$

It should make sense that $\frac{dy}{dt}$ is negative because the top of the ladder is sliding down.

Below are some of the equations you should know for the AP exam:

Common Formulas For Optimization/Related Rates Problems

Pythagorean Theorem

$$x^2 + y^2 = r^2$$

Area Formulas

Circle: $A = \pi r^2$ Rectangle: $A = lw$

Triangle: $A = \dfrac{1}{2}bh$ Trapezoid: $A = \dfrac{1}{2}h\left(b_1 + b_2\right)$

Volume Formulas

Sphere: $V = \dfrac{4}{3}\pi$ Right Prism: $V = Bh$ Cylinder: $V = \pi r^2 h$

Cone: $V = \dfrac{1}{3}\pi r^2 h$ Right Pyramid: $V = \dfrac{1}{3}Bh$ *Washer: $V = \pi(R^2 - r^2)h$

Surface Area Formulas

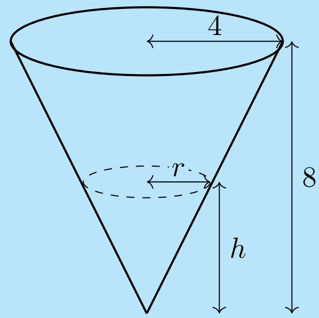
Sphere: $S = 4\pi r^2$ Cylinder: $S = 2\pi r^2 + 2\pi rh$

Cone: $S = \pi r^2 + \pi rl$ *Right Prism: $S = 2B + Ph$

N.B. The two equations with asterisks are less commonly used.

Another common related rates problem is where a tank of a particular shape is filling or draining.

Ex 1.8.4: A tank shaped like an inverted cone 8 feet in height and with a base diameter of 8 feet is filling at a rate of 10 ft^3/min. How fast is the height changing when the water is 6 feet deep?



Sol 1.8.4: The units on the rate of change tell us that this is a change in volume, or $\frac{dV}{dt}$. Therefore, we use the equation

$$V = \frac{1}{3}\pi r^2 h$$

But, there are too many variables in this equation to differentiate as it stands. Since the rate of the change of the height—$\frac{dh}{dt}$—is what we are looking for, we can eliminate r from the equation. By similar triangles, we have

$$\frac{r}{h} = \frac{4}{8}$$

$$r = \frac{1}{2}h$$

Substitution gives us a volume equation in terms of only height.

$$V = \frac{1}{3}\pi\left(\frac{1}{2}h\right)^2 h$$

$$V = \frac{\pi}{12}h^3$$

Differentiate and plug in our given values to solve for $\frac{dh}{dt}$.

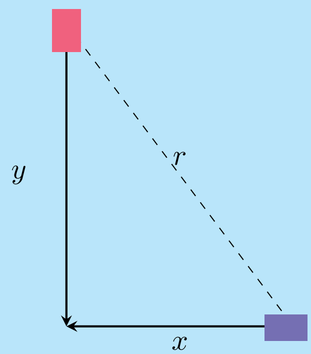
$$V = \frac{\pi}{12}h^3$$

$$\frac{dV}{dt} = \frac{\pi}{4}h^2\frac{dh}{dt}$$

$$10 = \frac{\pi}{4}(6)^2\frac{dh}{dt}$$

$$\frac{dh}{dt} = \boxed{\frac{10}{9\pi}\text{ ft/min}}$$

Ex 1.8.5: Two cars approach an intersection, one traveling south at 20 miles per hour and the other traveling west at 30 miles per hour. How fast is the direct distance between them decreasing when the westbound car is 0.6 miles and the southbound car is 0.8 miles away from the intersection?



Sol 1.8.5: As we can see in the picture, the distance between the two cars are related by the Pythagorean theorem.

$$x^2 + y^2 = r^2$$

We know several pieces of information. The southbound car is moving at 20 miles per hour; i.e. $\dfrac{dy}{dt} = -20$. By similar logic, we can deduce each of the following:

$$\frac{dy}{dt} = -20 \qquad \frac{dx}{dt} = -30$$

$$y = 0.8 \qquad x = 0.6$$

And, by the Pythagorean Theorem, $r = 1$. Now we take the derivative of the Pythagorean theorem to get

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 2r\frac{dr}{dt}$$

This is essentially an equation in six variables. But, we know five of those variables, so let's substitute and solve.

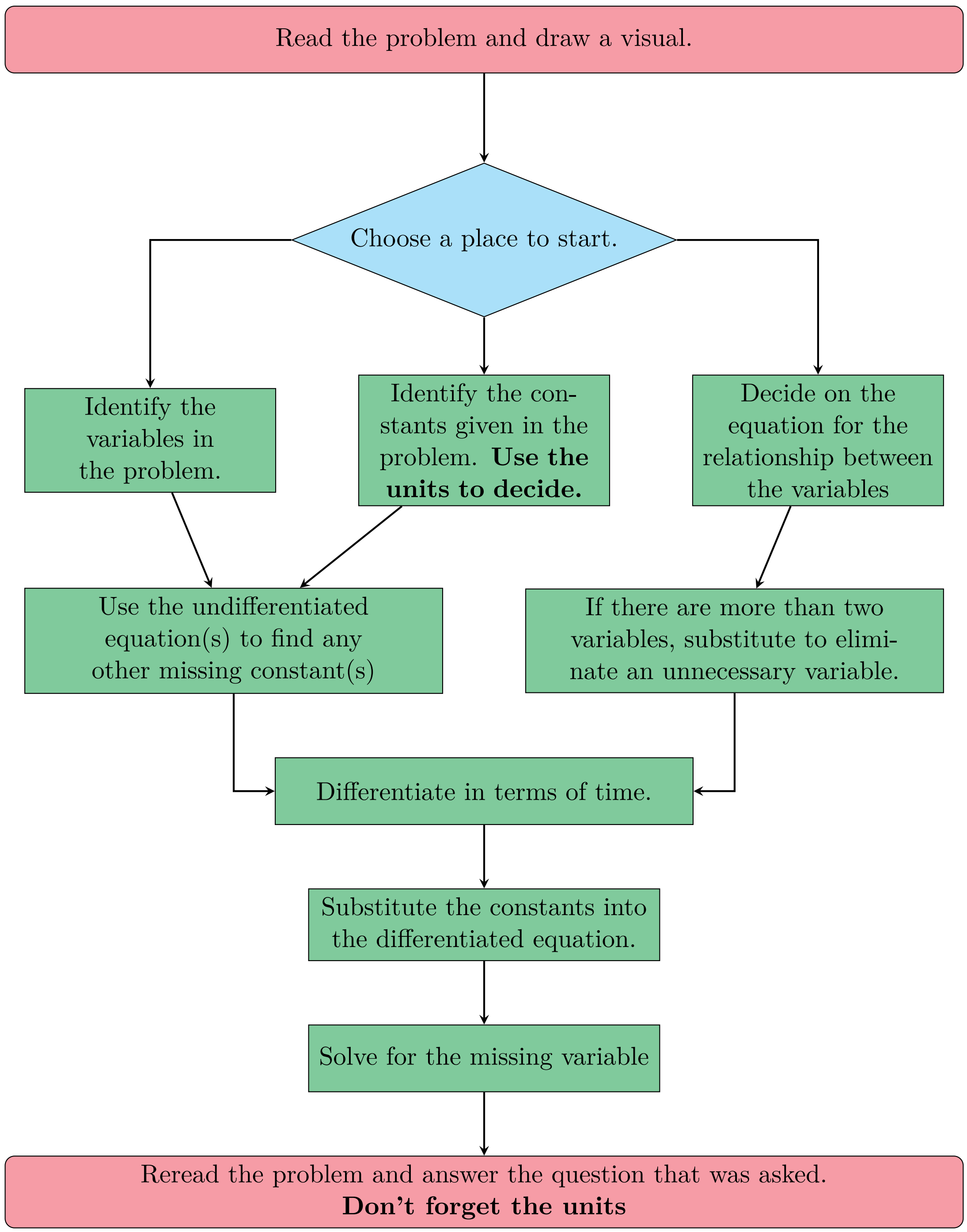
$$2(0.8)(-30) + (2)(0.6)(-20) = 2(1.0)\frac{dr}{dt}$$

$$\frac{dr}{dt} = \boxed{-36 \text{ mph}}$$

It should make sense that $\dfrac{dr}{dt}$ is negative since the two cars are approaching one another.

The units also make sense: since r is in miles and h is in hours, the final units should be miles per hour.

Strategy For Related Rates Problems



1.8 Free Response Homework

1. Two boats leave an island at the same time, one heading north and one heading east. The northbound boat is moving at 12 mph and the eastbound boat is moving at 5 mph. At $t = 0.2$ hours, the northbound boat is 1.4 miles away from the island and the eastbound boat is 1 mile away from the island.

(a) Draw a picture of the situation at any time t.

(b) What variables are present in the problem?

(c) What known quantities are given? What quantities can be determined directly from the given information? And what is the unknown for which to solve?

(d) What equation(s) relates the quantities? Which one will be differentiated?

(e) How fast is the distance between the two ships decreasing at $t = 0.2$ hours?

2. A railroad track and a road cross at right angles. An observer stands on the road 70 meters south of the intersection and watches an eastbound train traveling 60 m/sec.

(a) Draw a picture of the situation at any time t.

(b) What variables are present in the problem?

(c) What known quantities are given? What quantities can be determined directly from the given information? And what is the unknown for which to solve?

(d) What equation(s) relates the quantities? Which one will be differentiated?

(e) At how many meters per second is the train moving away from the observer 4 seconds after it passes the intersection?

3. A circular ink stain is spreading (i.e. the radius is changing) at half an inch per second.

(a) Draw a picture of the situation at any time t.

(b) What variables are present in the problem?

(c) What known quantities are given? What quantities can be determined directly from the given information? And what is the unknown for which to solve?

(d) What equation(s) relates the quantities? Which one will be differentiated?

(e) How fast is the area changing when the stain has a 1 inch diameter?

4. A screensaver has a rectangular logo that expands and contracts as it moves around the screen. The ratio of the sides stay constant, with the long side being 1.5 times the short side. At a particular moment, the long side is 3 cm, while the perimeter is changing by 0.25 cm/sec.

(a) Draw a picture of the situation at any time t.

(b) What variables are present in the problem?

(c) What known quantities are given? What quantities can be determined directly from the given information? And what is the unknown for which to solve?

(d) What equation(s) relates the quantities? Which one will be differentiated?

(e) How fast is the area of the screensaver changing at the given moment?

5. Sand is being dumped onto a pile at 30π ft/min. The pile forms a cone with the height always equal to the base diameter.

(a) Draw a picture of the situation at any time t.

(b) What variables are present in the problem?

(c) What known quantities are given? What quantities can be determined directly from the given information? And what is the unknown for which to solve?

(d) What equation(s) relates the quantities? Which one will be differentiated?

(e) How fast is the height changing when the pile is 5 feet high?

6. A cylindrical oil tank of height $30'$ and radius $10'$ is leaking at a rate of 300 ft^3/min.

(a) Draw a picture of the situation at any time t.

(b) What variables are present in the problem?

(c) What known quantities are given? What quantities can be determined directly from the given information? And what is the unknown for which to solve?

(d) What equation(s) relates the quantities? Which one will be differentiated?

(e) How fast is the oil level dropping?

7. Water is leaking out of an inverted conical tank at a rate of 5000 cm^3/min. The tank is 8 m tall and has a diameter of 4 m.

(a) Draw a picture of the situation at any time t.

(b) What variables are present in the problem?

(c) What known quantities are given? What quantities can be determined directly from the given information? And what is the unknown for which to solve?

(d) What equation(s) relates the quantities? Which one will be differentiated?

(e) Find the rate at which the height is decreasing when the water level is at 3 m.

(f) Find the rate of change of the radius at the same instant as part (e).

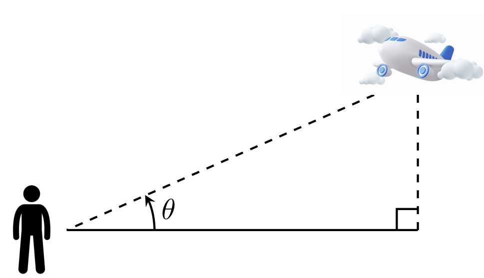
8. Sand is dumped onto a pile at 30π ft^3/min. The pile forms a cone with the height always equal to the base diameter. How fast is the base area changing when the pile is 10 feet high?

9. A spherical balloon is being inflated so that its volume is increasing at a rate of 6 ft^3/min. How fast is the radius changing when $r = 10$ ft?

10. The edge of a cube is expanding at a constant rate of 6 inches per second. What is the rate of the change of the volume, in inches cubed per second, when the total surface area of the cube is 54 in^2.

11. A 25-foot tall ladder is leaning against a wall. The bottom of the ladder is pushed toward the wall at 5 ft/sec. How fast is the top of the ladder moving up the wall when it is 7 feet up?

12. You are standing outside. A plane flies overhead, approaching you at constant altitude and a constant speed of 600 miles per hour. When the plane flies over a house 13 miles away from where you are standing, the angle of elevation is 0.647 radians. How quickly is the direct (diagonal) distance between you and the plane changing at that moment?



13. The altitude of a triangle is increasing at a rate of 2 cm/sec at the same time that the area of the triangle is increasing at a rate of 5 cm^2/sec. At what rate is the base increasing when the altitude is 12 cm and the area is 144 cm^2?

14. Two cars start moving away from the same point. One travels south at 60 mph, and the other travels west at 25 mph. At what rate is the distance between the cars increasing two hours later?

15. According to Boyle's Law, gas pressure varies directly with temperature and inversely with volume, as described by the following equation:

$$P = \frac{kT}{V}$$

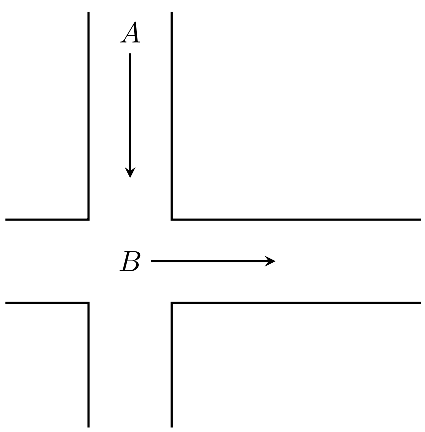
Suppose that the temperature is held constant while the pressure increases at at 20 kPa/min. What is the rate of change of the volume when the volume is 600 in^3 and the pressure is 150 kPa?

16. The Adiabatic Law for expansion of air can be represented with the equation

$$P \cdot V^{1.4} = \frac{4}{81},$$

where P represents pressure and V represents volume. If, at a specific instant, $P = 108$ lb/in^2 and is increasing at 27 lb/in^2 per second, what is the rate of change of the volume?

17. Person A is 220 feet north of an intersection and walking toward it at 10 ft/sec. Person B starts at the intersection and walks east at 5 ft/sec.



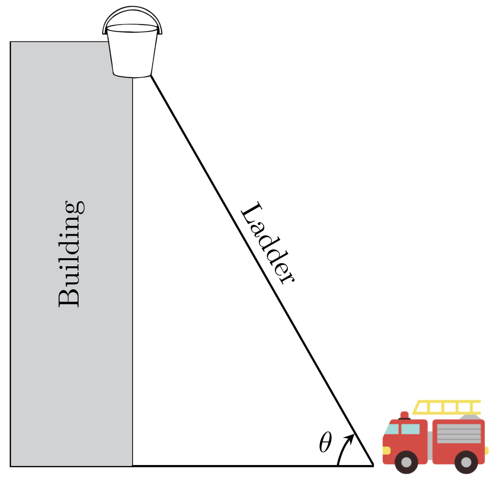
(a) At $t = 10$ seconds, how far is each person from the intersection?

(b) At $t = 10$ seconds, how far apart are the two people?

(c) How fast is the distance between the two people changing at $t = 10$ seconds?

(d) At time $t = 10$ seconds, if person A looks at person B, how fast is the angle between person A's line of sight to person B and the eastward direction changing?

18. A fire truck is parked 7 feet away from the base of a building and its ladder is extended to the top of the building. The ladder retracts at a rate of 0.5 feet per second, while the angle of the ladder changes such that the bucket at the end of the ladder comes down vertically.



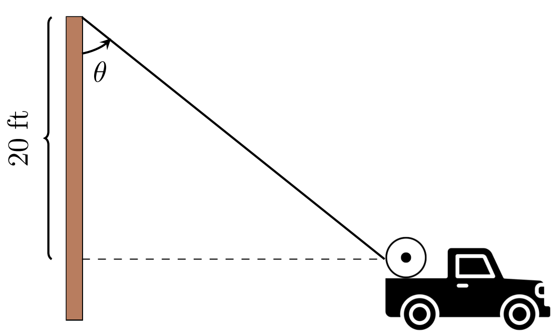
(a) How far is the ladder extended when the bucket is 10 feet above the ground?

(b) Find the rate at which the bucket is dropping vertically when the bucket is 10 feet above the ground.

(c) What is the relationship between the angle θ and the height of the bucket? Find θ, in radians, when the bucket is 10 feet above the ground.

(d) Find the rate, in radians per second, at which the angle the ladder forms with the ground is changing when the bucket is 10 feet above the ground.

19. A telephone crew is replacing a phone line from one telephone pole to the next. The line is on a spool on the back of a truck, and one end is attached to the top of a 25-foot pole. The vertical distance from the top of the pole to the level of the spool is 20 feet. The truck moves down the street at 20 ft/sec.



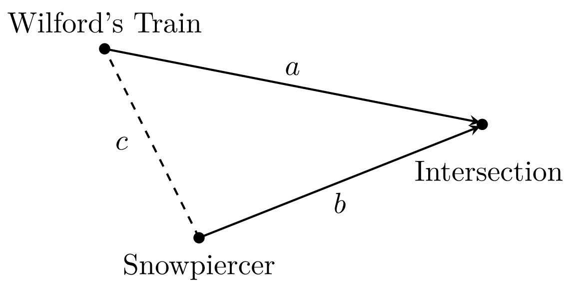
(a) Find the length of line that has been rolled out when $t = 15$ seconds.

(b) Find the rate at which the telephone line is coming off the spool when the truck is 50 feet from the pole.

(c) What is the relationship between the angle and the truck's distance from the pole? Find θ, in radians, when the truck is 40 feet from the pole.

(d) Find the rate, in radians per second, at which the angle the line forms with horizontal is changing when the truck is 40 feet from the pole.

20. At the end of the first season of the series *Snowpiercer*, a second train (Big Alice) controlled by the industrialist Mr. Wilford came down another track to intercept and stop Snowpiercer. If the two train tracks meet at a 30° then the Law of Cosines would apply such that $c^2 = a^2 + b^2 - 1.969ab$ as in the figure below. Snowpiercer is described as being two-and-a-half stories tall and $1,001$ cars long, with an average speed of 100 km/hour. Wilford's train Big Alice is much shorter, so it can average 120 km/hour.

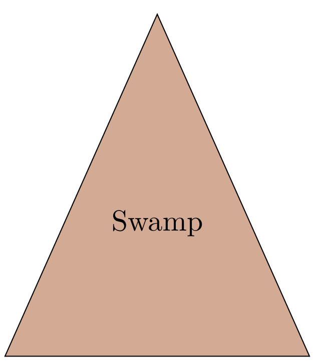


(a) If Snowpiercer is 38 km from the junction, how soon will it reach the junction?

(b) If Wilford's train is 45 km from the junction, which train will reach the junction first?

(c) How fast is the distance c between the two trains changing when Snowpiercer is 38 km from the junction and Big Alice is 45 km from the junction.

21. The triangle-shaped swamp on Oak Island has been proven to have been manmade sometime around 1250 AD, possibly by the Knights Templar. It has been drained several times, revealing a stone-paved wharf and paths, an ancient clay mine, and the remains of a galleon which had been destroyed by fire and sunk in the swamp to conceal it. The triangle is roughly isosceles, with legs measuring 730 feet and a base of 640 feet. The apex (top angle) of the triangle measures 0.79 radians.



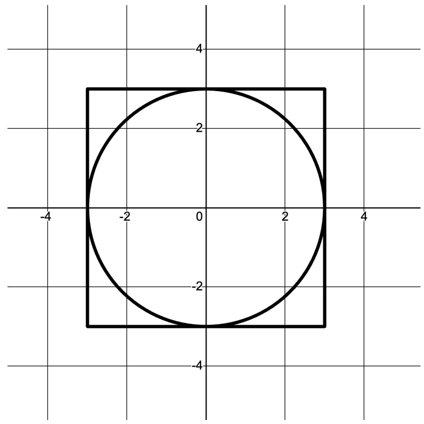
(a) Based on the Law of Sines, the area of a triangle can be determined by the equation $Area = \frac{1}{2}ab\sin(C)$, where a and b are the lengths of the legs and $\angle C$ is the measure of the angle in between the two legs. Find the surface area of the swamp before it was drained. Indicate the units.

(b) The length of third side of the triangle can be found using the Law of Cosine, $c^2 = a^2 + b^2 - 2ab\cos(C)$, where c is the third side and a, b, and $\angle C$ are the same as those of part (a). How long is the third side when the legs are each 370 feet?

(c) At $t = 24$ hours, the legs are 370 feet and their rate of change is -15.2 ft/hr. How fast is the third side of the triangle changing?

(d) Find the rate of change of the surface area. [Hint: Use the Law of Sines.] Indicate the units.

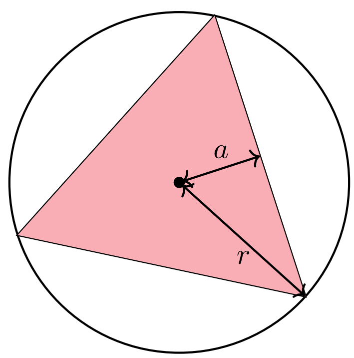
22. A circle is inscribed in a square as shown. The circumference of the circle is increasing at a constant rate of 4 inches per second. As the circle expands, the square expands to maintain the condition of tangency.



(a) Find the rate of change of the perimeter of the square.

(b) At the instant when the area of the circle is 16π square inches, find the rate at which the area *between the square and the circle* is increasing.

23. An equilateral triangle is inscribed in a circle. The circle's circumference is expanding at 6π in/sec and the triangle maintains the contact of its corners with the circle.



Given that the area of an equilateral triangle is equal to half the apothem a times the perimeter p, find out how fast the area inside the circle but outside the triangle is expanding when the area of the circle is 64π in. [Hint: Find p and a in terms of r.]

1.8 Multiple Choice Homework

1. The width of a square is increasing at a constant rate of 0.5 cm/sec. In terms of the perimeter P, what is the rate of change of the area of the square in centimeters squared

per second?

a) $-\frac{1}{2}P$ b) $-P$ c) $\frac{1}{4}P$ d) $\frac{1}{2}$ e) P

---

2. When $x = 18$, the rate at which $y = \sqrt{0.5x}$ is increasing is k times the rate at which x is increasing. What is the value of $\frac{1}{k}$?

a) $\frac{1}{12}$ b) $\frac{1}{6}$ c) 1 d) 6 e) 12

---

3. The side of a cube is expanding at a constant rate of 6 inches per second. What is the rate of change of the volume, in cubic inches per second, when the total surface area of the cube is 54 in^2/sec?

a) 324 b) 108 c) 18 d) 162 e) 54

---

4. If the volume of a cube is increasing at 20 cubic inches per second when each edge is 10 inches long, how fast is the surface area increasing?

a) $\frac{4}{3}$ b) 2 c) 4 d) 6 e) 8

---

5. When the height of a cylinder is 12 cm and the radius is 4 cm, the circumference is increasing at a rate of $\frac{\pi}{4}$

a) 4π b) 12π c) 20π d) 80π e) 100π

---

6. At what approximate rate (in cubic meters per minute) is the volume of a sphere changing at the instant when the surface area is 3 square meters and the radius is increasing at the rate of $\frac{1}{5}$ meters per minute?

a) 1.228 b) 1.905 c) 0.649 d) 0.600 e) 0.620

7. The radius of a sphere is decreasing at a rate of 2 centimeters per second. At the instant when the radius of the sphere is 3 cm, what is the rate of change, in square centimeters per second, of the surface area of the sphere? [Hint: The surface area S of a sphere with radius r is $4\pi r^2$.]

a) -108π b) -72π c) -48π d) -24π e) -16π

8. Water is flowing into a spherical tank with a 6-foot radius at the constant rate of 30π ft^3/hour. When the water is h feet deep, the volume of the water in the tank is given by the equation

$$V = \frac{\pi h^2}{3}(18 - h).$$

What is the rate at which the depth of the water in the tank is increasing the moment when the water is 2 feet deep?

a) 0.5 ft/hour b) 1.0 ft/hour c) 1.5 ft/hour d) 2.0 ft/hour e) 2.5 ft/hour

9. The radius of a circle is in increasing at a constant rate of 0.2 meters per second. What is the rate of increase in the area of the circle at the instant when the circumference of the circle is 20π meters?

a) 0.04π m^2/sec b) 0.4π m^2/sec c) 4π m^2/sec d) 40π m^2/sec e) 100π m^2/sec

10. If the rate of change of a number x with respect to time t, is x, what is the rate of change of the reciprocal of the number when $x = -\frac{1}{4}$?

a) -16 b) -4 c) $-\frac{1}{48}$ d) $\frac{1}{48}$ e) 4

11. Gravel is being dumped from a conveyor belt at a rate of 35 ft^3/min and its coarseness is such that it forms a pile in the shape of a cone whose base diameter and height are always equal. How fast is the height of the pile increasing when the pile is 15 ft high?

a) 0.27 ft/min b) 1.24 ft/min c) 0.14 ft/min d) 0.20 ft/min e) 0.60 ft/min

---

12. Two cars start moving from the same point. One travels south at 28 mi/h and the other travels west at 70 mi/h. At what rate is the distance between the cars increasing 5 hours later?

a) 75.42 mi/h b) 75.49 mi/h c) 76.40 mi/h d) 75.39 mi/h e) 75.38 mi/h

---

13. A Golden Rectangle is one where the ratio (called ϕ) of the length of the short side w to the long side l is equal to the ratio of the long side to the sum of the two sides. In other words, $l = 1.618w$. If a Golden Rectangle changes such that w is growing at 2 in/min, how fast is the area changing when w is 5 inches?

a) 1.618 in^2/min b) 16.18 in^2/min c) 3.236 in^2/min d) 32.36 in^2/min e) 37.78 in^2/min

---

1.9: Logarithmic Differentiation

With implicit differentiation and the Chain Rule, we learned some powerful tools for differentiating functions and relations. The Product and Quotient Rules also allowed us to take derivatives of certain functions that would otherwise be impossible to differentiate. Sometimes, however, with very complex functions, it becomes easier to manipulate an equation so that it is easier to take the derivative. This is where logarithmic differentiation comes in.

OBJECTIVES

Determine When It Is Appropriate to Use Logarithmic Differentiation.
Use Logarithmic Differentiation to Take The Derivatives of Complicated Functions.

Before we begin, it would be helpful to look at a few exponent and logarithm rules that we should recall from algebra and precalculus.

$a^x a^y = a^{x+y}$	$\log_a x + \log_a y = \log_a xy$
$\dfrac{a^x}{a^y} = a^{x-y}$	$\log_a x - \log_a y = \log_a \dfrac{x}{y}$
$(a^x)^y = a^{xy}$	$\log_a x^n = n \log_a x$

Since logarithms are exponents expressed in a different form, all of the above rules are derived from those of exponents.and you can see the corresponding exponential rule. Because of our algebraic rules, we can do whatever we want to both sides of an equation. In algebra, we usually used this to solve for a variable. In calculus, we can use this principle to make many derivative problems significantly easier.

Ex 1.9.1: Find the derivative of $y = \left(x^2 + 7x - 3\right)(\sin(x))$.

Sol 1.9.1: Traditionally, we would use the Product Rule to take the derivative of this function.

$$\frac{d}{dx}\left[y = \left(x^2 + 7x - 3\right)(\sin(x))\right]$$

$$\frac{dy}{dx} = \boxed{\left(x^2 + 7x - 3\right)\cos(x) + (2x+7)(\sin(x))}$$

Obviously, this is a straightforward problem that can be easily done using the product rule. If, however, I took the natural log of both sides of the equation, I can achieve the same results, and never use the product rule.

$$\ln y = \ln\left[\left(x^2+7x-3\right)(\sin(x))\right]$$

Let's simplify using our log rules.

$$\ln y = \ln\left(x^2+7x-3\right) + \ln(\sin(x))$$

$$\frac{d}{dx}\left[\ln(y) = \ln\left(x^2+7x-3\right) + \ln(\sin(x))\right]$$

$$\frac{1}{y}\left(\frac{dy}{dx}\right) = \frac{2x+7}{x^2+7x-3} + \frac{\cos(x)}{\sin(x)}$$

$$\frac{dy}{dx} = \left(\frac{2x+7}{x^2+7x-3} + \frac{\cos(x)}{\sin(x)}\right)(y)$$

Now just substitute y back in and simplify.

$$\frac{dy}{dx} = \left(\frac{2x+7}{x^2+7x-3} + \frac{\cos(x)}{\sin(x)}\right)\left(\left(x^2+7x-3\right)(\sin(x))\right)$$

$$\frac{dy}{dx} = \boxed{\left(x^2+7x-3\right)\cos(x) + (2x+7)(\sin(x))}$$

Clearly, we got the same answer that we got from the product rule, but with significantly more effort.

Logarithmic differentiation is a tool we can use, but we have to use it judiciously, as we don't want to make problems more difficult than they have to be. Where logarithmic differentiation has the potential to be really useful is with functions that are excessively painful to work with (or impossible to take the derivative of any other way) because of multiple operations. Consider the following example:

Ex 1.9.2: Find $\dfrac{dy}{dx}$ for $y = \dfrac{(x^2+5)\sin\left(3x^3\right)}{\tan(5x+2)}$.

Sol 1.9.2: We could take the derivative by applying the Chain Rule, Quotient Rule, and Product Rule, but that would be a time-consuming and tedious process. It's much easier to take the natural log of both sides, simplify and then take the derivative.

$$\ln y = \ln\left[\frac{(x^2+5)\sin\left(3x^3\right)}{\tan(5x+2)}\right]$$

$$\ln y = \ln\left(x^2+5\right) + \ln\left(\sin\left(3x^3\right)\right) - \ln(\tan(5x+4))$$

$$\frac{d}{dx}\left[\ln y = \ln\left(x^2+5\right) + \ln\left(\sin\left(3x^3\right)\right) - \ln(\tan(5x+4))\right]$$

$$\frac{1}{y}\left(\frac{dy}{dx}\right) = \frac{2x}{x^2+5} + \frac{9x^2\cos\left(3x^3\right)}{\sin\left(3x^3\right)} - \frac{5\sec^2(5x+2)}{\tan(5x+2)}$$

$$\frac{dy}{dx} = \left(\frac{2x}{x^2+5} + \frac{9x^2\cos\left(3x^3\right)}{\sin\left(3x^3\right)} - \frac{5\sec^2(5x+2)}{\tan(5x+2)}\right)(y)$$

$$\frac{dy}{dx} = \boxed{\left(\frac{2x}{x^2+5} + \frac{9x^2\cos\left(3x^3\right)}{\sin\left(3x^3\right)} - \frac{5\sec^2(5x+2)}{\tan(5x+2)}\right)\left(\frac{(x^2+5)\sin\left(3x^3\right)}{\tan(5x+2)}\right)}$$

Now that may seem long and messy, but try it any other way, and you might end up taking a lot more time, with a lot more algebra and a lot more potential spots to make mistakes.

Ex 1.9.3: Find $f'(\pi)$ for $f(z) = z^{\cos(z)}$.

Sol 1.9.3:

$$\ln(f(z)) = \ln\left(z^{\cos(z)}\right)$$

$$\frac{d}{dz} = \left[\ln(f(z)) = \ln\left(z^{\cos(z)}\right)\right]$$

$$\frac{f'(z)}{f(z)} = \frac{\cos(z)}{z} - (\ln z)(\sin(z))$$

$$f'(z) = \left(\frac{\cos(z)}{z} - (\ln z)(\sin(z))\right)(f(z))$$

$$f'(z) = \left(\frac{\cos(z)}{z} - (\ln z)(\sin(z))\right)\left(z^{\cos(z)}\right)$$

$$f'(\pi) = \left(\frac{\cos(\pi)}{\pi} - (\ln \pi)(\sin(\pi))\right)\left(\pi^{\cos(\pi)}\right) = \boxed{-\frac{1}{\pi^2}}$$

We could have also done this problem using the change of base property that we learned in precalculus, and we would get the same answer in roughly the same number of steps.

Again, there are often more than one way to do a specific problem, and part of what we do as mathematicians is decide on the simplest **correct** method to solving a problem. The issue many people have when learning more difficult mathematical concepts is that they try to oversimplify a problem and end up getting it wrong as a result.

1.9 Free Response Homework

Find the derivatives of the following functions. Only use logarithmic differentiation **when appropriate**.

1. $y = (2x+1)^4 \left(x^3 - 3\right)^5$
2. $z = \left(y^3 - 3\right) e^{(2y+1)}$
3. $y = \dfrac{\sin^2(x)\tan^4(x)}{\left(x^2+5\right)^2}$
4. $g(t) = t \ln t$
5. $y = \ln^x x$
6. $p(v) = v^{e^v}$

Complete the following

7. Find $\dfrac{dt}{dt}$ if $t^u = u^t$.

8. Find $\dfrac{dy}{dx}$ for the function $y = \left(e^{17x^4}\right)\left(\sin^7(x)\right)(5x-17)^{12}(\cot(5x))$.

9. Use logarithmic differentation to find $\dfrac{dq}{dt}$ if $q = \dfrac{e^{t^4-15}\sin^5(3t)}{(\ln t)^{10}}$.

10. Use logarithmic differentation to find $\dfrac{dy}{dx}$ if $y = e^{150x-19}\ln^{100}(\sin(x))\sqrt{x^2-1}$.

11. Use logarithmic differentiation to prove the Product Rule.

12. Use logarithmic differentiation to prove the Quotient Rule.

Full Name: AP Calculus BC

Date: Chapter 1 Practice Test

Multiple Choice Section **20 Minutes; No Calculator**

Show All Work

1. Which of the following equations is true?

a) $\dfrac{d}{dx}\left[\cot^{-1}(4x)\right] = -\dfrac{4}{16x^2+1}$

b) $\dfrac{d}{dx}\left[e^{\cos(x)}\right] = -e^{-\sin(x)}\cos(x)$

c) $\dfrac{d}{dx}\left[\ln^3\left(1-x^2\right)\right] = \dfrac{3\ln^2\left(1-x^2\right)}{1-x^2}$

d) $\dfrac{d}{dx}\left[e^{2x}\cos^{-1}(x)\right] = \dfrac{2e^{2x}}{\sqrt{1-xe^2}}$

---

2. If $f(x) = \cot^{-1}(\sin(x))$, then $\dfrac{d}{dx}[f(x)] =$

a) $-\csc^2(\sin(x))\cos^2(x)$ b) $\sec(x)$ c) $\dfrac{\cos(x)}{1+\sin^2(x)}$ d) $-\dfrac{\cos(x)}{1+\sin^2(x)}$ e) -1

---

3. Let $y = f(x)$ be a solution to the differential equation $\dfrac{dy}{dx} = x - y^2$ with the initial condition $f(0) = 1$. What is the best approximation for $f(2)$ if Euler's Method is used, starting at $x = 0$ with a step size of 1.0

a) -1 b) 0 c) 1 d) 2 e) 3

---

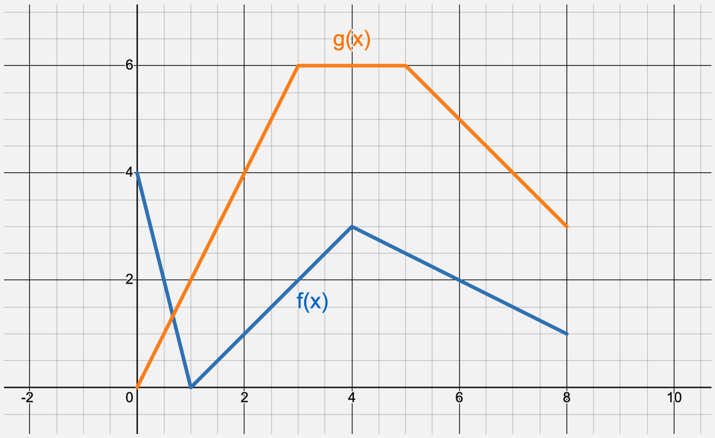
4. Selected values of f, g, and their derivatives are indicated in the table below. Let $h(x) = g\left(f\left(\sqrt{x}\right)\right)$. What is the value of $h'(4)$?

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	4	3	4	-1
4	2	-1	7	8
16	1	2	2	1

a) 6 b) $\frac{1}{4}$ c) 1 d) 2 e) 24

---

5. The figure below shows the graph of the functions f and g. If $B(x) = \frac{g(x)}{f(x)}$. what is $B'(6)$?



a) $-\frac{9}{2}$ b) $\frac{1}{8}$ c) -2 d) 0 e) DNE

---

6. If $f(x) = 3\sin(x) + 4\cos^2(x)$, then $f''\left(\frac{\pi}{2}\right)$

a) 3 b) 0 c) 5 d) -8 e) -3

---

7. Which of the following is an equation of the line tangent to the graph of $f(x) = x^6 + x^5 + x^2$ at the point where $f'(x) = -1$?

a) $-3x - 2$ b) $-3x + 4$ c) $-x + 0.905$
d) $-x + 0.271$ e) $-x - 0.271$

---

8. A biologist is tracking the growth of a circular colony of bacteria in a Petri dish. She observes that the colony is expanding at rate of 15 mm^2/hour. Find the rate at which the radius is increasing when the diameter is 5 mm.

a) 3 b) 3π c) $\dfrac{3}{\pi}$ d) $\dfrac{3}{2\pi}$ e) $\dfrac{3\pi}{2}$

---

9. What is the slope of the line tangent to the curve $y^2 + x = -2xy - 5$ at the point $(2, 1)$?

a) $-\dfrac{4}{3}$ b) $-\dfrac{3}{4}$ c) $-\dfrac{1}{2}$ d) $-\dfrac{1}{4}$ e) 0

Free Response Section **45 Minutes; No Calculator**

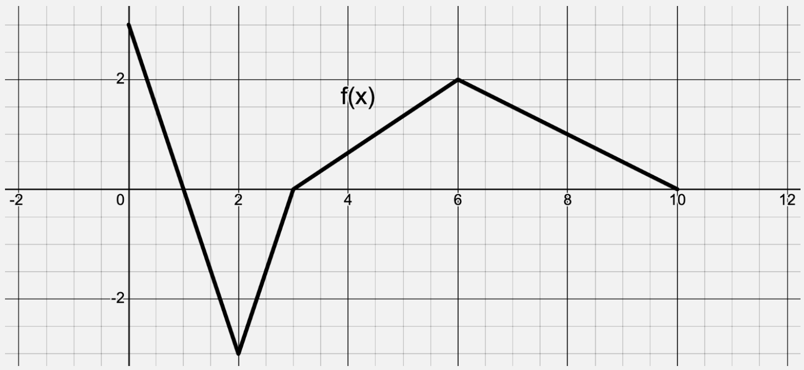
Show All Work

1. Compute the following derivatives.

   (a) $\dfrac{d}{dx}\left[\sin^{-1}\left(e^{2x}\right)\right]$

   (b) $\dfrac{d}{dx}\left[\ln\left(\cot\left(\sqrt{x}\right)\right)\right]$

2. Let $f(x)$ be the function defined by $f(x) = \sin\left(\frac{\pi}{2}\right)x$, let $g(x)$ be a differentiable function with selected values for $g(x)$ and $g'(x)$ given in the table below, and let $h(x)$ be a differentiable function whose graph is given below.



x	$g(x)$	$g'(x)$
0	-2	12
2	0	-3
4	5	5
6	3	8
8	-4	11

(a) Find the equation of the line tangent to $h(x)$ at $x = 8$.

(b) Let K be the function defined by $K(x) = g(f(x))$. Find $K'(6)$.

(c) Let M be the function defined by $M(x) = g(x) \cdot f(x)$. Find $M'(4)$.

(d) Let J be the function defined by $J(x) = \dfrac{h(x)}{g\left(\frac{1}{2}x\right)}$. Find $J'(8)$.

---

3. If $F(x) = \ln\left(3x^2 - 2x + 1\right)$, find $F''(x)$.

4. Find the equations of the lines tangent and normal to $g(x) = e^{4x}\cos(x)$ at $x = 0$ and use it to approximate $g(-0.2)$

5. Consider the curve given by $2y - x + xy = 8$

(a) Show that $\dfrac{dy}{dx} = \dfrac{1-y}{2+x}$.

(b) Find the coordinates of the point(s) where the tangent line is horizontal or prove why there is no such point(s).

(c) Find the value of $\frac{d^2y}{dx^2}$ at $(1, 3)$. Does the curve have a relative maximum, a relative minimum, or neither at $(0, 1)$? Justify your answer.

6. Two people on bikes are at the same place. One of the bikers starts riding directly north at a rate of 8 m/sec. Five seconds after the first biker started riding north, the second starts to ride directly east at a rate of 5 m/sec At what rate is the distance between the two riders increasing 20 seconds after the *second* person started riding?

Chapter 1 Answer Key

1.1 Free Response Answers

1. $f'(x) = \boxed{2x+3}$

2. $f'(t) = \boxed{t^3}$

3. $y' = \boxed{-\frac{2}{3}x^{-\frac{5}{3}}}$

4. $y' = \boxed{5e^x}$

5. $v'(r) = \boxed{4\pi r^2}$

6. $g'(x) = \boxed{2x - 2x^{-3}}$

7. $y' = \boxed{\frac{3}{2}x^{\frac{1}{2}} + 2x^{-\frac{1}{2}} - \frac{3}{2}x^{-\frac{3}{2}}}$

8. $u' = \boxed{\frac{2}{3}t^{-\frac{1}{3}} + 3t^{\frac{1}{2}}}$

9. $z' = \boxed{-\frac{10A}{y^{11}} + Be^y}$

10. $y' = \boxed{e^{x+1}}$

11. $\boxed{7x^6 - \frac{7}{2}x^{-\frac{1}{8}} + 7^x \ln 7 + \frac{4}{7}x^{-\frac{11}{7}} - \frac{1}{5}x^{-2}}$

12. $\boxed{6x^5 - \frac{7}{2}x^{\frac{1}{6}} + 5^x \ln 5 + \frac{5}{3}x^{-\frac{8}{3}} - \frac{1}{8}x^{-2}}$

13. $\boxed{4x^3 - 18x^{\frac{2}{7}} + 8^x \ln 8 + \frac{7}{3}x^{-\frac{10}{3}} - \frac{1}{8}x^{-2}}$

14. $\boxed{\frac{3}{2}x^{\frac{1}{2}} - \frac{1}{2}x^{-\frac{1}{2}}}$

15. $\boxed{\frac{7}{3}z^{\frac{4}{3}} - \frac{4}{3}z^{-\frac{2}{3}}}$

16. $\boxed{\frac{11}{5}x^{\frac{6}{5}} - \frac{24}{5}x^{\frac{1}{5}} + \frac{3}{5}x^{-\frac{4}{5}}}$

17. $\boxed{60t^4 + 9t^2 + 56t}$

18. $\boxed{\frac{-7\left(3x^2+4\right)}{\left(x^3+4x-\pi\right)^8}}$

19. $\boxed{\frac{3x-2}{\left(3x^2-4x+9\right)^2}}$

20. $\boxed{\frac{3x^2-2}{7\left(x^3-2x\right)^{\frac{6}{7}}}}$

21. $\boxed{10y^{\frac{3}{2}} - 3y^{\frac{1}{2}} - \frac{5}{2}y^{-\frac{1}{2}}}$

22. $\boxed{\frac{3}{4}v^{\frac{1}{2}} - v^{-\frac{1}{2}} - \frac{7}{4}v^{-\frac{3}{2}}}$

23. $\boxed{-\frac{7}{5}w^{-2} + \frac{8}{5}w^{-3} - \frac{3}{5}w^{-4}}$

24. $\boxed{\frac{3}{7}w^{-2} + \frac{8}{7}w^{-3}}$

25. $f'(x) = \boxed{\frac{2+3x^2}{4\left(1+2x+x^3\right)^{\frac{3}{4}}}}$

26. $f'(x) = \boxed{\frac{3\left(-x^{-2}+2+e^x\right)}{5\left(x^{-1}+2x+e^x\right)^{\frac{2}{5}}}}$

27. $f'(x) = \boxed{37\left(3x^2+2\right)\left(x^3+2x\right)^{36}}$

28. $f'(x) = \boxed{15x^4 - 15x^2}$

29. $f'(2) = \boxed{-4e^3}$

30. $\dfrac{dy}{dx} = \boxed{\dfrac{e^{\sqrt{x}}}{2\sqrt{x}}}$

31. $f'\left(\sqrt{5}\right) = \boxed{-\dfrac{\sqrt{5}}{3}}$

32. $f'(x) = \boxed{-\dfrac{xe^{\sqrt{9-x^2}}}{\sqrt{9-x^2}}}$

33. $\boxed{v'(t) = \dfrac{1}{2}\left[\left(\dfrac{E(t)}{3} + 3t\right)^{\frac{3}{7}} - 4\right]^{-\frac{1}{2}} \left[\dfrac{3}{7}\left(\dfrac{E(t)}{3} + 3t\right)^{-\frac{4}{7}}\right]\left(\dfrac{1}{3}E'(t) + 3\right)}$

34. $\boxed{v'(t) = \dfrac{1}{3}\left[\left(\dfrac{C(t)}{7} + 4t^2\right) - 1\right]^{-\frac{2}{3}} \left[\dfrac{5}{7}\left(\dfrac{C(t)}{7} + 4t^2\right)^{-\frac{2}{7}}\right]\left(\dfrac{1}{7}C'(t) + 8t\right)}$

1.1 Multiple Choice Answers

1	2	3	4	5	6	7	8
C	**B**	**A**	**E**	**D**	**E**	**B**	**E**

1.2 Free Response Set A Answers

1. $y' = \boxed{4\cos(4x)}$

2. $y' = \boxed{20x^4 \sec\left(x^5\right) \tan\left(x^5\right)}$

3. $f'(t) = \boxed{\dfrac{\sec^2(t)}{3\left(\tan(t) + 1\right)^{\frac{2}{3}}}}$

4. $f'(\theta) = \boxed{-\tan(\theta)}$

5. $y' = \boxed{-3\cos^2(x)\sin(x)}$

6. $y' = \boxed{-3x^2 \sin\left(a^3 + x^3\right)}$

7. $f'(x) = \boxed{-\frac{\sin(\ln x)}{x}}$

8. $f'(x) = \boxed{\frac{1}{5x(\ln x)^{\frac{4}{5}}}}$

9. $f'(x) = \boxed{\frac{\cos(x)}{(\ln(10))(\sin(x)+2)}}$

10. $f'(x) = \boxed{-\frac{3}{(\ln 2)(1-3x)}}$

11. $y' = \boxed{\frac{e^x}{\sqrt{1-e^{2x}}}}$

12. $y' = \boxed{\frac{1}{2\sqrt{x}(x+1)}}$

13. $\boxed{\frac{3e^{3x}}{\sqrt{1-e^{6x}}}}$

14. $\boxed{-\frac{2e^{2x}}{e^{4x}+1}}$

15. $\boxed{\frac{2x}{x^4+1}}$

16. $\boxed{0}$

17. $\boxed{3(2x+2)e^{x^2+2x}}$

18. $\boxed{-3(2x+2)\sin\left(x^2+2x\right)}$

19. $\boxed{\frac{x^2}{(16+x^3)^{\frac{2}{3}}}}$

20. $\boxed{\frac{1}{x^3\sqrt{1-\frac{1}{4}x^{-4}}}}$

21. $\boxed{35e^{\tan(7x)}\sec^2(7x)}$

22. $\boxed{-\frac{x\sin\left(x^2-1\right)}{\sqrt{\cos\left(x^2-1\right)}}}$

23. $\boxed{\frac{6x\ln^2\left(x^2+1\right)}{x^2+1}}$

24. $\boxed{3x^2\cot\left(x^3\right)}$

25. $\boxed{\tan(x)}$

26. $\boxed{-2x\sin\left(x^2\right)}$

27. $f'(x) = \boxed{\frac{2x}{x^2+3}}$

28. $g'(x) = \boxed{\frac{2x-4}{x^2-4x+4}}$

29. $h'(x) = \boxed{\frac{x}{\sqrt{x^2+5}}}$

30. $F'(x) = \boxed{\frac{6x-6}{3\left(3x^2-6x+1\right)^{\frac{2}{3}}}}$

31. $y' = \boxed{-\frac{\sin(x)}{\sqrt{1-\cos^2(x)}}}$

32. $y' = \boxed{-\frac{x}{\sqrt{1-x^2}}}$

33. $y' = \boxed{6\sec^2(3\theta)\tan(3\theta)}$

34. $y' = \boxed{-7\cot^6(\sin(x))\csc^2(\sin(x))\cos(x)}$

35. $y' = \boxed{\dfrac{\sqrt{2}}{\sqrt{1-2x^2}}}$

36. $y' = \boxed{\dfrac{2}{\sqrt{1-(2x+1)^2}}}$

For **problem 31**, a common trap is to use the Pythagorean Identity to simplify $\sqrt{1-\cos^2(x)}$ to $\sin(x)$, and then simplify $-\dfrac{\sin(x)}{\sin(x)}$ to -1. However, one must note that the result of a square root must be a positive value. Therefore, the simplification really becomes $-\dfrac{\sin(x)}{|\sin(x)|}$. This structure is also known as the negative **signum** function for sine, also denoted as $-\operatorname{sgn}(\sin(x))$. For more information about the signum function, take a look at this article.

1.2 Free Response Set B Answers

1. $y' = \boxed{-\dfrac{3e^{3z}}{\sqrt{1-e^{6z}}}}$

2. $y' = \boxed{\dfrac{2x}{\left(x^2-1\right)^2+1}}$

3. $y' = \boxed{\dfrac{1}{|x|\sqrt{16x^2-1}} - \dfrac{4}{\sqrt{1-16x^2}}}$

4. $f'(x) = \boxed{\dfrac{5}{(25x^2+1)\tan^{-1}(5x)}}$

5. $g'(w) = \boxed{0}$

6. $f'(t) = \boxed{\dfrac{t}{(t^2+9)\sqrt{t^2+8}}}$

7. $\boxed{-e^{\csc(\theta)}\cot(\theta)\csc(\theta) - \dfrac{2\theta\csc^2\left(\theta^2\right)}{\cot\left(\theta^2\right)} - \sec(\theta)\tan(\theta)}$

8. $\boxed{3\left(3x^2+\dfrac{5}{x}\right)\tan\left(5\ln x + x^3 + 7\right)}$

9. $\boxed{\dfrac{3\left(2x+5e^x\right)\sec^2\left(x^2+5e^x+7\right)}{\tan\left(x^2+5e^x+7\right)}}$

10. $\boxed{-\dfrac{2\csc^2\left(\ln\left(5x^2\right)\right)}{x}}$

11. $\boxed{\dfrac{2x+4}{2\left(x^2+4x-5\right)}}$

12. $\boxed{\dfrac{35\sin^4(\ln(7t+3))\cos(\ln(7t+3))}{7t+3}}$

13. $\boxed{-\dfrac{(14x+1)\cot\left(\ln\left(7x^2+x\right)\right)\csc\left(\ln\left(7x^2+x\right)\right)}{7x^2+x}}$

14. $\boxed{4t}$

15. $\boxed{\dfrac{9-54x-15x^{-4}}{2\sqrt{9x-27x^2+5x^{-1}}}}$

16. $\boxed{5\sec(5x)\tan(5x) - e^x\csc^2\left(e^x\right) - \dfrac{10}{x}}$

17. $\dfrac{dz}{dt} = \boxed{-\tan(t) + e^t\sec\left(e^t\right)\tan\left(e^t\right)}$

18. $\dfrac{dz}{dt} = \boxed{\dfrac{\sec^2(t)}{\tan(t)} + e^t\cos\left(e^t\right)}$

19. $\dfrac{dz}{d\theta} = \boxed{-\dfrac{\csc^2(\theta)}{\cot(\theta)} + \dfrac{\sec(\ln(\theta))\tan(\ln(\theta))}{\theta}}$

20. $\dfrac{dz}{d\theta} = \boxed{-\tan(\theta) + \dfrac{\cos(\ln(\theta))}{\theta}}$

21. $f'(3) = \boxed{\dfrac{81\pi}{4}}$

1.2 Multiple Choice Answers

1	2	3	4	5	6	7	8	9	10
E	**E**	**C**	**B**	**D**	**A**	**B**	**B**	**D**	**C**

1.3 Free Response Set A Answers

1. $y' = \boxed{-t^3\sin(t) + 3t^2\cos(t)}$

2. $y' = \boxed{-48x(2x-5)^4\left(8x^2-5\right)^{-4} + 8(2x-5)^3\left(8x^2-5\right)^{-3}}$

3. $y' = \boxed{\dfrac{\sec^3(x) - (\tan(x)-1)(\sec(x)\tan(x))}{\sec^2(x)}}$

4. $y' = \boxed{\dfrac{x^2\cos(x) - 2x\sin(x)}{x^4}}$

5. $y' = \boxed{-2x^2e^{-x^2} + e^{-x^2}}$

6. $y' = \boxed{\dfrac{\left(r^2+1\right)^{\frac{1}{2}} - r^2\left(r^2+1\right)^{-\frac{1}{2}}}{r^2+1}}$

7. $y' = \boxed{e^{x\cos(x)}\left(-x\sin(x) + \cos(x)\right)}$

8. $y' = \boxed{-3e^{-5x}\sin(3x) - 5e^{-5x}\cos(3x)}$

9. $y' = \boxed{-\dfrac{\cos\left(x^{-1}\right)}{x} + \sin\left(x^{-1}\right)}$

10. $y' = \boxed{-\dfrac{xe^{-x}}{e^{-x} + xe^{-x}}}$

11. $y' = \boxed{\dfrac{\frac{x}{|x|\sqrt{x^2-1}} - \sec^{-1}(x)}{x^2}}$

12. $y' = \boxed{\dfrac{x^3\sec(x)\tan(x) - 3x^2\sec(x)}{x^6}}$

13. $y' = \boxed{1 + 2x\tan^{-1}(x)}$

14. $y' = \boxed{\dfrac{2x}{x^2+4} - \dfrac{x}{2\left(1+\frac{x^2}{4}\right)} - \tan^{-1}\left(\dfrac{x}{2}\right)}$

15. $f'(x) = \boxed{\dfrac{1}{2\sqrt{\ln(x)}} + \sqrt{\ln(x)}}$

16. $g'(x) = \boxed{8(-2x+1)\left(3+x-x^2\right)^7 + 20(4x+1)^4\left(3+x-x^2\right)^8}$

17. $f'(x) = \boxed{\cos^{-1}(x)}$

18. $g'(x) = \boxed{-\sqrt{1-x^2} + \dfrac{x^2}{\sqrt{1-x^2}} - \dfrac{1}{\sqrt{1-x^2}}}$

19. $\boxed{\dfrac{\left(x^2-9\right)(6x+4) - \left(3x^2+4x-3\right)(2x)}{\left(x^2-9\right)^2}}$

20. $\boxed{\dfrac{(x+2)\left(3x^2-4x-5\right) - \left(x^3-2x^2-5x+6\right)}{(x+2)^2}}$

21. $\boxed{\dfrac{\left(3x^3\right)\left(5x^4-36x^2-19\right) - \left(x^5-12x^3-19x\right)\left(9x^2\right)}{9x^6}}$

22. $\boxed{\dfrac{\left(x^3+1\right)(3) - (3x)\left(3x^2\right)}{\left(x^3+1\right)^2}}$

23. $\boxed{\dfrac{\left(x^2-9x+20\right) - (x-4)(2x-9)}{\left(x^2-9x+20\right)^2}}$

24. $\boxed{\dfrac{(\sin(x))(\sec^2(x)) - (\tan(x)+5)(\cos(x))}{\sin^2(x)}}$

25. $\boxed{\dfrac{(1-\cos(x))(\cos(x)) - \sin^2(x)}{(1-\cos(x))^2}}$

26. $\boxed{\dfrac{2x\cos(x) + x^2\sin(x)}{\cos^2(x)}}$

27. $\dfrac{dy}{dx} = \boxed{\dfrac{\left(x^2-4\right)(2x) - \left(x^2-3\right)(2x)}{\left(x^2-4\right)^2}}$

28. $f'(x) = \boxed{\dfrac{\left(x^2-x-3\right)(2x+2) - \left(x^2+2x-8\right)(2x-1)}{\left(x^2-x-3\right)^2}}$

29. $y' = \boxed{\dfrac{(x-4)(2x+2) - \left(x^2+2x-3\right)(x-4)}{(x-4)^2}}$

30. $f'(x) = \boxed{\dfrac{\ln(x)-1}{\ln^2(x)}}$

31. $h'(t) = \boxed{17\left(\dfrac{1+x^2}{1-x^2}\right)^{16}\left(\dfrac{\left(1-x^2\right)(2x) - \left(1+x^2\right)(-2x)}{\left(1-x^2\right)^2}\right)}$

32. $\dfrac{dy}{dx} = \boxed{\dfrac{(\cos(x)-3)\left(\sec^2(x)\right) - (\tan(x))(-\sin(x))}{(\cos(x)-3)^2}}$

33. $f'(x) = \boxed{5\left(x\sin(2x) + \tan^4\left(x^7\right)\right)^4\left(2x\cos(2x) + \sin(2x) + 28\tan^3\left(x^7\right)\sec^2\left(x^7\right)\right)}$

34. $f'(x) = \boxed{e^x - \dfrac{x^2}{x^2+1} - 2x\arctan(x)}$

35. $f'\left(\dfrac{\pi}{4}\right) = \boxed{\dfrac{1}{2}}$

36. $y'(1) = \boxed{\dfrac{7}{2}}$

1.3 Free Response Set B Answers

1. $x'(t) = \boxed{e^{t^2}\cos\left(t^2-5t^4\right)\left(2t-20t^3\right)+2te^{t^2}\sin\left(t^2-5t^4\right)}$

2. $x'(t) = \boxed{12t^3e^{5t}\sec^2\left(3t^4\right)+5e^{5t}\tan\left(3t^4\right)}$

3. $y' = \boxed{\dfrac{(x-3)(2x+2)-(x^2+2x-15)}{(x-3)^2}}$

4. $x'(t) = \boxed{e^t\left(2t-20t^3\right)+\left(t^2-5t^4\right)e^t}$

5. $\boxed{\dfrac{\left(\sin\left(x^3\right)\right)\left(e^x+14x\right)-\left(e^x+7x^2+5\right)\left(3x^2\cos\left(x^3\right)\right)}{\sin^2\left(x^3\right)}}$

6. $\boxed{-\dfrac{e^{\sin(x)}e^x\csc^2\left(e^x\right)}{\cot\left(e^x\right)}+e^{\sin(x)}\cos(x)\ln\left(\cot\left(e^x\right)\right)}$

7. $\boxed{2x^3\cos\left(x^2\right)+2x\sin\left(x^2\right)+\dfrac{\ln x-\frac{x+1}{x}}{\ln^2 x}}$

8. $\boxed{-2x^3\sin\left(x^2\right)+2x\cos\left(x^2\right)+\dfrac{xe^x-e^x}{x^2}}$

9. $\boxed{\dfrac{5x^5}{5x+4}+5x^4\ln(5x+4)+\dfrac{\ln x-1}{\ln^2 x}}$

10. $\boxed{\dfrac{-2xe^{-5x}\sin\left(x^2-3\right)+5e^{-5x}\cos\left(x^2-3\right)}{e^{-10x}}}$

11. $\boxed{-e^{x^2}\sin(x)+2x\cos(x)}$

12. $\boxed{\dfrac{\ln(4x)\sec^2(x)-\frac{1}{x}(1+\tan(x))}{\ln^2(4x)}}$

13. $\boxed{\sin(x)\sec^2(x)+\tan(x)\cos(x)}$

14. $\boxed{\dfrac{\frac{1}{x}\csc(x)+(1+\ln x)\csc(x)\cot(x)}{\csc^2(x)}}$

15. $\boxed{e^{5x^4}\cot(x)+20x^3e^{5x^4}\ln(\sin(x))}$

16. $\boxed{5x\cos(x)+5\sin(x)+2e^{2x}-\dfrac{6x}{3x^2+1}+\dfrac{\left(x^2+1\right)-2x^2}{\left(x^2+1\right)^2}}$

17. $\boxed{\tan\left(e^x\right)\left(4x^3-15x^2+1\right)+e^x\left(x^4-5x^3+x\right)\sec^2\left(e^x\right)}$

18. $\boxed{\dfrac{5\ln(3x+7)-\frac{3(5x+2)}{3x+7}}{\ln^2(3x+7)}}$

19. $\boxed{\dfrac{\left(3x^3\right)\left(5x^4-36x^2-19\right)-\left(x^5-12x^3-19x\right)\left(9x^2\right)}{9x^6}}$

20. $\boxed{32\cos^2(4x+2)-32\sin^2(4x+2)}$

21. $g'(z) = \boxed{118\left(\dfrac{e^{5z}}{1+\ln z}\right)^{117}\left(\dfrac{(\ln z+1)\left(5e^{5z}\right)-e^{5z}\left(\frac{1}{z}\right)}{(1+\ln z)^2}\right)}$

22. $g'(t) = \boxed{15\left(\dfrac{t^2-4}{1-t^2}\right)^{14}\left(\dfrac{\left(1-t^2\right)(2t)-\left(t^2-4\right)(-2t)}{\left(1-t^2\right)^2}\right)}$

23. $y' = \boxed{\left(\dfrac{1}{\left(\frac{2e^x}{1-e^{2x}}\right)^2+1}\right)\left(\dfrac{\left(1-e^{2x}\right)\left(2e^x\right)-\left(2e^x\right)\left(-2e^{2x}\right)}{\left(1-e^{2x}\right)^2}\right)}$

24. $f'(x) = \boxed{x^2\left(-\dfrac{1}{\sqrt{1-x^2}}\right)+\arccos(x)(2x)}$

25. $\dfrac{dy}{du} = \boxed{\dfrac{2u}{u^2+1} - \dfrac{u}{u^2+1} + \cot^{-1}(u)}$

26. $\boxed{\dfrac{\frac{(x+1)-(x-1)}{(x+1)^2}}{\sqrt{1-\left(\frac{x-1}{x+1}\right)^2}}}$

27. $f'(t) = \boxed{\dfrac{1}{\sqrt{1-\left(\frac{t}{c}\right)^2}} + \dfrac{t}{\sqrt{c^2-t^2}}}$

28. $y' = \boxed{\dfrac{2}{\sqrt{1-\frac{x^2}{4}}} - \dfrac{x^2}{\sqrt{4-x^2}} + \sqrt{4-x^2}}$

29. $f'(1) = \boxed{\dfrac{7503}{5} + \ln 5}$

1.3 Multiple Choice Answers

1	2	3	4	5	6	7	8
D	**E**	**B**	**E**	**A**	**D**	**A**	**B**

1.4 Free Response Homework

1. $f''(x) = \boxed{20x^3 + 12}$

2. $h''(x) = \boxed{60x^2 + 54x - 8}$

Chapter 2:

Intro To Anti-Derivatives

Chapter 2 Overview: Anti-Derivatives

As noted in the introduction, Calculus is essentially comprised of four operations:

- Limits
- Derivatives
- Indefinite Integrals (Or Anti-Derivatives)
- Definite Integrals

As mentioned above, there are two types of integrals — the definite integral and the indefinite integral. The definite integral was explored first as a way to determine the area bounded by a curve, rather than bounded by a polygon. The summation of infinite rectangles is

$$A = \sum_{i=1}^{n} f\left(x_i\right) \cdot \Delta x,$$

and the representation

$$\int_a^b f(x)\, dx$$

is the exact amount, with $\displaystyle\int$ being an elongated and stylized s for "sum".

Newton and Leibnitz made the connection between the definite integral and the antiderivative, showing that the process of reversing the derivative results in an infinite summation. The antiderivative and indefinite integral are inverses of each other, just as squares and square roots or exponential and log functions. In this chapter, we will consider how to reverse the differentiation process. In a later chapter, we will dive deeper into the definite integral. Let's start by reviewing our derivative rules, as they will be necessary for us to take the antiderivative.

You must know the derivative rules in order to know the antiderivative rules!

The Power Rule: $\dfrac{d}{dx}\left[u^n\right] = nu^{n-1}\dfrac{du}{dx}$

The Product Rule: $\dfrac{d}{dx}\left[u \cdot v\right] = u \cdot \dfrac{dv}{dx} + v \cdot \dfrac{du}{dx}$

The Quotient Rule: $\dfrac{d}{dx}\left[\dfrac{u(x)}{v(x)}\right] = \dfrac{v \cdot \dfrac{du}{dx} - u \cdot \dfrac{dv}{dx}}{v^2}$

The Chain Rule: $\dfrac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x)$

$$\frac{d}{dx}[\sin(u)] = (\cos(u))\frac{du}{dx} \qquad \frac{d}{dx}[\csc(u)] = (-\csc(u)\cot(u))\frac{du}{dx}$$

$$\frac{d}{dx}[\cos(u)] = (-\sin(u))\frac{du}{dx} \qquad \frac{d}{dx}[\sec(u)] = (\sec(u)\tan(u))\frac{du}{dx}$$

$$\frac{d}{dx}[\tan u] = \left(\sec^2(u)\right)\frac{du}{dx} \qquad \frac{d}{dx}[\cot(u)] = \left(-\csc^2(u)\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[e^u\right] = \left(e^u\right)\frac{du}{dx} \qquad \frac{d}{dx}[\ln u] = \left(\frac{1}{u}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[a^u\right] = \left(a^u \cdot \ln u\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\log_a u\right] = \left(\frac{1}{u \cdot \ln a}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[\sin^{-1}(u)\right] = \left(\frac{1}{\sqrt{1-u^2}}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\csc^{-1}(u)\right] = \left(\frac{-1}{|u|\sqrt{u^2-1}}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[\cos^{-1}(u)\right] = \left(\frac{-1}{\sqrt{1-u^2}}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\sec^{-1}(u)\right] = \left(\frac{1}{|u|\sqrt{u^2-1}}\right)\frac{du}{dx}$$

$$\frac{d}{dx}\left[\tan^{-1}(u)\right] = \left(\frac{1}{u^2+1}\right)\frac{du}{dx} \qquad \frac{d}{dx}\left[\cot^{-1}(u)\right] = \left(\frac{-1}{u^2+1}\right)\frac{du}{dx}$$

2.1: The Anti-Power Rule

As we have seen, we can deduce things about a function if its derivative is known. It would be valuable to have a formal process to determine the original function from its derivative accurately. The process is called antidifferentiation, or integration.

Symbol for the Integral

$\int f(x)\,dx$ "the integral of f of x, d-x"

The dx is called the differential. For now, we will treat it as part of the integral symbol. It tells us the independent variable of the function [usually, but not always, x]. It does have a meaning on its own, but we will explore that later.

Looking at the integral as an antiderivative, we should be able to figure out the basic process. Remember:

$$\frac{d}{dx}\left[x^n\right] = nx^{n-1}$$

and

$$\frac{d}{dx}[\text{constant}] = 0$$

It follows that if we are starting with the derivative and want to reverse the process, the power must increase by one and we should divide by this new power. Formally,

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \text{ for } n \neq -1$$

The $+C$ is to account for any constant that might've been in the equation before the derivative was taken. Note that $n = -1$ does not work with this rule because it results in a division by zero. However, we know from our derivative rules that the derivative of $\ln x$ yields x^{-1}. Therefore, we can append our anti-power rule.

The Complete Anti-Power Rule

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \text{ for } n \neq -1$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

Since $\dfrac{d}{dx}[f(x)+g(x)] = \dfrac{d}{dx}[f(x)] + [g(x)]$ and $\dfrac{d}{dx}[cx^n] = c\dfrac{d}{dx}[x^n]$, it follows that:

$$\int (f(x)+g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx$$

$$\int c(f(x))\,dx = c\int f(x)\,dx$$

These allow us to integrate a polynomial by integrating each term separately.

OBJECTIVES

Find the Anti-Derivative of a Polynomial.
Integrate Functions Using Transcendental Operations
Use Integration to Solve Rectilinear Motion Problems

Ex 2.1.1: $\displaystyle\int \left(3x^2+4x+5\right)\,dx$

Sol 2.1.1:

$$\int \left(3x^2+4x+5\right)\,dx = 3\frac{x^{2+1}}{2+1} + 4\frac{x^{1+1}}{1+1} + 5\frac{x^{0+1}}{0+1} + C$$

$$= \frac{3x^3}{3} + \frac{4x^2}{2} + \frac{5x^1}{1} + C$$

$$= \boxed{x^3+2x^2+5x+C}$$

Ex 2.1.2: $\displaystyle\int \left(x^4+4x^2+5+\frac{1}{x}-\frac{1}{x^5}\right)\,dx$

Sol 2.1.2:

$$\int \left(x^4+4x^2+5+\frac{1}{x}-\frac{1}{x^5}\right)\,dx = \frac{x^{4+1}}{4+1} + \frac{4x^{2+1}}{2+1} + \frac{5x^{0+1}}{0+1} + \ln|x| - \frac{x^{-5+1}}{-5+1} + C$$

$$= \boxed{\frac{1}{5}x^5 + \frac{4}{3}x^3 + 5x + \ln|x| + \frac{1}{4x^4} + C}$$

Ex 2.1.3: $\displaystyle\int \left(x^2 + \sqrt[3]{x} - \frac{4}{x}\right)\,dx$

Sol 2.1.3:

$$\begin{aligned}\int \left(x^2 + \sqrt[3]{x} - \frac{4}{x}\right)\,dx &= \int \left(x^2 + x^{\frac{1}{3}} - \frac{4}{x}\right)\,dx \\ &= \frac{x^{2+1}}{2+1} + \frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1} - 4\ln|x| + C \\ &= \boxed{\frac{1}{3}x^3 - \frac{3}{4}x^{\frac{4}{3}} + 4\ln|x| + C}\end{aligned}$$

Integrals of products and quotients can be done easily IF they can be turned into a polynomial.

Ex 2.1.4: $\displaystyle\int \left(x^2 + \sqrt[3]{x}\right)(2x+1)\,dx$

Sol 2.1.4:

$$\begin{aligned}\int \left(x^2 + \sqrt[3]{x}\right)(2x+1)\,dx &= \int \left(2x^3 + 2x^{\frac{4}{3}} + x^2 + x^{\frac{1}{3}}\right)\,dx \\ &= \frac{2x^4}{4} + \frac{2x^{\frac{7}{3}}}{\frac{7}{3}} + \frac{x^3}{3} + \frac{x^{\frac{4}{3}}}{\frac{4}{3}} \\ &= \boxed{\frac{1}{2}x^4 + \frac{6}{7}x^{\frac{7}{3}} + \frac{1}{3}x^3 + \frac{3}{4}x^{\frac{4}{3}} + C}\end{aligned}$$

The next example is called an initial value problem. It has an ordered pair (or initial value pair) that allows us to solve for C.

Ex 2.1.5: $f'(x) = 4x^3 - 6x + 3$. Find $f(x)$ if $f(0) = 13$.

Sol 2.1.5:

$$\begin{aligned}f(x) &= \int \left(4x^3 - 6x + 3\right)\,dx \\ &= x^4 - 3x^2 + 3x + C\end{aligned}$$

$$f(0) = 0^4 - 3(0)^2 + 3(0) + C$$

$$= 13 \therefore C = 13$$

$$\therefore \boxed{f(x) = x^4 - 3x^2 + 3x + 13}$$

Now, let's take a look at a type of problem called a *rectilinear motion* problem. In these problems, we study the motion of an object moving along a straight line—its position, velocity, and acceleration.

Ex 2.1.6: The acceleration of particle is described by $a(t) = 3t^2 + 8t + 1$. Find the distance equation for $x(t)$ if $v(0) = 3$ and $a(0) = 1$.

Sol 2.1.6:

$$v(t) = \int a(t)\,dt$$

$$= \int \left(3t^2 + 8t + 1\right)\,dt$$

$$= t^3 + 4t^2 + t + C_1$$

$$3 = (0)^3 + 4(0)^2 + (0) + C_1 \therefore 3 = C_1$$

$$v(t) = t^3 + 4t^2 + t + 3$$

$$x(t) = \int v(t)\,dt$$

$$= \int \left(t^3 + 4t^2 + t + 3\right)\,dt$$

$$= \frac{1}{4}t^4 + \frac{4}{3}t^3 + \frac{1}{2}t^2 + 3t + C_2$$

$$1 = \frac{1}{4}(0)^4 + \frac{4}{3}(0)^3 + \frac{1}{2}(0)^2 + 3(0) + C_2 \therefore 1 = C_2$$

$$\boxed{x(t) = \frac{1}{4}t^4 + \frac{4}{3}t^3 + \frac{1}{2}t^2 + 3t + 1}$$

Ex 2.1.7: The acceleration of a particle is described by $a(t) = 12t^2 - 6t + 4$. Find the distance equation for $x(t)$ if $v(1) = 0$ and $x(1) = 3$.

Sol 2.1.7:

$$v(t) = \int a(t)\,dt$$

$$= \int \left(12t^2 - 6t + 4\right)$$

$$= 4t^3 - 3t^2 - 4t + C_1$$

$$0 = 4(1)^3 - 3(1)^2 + 4(1) + C_1 \therefore -5 = C_1$$

$$v(t) = 4t^3 - 3t^2 - 4t - 5$$

$$x(t) = \int v(t)\,dt$$

$$= \int \left(4t^3 - 3t^2 - 4t - 5\right)\,dt$$

$$= t^4 - t^3 - 2t^2 - 5t + C_2$$

$$3 = (1)^4 - (1)^3 - 2(1)^2 - 5(1) + C_2 \therefore 6 = C_2$$

$$\boxed{x(t) = t^4 - t^3 - 2t^2 - 5t + 6}$$

The proof of all the transcendental integral rules can be left to a more formal Calculus course. But, since the integral is the inverse of the derivative, the discovery of the rules should be obvious from looking at the comparable derivative rules.

Transcendental Integral Rules

$$\int \cos(u)\,du = \sin(u) + C \qquad \int \csc(u)\cot(u)\,du = -\csc(u) + C$$

$$\int \sin(u)\,du = -\cos(u) + C \qquad \int \sec(u)\tan(u)\,du = \sec(u) + C$$

$$\int \sec^2(u)\,du = \tan(u) + C \qquad \int \csc^2(u)\,du = -\cot(u) + C$$

$$\int e^u\,du = e^u + C \qquad \int \frac{1}{u}\,du = \ln|u| + C$$

$$\int a^u\,du = \frac{a^u}{\ln|a|} + C \qquad \int \frac{1}{\sqrt{1-u^2}}\,du = \sin^{-1}(u) + C$$

$$\int \frac{1}{1+u^2}\,du = \tan^{-1}(u) + C \qquad \int \frac{1}{u\sqrt{u^2-1}}\,du = sec^{-1} + C$$

Note that there are only three integrals that yield inverse trig functions, but there were six inverse trig derivatives. This is because the other three derivative rules are just the negatives of the first three.

Ex 2.1.8: $\int (\sin(x) + 3\cos(x))\ dx$

Sol 2.1.8:

$$\begin{aligned}\int (\sin(x) + 3\cos(x))\ dx &= \int \sin(x)\,dx + 3\int \cos(x)\,dx \\ &= \boxed{-\cos(x) + 3\sin(x) + C}\end{aligned}$$

Ex 2.1.9: $\int \left(e^x + 4 + 3\csc^2(x)\right)\ dx$

Sol 2.1.9:

$$\int \left(e^x + 4 + 3\csc^2(x)\right)\ dx = \int e^x\,dx + 4\int\ dx + 3\int \csc^2(x)\,dx$$

$$= \boxed{e^x + 4x - 3\cot(x) + C}$$

Now, let's take a look at some more complex integrals that yield inverse trig functions. These more general forms extend the earlier rules by introducing a constant a, and they are especially useful when working with substitutions or integrals that don't simplify neatly to the unit case.

Trig Inverse Integral Rules

$$\int \frac{1}{\sqrt{a^2 - u^2}}\, du = \sin^{-1}\left(\frac{u}{a}\right) + C$$

$$\int \frac{1}{u^2 + a^2}\, du = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$$

$$\int \frac{1}{u\sqrt{u^2 - a^2}}\, du = \frac{1}{a}\sec^{-1}\left(\frac{u}{a}\right) + C$$

Ex 2.1.10: Find $\displaystyle\int \frac{1}{x^2 + 4}\, dx$

Sol 2.1.10: All we need to do is apply our formula above.

$$\int \frac{1}{x^2 + 4}\, dx = \boxed{\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C}$$

Ex 2.1.11: If $\dfrac{dy}{dx} = \sec(x)(\sec(x)\tan(x))$, find $y(x)$ if $y(0) = 0$.

Sol 2.1.11:

$$\begin{aligned} y &= \int (\sec(x)(\sec(x)\tan(x)))\, dx \\ &= \int \left(\sec^2(x)\right) dx + \int (\sec(x)\tan(x))\, dx \\ &= \tan(x) + \sec(x) + C \end{aligned}$$

$$0 = \tan(0) + \sec(0) + C$$

$$0 = 0 + 1 + C \therefore C = -1$$

$$\boxed{y = \tan(x) + \sec(x) - 1}$$

2.1 Free Response Homework

Perform the antidifferentiation.

1. $\displaystyle\int \left(6x^2 - 2x + 3\right)\, dx$

2. $\displaystyle\int \left(x^3 + 3x^2 - 2x + 4\right)\, dx$

3. $\displaystyle\int \frac{2}{\sqrt[3]{x}}\, dx$

4. $\displaystyle\int \left(8x^4 - 4x^3 + 9x^2 + 2x + 1\right)\, dx$

5. $\displaystyle\int x^3\left(4x^2 + 5\right)\, dx$

6. $\displaystyle\int (4x-1)(3x+8)\, dx$

7. $\displaystyle\int \left(\sqrt{x} - \frac{6}{\sqrt{x}}\right)\, dx$

8. $\displaystyle\int \frac{x^2 + \sqrt{x} + 3}{x}\, dx$

9. $\displaystyle\int (x+1)^3\, dx$

10. $\displaystyle\int (4x-3)^2\, dx$

11. $\displaystyle\int \left(\sqrt{x} + 3\sqrt{3} - \frac{6}{\sqrt{x}}\right)\, dx$

12. $\displaystyle\int \frac{4x^3 + \sqrt{x} + 3}{x^2}\, dx$

13. $\displaystyle\int \left(x^2 + 5x + 6\right)\, dx$

14. $\displaystyle\int \frac{x^2 - 4x + 7}{x}\, dx$

15. $\displaystyle\int \frac{x^5 - 7x^3 + 2x - 9}{2x}\, dx$

16. $\displaystyle\int \frac{x^3 + 3x^2 + 3x + 1}{x+1}\, dx$

17. $\displaystyle\int \left(y^2 + 5\right)^2\, dy$

18. $\displaystyle\int \left(4t^2 + 1\right)\left(3t^3 + 7\right)\, dt$

Complete the following problems.

19. $f'(x) = 3x^2 - 6x + 3$. Find $f(x)$ if $f(0) = 2$.

20. $f'(x) = x^3 + x^2 - x + 3$. Find $f(x)$ if $f(1) = 0$.

21. $f'(x) = (\sqrt{x} - 2)(3\sqrt{x} + 1)$. Find $f(x)$ if $f(4) = 1$.

22. The acceleration of a particle is described by $a(t) = 36t^2 - 12t + 8$. Find the distance equation for $x(t)$ if $v(1) = 1$ and $x(1) = 3$.

23. The acceleration of a particle is described by $a(t) = t^2 - 2t + 4$. Find the distance equation for $x(t)$ if $v(0) = 2$ and $x(0) = 4$.

2.1 Multiple Choice Homework

1. $\int \frac{1}{x^2}\, dx =$

a) $\ln\left(x^2\right) + C$ b) $-\ln\left(x^2\right) + C$ c) $\frac{1}{x} + C$ d) $-\frac{1}{x} + C$ e) $-\frac{2}{x^3} + C$

---

2. $\int x\left(10 + 8x^4\right)\, dx =$

a) $5x^2 + \frac{4}{3}x^6 + C$ b) $5x^2 + \frac{8}{5}x^5 + C$ c) $10x + \frac{4}{3}x^6 + C$

d) $5x^2 + 8x^6 + C$ e) $5x^2 + \frac{8}{7}x^6 + C$

---

3. $\int x\sqrt{3x}\, dx$

a) $\frac{2\sqrt{3}}{5}x^{\frac{5}{2}} + C$ b) $\frac{5\sqrt{3}}{2}x^{\frac{5}{2}} + C$ c) $\frac{\sqrt{3}}{2}x^{\frac{1}{2}} + C$ d) $2\sqrt{3x} + C$ e) $\frac{5\sqrt{3}}{2}x^{\frac{3}{2}} + C$

---

4. $\int (x-1)\sqrt{x}\, dx =$

a) $\frac{3}{2}\sqrt{x} - \frac{1}{\sqrt{x}} + C$ b) $\frac{2}{3}x^{\frac{2}{3}} + \frac{1}{2}x^{\frac{1}{2}} + C$ c) $\frac{1}{2}x^2 - x + C$

d) $\frac{2}{5}x^{\frac{5}{2}} - \frac{2}{3}x^{\frac{3}{2}} + C$ e) $\frac{1}{2}x^2 + 2x^{\frac{2}{3}} + C$

---

5. A particle is moving upward along the y-axis until it reaches the origin and then it moves downward such that $v(t) = 8 - 2t$ for $t \geq 0$. The position of the particle at time t is given by

a) $y(t) = -t^2 + 8t - 16$ b) $y(t) = -t^2 + 8t + 16$ c) $y(t) = 2t^2 - 8t - 16$

d) $8t - t^2$ e) $8t - 2t^2$

---

6. If a particle's acceleration is given by $a(t) = 12t + 4$, and $v(1) = 5$ and $y(0) = 2$, then

$y(2) =$

a) 20 b) 10 c) 4 d) 16 e) 12

2.2: Integration by U-Substitution

Reversing the Power Rule was fairly easy. The other three core derivative rules—the Product Rule, the Quotient Rule, and the Chain Rule—are a little more complicated to undo. This is because they yield a more complicated function as a derivative, one which has several algebraic simplification steps. The integral of a rational function is particularly difficult to unravel because, as we have seen, rational derivatives can be obtained by differentiating a composite function with a log or a radical, or by differentiating another rational function. The same goes for reversing the Product Rule.

Key Idea: There is no single Product or Quotient Rule for integrals.

Instead, there are several techniques that apply in different situations, and it is not always obvious at the outset which one will be most effective. The choice depends on the algebraic manipulations that produced the product or quotient in the first place.

Products can be a result of:	Quotients can be the result of:
• The Chain Rule • Differentiating a product • Differentiating some trig functions	• Common denominators • Differentiating a quotient • Differentiating a log with a composite function • Differentiating some trig inverse functions

Composite functions are among the most pervasive functions in math. Therefore, we will start with undoing products and quotients that involve composites.

Remember:

$$\text{The Chain Rule: } \frac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x)$$

The derivative of a composite function often becomes a product of two functions: one part still composite and the other not. So, when we see a product in an integral, it may have originated from the Chain Rule. Unlike differentiation, however, integration in this case does not follow a fixed formula. Instead, it involves a process of substitution that sometimes works and sometimes does not. We make an informed guess and check whether it simplifies the integral. In later parts of Calculus, you will learn additional techniques to use when this approach is not successful.

Steps to Integration by U-Substitution

1. Make sure that you are integrating a product or quotient.

2. Identify the inner function of the composite and set u equal to it.

3. Differentiate u to find du in terms of dx.
4. Adjust the integral by multiplying and dividing by a constant if needed so a factor matches du. [See **Ex 2.2.2**]
5. Rewrite the integral entirely in terms of u and du.
6. Integrate using the power rule or other appropriate rules.
7. Substitute back the original x-expression for u.

This is one of those mathematical processes that makes little sense when first seen. But, after seeing several examples, the meaning should become clear. *Be patient!*

OBJECTIVES

Use U-Substitution to Integrate Composite Expressions.

Ex 2.2.1: $\displaystyle\int 3x^2\left(x^3+5\right)^{10}\,dx$

Sol 2.2.1:

$\left(x^3+5\right)$ is the inner function.

$\hookrightarrow u = x^3+5$

$\hookrightarrow du = 3x^2\,dx$

$$\begin{aligned}\int 3x^2\left(x^3+5\right)^{10}\,dx &= \int u^{10}\,du\\ &= \frac{u^{11}}{11}+C\\ &= \boxed{\frac{1}{11}\left(x^3+5\right)^{11}}\end{aligned}$$

Ex 2.2.2: $\displaystyle\int x\left(x^2+5\right)^{3}\,dx$

Sol 2.2.2:

$$\left(x^2+5\right) \text{ is the inner function.}$$

$$\hookrightarrow\ u = x^2+5$$

$$\hookrightarrow\ du = 2x\,dx$$

$$\begin{aligned}\int x\left(x^2+5\right)^3\,dx &= \frac{1}{2}\int (2x)\left(x^2+5\right)^3\,dx \\ &= \frac{1}{2}\int u^3\,du \\ &= \frac{1}{2}\cdot\frac{u^4}{4}+C \\ &= \boxed{\frac{1}{8}\left(x^2+5\right)^4+C}\end{aligned}$$

Notice how the factor of 2 from $du = 2x\,dx$ is accounted for by multiplying by $\frac{1}{2}$ when substituting. This ensures the integral is correctly expressed in terms of u.

Ex 2.2.3: $\int \left(x^3+x\right)\sqrt[4]{x^4+2x^2-5}\,dx$

Sol 2.2.3:

$$\sqrt[4]{x^4+2x^2-5} \text{ is the inner function.}$$

$$\hookrightarrow\ u = x^4+2x^2-5$$

$$\hookrightarrow\ du = \left(4x^3+4x\right)\,dx = 4\left(x^3+x\right)\,dx$$

$$\begin{aligned}\int \left(x^3+x\right)\sqrt[4]{x^4+2x^2-5}\,dx &= \frac{1}{4}\int 4\left(x^3+x\right)\sqrt[4]{x^4+2x^2-5}\,dx\\ &= \frac{1}{4}\int \sqrt[4]{u}\,du\\ &= \frac{1}{4}\cdot\frac{4u^{\frac{5}{4}}}{5}+C\\ &= \boxed{\frac{1}{5}\left(x^4+2x^2-5\right)^{\frac{5}{4}}+C}\end{aligned}$$

Ex 2.2.4: $\displaystyle\int \frac{3x^2+4x-5}{\left(x^3+2x^2-5x+2\right)^3}\,dx$

Sol 2.2.4:

x^3+2x^2-5x+2 is the inner function.

$\hookrightarrow\ u = x^3+2x^2-5x+2$

$\hookrightarrow\ du = \left(3x^2+4x-5\right)\,dx$

$$\begin{aligned}\int \frac{3x^2+4x-5}{\left(x^3+2x^2-5x+2\right)^3}\,dx &= \int \frac{1}{u^3}\,du\\ &= -\frac{1}{2}u^{-2}+C\\ &= \boxed{-\frac{1}{2}\left(x^3+2x^2-5x+2\right)^{-2}+C}\end{aligned}$$

Of course, u-substitution will apply to the transcendental functions as well.

Ex 2.2.5: $\displaystyle\int \sin(5x)\,dx$

Sol 2.2.5:

$\hookrightarrow\ u = 5x$

$\hookrightarrow\ du = 5\,dx$

$$\int \sin(5x)\,dx = \frac{1}{5}\int 5\sin(5x)\,dx$$
$$= \frac{1}{5}\int \sin(u)\,du$$
$$= \frac{1}{5}\cdot(-\cos(u)) + C$$
$$= \boxed{-\frac{1}{5}\cos(5x) + C}$$

Ex 2.2.6: $\int \sin^6(x)\cos(x)\,dx$

Sol 2.2.6:

$\hookrightarrow u = \sin(x)$

$\hookrightarrow du = \cos(x)\,dx$

$$\int \sin^6(x)\cos(x)\,dx = \int u^6\,du$$
$$= \frac{1}{7}u^7 + C$$
$$= \boxed{\frac{1}{7}\sin^7(x) + C}$$

Ex 2.2.7: $\int x^5 \sin\left(x^6\right)\,dx$

Sol 2.2.7:

$\hookrightarrow u = x^6$

$\hookrightarrow du = 6x^5\,dx$

$$
\begin{aligned}
\int x^5 \sin\left(x^6\right)\, dx &= \frac{1}{6}\int 6x^5 \sin\left(x^6\right)\, dx \\
&= \frac{1}{6}\int \sin(u)\, du \\
&= -\frac{1}{6}\cos(u) + C \\
&= \boxed{-\frac{1}{6}\cos\left(x^6\right) + C}
\end{aligned}
$$

Ex 2.2.8: $\int \cot^3(x)\csc^2(x)\, dx$

Sol 2.2.8

$\hookrightarrow\ u = \cot(x)$

$\hookrightarrow\ du = -\csc^2(x)\, dx$

$$
\begin{aligned}
\int \cot^3(x)\csc^2(x)\, dx &= -\int \cot^3(x)\left(-\csc^2(x)\right)\, dx \\
&= -\int u^3\, du \\
&= -\frac{1}{4}u^4\, du + C \\
&= \boxed{-\frac{1}{4}\cot^4(x) + C}
\end{aligned}
$$

Ex 2.2.9: $\int \frac{\cos(\sqrt{x})}{\sqrt{x}}\, dx$

Ex 2.2.9:

$\hookrightarrow\ u = \sqrt{x}$

$\hookrightarrow\ du = \frac{1}{2}x^{-\frac{1}{2}}$

$$\int \frac{\cos(\sqrt{x})}{\sqrt{x}}\,dx = 2\int(\cos(\sqrt{x}))\left(\frac{1}{2}x^{-\frac{1}{2}}\right)\,dx$$

$$= 2\int \cos(u)\,du$$

$$= 2\sin(u) + C$$

$$= \boxed{2\sin(\sqrt{x}) + C}$$

Ex 2.2.10: $\int xe^{x^2+1}\,dx$

Sol 2.2.10:

$$\hookrightarrow\; u = x^2 + 1$$

$$\hookrightarrow\; du = 2x\,dx$$

$$\int xe^{x^2+1}\,dx = \frac{1}{2}\int (2x)e^{x^2+1}\,dx$$

$$= \frac{1}{2}\int e^u\,du$$

$$= \frac{1}{2}e^u + C$$

$$= \boxed{\frac{1}{2}e^{x^2+1} + C}$$

Ex 2.2.11: $\int \frac{x}{\sqrt{1-x^4}}\,dx$

Sol 2.2.11:

$$\hookrightarrow\; u = x^2$$

$$\hookrightarrow\; du = 2x\,dx$$

$$\begin{aligned}
\int \frac{x}{\sqrt{1-x^4}}\,dx &= \frac{1}{2}\int (2x)\frac{1}{\sqrt{1-(x^2)^2}}\,dx \\
&= \frac{1}{2}\int \frac{1}{\sqrt{1-u^2}}\,du \\
&= \frac{1}{2}\sin^{-1}(u) + C \\
&= \boxed{\frac{1}{2}\sin^{-1}\left(x^2\right) + C}
\end{aligned}$$

Ex 2.2.12: $\int \left(xe^{x^2} + 4x^2 - 3\sin(5x)\right)\,dx$

Ex 2.2.12:

$$\int \left(xe^{x^2} + 4x^2 - 3\sin(5x)\right)\,dx = \int xe^{x^2}\,dx + \int 4x^2\,dx - \int 3\sin(5x)\,dx$$

$$\hookrightarrow u_1 = x^2 \qquad\qquad \hookrightarrow u_2 = 5x$$

$$\hookrightarrow du_1 = 2x\,dx \qquad\qquad \hookrightarrow du_2 = 5\,dx$$

$$\begin{aligned}
\int xe^{x^2}\,dx + \int 4x^2\,dx - \int 3\sin(5x)\,dx &= \frac{1}{2}\int (2x)e^{x^2}\,dx + 4\int x^2\,dx - \frac{3}{5}\int 5\sin(5x)\,dx \\
&= \frac{1}{2}\int e^{u_1}\,du_1 + 4\int x^2\,dx - \frac{3}{5}\int \sin\left(u_2\right)\,du_2 \\
&= \frac{1}{2}e^{u_1} + 4\frac{x^3}{3} - \frac{3}{5}(-\cos(u_2)) + C \\
&= \boxed{\frac{1}{2}e^{x^2} + \frac{4}{3}x^3 + \frac{3}{5}\cos(5x) + C}
\end{aligned}$$

2.2 Free Response Homework Set A

Perform the antidifferentiation.

1. $\int (5x+3)^3 \, dx$

2. $\int \left(x^3\left(x^4+5\right)\right)^{24} \, dx$

3. $\int \left(1+x^3\right)^2 \, dx$

4. $\int (2-x)^{\frac{2}{3}} \, dx$

5. $\int x\sqrt{2x^2+3}\, dx$

6. $\int \frac{1}{(5x+2)^3} \, dx$

7. $\int \frac{x^3}{\sqrt{1+x^4}} \, dx$

8. $\int \frac{x+1}{\sqrt[3]{x^2+2x+3}} \, dx$

9. $\int \left(x^5 - \sin(3x) + xe^{x^2}\right) \, dx$

10. $\int \left(x^2 \sec^2\left(x^3\right) + \frac{\ln^3 x}{x}\right) \, dx$

11. $\int x^4 \cos\left(x^5\right) \, dx$

12. $\int \sin(7x+1)\, dx$

13. $\int \sec^2(3x-1)\, dx$

14. $\int \frac{\sin(\sqrt{x})}{\sqrt{x}} \, dx$

15. $\int \tan^4(x)\sec^2(x)\, dx$

16. $\int \frac{\ln x}{x} \, dx$

17. $\int e^{6x} \, dx$

18. $\int \frac{\cos(2x)}{\sin^3(2x)} \, dx$

19. $\int \frac{x\ln\left(x^2+1\right)}{x^2+1} \, dx$

20. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx$

21. $\int \sqrt{\cot(x)}\csc^2(x)\, dx$

22. $\int \frac{1}{x^2} \sin\left(\frac{1}{x}\right) \cos\left(\frac{1}{x}\right) \, dx$

23. $\int \frac{x}{1+x^4} \, dx$

24. $\int \frac{\cos(x)}{\sqrt{1-\sin^2(x)}} \, dx$

2.2 Free Response Homework Set B

Perform the antidifferentiation.

1. $\int (2x+5)\left(x^2+5x+6\right)^6 \, dx$

2. $\int 3t^2\left(t^3+1\right)^5 \, dt$

3. $\int \frac{10m+15}{\sqrt[4]{m^2+3m+1}} \, dm$

4. $\int \frac{3x^2}{\left(1+x^3\right)^5} \, dx$

5. $\int (4s+1)^5 \, ds$

6. $\int \frac{5t}{t^2+1} \, dt$

7. $\int \frac{3m^2}{m^3+8} \, dm$

8. $\int (181x+1)^5 \, dx$

9. $\int \frac{v^2}{5-v^3} \, dv$

10. $\int \left(x^7 - \cot(5x) + xe^{x^2}\right) \, dx$

11. $\int \frac{\cos(x)}{1+\sin(x)} \, dx$

12. $\int \left(x^2 \sec^2\left(4x^3\right) + 2xe^{x^2}\right) \, dx$

13. $\int \sec^2(2x) \, dx$

14. $\int \frac{\csc^2\left(e^{-x}\right)}{e^x} \, dx$

15. $\int \frac{\sec(\ln x)\tan(\ln x)}{3x} \, dx$

16. $\int \left(x^5 + \frac{7}{x^2} - e^{2x} + \sec^2(x)\right) \, dx$

17. $\int e^x \csc\left(e^x\right) \cot\left(e^x\right) \, dx$

18. $\int \left(e^x - 2\right)\left(e^x - 1\right) \, dx$

19. $\int x^2 \sin\left(x^3\right) \, dx$

20. $\int te^{5t^2+1} \, dt$

21. $\int \left(e^y + 1\right)^2 \, dy$

22. $\int x \sec^2\left(x^2\right) \sqrt{\tan\left(x^2\right)} \, dx$

23. $\int \sin(3t)\cos^5(3t) \, dt$

24. $\int x \cos\left(x^2\right) e^{\sin\left(x^2\right)} \, dx$

25. $\int \tan(\theta)\ln(\sec(\theta)) \, d\theta$

26. $\int \left(e^{4y} + 2y^2 - 7\cos(3y)\right) \, dy$

27. $\int \frac{\sin(x+4)}{\cos^7(x+4)} \, dx$

28. $\int \left(\frac{2x}{x^2+5} - \sec^2(3x) + xe^{x^2} - \pi\right) \, dx$

29. $\int e^{2t} \sec^2\left(e^{2t}\right) \, dt$

30. $\int \frac{18\ln m}{m} \, dm$

31. $\int \sec^2(y)\tan^5(y)$. Verify your answer by taking the derivative.

32. $\int \left(\cos(\theta)e^{\sin(\theta)} + \frac{\theta}{\theta^2+1}\right) \, d\theta$. Verify your answer by taking the derivative.

33. $\int t \sec^2\left(4t^2\right) \sqrt{\tan\left(4t^2\right)} \, dt$. Verify your answer by taking the derivative.

34. $\int \frac{2y\cos\left(y^2\right)}{\sin^4\left(y^2\right)} \, dy$. Verify your answer by taking the derivative.

2.2 Multiple Choice Homework

1. $\displaystyle\int \frac{x}{x^2-4}\,dx =$

a) $-\dfrac{1}{4\left(x^2-4\right)^2}+C$ b) $\dfrac{1}{2\left(x^2-4\right)}+C$ c) $\dfrac{1}{2}\ln\left|x^2-4\right|+C$

d) $2\ln\left|x^2-4\right|+C$ e) $\dfrac{1}{2}\arctan\left(\dfrac{x}{2}\right)+C$

---

2. $\displaystyle\int \frac{e^{\sqrt{x}}}{2\sqrt{x}}\,dx =$

a) $\ln\left(\sqrt{x}\right)+C$ b) $x+C$ c) e^x+C d) $\dfrac{1}{2}e^{2\sqrt{x}}+C$ e) $e^{\sqrt{x}}+C$

---

3. When using the substitution $u=\sqrt{1+x}$, an antiderivative of $\displaystyle\int 60x\sqrt{1+x}\,dx$ is.

a) $20u^3-60u+C$ b) $15u^4-30u^2+C$ c) $30u^4-60u^2+C$

d) $24u^5-40u^3+C$ e) $12u^6-20u^4+C$

---

4. $\displaystyle\int \frac{3x^2}{\sqrt{x^3+3}}\,dx =$

a) $2\sqrt{x^3+3}+C$ b) $\dfrac{3}{2}\sqrt{x^3+3}+C$ c) $\sqrt{x^3+3}+C$

d) $\ln\left(\sqrt{x^3+3}\right)+C$ e) $\ln\left(x^3+3\right)+C$

---

5. $\displaystyle\int x\left(x^2-1\right)^4\,dx =$

a) $\dfrac{1}{10}x^2\left(x^2-1\right)^5+C$ b) $\dfrac{1}{10}\left(x^2-1\right)^5+C$ c) $\dfrac{1}{5}\left(x^3-x\right)^5+C$

d) $\dfrac{1}{5}\left(x^2-1\right)^5+C$ e) $\dfrac{1}{5}\left(x^2-x\right)^5+C$

---

6. $\displaystyle\int 4x^2\sqrt{3+x^3}\,dx =$

a) $\dfrac{16\left(3+x^3\right)^{\frac{3}{2}}}{9}+C$ b) $\dfrac{8\left(3+x^3\right)^{\frac{3}{2}}}{9}+C$ c) $\dfrac{8\left(3+x^3\right)^{\frac{3}{2}}}{3}+C$

d) $\dfrac{4}{3\left(3+x^3\right)^{\frac{1}{2}}}+C$ e) $\dfrac{8}{3\left(3+x^3\right)^{\frac{1}{2}}}+C$

---

7. $\displaystyle\int \left(x^3+2+\frac{1}{x^2+1}\right)dx =$

a) $\dfrac{x^4}{4}+2x+\tan^{-1}(x)+C$ b) $x^4+2+\tan^{-1}(x)+C$ c) $\dfrac{x^4}{4}+2x+\dfrac{3}{x^3+3}+C$

d) $\dfrac{x^4}{4}+2x+\tan^{-1}\left(2x^2\right)+C$ e) $4+2x+\tan^{-1}(x)+C$

---

8. $\displaystyle\int \cos(3-2x)\,dx =$

a) $\sin(3-2x)+C$ b) $-\sin(3-2x)+C$ c) $\dfrac{1}{2}\sin(3-2x)+C$

d) $-\sin(3-2x)+C$ e) $-\dfrac{1}{5}\sin(3-2x)+C$

---

9. $\displaystyle\int \frac{x-2}{x-1}\,dx =$

a) $-\ln|x-1|+C$ b) $x+\ln|x-1|+C$ c) $x-\ln|x-1|+C$

d) $x+\sqrt{x-1}+C$ e) $x-\sqrt{x-1}+C$

---

10. $\displaystyle\int \frac{e^{x^2}-2x}{e^{x^2}}\,dx =$

a) $x-e^{x^2}+C$ b) $x-e^{-x^2}+C$ c) $x+e^{-x^2}+C$ d) $-e^{x^2}+C$ e) $-e^{-x^2}+C$

---

11. $\displaystyle\int 6\sin(x)\cos^2(x)\,dx =$

a) $2\sin^3(x)+C$ b) $-2\sin^3(x)+C$ c) $2\cos^3(x)+C$

d) $-2\cos^3(x)+C$ e) $3\sin^2(x)\cos^2(x)+C$

---

12. $\displaystyle\int \frac{4x}{1+x^2}\,dx =$

a) $4\arctan(x)+C$ b) $\dfrac{4}{x}\arctan(x)+C$ c) $\dfrac{1}{2}\ln\left(1+x^2\right)+C$

d) $2\ln\left(1+x^2\right)+C$ e) $2x^2+4\ln|x|+C$

---

13. $\displaystyle\int \frac{x}{4+x^2}\,dx =$

a) $\tan^{-1}\left(\dfrac{x}{2}\right)+C$ b) $\ln\left(4+x^2\right)+C$ c) $\tan^{-1}(x)+C$

d) $\dfrac{1}{2}\ln\left(4+x^2\right)+C$ e) $\dfrac{1}{2}\tan^{-1}\left(\dfrac{x}{2}\right)+C$

---

14. $\displaystyle\int \left(2^x-4e^{2\ln x}\right)\,dx =$

a) $2^x\ln 2-\dfrac{4}{3}e^{2\ln x}+C$ b) $x2^{x-1}-\dfrac{4}{3}x^3+C$ c) $\dfrac{2^x}{\ln 2}-\dfrac{4}{3}e^{2\ln x}+C$

d) $x2^{x-1}-\dfrac{4}{3}e^{2\ln x}+C$ e) $\dfrac{2^x}{\ln 2}-\dfrac{4}{3}x^3+C$

---

15. The antiderivative of $2\tan(x)$ is:

a) $2\ln|\sec(x)|+C$ b) $2\sec^2(x)+C$ c) $\ln|\sec^2(x)|+C$

d) $2\ln|\cos(x)|+C$ e) $\ln|2\sec(x)|+C$

---

16. Which of the following statements are true?

I. $\displaystyle\int x^4\sin\left(x^5\right)\,dx = -\frac{1}{5}\cos\left(x^5\right)+C$

II. $\displaystyle\int \tan(x)\, dx = \sec^2(x) + C$

III. $\displaystyle\int \left(x^3 + x\right) \sqrt[4]{x^4 + 2x^2 - 5}\, dx = \frac{1}{5}\left(x^4 + 2x^2 - 5\right)^{\frac{5}{4}} + C$

a) I only b) II only c) III only d) I and II only

e) II and III only f) I and III only g) I, II, and III h) None of these

---

17. If $x'(t) = 2t\cos\left(t^2\right)$, find $x(t)$ when $x\left(\sqrt{\frac{\pi}{2}}\right) = 3$

a) $x(t) = -4t^2 \sin\left(t^2\right)$ b) $x(t) = -4t^2 \sin\left(t^2\right) + 2\cos\left(t^2\right)$ c) $x(t) = \sin\left(t^2\right) + 3$

d) $x(t) = -\sin\left(t^2\right) + 4$ e) $x(t) = \sin\left(t^2\right) + 2$

---

18. A particle moves along the y-axis so that at any time $t \geq 0$, its velocity is given $v(t) = \sin(2t)$. If the position of the particle at time $t = \frac{\pi}{2}$ is $y = 3$. What is the particle's position at time $t = 0$?

a) -4 b) 2 c) 3 d) 4 e) 6

2.3 Separable Differential Equations

Differential Equation → Definition: An equation that contains a derivative.

General Solution → Definition: All of the y-equations that would have the given equation as their derivative. Note the $+C$ which gives multiple equations.

Initial Condition → Definition: Constraint placed on a differential equation; sometimes called an initial value.

Particular Solution → Definition: Solution obtained from solving a differential equation when an initial condition allows you to solve for C.

Separable Differential Equation → Definition: A differential equation in which all terms with y's can be moved to the left side of an equals sign (=), and in which all terms with x's can be moved to the right side of an equals sign (=), by multiplication and division only.

Let's take a look at some examples of separable differential equations.

Ex 2.3.1: Separate the variables in the following differential equations:

$$\text{a) } \frac{dy}{dx} = -\frac{x}{y}$$

$$\text{b) } \frac{dy}{dx} = x\sec(y)$$

$$\text{c) } y' = 2xy - 3y$$

Sol 2.3.1:
a)

$$(y)\frac{dy}{dx} = -\frac{x}{y}(y)$$

$$(dx)\frac{y\,dy}{dx} = -x(dx)$$

$$\boxed{y\,dy = -x\,dx}$$

b)

$$\left(\frac{1}{\sec(y)}\right)\frac{dy}{dx} = x\sec(y)\left(\frac{1}{\sec(y)}\right)$$

$$(dx)\cos(y)\frac{dy}{dx} = x(dx)$$

$$\boxed{\cos(y)\,dy = x\,dx}$$

c)

$$\frac{dy}{dx} = (2x-3)y$$

$$\left(\frac{1}{y}\right)\frac{dy}{dx} = (2x-3)y\left(\frac{1}{y}\right)$$

$$(dx)\left(\frac{1}{y}\right)\frac{dy}{dx} = (2x-3)(dx)$$

$$\boxed{\frac{1}{y}\,dy = (2x-3)\,dx}$$

OBJECTIVES

Given a Separable Differential Equation, Find the General Solution.
Given a Separable Differential Equation and an Initial Condition, Find a Particular Solution.

Steps to Solving Differential Equations

1. Separate the variables. Move all terms involving y (and dy) to one side and all terms involving x (and dx) to the other. Keep any constants on the right side of the equation.

2. Integrate both sides. Keep $+C$ only on the right side of the equation.

3. Solve for y, if possible. If the integration produces a natural log, isolate y. If not, solve for C. Note: $e^{\ln|y|} = y$, as e raised to any power is positive.

4. Apply initial conditions (if given). Substitute initial values to solve for C.

Ex 2.3.2: Find the general solution to the differential equation $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

Sol 2.3.2:

$\frac{dy}{dx} = -\frac{x}{y}$ — Start here.

$y\,dy = -x\,dx$ — Separate all the y terms to the left side of the equation and all of the x terms to the right side.

$\int y\,dy = \int -x\,dx$ — Integrate both sides.

$\frac{1}{2}y^2 = -\frac{1}{2}x^2 + C$ — You only need C on one side of the equation.

$y^2 = -x^2 + C$ — Multiply both sides by 2. Note that $2C$ is still a constant, so we'll continue to denote it as C.

$x^2 + y^2 = C$ — This is the family of circles with radius $\sqrt{C}$ centered at the origin.

$y = \pm\sqrt{C - x^2}$ — Isolate y.

Since we usually solve our equation for y, our solution will be $\boxed{y = \pm\sqrt{C - x^2}}$.

Also, note that we can check our solution by taking its derivative.

$$x^2 + y^2 = C$$

$$\frac{d}{dx}\left[x^2 + y^2\right] = \frac{d}{dx}[C]$$

$$2x + 2y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y} \quad \checkmark$$

Ex 2.3.3: Find the general solution to the differential equation $\frac{dm}{dt} = mt$

Sol 2.3.2:

$\frac{dm}{dt} = mt$ — Start here.

$\frac{1}{m}\,dm = t\,dt$	Separate all the m terms to the left side of the equation and all of the t terms to the right side of the equation.
$\int \frac{1}{m}\,dm = \int t\,dt$	Integrate both sides.
$\ln\lvert m\rvert = \frac{1}{2}t^2 + C$	You only need C on one side of the equation.
$e^{\ln\lvert m\rvert} = e^{\frac{1}{2}t^2+C}$	e both sides of the equation to solve for y.
$m = e^{\frac{1}{2}t^2}e^{C}$	Pull out the constants from the equation.
$m = Ke^{\frac{1}{2}t^2}$	e^C is still a constant, which we will just denote as K.

Ex 2.3.4: Find the particular solution to $y' = 2xy - 3y$, given $y(3) = 2$.

Sol 2.3.4: To find the particular solution, recall that we first need to find the general solution.

$$\frac{dy}{dx} = (2x-3)y$$

$$\int \frac{1}{y}\,dy = \int (2x-3)\,dx$$

$$\ln\lvert y\rvert = x^2 - 3x + C$$

$$y = e^{x^2-3x+C}$$

$$= e^{x^2-3x}e^{C}$$

$$= Ke^{x^2-3x}$$

Now, let's plug in our initial value to solve for K.

$$y(3) = 2 \therefore Ke^{3^2-3(3)} = 2 \therefore Ke^0 = 2 \therefore K = 2$$

So, our particular solution is $\boxed{y = 2e^{x^2-3x}}$.

Ex 2.3.5: Find the particular solution to $\dfrac{dy}{dx} = x^2y$, given $y(0) = -2$.

Sol 2.3.5: Once again, let's find the general solution first.

$$\frac{dy}{dx} = x^2y$$

$$\int \frac{1}{y} = \int x^2\,dx$$

$$\ln|y| = \frac{1}{3}x^3 + C$$

$$y = e^{\frac{1}{3}x^3+C}$$

$$= e^{\frac{1}{3}}x^3e^C$$

$$= Ke^{\frac{1}{3}}$$

Now, let's find K.

$$y(0) = -2 \therefore Ke^{\frac{1}{3}(0)} = -2 \therefore Ke^0 = -2 \therefore K = -2$$

So, our particular solution is $\boxed{y = -2e^{\frac{1}{3}x^3}}$.

Ex 2.3.6: Find the particular solution to $\dfrac{dy}{dx} = x^2y^3$, given $y(0) = 1$.

Sol 2.3.6:

$$\frac{dy}{dx} = x^2y^3$$

$$\int \frac{1}{y^3}\,dy = \int x^2\,dx$$

$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$

Usually, at this step, we would isolate y to find the general solution. However, notice that it's easier to solve for C here, because solving for y will likely result in nested fractions, which can be difficult to work with.

$$y(0) = 1 \therefore -\frac{1}{2(1)^2} = \frac{0^2}{2} + C \therefore C = -\frac{1}{2}$$

Now, we can continue solving for the particular solution.

$$-\frac{1}{2y^2} = \frac{x^2}{2} - \frac{1}{2}$$

$$\frac{1}{y^2} = -x^2 + 1$$

$$y^2 = \frac{1}{1-x^2}$$

$$y = \frac{1}{\pm\sqrt{1-x^2}}$$

Since $x = 0$ gave us $y = +1$, our particular solution must be:

$$\boxed{y = \frac{1}{\sqrt{1-x^2}}}$$

Ex 2.3.7: Let $y = f(x)$ be a differentiable function such that $\frac{dy}{dx} = \frac{y+1}{x^2+9}$ and suppose the point $(0, -3)$ is on the graph of $y = f(x)$.

(a) Use implicit differentiation to find $\frac{d^2y}{dx^2}$.

(b) Determine if the point $(0, -3)$ is at a maximum, a minimum, or neither.

(c) Find the particular solution to $\frac{dy}{dx} = \frac{y+1}{x^2+9}$ at $(0, -3)$.

Sol 2.3.7:
a)

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{y+1}{x^2+9}\right]$$

$$= \frac{(x^2+9)\frac{dy}{dx} - (y+1)(2x)}{(x^2+9)^2}$$

$$= \frac{(x^2+9)\left(\dfrac{y+1}{x^2+9}\right) - (y+1)(2x)}{(x^2+9)}$$

$$= \frac{(y+1) - (y+1)(2x)}{(x^2+9)^2}$$

$$= \boxed{\frac{(y+1)(1-2x)}{(x^2+9)}}$$

b)

$$\left.\frac{d^2y}{dx^2}\right|_{(0,-3)} = \frac{(-3)+1}{(0)^2+9}$$

$$= -\frac{2}{9}$$

Because the derivative at the point $(0, -3)$ is not equal to zero, the point is $\boxed{\text{neither a maximum nor a minimum}}$.

c)

$$\frac{dy}{dx} = \frac{y+1}{x^2+9}$$

$$\int \frac{1}{y+1}\,dy = \int \frac{1}{x^2+9}\,dx$$

$$\ln|y+1| = \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + C$$

$$y+1 = e^{\frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)+C}$$

$$y = Ke^{\frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)} - 1$$

$$y(0) = -3 \therefore -3 = Ke^{\frac{1}{3}\tan^{-1}\left(\frac{(0)}{3}\right)} - 1 \therefore K = -2$$

$$\boxed{y = -2e^{\frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)} - 1}$$

2.3 Free Response Homework

Separate the variables for the following differential equations.

1. $\dfrac{dy}{dx} = \dfrac{y}{x}$

2. $\dfrac{dy}{dx} = xy^2$

3. $\left(x^2+1\right)\dfrac{dy}{dx} = xy$

4. $\dfrac{dy}{dt} = \dfrac{\sec^2(t)}{\sec(y)\tan(y)}$

5. $\dfrac{dv}{dt} = 2+2v+t+tv$

6. $\dfrac{dy}{dx} = \dfrac{x^2+1}{\sec(y)\tan(y)}$

7. $\dfrac{dy}{dt} = \dfrac{t}{y\sqrt{y^2+1}}$

8. $\dfrac{dy}{dx} = \dfrac{y^2+1}{xy}$

Find the particular solutions to these differential equations with the given initial conditions.

9. $\dfrac{dy}{dx} = \dfrac{3y^2}{1+x^2} \quad \Big| \quad y(1)=5$

10. $\dfrac{dy}{dx} = \dfrac{x^2\sqrt{x^3-2}}{y^2} \quad \Big| \quad y(3)=0$

11. $\dfrac{dy}{dx} = \dfrac{e^{2x}}{4y^3} \quad \Big| \quad y(0)=1$

12. $\dfrac{dy}{dx} = y^2\cos(x) \quad \Big| \quad y\left(\dfrac{\pi}{2}\right)=1$

13. $\dfrac{dy}{dx} = 4xy^3 \quad \Big| \quad y(0)=-2$

14. $\dfrac{d\theta}{dr} = \dfrac{1+\sqrt{r}}{\sqrt{\theta}} \quad \Big| \quad \theta(4)=9$

15. $\dfrac{dy}{dx} = 8xy \quad \Big| \quad y(0)=5$

16. $\dfrac{dy}{dx} = 3y(2x+1) \quad \Big| \quad y(0)=1$

17. $\dfrac{dy}{dx} = 6x^2y^2 \quad \Big| \quad y(1)=3$

18. $\dfrac{dy}{dx} = y^2+1 \quad \Big| \quad y(1)=0$

19. $\dfrac{dy}{dx} = \dfrac{2x^3}{3y^2} \quad \Big| \quad y\left(\sqrt{2}\right)=0$

20. $\dfrac{dy}{dx} = \dfrac{\cos(x)}{y} \quad \Big| \quad y\left(\dfrac{\pi}{2}\right)=3$

21. $\dfrac{dy}{dx} = xy^2 \quad \Big| \quad y(0)=5$

22. $\dfrac{dy}{dx} = \dfrac{\sec(y)}{x} \quad \Big| \quad y(1)=\dfrac{\pi}{2}$

23. $\dfrac{dy}{dx} = y^2+1 \quad \Big| \quad y(1)=0$

24. $\dfrac{dy}{dx} = \dfrac{2x}{y} \quad \Big| \quad y(0)=1$

25. $\dfrac{du}{dt} = \dfrac{2t+\sec^2(t)}{2u} \quad \Big| \quad u(0)=-5$

26. $\dfrac{dy}{dx}\left(x^2+1\right)(2-y) \quad \Big| \quad y(1)=3$

27. $\dfrac{dy}{dx} = \dfrac{y^2+1}{xy} \quad \Big| \quad y(0)=-1$

28. $\dfrac{dy}{dx} = \dfrac{xy^2+x}{y} \quad \Big| \quad y(0)=-1$

29. $\dfrac{dy}{dx} = \dfrac{\sin(x)}{\sin(y)}$ $\Big|$ $y(0) = \dfrac{\pi}{2}$

30. $\dfrac{dy}{dx} = yx - y\sin(x)$ $\Big|$ $y(0) = 5e$

31. Solve the equation $e^{-y}\dfrac{dy}{dx} + \cos(x) = 0$.

32. Find an equation of the curve that satisfies $\dfrac{dy}{dx} = 4x^3y$ and whose y-intercept is 7.

33. Let $y = f(x)$ be a differentiable function such that $\dfrac{dy}{dx} = y^2(6 - 2x)$, and suppose the point $\left(3,\ -\dfrac{1}{3}\right)$ is on the graph of $y = f(x)$.

(a) Use implicit differentiation to find $\dfrac{d^2y}{dx^2}$.

(b) Use the solution to a) to determine if the point $\left(3,\ -\dfrac{1}{3}\right)$ is at a maximum, a minimum, or neither.

(c) Find the particular solution to $\dfrac{dy}{dx} = y^2(6 - 2x)$ at $\left(3\ -\dfrac{1}{3}\right)$.

34. Let $y = f(x)$ be a differentiable function such that $\dfrac{dy}{dx} = xy + y$, and suppose the point $(-1, 2)$ is on the graph of $y = f(x)$.

(a) Use implicit differentiation to find $\dfrac{d^2y}{dx^2}$.

(b) Use the solution to a) to determine if the point $(-1,\ 2)$ is at a maximum, a minimum, or neither.

(c) Find the particular solution to $\dfrac{dy}{dx} = xy + y$ at $(-1,\ 2)$.

35. Let $y = f(x)$ be a differentiable function such that $\dfrac{dy}{dx} = \dfrac{3x^2}{y+2}$, and suppose the point $(-1, 2)$ is on the graph of $y = f(x)$.

(a) Use implicit differentiation to find $\dfrac{d^2y}{dx^2}$.

(b) Use the solution to a) to determine if the point $(0,\ 1)$ is at a maximum, a minimum, or neither.

(c) Find the particular solution to $\dfrac{dy}{dx} = \dfrac{3x^2}{y+2}$ at $(0,\ 1)$.

36. Let $y = f(x)$ be a differentiable function such that $\dfrac{dy}{dx} = (x-1)(y+2)$, and suppose the point $(1, 0)$ is on the graph of $y = f(x)$.

(a) Use implicit differentiation to find $\dfrac{d^2y}{dx^2}$.

(b) Use the solution to a) to determine if the point (1, 3) is at a maximum, a minimum, or neither.

(c) Find the particular solution to $\dfrac{dy}{dx} = (x-1)(y+2)$ at (1, 3).

2.3 Multiple Choice Homework

1. If $\dfrac{dy}{dx} = \sin(x)\cos^3(x)$ and if $y = 1$ when $x = \pi$, what is the value of y when $x = 0$?

a) 0 b) 1 c) 1.5 d) 2 e) 2.5

---

2. If $\dfrac{dy}{dx} = \cos(x)\sin^2(x)$ and if $y = 0$ when $x = \pi$, what is the value of y when $x = 0$?

a) -1 b) $-\dfrac{1}{3}$ c) 0 d) $\dfrac{1}{3}$ e) 1

---

3. The solution to the differential equation $\dfrac{dy}{dx} = 8xy$ with initial condition $y(0) = 5$ is

a) $\ln\left(4x^2+5\right)$ b) $e^{4x^2}+5$ c) $e^{4x^2}+4$ d) $5\ln\left(4x^2\right)$ e) $5e^{4x^2}$

---

4. Identify the first mistake (if any) in this process:

Problem:	$\dfrac{dy}{dx} = xy + x$
Step 1:	$\dfrac{1}{y+1}\,dy = x\,dx$
Step 2:	$\ln\lvert y+1\rvert = x^2 + C$
Step 3:	$y + 1 = e^{x^2} + C$
Step 4:	$y = e^{x^2} + C$

a) Step 1   b) Step 2   c) Step 3   d) Step 4   e) No mistake

---

5. Identify the first mistake (if any) in this process:

Problem:	$\dfrac{dy}{dx} = 6x^2y^2$
Step 1:	$\dfrac{1}{y^2}\,dy = 6x^2\,dx$
Step 2:	$\ln\lvert y^2\rvert = 2x^3 + C$
Step 3:	$y^2 = e^{2x^3+C}$
Step 4:	$y = \pm\sqrt{Ke^{2x^3}}$

a) Step 1   b) Step 2   c) Step 3   d) Step 4   e) No mistake

---

2.4 Integration by Back-Substitution

Sometimes when applying the Chain Rule, the other factor is not the du, or there are extra $x's$ that must be replaced with some form of u. The method of choosing u to equal the inside of the composite function remains the same, but there is more substitution necessary.

Steps to Integration by Back-Substitution

1. Find u and du, just as with u-substitution.
2. Handle extra x's. Identify any remaining "extra" x terms, and express x in terms of u.
3. Replace all x-expressions in the integral with u and du.
4. Integrate appropriately, and replace u with the original x-expression to get the final answer

All of this is best understood with some examples.

Ex 2.4.1: $\displaystyle\int x^3\left(x^2+4\right)^{\frac{3}{2}}\,dx$

Sol 2.4.1:

$$\hookrightarrow u = x^2+4 \therefore x^2 = u-4$$
$$\hookrightarrow du = 2x\,dx$$

$$\begin{aligned}
\int x^3\left(x^2+4\right)^{\frac{3}{2}}\,dx &= \frac{1}{2}\int (2x)x^2\left(x^2+4\right)^{\frac{3}{2}}\,dx \\
&= \frac{1}{2}\int (u-4)u^{\frac{3}{2}}\,du \\
&= \frac{1}{2}\int \left(u^{\frac{5}{2}}-4u^{\frac{3}{2}}\right)du \\
&= \frac{1}{2}\left(\frac{2}{7}u^{\frac{7}{2}}-\frac{8}{5}u^{\frac{5}{2}}\right)+C \\
&= \boxed{\frac{1}{7}\left(x^2+4\right)^{\frac{7}{2}}-\frac{4}{5}\left(x^2+4\right)^{\frac{5}{2}}+C}
\end{aligned}$$

Notice how when we attempted u-substitution, an x^2 remained in the equation. That is the reason why we expressed x^2 in terms of u.

Ex 2.4.2: $\int (x+1)\sqrt{x-1}\,dx$

Sol 2.4.2:

$\hookrightarrow u = x - 1 \therefore x = u + 1$

$\hookrightarrow du = dx$

$$\begin{aligned}
\int (x+1)\sqrt{x-1}\,dx &= \int ((u+1)+1)\sqrt{u}\,du \\
&= \int (u+2)\sqrt{u}\,du \\
&= \left(u^{\frac{3}{2}} + 2u^{\frac{1}{2}}\right)\,du \\
&= \frac{2}{5}u^{\frac{5}{2}} + \frac{4}{3}u^{\frac{3}{2}} + C \\
&= \boxed{\frac{2}{5}(x-1)^{\frac{5}{2}} + \frac{4}{3}(x-1)^{\frac{3}{2}} + C}
\end{aligned}$$

Ex 2.4.3: $\int (x+2)(x-3)^4\,dx$

Ex 2.4.3:

$\hookrightarrow u = x - 3 \therefore x = u + 3$

$\hookrightarrow du = dx$

$$\int (x+2)(x-3)^4\,dx = ((u+3)+2)u^4\,du$$
$$= \int (u+5)u^4\,du$$
$$= \int u^5 + 5u^4\,du$$
$$= \frac{1}{6}u^6 + u^5 + C$$
$$= \boxed{\frac{1}{6}(x-3)^6 + (x-3)^5 + C}$$

Ex 2.4.4: $\displaystyle\int \frac{x^2+4}{x+2}\,dx$

Sol 2.4.4: There are two ways to approach this problem. One could use polynomial long division to simplify before integrating:

$$\int \frac{x^2+4}{x+2}\,dx = \int \left(x - 2 + \frac{8}{x+2}\right)\,dx$$

Then

$\hookrightarrow$ $u = x+2$

$\hookrightarrow$ $du = dx$

$$\int \left(x - 2 + \frac{8}{x+2}\right)\,dx = \int (x-2)\,dx + 8\int \frac{1}{u}\,du$$
$$= \frac{1}{2}x^2 - 2x + 8\ln|u| + C$$
$$= \boxed{\frac{1}{2}x^2 - 2x + 8\ln|x+2| + C}$$

An alternative method to solving this problem would be to make the denominator u and use back-substitution:

$\hookrightarrow$ $u = x+2 \therefore x = u = 2$

$\hookrightarrow$ $du = dx$

$$
\begin{aligned}
\int \frac{x^2+4}{x+2}\,dx &= \int \frac{(u-2)^2+4}{u}\,du \\
&= \int \frac{u^2-4u+4+4}{u}\,du \\
&= \int \frac{u^2-4u+8}{u}\,du \\
&= \int \left(u-4+\frac{8}{u}\right)\,du \\
&= \frac{1}{2}u^2-4u+8\ln|u|+C \\
&= \boxed{\frac{1}{2}(x+2)^2-4(x+2)+8\ln|x+2|+C}
\end{aligned}
$$

From visual inspection, these answers may appear different. However, with FOILing and adding like terms, it can be shown that these are in fact the same answers.

As a rule of thumb, doing algebraic simplification before calculus will generally make the problem shorter and simpler. In this case, polynomial long division made the problem easier than back-substitution.

2.4 Free Response Homework

Perform the antidifferentiation.

1. $\int x\sqrt{4-x}\,dx$

2. $\int x^5\sqrt{x^3+4}\,dx$

3. $\int \frac{x+5}{2x+3}\,dx$

4. $\int x^3\left(x^2+1\right)^{12}$

5. $\int \frac{(3+\ln x)^2(2-\ln x)}{x}\,dx$

6. $\int \sqrt{4-\sqrt{x}}\,dx$

7. $\int x^5\left(x^2+4\right)^2\,dx$

8. $\int \sqrt{x+3}(x+1)^2\,dx$

9. $\int (t-1)(2t+4)^5\,dt$

10. $\int (z-3)(3z-1)^3\,dz$

11. $\int \frac{y^5}{\sqrt{y^3+5}}\,dy$

12. $\int \frac{w^5}{w^2+4}\,dw$

13. $\int \frac{x^5}{(x^2-1)^{\frac{5}{2}}}\,dx$

14. $\int \frac{x^7}{(x^4+4)^{\frac{3}{2}}}\,dx$

15. $\int (x+2)\sqrt[3]{x-1}\,dx$

16. $\int \sqrt{4-x}(2x+5)\,dx$

2.4 Multiple Choice Homework

1. $\int x^3\sqrt{1+x^2}\,dx =$

a) $\frac{x^4}{2}\cdot\frac{(1+x^2)^{\frac{3}{2}}}{3}+C$

b) $\frac{1}{2}\left(1+x^2\right)^{\frac{1}{2}}+\frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}}+C$

c) $-\frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}}+\frac{1}{5}\left(1+x^2\right)^{\frac{5}{2}}+C$

d) $\frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}}-\frac{1}{5}\left(1+x^2\right)^{\frac{5}{2}}+C$

e) $\frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}}+C$

---

2. $\int x^5\sqrt{1+x^2}\,dx =$

a) $\frac{x^6\,(1+x^2)^{\frac{3}{2}}}{18}+C$

b) $\frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}}+\frac{2}{7}\left(1+x^2\right)^{\frac{7}{2}}+C$

c) $\frac{1}{7}\left(1+x^2\right)^{\frac{7}{2}}-\frac{2}{5}\left(1+x^2\right)^{\frac{5}{2}}+\frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}}+C$

d) $\frac{2}{7}\left(1+x^2\right)^{\frac{7}{2}}-\frac{4}{5}\left(1+x^2\right)^{\frac{5}{2}}+\frac{2}{3}\left(1+x^2\right)^{\frac{3}{2}}+C$

e) $\frac{1}{7}\left(1+x^2\right)^{\frac{7}{2}}+\frac{2}{5}\left(1+x^2\right)^{\frac{5}{2}}+\frac{1}{3}\left(1+x^2\right)^{\frac{3}{2}}+C$

---

3. $\displaystyle\int \frac{x^3}{9-x^2}\,dx =$

a) $\frac{x^3}{3}\cdot\frac{(9-x^2)^{\frac{3}{2}}}{9}+C$

b) $\frac{9}{2}\ln|9-x^2|+\frac{1}{2}\left(9-x^2\right)+C$

c) $\frac{9}{2}\left(9-x^2\right)^2-\frac{1}{2}\left(9-x^2\right)+C$

d) $\frac{9}{2}\left(9-x^2\right)+\frac{1}{2}\left(9-x^2\right)^2+C$

e) $\frac{x^4}{36}-\frac{x^2}{2}+C$

---

4. $\displaystyle\int \frac{x^5}{x^2+5}\,dx =$

a) $\frac{1}{4}\left(x^2+5\right)^2-5\left(x^2+5\right)+\frac{25}{2}\ln|x^2+5|+C$

b) $\frac{1}{4}\left(x^2+5\right)^2-5\left(x^2+5\right)+\frac{25}{2}\tan^{-1}\left(x^2+5\right)+C$

c) $\frac{1}{4}\left(x^2+5\right)^2+5\left(x^2+5\right)+\frac{25}{2}\ln|x^2+5|+C$

d) $\frac{1}{4}\left(x^2+5\right)^2+5\left(x^2+5\right)+\frac{25}{2}\tan^{-1}\left(x^2+5\right)+C$

e) None of the above

---

5. $\displaystyle\int e^{2x}\sqrt{e^x+1}\,dx =$

a) $e^{2x}\left(e^x+1\right)^{\frac{3}{2}}+C$

b) $\frac{2}{5}\left(e^x+1\right)^{\frac{5}{2}}-3\left(e^x+1\right)^{\frac{3}{2}}+C$

c) $\frac{2}{5}(e^x+1)^{\frac{5}{2}} - \frac{2}{3}(e^x+1)^{\frac{3}{2}} + C$

d) $\frac{2}{5}(e^x+1)^{\frac{5}{2}} + 3(e^x+1)^{\frac{3}{2}} + C$

e) $\frac{2}{5}e^{\frac{5x}{2}} - 5e^{\frac{3x}{2}} + C$

---

6. $\frac{x^3}{\sqrt{4-x^2}}\,dx =$

a) $\frac{1}{3}\left(4-x^2\right)^{\frac{3}{2}} - 4\left(4-x^2\right)^{\frac{1}{2}} + C$

b) $\frac{2}{3}\left(4-x^2\right)^{\frac{3}{2}} - 2\left(4-x^2\right)^{\frac{1}{2}} + C$

c) $\frac{1}{3}\left(4-x^2\right)^{\frac{3}{2}} - 4\sin^{-1}\left(\frac{x}{2}\right) + C$

d) $\frac{2}{3}\left(4-x^2\right)^{\frac{3}{2}} - 2\sin^{-1}\left(\frac{x}{2}\right) + C$

e) $\frac{1}{3}\left(4-x^2\right)^{\frac{3}{2}} - \sin^{-1}\left(\frac{x}{2}\right) + C$

---

7. $\frac{4-x}{\sqrt{4-x^2}}\,dx =$

a) $4\sin^{-1}\left(\frac{x}{2}\right) + \left(4-x^2\right)^{\frac{1}{2}} + C$

b) $2\sin^{-1}\left(\frac{x}{2}\right) + \left(4-x^2\right)^{\frac{1}{2}} + C$

c) $4\left(4-x^2\right)^{\frac{1}{2}} + C$

d) $\sin^{-1}\left(\frac{x}{2}\right) - \left(4-x^2\right)^{\frac{1}{2}} + C$

e) $\frac{2}{3}\left(4-x^2\right)^{\frac{3}{2}} + C$

2.5 Powers of Trig Functions: Sine and Cosine

Another instance of back-substitution involves the trig functions and the Pythagorean Identities. As we saw in the previous section, since

$$\frac{d}{dx}[\cos(x)] = -\sin(x) \text{ and } \frac{d}{dx}[\sin(x)] = \cos(x),$$

one of these functions can serve as the du while the other serves as u. But, what about when higher exponents are involved? In general, what about integrals in the form

$$\int \sin^m(x)\cos^n(x)\,dx?$$

Remember:

$$\boxed{\sin^2(\theta) + \cos^2(\theta) = 1}$$

OBJECTIVES

Use Integration by Substitution to Integrate Integrands Involving Sine and Cosine.

There are two cases of integration of this kind of integrand, depending on the powers m and n.

Case 1

The simpler (and more common on the AP Test) case is when either m, n, or both m and n are odd numbers. One of whichever function has the odd power will be the du and the rest of those functions can convert to the other trig function by means of the Pythagorean Identities.

Ex 2.5.1: $\int \sin^4(x)\cos^3(x)\,dx$

Sol 2.5.1: Since $\cos(x)$ has the odd power, we will make that our du.

$\hookrightarrow u = \sin(x)$

$\hookrightarrow du = \cos(x)\,dx$

$$\begin{aligned}
\int \sin^4(x)\cos^3(x)\,dx &= \int \sin^4(x)\cos^2(x)\cos(x)\,dx \\
&= \int \sin^4(x)\left(1-\sin^2(x)\right)\cos(x)\,dx \\
&= \int u^4\left(1-u^2\right)\,du \\
&= \int \left(u^4-u^6\right)\,du \\
&= \frac{1}{5}u^5 - \frac{1}{7}u^7 + C \\
&= \boxed{\frac{1}{5}\sin^5(x) - \frac{1}{7}\sin^7(x) + C}
\end{aligned}$$

Ex 2.5.2: $\displaystyle\int \sin^5(x)\cos^2(x)\,dx$

Sol 2.5.2: Since $\sin(x)$ has the odd power, we will make that our du.

$\hookrightarrow$ $u = \cos(x)$

$\hookrightarrow$ $du = -\sin(x)\,dx$

$$
\begin{aligned}
\int \sin^5(x)\cos^2(x) &= -\int \sin^4(x)(-\sin(x))\cos^2(x)\,dx \\
&= -\int \left(\sin^2(x)\right)^2(-\sin(x))\cos^2(x)\,dx \\
&= -\int \left(1-\cos^2(x)\right)^2(-\sin(x))\cos^2(x)\,dx \\
&= -\int \left(1-u^2\right)^2 u^2\,du \\
&= -\int \left(1-2u^2+u^4\right)u^2\,du \\
&= -\int \left(u^2-2u^4+u^6\right)\,du \\
&= -\left(\frac{1}{3}u^3-\frac{2}{5}u^5+\frac{1}{7}u^7\right)+C \\
&= \boxed{-\frac{1}{3}\cos^3(x)+\frac{2}{5}\cos^5(x)-\frac{1}{7}\cos^7(x)+C}
\end{aligned}
$$

If both powers are odd, either function can serve as the u. However, it is generally easier to choose $u=\sin(x)$ as there is no negative sign to deal with.

Ex 2.5.3: $\displaystyle\int \tan(x)\,dx$

Sol 2.5.3: At first, this does not appear to be a sine or cosine integral, but a basic substitution reveals that it is.

$$
\int \tan(x)\,dx = \int \frac{\sin(x)}{\cos(x)}\,dx
$$

$$
\hookrightarrow u=\cos(x)
$$

$$
\hookrightarrow du=-\sin(x)\,dx
$$

$$\begin{aligned}
\int \frac{\sin(x)}{\cos(x)}\,dx &= -\int (-\sin(x))\frac{1}{\cos(x)}\,dx \\
&= -\int \frac{1}{u}\,du \\
&= -\ln|u| + C \\
&= -\ln|\cos(x)| + C \\
&= \boxed{\ln|\sec(x)| + C}
\end{aligned}$$

It may not be immediately apparent why $-\ln|\cos(x)|$ can be rewritten as $\ln|\sec(x)|$. The reason lies within the log rule $\log\left(a^b\right) = b\log a$:

$$\begin{aligned}
-\ln|\cos(x)| &= -\ln\left|\left(\frac{1}{\cos(x)}\right)^{-1}\right| \\
&= -1\cdot -\ln\left|\frac{1}{\cos(x)}\right| \\
&= \ln\left|\frac{1}{\cos(x)}\right| \\
&= \ln|\sec(x)|
\end{aligned}$$

This gives us two more integral rules:

$$\int \tan(u)\,du = \ln|\sec(u)| + C \qquad\qquad \int \cot(u)\,du = \ln|\sin(u)| + C$$

Case 2

The more difficult situation is when both powers are even. In this case, variations on the half angle argument rules come into play.

Remember:

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \qquad \cos^2(x) = \frac{1}{2}(1 + \cos(2x))$$

Ex 2.5.4: $\displaystyle\int \cos^2(x)\,dx$

Sol 2.5.4:

$$\int \cos^2(x) = \int \frac{1}{2}(1 + \cos(x))\,dx$$

$\hookrightarrow u = 2x$

$\hookrightarrow du = 2\,dx$

$$\int \frac{1}{2}(1 + \cos(x))\,dx = \frac{1}{2} \cdot \frac{1}{2} \int (1 + \cos(x))2\,dx$$

$$= \frac{1}{4} \cdot (1 + \cos(u))\,du$$

$$= \frac{1}{4}u + \frac{1}{4}\sin(u) + C$$

$$= \boxed{\frac{1}{2}x + \frac{1}{4}\sin(2x) + C}$$

This example alludes to two more integral equations that are helpful to know:

$$\int \cos^2(u)\,du = \frac{1}{2}u + \frac{1}{4}\sin(2u) + C \qquad \int \sin^2(u)\,du = \frac{1}{2}u - \frac{1}{4}\sin(2u) + C$$

Ex 2.5.5: $\displaystyle\int \sin^4(x)\cos^2(x)\,dx$

Sol 2.5.5:

$$\int \sin^4(x)\cos^2(x)\,dx = \int \left(\frac{1}{2}(1 - \cos(2x))\right)^2 \left(\frac{1}{2}(1 + \cos(2x))\right)\,dx$$

$$= \frac{1}{8}\int \left(1 - 2\cos(2x) + \cos^2(2x)\right)(1 + \cos(2x))\,dx$$

$$= \frac{1}{8}\int \Big(1 - \cos(2x) - \cos^2(2x) + \cos^3(2x)\Big)\ dx$$

$$\hookrightarrow u = 2x$$

$$\hookrightarrow du = 2\,dx$$

$$= \frac{1}{2}\cdot\frac{1}{8}\int \Big(1 - \cos(2x) - \cos^2(2x) + \cos^3(2x)\Big)\,2\,dx$$

$$= \frac{1}{16}\int \Big(1 - \cos(u) - \cos^2(u) + \cos^3(u)\Big)\ du$$

$$= \frac{1}{16}\int du - \frac{1}{16}\int \cos(u)\,du - \frac{1}{16}\int \cos^2(u)\,du + \frac{1}{16}\int \cos^3(u)\,du$$

$$= \frac{1}{16}u - \frac{1}{16}\sin(u) - \frac{1}{16}\left(\frac{1}{2}u - \frac{1}{4}\sin(2u)\right)$$
$$+ \frac{1}{16}\int \cos^2(u)\cos(u)\,du + C$$

$$\hookrightarrow v = \sin(u)$$

$$\hookrightarrow dv = \cos(u)\,dx$$

$$= \frac{1}{16}u - \frac{1}{16}\sin(u) - \frac{1}{16}\left(\frac{1}{2}u - \frac{1}{4}\sin(2u)\right)$$
$$+ \frac{1}{16}\int \Big(1 - \sin^2(u)\Big)\cos(u)\,du + C$$

$$= \frac{1}{16}u - \frac{1}{16}\sin(u) - \frac{1}{32}u + \frac{1}{64}\sin(2u) + \frac{1}{16}\int \Big(1 - v^2\Big)\ dv + C$$

$$= \frac{1}{16}u - \frac{1}{16}\sin(u) - \frac{1}{32}u + \frac{1}{64}\sin(2u) + \frac{1}{16}\left(v - \frac{1}{3}v^3\right) + C$$

$$= \frac{1}{16}(2x) - \frac{1}{16}\sin(2x) - \frac{1}{32}(2x) + \frac{1}{64}\sin(2(2x))$$
$$+ \frac{1}{16}\left(\sin(2x) - \frac{1}{3}\sin^3(2x)\right) + C$$

$$= \frac{1}{8}x - \frac{1}{16}\sin(2x) - \frac{1}{16}x + \frac{1}{64}\sin(4x) + \frac{1}{16}\sin(2x)$$
$$- \frac{1}{48}\sin^3(2x) + C$$

$$= \boxed{\frac{1}{16}x + \frac{1}{64}\sin(4x) - \frac{1}{48}\sin^3(2x) + C}$$

2.5 Free Response Homework

Perform the antidifferentiation.

1. $\int \sin^3(x)\cos^2(x)\,dx$
2. $\int \sin^4(x)\cos^5(x)\,dx$
3. $\int \sin^2(x)\cos^7(x)\,dx$
4. $\int \sin^5(x)\cos^6(x)\,dx$
5. $\int \sin(x)\cos^5(x)\,dx$
6. $\int \sin^5(x)\cos^5(x)\,dx$
7. $\int \sin^2(x)\cos^2(x)\,dx$
8. $\int \sin^2(x)\cos^4(x)\,dx$

2.5 Multiple Choice Homework

1. For $\int \sin^3(x)\cos^5(x)\,dx$, the correct u-substitution is

   a) $u = \sin(x)$ b) $u = \cos(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

2. For $\int \sin^3(5x)\cos^2(5x)\,dx$, the correct u-substitution is

   a) $u = \sin(x)$ b) $u = \cos(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

3. For $\int \sin^4(4x)\cos^5(4x)\,dx$, the correct u-substitution is

   a) $u = \sin(x)$ b) $u = \cos(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

4. For $\int \sin^2(x)\cos^4(x)\,dx$, the correct u-substitution is

   a) $u = \sin(x)$ b) $u = \cos(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

5. $\int \cos^2(2x)\,dx =$

a) $\sin(4x)+C$ b) $\frac{1}{2}x+\frac{1}{8}\sin(4x)+C$ c) $\frac{1}{2}x-\frac{1}{8}\sin(4x)+C$

d) $x+\frac{1}{4}\sin(4x)+C$ e) $x+\frac{1}{8}\cos(4x)+C$

---

6. $\int \cos^2\left(\frac{1}{2}x\right)\,dx =$

a) $\sin(4x)+C$ b) $\frac{1}{2}x+\frac{1}{4}\sin(x)+C$ c) $\frac{1}{2}x-\frac{1}{4}\sin(x)+C$

d) $\frac{1}{4}x+\frac{1}{2}\sin(x)+C$ e) $\frac{1}{4}x+\frac{1}{2}\cos(x)+C$

---

7. Identify the first mistake (if any) in this process:

Problem: $\int \sin^3(2x)\cos^4(2x) =$

Step 1: $= -\frac{1}{2}\int \sin^2(2x)\cos^4(2x)(-\sin(2x))2\,dx$

Step 2: $= -\frac{1}{2}\int \left(1-u^2\right)u^4\,du$

Step 3: $= -\frac{1}{2}\int \left(u^4-u^6\right)\,du$

Step 4: $= -\frac{1}{2}\left(\frac{1}{5}u^5-\frac{1}{7}u^7+C\right)$

Step 5: $= -\frac{1}{10}\sin^5(4x)+\frac{1}{14}\sin^7(4x)+C$

a) Step 1 b) Step 2 c) Step 3 d) Step 4 e) No mistake

2.6 Powers of Trig Functions: Secant, Tangent, and Beyond

As with sine and cosine, secant and tangent work together in a Pythagorean Identity, as well as cosecant and cotangent. So, we will be considering integrals of the form

$$\int \sec^m(x)\tan^n(x)\,dx \quad \text{and} \quad \int \csc^m(x)\cot^n(x)\,dx$$

to be cases of integration by u-substitution.

Remember:

$$\tan^2(\theta)+1=\sec^2(\theta) \qquad \cot^2(\theta)+1=\csc^2(\theta)$$

$$\frac{d}{dx}[\tan(u)]=\left(\sec^2(u)\right)\frac{du}{dx} \qquad \frac{d}{dx}[\sec(u)]=(\sec(u)\tan(u))\frac{du}{dx}$$

$$\frac{d}{dx}[\cot(u)]=\left(-\csc^2(u)\right)\frac{du}{dx} \qquad \frac{d}{dx}[\csc(u)]=(-\csc(u)\cot(u))\frac{du}{dx}$$

OBJECTIVES

Use Integration by Substitution to Integrate Integrands Involving Secant, Tangent, Cosecant, and Cotangent.

There are three cases of integration of this kind of integrand, depending on the powers m and n.

Case 1

The first case is when the secant's (or cosecant's) power is even. In this case, our u would be $\tan(x)$ or $\cot(x)$ and our du would be $\sec^2(x)\,dx$ or $-\csc^2(x)\,dx$.

Ex 2.6.1: $\int \sec^4(x)\tan^2(x)\,dx$

Sol 2.6.1:

$\hookrightarrow u=\tan(x)$

$\hookrightarrow du=\sec^2(x)$

$$\begin{aligned}
\int \sec^4(x)\tan^2(x) &= \int \sec^2(x)\tan^2(x)\sec^2(x)\,dx \\
&= \int \left(1+\tan^2(x)\right)\tan^2(x)\sec^2(x)\,dx \\
&= \int \left(1+u^2\right)u^2\,du \\
&= \int \left(u^2+u^4\right)\,du \\
&= \frac{1}{3}u^3+\frac{1}{5}u^5+C \\
&= \boxed{\frac{1}{3}\tan^3(x)+\frac{1}{5}\tan^5(x)+C}
\end{aligned}$$

Case 2

The second case is when the tangent's (or cotangent's) power is odd. In this case, our u would be $\sec(x)$ or $\csc(x)$ and our du would be $\sec(x)\tan(x)\,dx$ or $-\csc(x)\cot(x)\,dx$.

Ex 2.6.2: $\displaystyle\int \csc^3(x)\cot^3(x)\,dx$

Sol 2.6.2:

$\hookrightarrow u = \csc(x)$

$\hookrightarrow du = -\csc(x)\cot(x)\,dx$

$$\begin{aligned}
\int \csc^3(x)\cot^3(x)\,dx &= -\int \csc^2(x)\cot^2(x)(-\csc(x)\cot(x))\,dx \\
&= -\int u^2\left(u^2-1\right)\,du \\
&= -\int \left(u^4-u^2\right)\,du \\
&= -\frac{1}{5}u^5+\frac{1}{3}u^3+C \\
&= \boxed{-\frac{1}{5}\csc^5(x)+\frac{1}{3}\csc^3(x)+C}
\end{aligned}$$

If both cases are present (that is, the tangent's/cotangent's power is odd AND the secant's/cosecant's power is even), then any of the functions can serve as u.

Case 3

The third and final case is when the tangent's/cotangent's power is even AND the secant's/cosecant's power is odd. This case is no longer doable by integration by substitution, and requires a technique known as *integration by parts*. We will need to wait until Chapter 8 to approach this case.

Quick Summary of 2.5-2.6:

I. $\displaystyle\int \sin^m(x)\cos^n(x)\,dx$

   a) The odd power determines du. The other function is u.

   b) If both powers are even, use the half-angle formulas and simplify.

II. $\displaystyle\int \sec^m(x)\tan^n(x)\,dx$ or $\displaystyle\int \csc^n(x)\cot^m(x)\,dx$

   a) If both powers are even, $u = \tan(x)$ or $\cot(x)$ and $du = \sec^2(x)\,dx$ or $-\csc^2(x)\,dx$.

   b) If both powers are odd, $u = \sec(x)$ or $\csc(x)$ and $du = \sec(x)\tan(x)\,dx$ or $-\csc(x)\cot(x)\,dx$.

   c) If n is even and m is odd, either (a) or (b) will work.

   d) If n is odd and m is even, neither (a) nor (b) will work.

III. For any other mix of trig functions, convert all to sine and cosine and use I. above.

2.6 Free Response Homework

Perform the antidifferentiation.

1. $\int \sec^2(x)\tan^5(x)\,dx$
2. $\int \sec^6(x)\tan^4(x)\,dx$
3. $\int \sec^5(x)\tan^7(x)\,dx$
4. $\int \sec^2(x)\tan^6(x)\,dx$
5. $\int \sec^6(x)\tan^3(x)\,dx$
6. $\int \csc^2(x)\cot^5(x)\,dx$
7. $\int \csc^4(x)\cot(x)\,dx$
8. $\int \csc^7(x)\cot^5(x)\,dx$

2.6. Multiple Choice Homework

1. For $\int \csc^3(x)\cot^5(x)\,dx$, the correct u-substitution is

a) $u = \csc(x)$ b) $u = \cot(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

2. For $\int \csc^4(x)\cot^4(x)\,dx$, the correct u-substitution is

a) $u = \csc(x)$ b) $u = \cot(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

3. For $\int \sec^4(x)\tan^5(x)\,dx$, the correct u-substitution is

a) $u = \sec(x)$ b) $u = \tan(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

4. For $\int \sec^5(x)\tan^4(x)\,dx$, the correct u-substitution is

a) $u = \sec(x)$ b) $u = \tan(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

5. Which of the following statements are true?

I. $\int \sec(x)\,dx = \ln|\sec(x) + \tan(x)| + C$

II. $\displaystyle\int \tan(x)\,dx = \sec^2(x) + C$

III. $\displaystyle\int x^2 \cot\left(x^3\right)\,dx = \frac{1}{3}\ln\left|\sin\left(x^3\right)\right| + C$

a) I only b) II only c) III only d) I and II only e) I and III only

---

6. Which of the following statements are **false**?

I. $\displaystyle\int x^5 \sec\left(x^6\right) = \frac{1}{6}\ln\left|\sec\left(x^6\right) + \tan\left(x^6\right)\right| + C$

II. $\displaystyle\int \cot(x)\,dx = -\csc^2(x) + C$

III. $\displaystyle\int \csc(x)\,dx = \ln|\csc(x) - \cot(x)| + C$

a) I only b) II only c) III only d) I and II only e) I and III only

---

7. Identify the first mistake (if any) in this process:

Problem:	$\displaystyle\int \sec^4(2x)\tan^3(2x) =$
Step 1:	$\displaystyle= \frac{1}{2}\int \sec^2(2x)\tan^3(2x)\sec^2(2x)2\,dx$
Step 2:	$\displaystyle= \frac{1}{2}\int \left(1-u^2\right)u^3\,du$
Step 3:	$\displaystyle= \frac{1}{2}\int \left(u^3-u^5\right)\,du$
Step 4:	$\displaystyle= \frac{1}{2}\left(\frac{1}{4}u^4 - \frac{1}{6}u^6 + C\right)$
Step 5:	$\displaystyle= -\frac{1}{8}\tan^4(2x) - \frac{1}{12}\tan^6(2x) + C$

a) Step 1 b) Step 2 c) Step 3 d) Step 4 e) No mistake

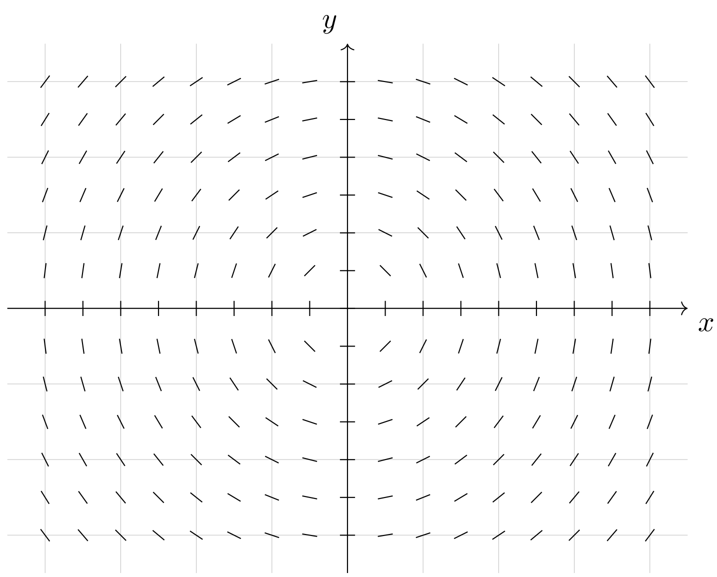
2.7: Intro to AP: Slope Fields

This section is very much an AP-driven section. Recall that one of the big emphasis of CollegeBoard regarding calculus was that it should be understood and explained in many different modes. The topic of differential equations fits nicely into this paradigm in that the visual is a graphical representation and the connection between the equation and the slopes is a numerical process.

Slope Field → Definition: Given any function f, a slope field is drawn by taking evenly spaced points on the Cartesian coordinate system (usually points of integer coordinates) and, at each point, drawing a small line with the slope of f.

IMPORTANT: For this section, assume the grids of all graphs are 1 unit in length and height

Here is an example of a slope field of the differential equation $\dfrac{dy}{dx} = \dfrac{x}{y}$:



OBJECTIVES

Given a Differential Equation, Sketch its Slope Field.
Given a Slope Field, Sketch a Particular Solution Curve.
Given a Slope Field, Determine the Family of Functions to Which the Solution Curves Belong.
Given a Slope Field, Determine the Differential Equation that it Represents.

The are four ways that the AP Exam usually approaches slope fields:

1. Draw a slope field (free response).

2. Sketch the solution to a slope field (free response).

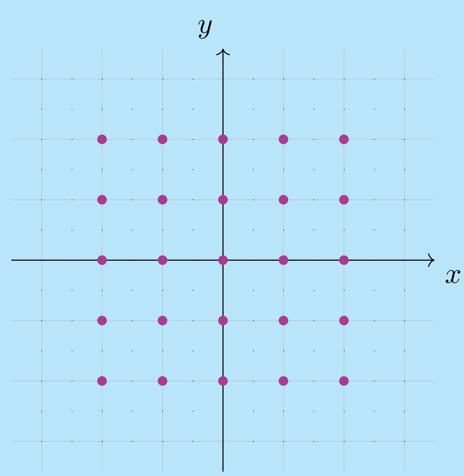
3. Identify the differential equation for a slope field (multiple choice).

4. Identify the solution equation to a slope field (multiple choice).

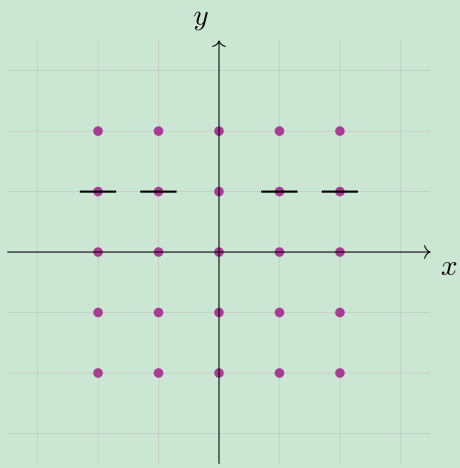
Two of these types of questions (items one and three) are numerically based and two (items two and four) are graphically oriented.

Slope Fields Numerically (FRQs)

Ex 2.7.1: Sketch the slope field for $\dfrac{dy}{dx} = \dfrac{1-y}{x}$ at the points indicated.



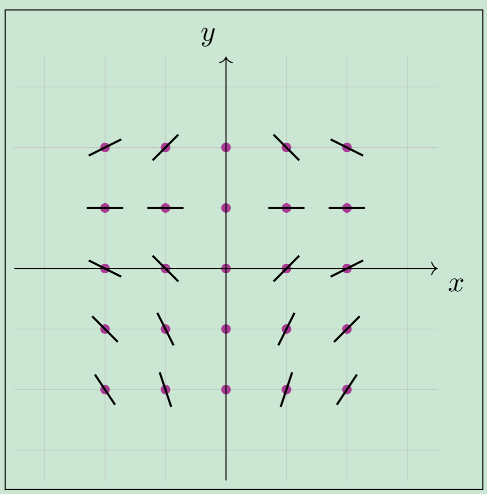
Sol 2.7.1: Note that wherever $y = 1$, $\dfrac{dy}{dx} = 0$. So, the segments at $y = 1$ will be horizontal. Also, where $x = 0$, the derivative does not exist (due to division by zero). So, there will be no segments on the x-axis.



To find the slant of the rest of the line segments, we can plug the numerical values of the points into $\frac{dy}{dx} = \frac{1-y}{x}$.

$(-2,2) \to \frac{dy}{dx} = \frac{1}{2}$	$(-1,2) \to \frac{dy}{dx} = 1$	$(1,2) \to \frac{dy}{dx} = -1$	$(2,2) \to \frac{dy}{dx} = -\frac{1}{2}$
$(-2,0) \to \frac{dy}{dx} = -\frac{1}{2}$	$(-1,0) \to \frac{dy}{dx} = -1$	$(1,0) \to \frac{dy}{dx} = 1$	$(2,0) \to \frac{dy}{dx} = \frac{1}{2}$
$(-2,-1) \to \frac{dy}{dx} = -1$	$(-1,-1) \to \frac{dy}{dx} = -2$	$(1,-1) \to \frac{dy}{dx} = 2$	$(2,-1) \to \frac{dy}{dx} = 1$
$(-2,-2) \to \frac{dy}{dx} = -\frac{3}{2}$	$(-1,-2) \to \frac{dy}{dx} = -3$	$(1,-2) \to \frac{dy}{dx} = 3$	$(2,-2) \to \frac{dy}{dx} = \frac{3}{2}$

These values give us our final slope field:



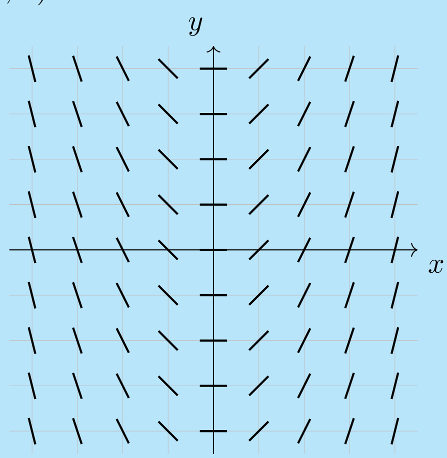
We can use the steps we took in this example to generalize some steps to sketching slope fields:

Steps to Sketching a Slope Field:

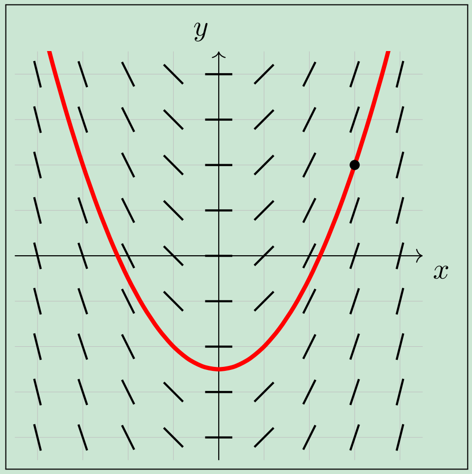
1. Determine the grid of points for which you need to sketch (generally, the points are given).
2. Pick your first point. note its x and y coordinate. Plug these numbers into the differential equation; the output represents the slope at that point.
3. Find the point on the graph. Make a little line/dash at that point whose slope represents the slope that you found in Step 2.
4. Repeat this process for all the points needed.

Slope Fields Graphically (FRQ)

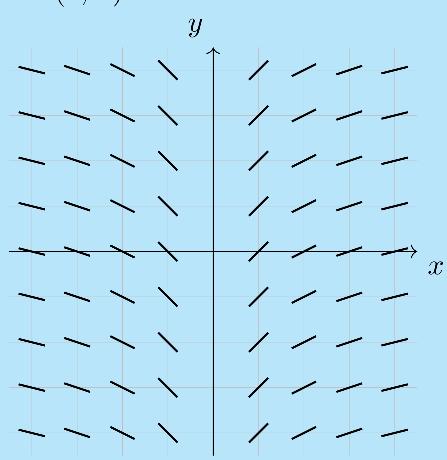
Ex 2.7.2: Given the slope field for $\dfrac{dy}{dx} = x$ below, sketch the particular solution given the initial condition of (3, 2).



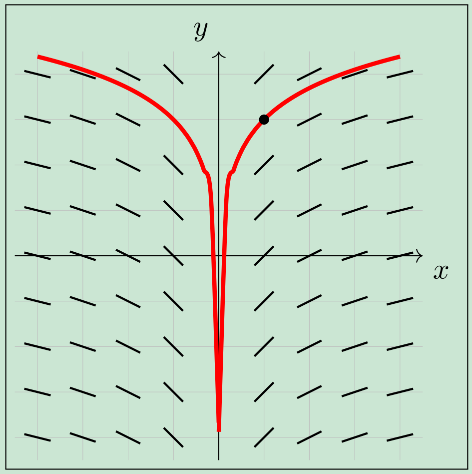
Sol 2.7.2: To find the particular solution, we simply need to start at (3, 2) and follow the slope segments:



Ex 2.7.3: Given the slope field for $x\dfrac{dy}{dx} = 1$ below, sketch the particular solution given the initial condition of (1, 3).



Sol 2.7.3: Once again, we simply need to start at (1, 3) and follow the slope segments:



Note that there appears to be a vertical asymptote at $y = 0$.

Slope Fields Numerically (MCQ)

Let's summarize what we know about slopes of lines in terms of numbers.

1. Horizontal lines have $\dfrac{dy}{dx} = 0$.

2. Vertical lines have an undefined derivative.

3. Lines with positive slopes go up from left to right.

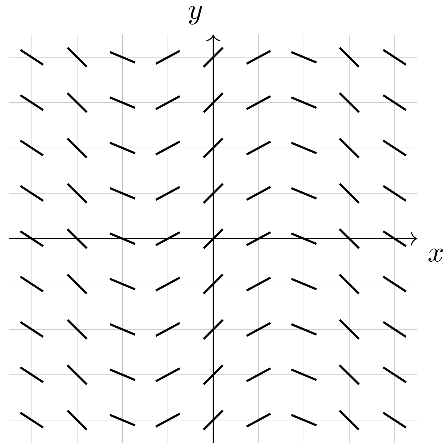
4. Lines with negative slopes go down from left to right.

Two other facts are obvious from viewing a slope field and its associated differential equation:

5. If all dashes in each **column** are parallel to each other, then $\dfrac{dy}{dx}$ has **no** y.

6. If all dashes in each **row** are parallel to each other, then $\dfrac{dy}{dx}$ has **no** x.

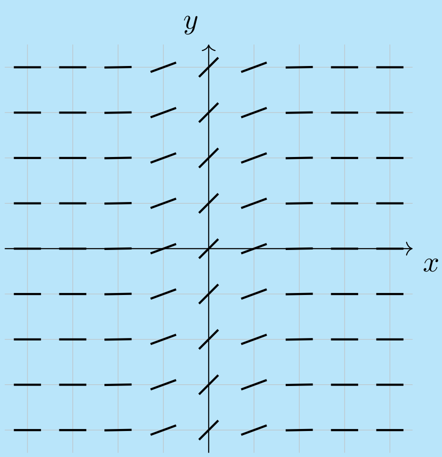
To make points five and six clearer, take a look at the following slope field:

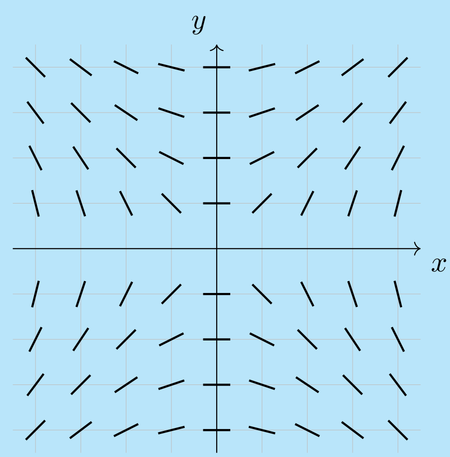


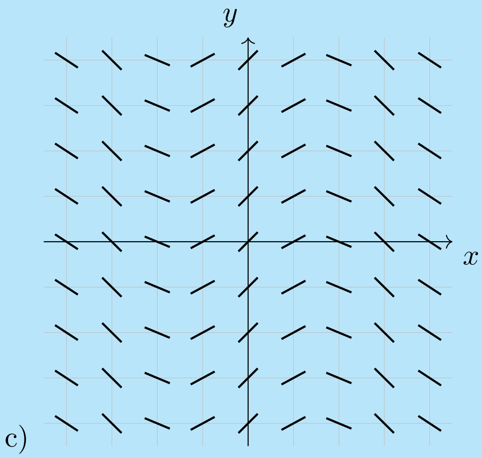
The differential equation that represents this slope field would not have a y in the equation, because the segments in each column are parallel with each other.

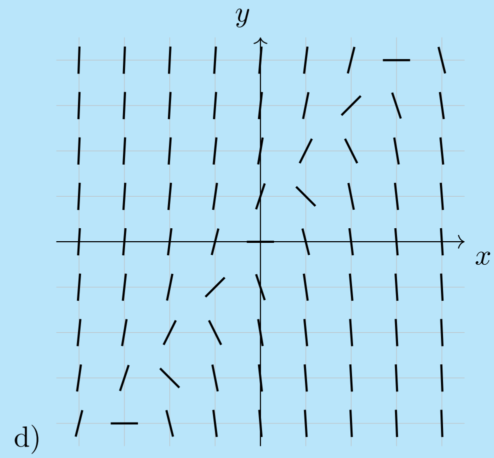
For those curious, the equation for this slope field is $\dfrac{dy}{dx} = \cos(x)$, which in fact does not contain a y!

Ex 2.7.4: Which of the following slope fields matches $\dfrac{dy}{dx} = 3y - 4x$?

a) 

b) 





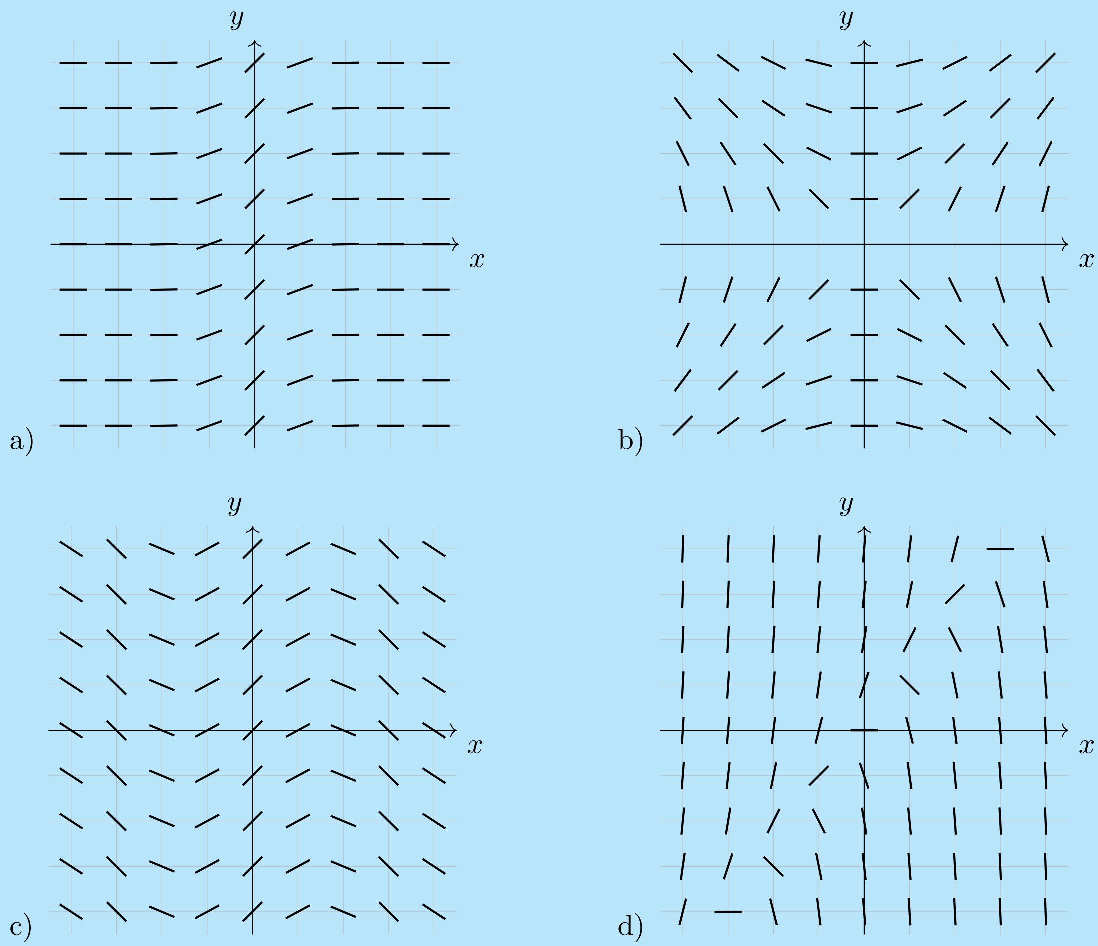
Sol 2.7.4: These types of problems are best done by a combination of process of elimination and substituting numbers. Options (a) and (c) all have slopes parallel to one another in each column. Therefore, they cannot be the slope field for the given differential equation, since the given equation has both x and y.

Now, since we cannot eliminate any other obvious choices, we can move to substitution. Option (b) appears to have horizontal slopes when $x = 0$, so let's plug in a point on the y axis into our differential equation to see if it is equal to zero.

$$\left.\frac{dy}{dx}\right|_{(0,3)} = 3(3) - 4(0) = 9 \neq 0$$

Clearly, option (b) is not our desired slope field. Therefore, the correct answer is option $\boxed{\text{(d)}}$.

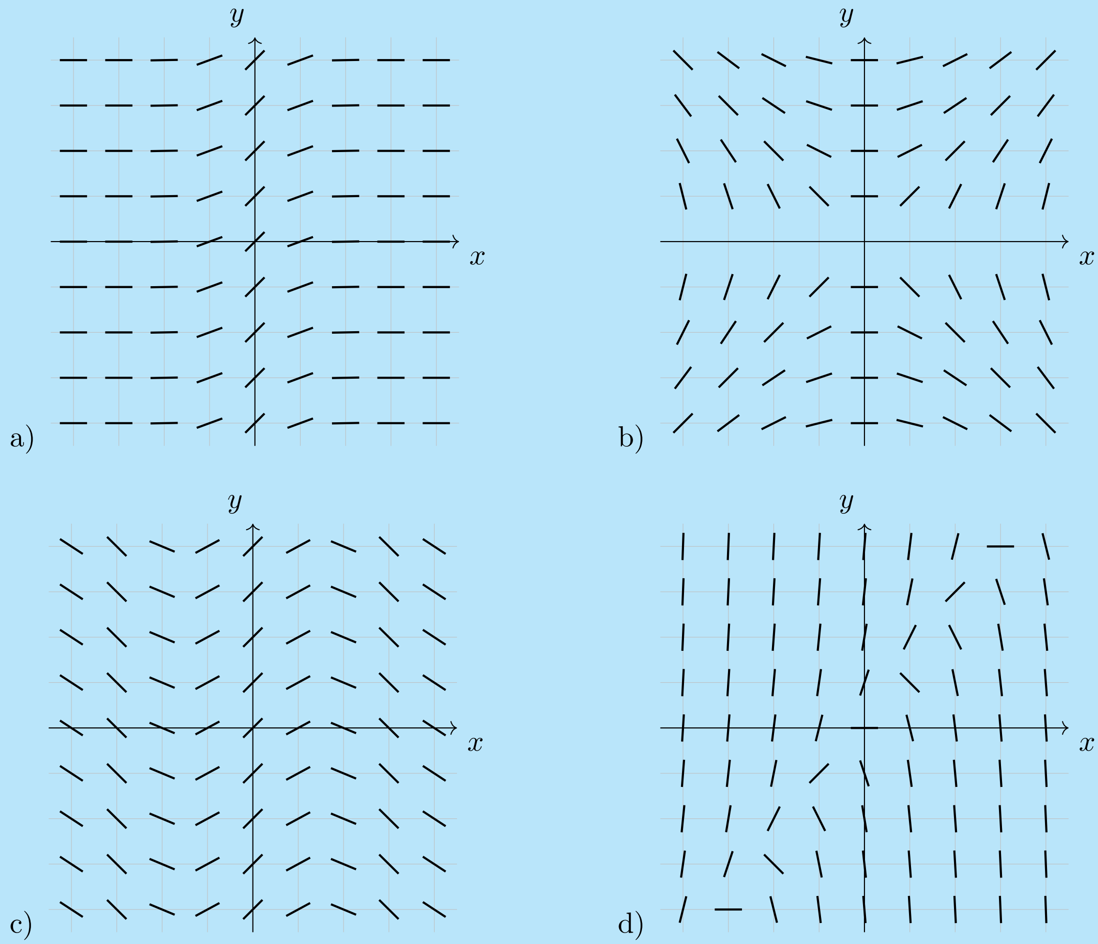
Ex 2.7.5: Which of the following slope fields matches $\dfrac{dy}{dx} = e^{-x^2}$?



Sol 2.7.5: Once again, let's begin with elimination. Since our differential equation only has one variable, we know that our slope field must consist of columns of parallel dashes. This eliminates options (b) and (d).

Now, lets look at the equation. We know that raising e to the power of anything will produce a positive number, so all our slopes must go up from left to right. Therefore, it's obvious that our answer is option $\boxed{\text{(a)}}$.

Ex 2.7.6: Which of the following slope fields matches $\dfrac{dy}{dx} = \dfrac{x}{y}$?



Sol 2.7.6: Using the techniques in **Ex 2.7.4** and **Ex 2.7.5**, we can directly eliminate options (a) and (c).

Now, yet again, we look at the equation. Specifically, we look at what happens when $x = 0$:

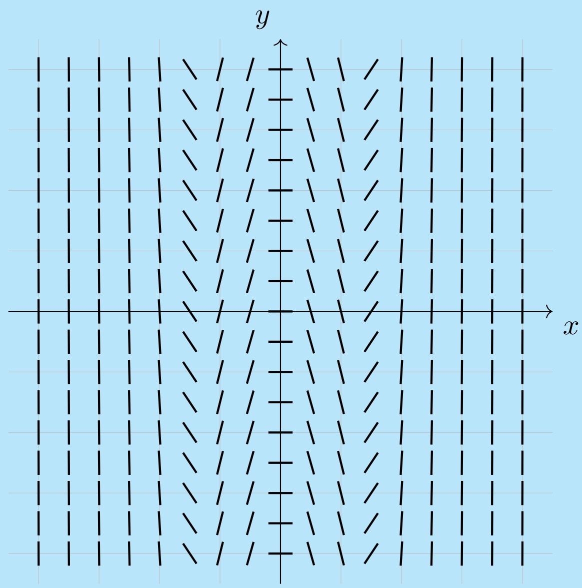
$$\left.\frac{dy}{dx}\right|_{(0,\,c)} = \frac{0}{c} = 0$$

So, our slope field must have a slope of zero when $x = 0$. This means that option $\boxed{\text{(b)}}$ is the correct slope field for this differential equation.

Slope Fields Graphically (FRQ)

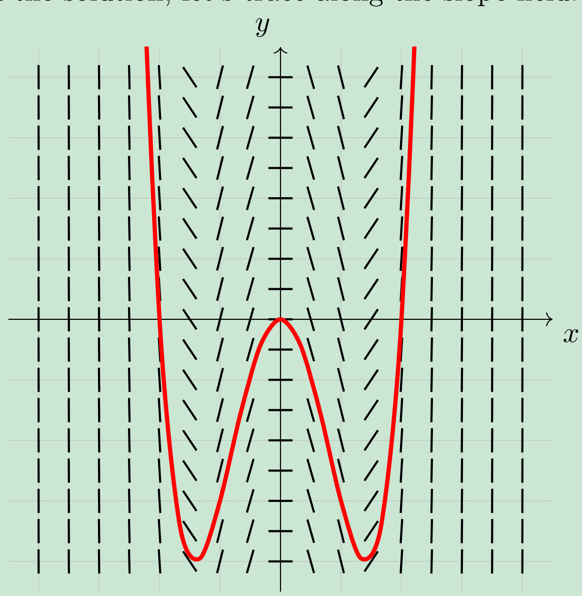
For these types of problems, one would typically sketch a solution and decide from among the options based on what was learned about families of functions in precalculus. For a refresher on families of functions, please see Chapter 0.

Ex 2.7.7: Which of the following equations might be the solution to the slope field shown in the figure below?



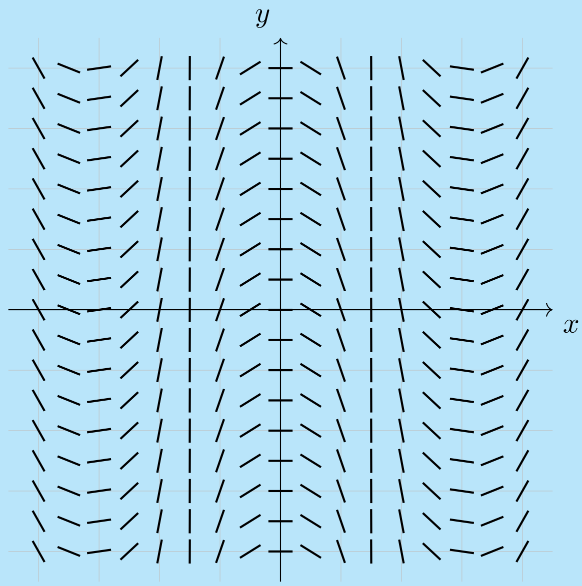
a) $y = x^4 - 4x^2$ b) $y = 4x - x^3$ c) $y = -\cos(x)$ d) $y = -\sec(x)$

Sol 2.7.7: To find the solution, let's trace along the slope field.



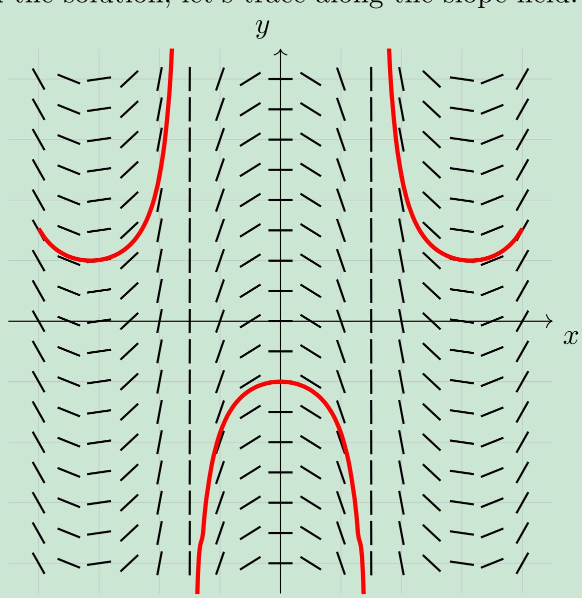
This is a quartic function, so our answer is option $\boxed{\text{a) } y = x^4 - 4x^2}$.

Ex 2.7.8: Which of the following equations might be the solution to the slope field shown in the figure below?



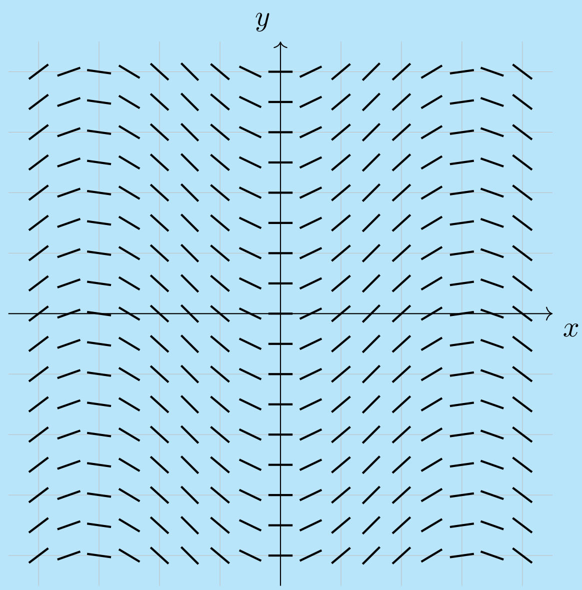
a) $y = x^4 - 4x^2$ b) $y = 4x - x^3$ c) $y = -\cos(x)$ d) $y = -\sec(x)$

Sol 2.7.8: To find the solution, let's trace along the slope field.



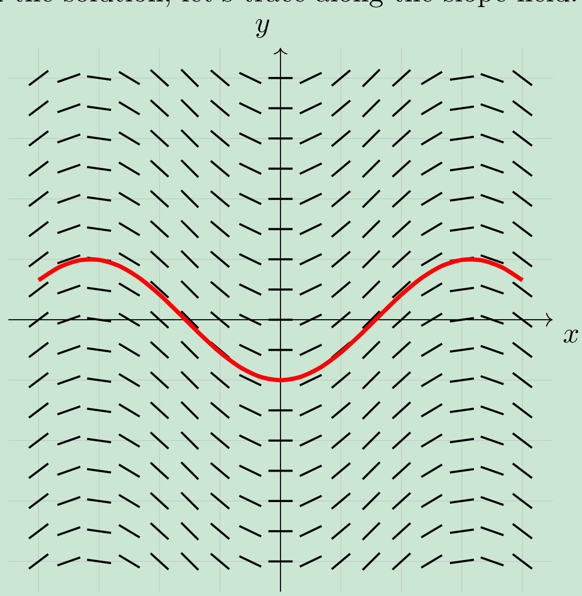
This is very clearly a secant function, so our answer is option $\boxed{\text{d) } y = -\sec(x)}$.

Ex 2.7.9: Which of the following equations might be the solution to the slope field shown in the figure below?



a) $y = x^4 - 4x^2$ b) $y = 4x - x^3$ c) $y = -\cos(x)$ d) $y = -\sec(x)$

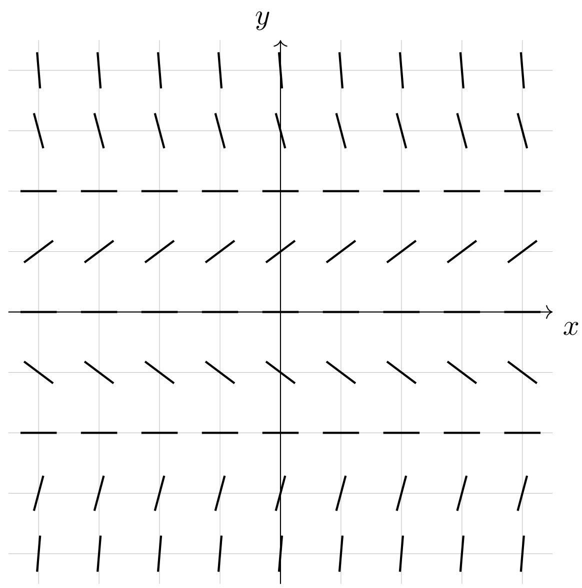
Sol 2.7.9: To find the solution, let's trace along the slope field.



This appears to be some sort of trig function. By noting that y appears to be -1 when $x = 0$, we can deduce that this is a negative cosine function. Therefore, our answer is option $\boxed{\text{c) } y = -\cos(x)}$.

2.7 Free Response Homework

1. A slope field for the differential equation $y' = y\left(1 - \frac{1}{4}y^2\right)$ is shown.



Sketch the graphs of the solutions that satisfy the given initial conditions:

(a) $y(0) = 1$

(b) $y(0) = -1$

(c) $y(0) = -3$

(d) $y(0) = 3$

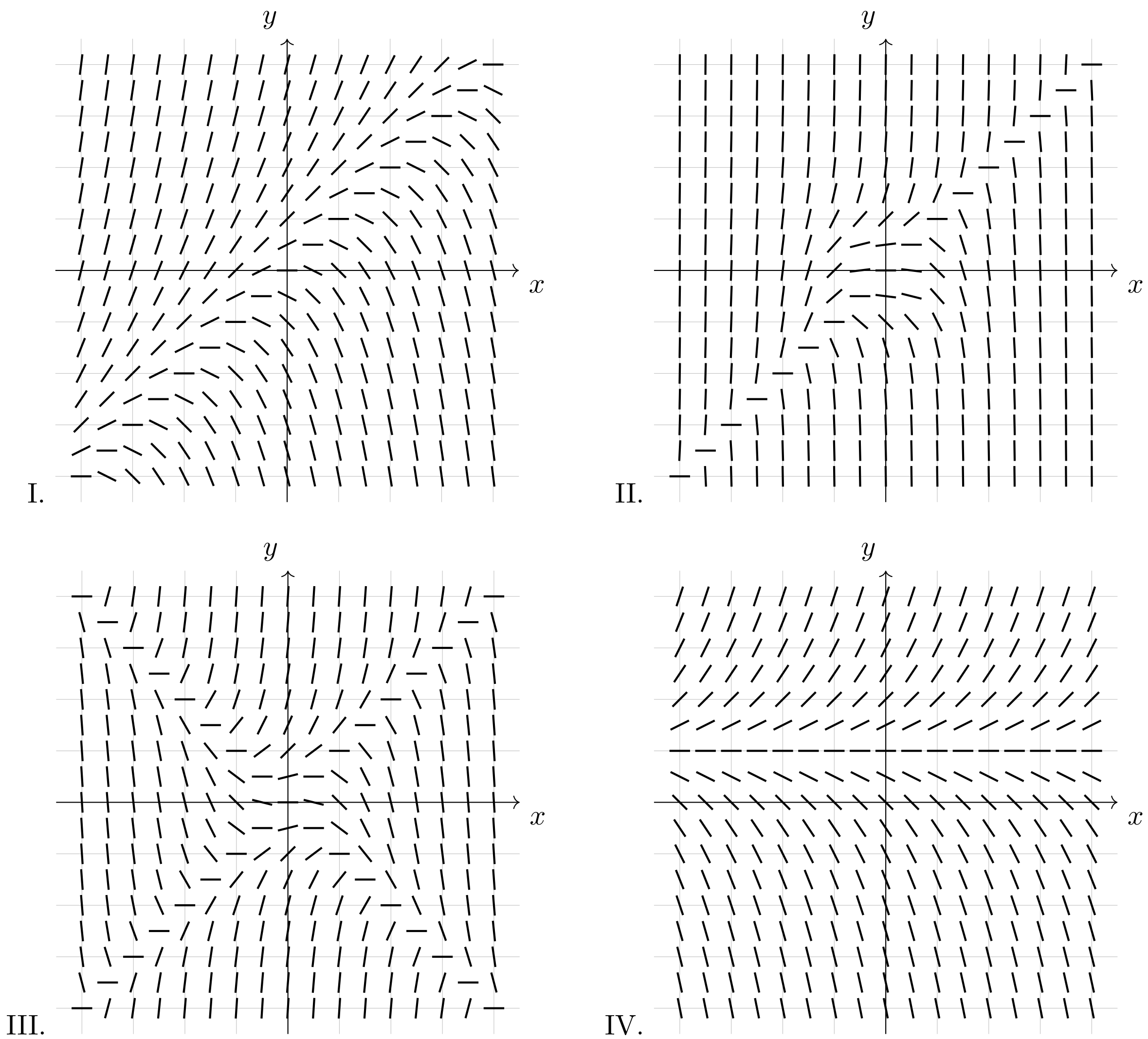
Match each differential equation with its slope field (labeled I-IV). Give reasons for your answer.

2. $\dfrac{dy}{dx} = y - 1$

3. $\dfrac{dy}{dx} = y - x$

4. $\dfrac{dy}{dx} = y^2 - x^2$

5. $\dfrac{dy}{dx} = y^3 - x^3$



6. Use the slope field labeled I. above to sketch the graph of the solutions that satisfy the given initial conditions.

(a) $y(0) = 1$

(b) $y(0) = 0$

(c) $y(0) = -1$

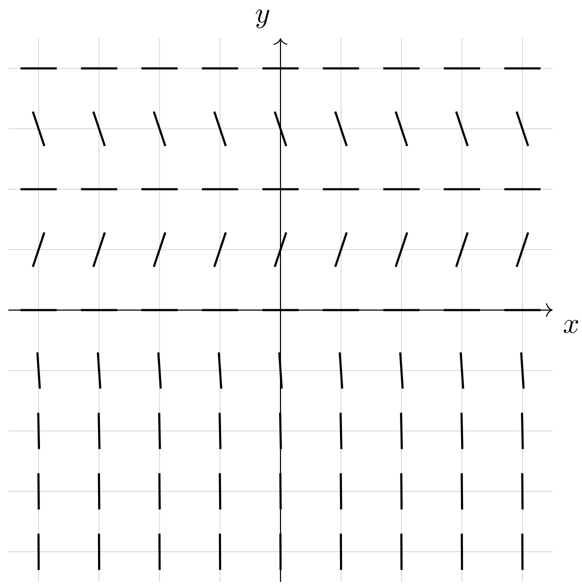
7. Sketch a slope field for the differential equation below. Then, use it to sketch three different solution curves.

$$y' = 1 + y$$

8. Sketch a slope field for the differential equation below. Then, use it to sketch a solution curve that passes through the point $(1, 0)$.

$$y' = y - 2x$$

9. A slope field for the differential equation $y' = y(y-2)(y-4)$ is shown.



For parts (a) through (d), sketch the graphs of the solutions that satisfy the given initial conditions.

(a) $y(0) = -0.3$

(b) $y(0) = 1$

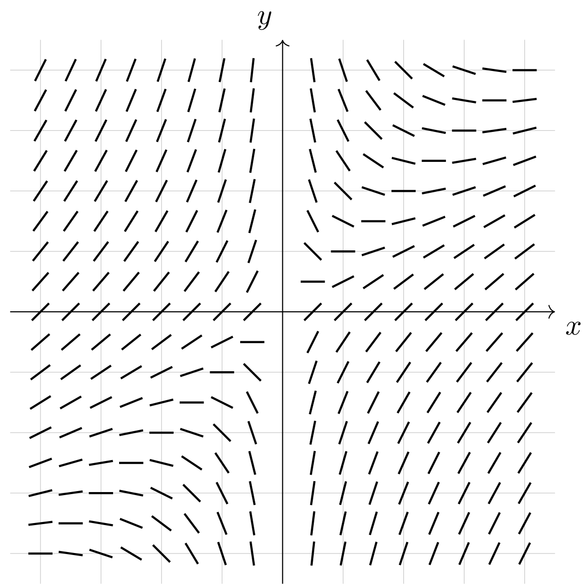
(c) $y(0) = 3$

(d) $y(0) = 4.3$

(e) If the initial condition is $y(0) = c$, for what values of c is $\lim_{t \to \infty} y(t)$ finite?

2.7 Multiple Choice Homework

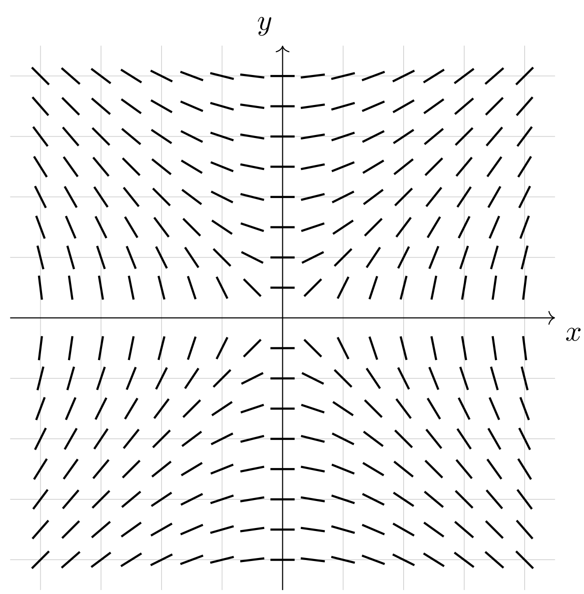
1. Which of the following differential equations corresponds to the slope field shown in the figure below?



a) $\dfrac{dy}{dx} = -\dfrac{y^2}{x}$ b) $\dfrac{dy}{dx} = 1 - \dfrac{y}{x}$ c) $\dfrac{dy}{dx} = y^3$ d) $\dfrac{dy}{dx} = x - \dfrac{1}{2}x^3$ e) $\dfrac{dy}{dx} = x + y$

---

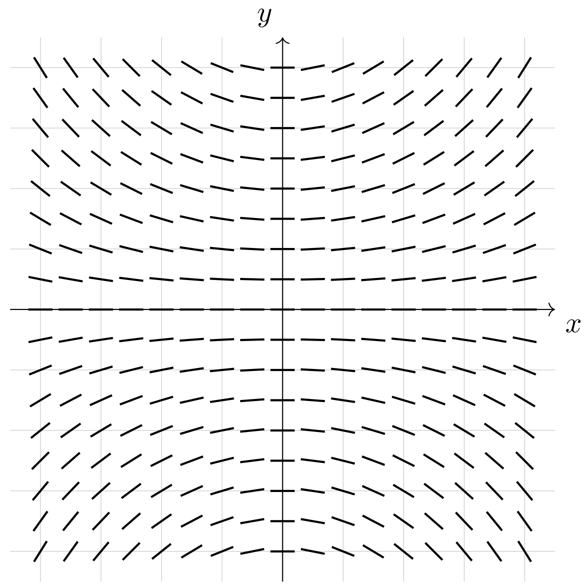
2. Which of the following differential equations corresponds to the slope field shown below?



a) $\dfrac{dy}{dx} = -\dfrac{y}{x}$ b) $\dfrac{dy}{dx} = 5xy$ c) $\dfrac{dy}{dx} = \dfrac{xy}{10}$ d) $\dfrac{dy}{dx} = \dfrac{y}{x}$ e) $\dfrac{dy}{dx} = \dfrac{x}{y}$

---

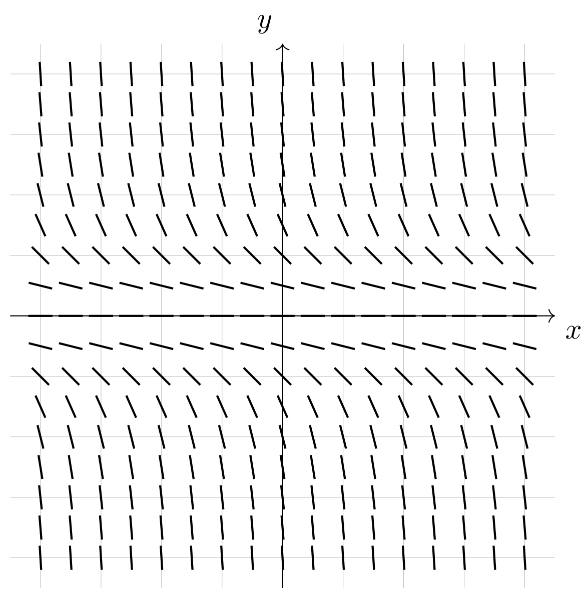
3. Which of the following differential equations corresponds to the slope field shown in the figure below?



a) $\dfrac{dy}{dx} = -\dfrac{y}{x}$ b) $\dfrac{dy}{dx} = 5xy$ c) $\dfrac{dy}{dx} = \dfrac{xy}{10}$ d) $\dfrac{dy}{dx} = \dfrac{y}{x}$ e) $\dfrac{dy}{dx} = \dfrac{x}{y}$

---

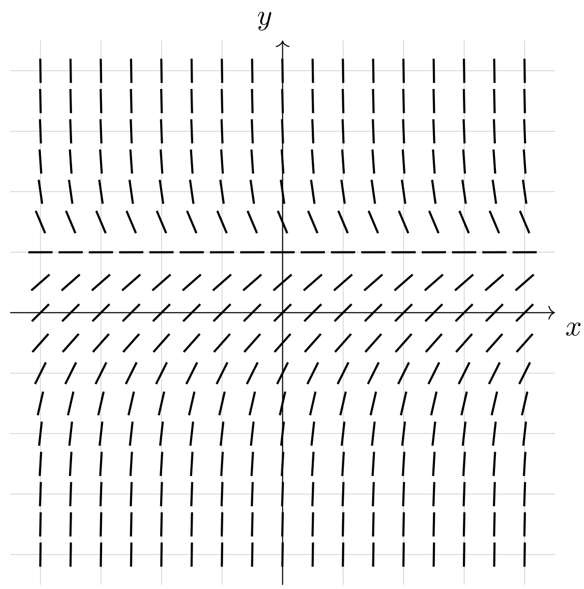
4. Which of the following equations might be the solution to the slope field shown in the figure below?



a) $y = 12x - x^3$ b) $-\cos(x)$ c) $\sec(x)$ d) $x = -y^2$ e) $x = -y^3$

---

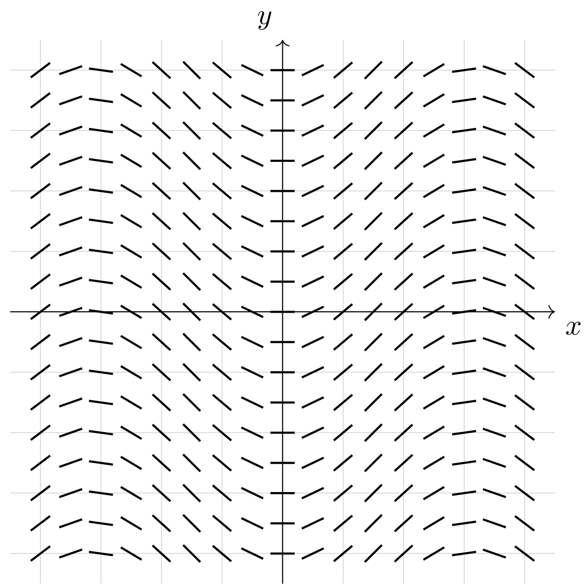
5. Which of the following differential equations corresponds to the slope field shown in the figure below?



a) $\dfrac{dy}{dx} = 1 - y^3$ b) $\dfrac{dy}{dx} = y^2 - 1$ c) $\dfrac{dy}{dx} = -\dfrac{x^2}{y^2}$ d) $\dfrac{dy}{dx} = x^2 y$ e) $\dfrac{dy}{dx} = x + y$

---

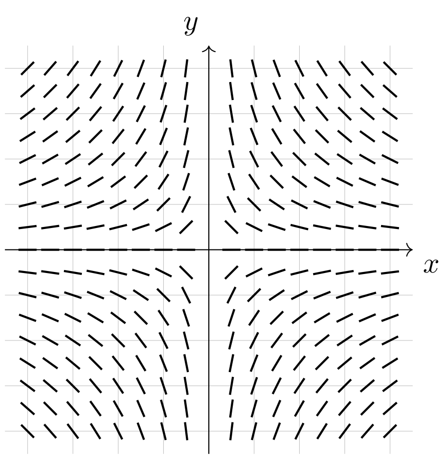
6. Which of the following differential equations corresponds to the slope field shown in the figure below?



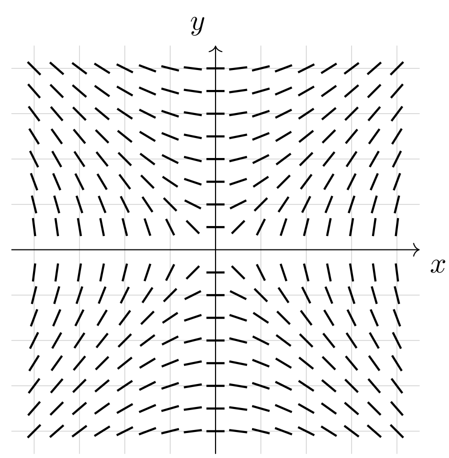
a) $y = 4x - x^3$ b) $y = -\cos(x)$ c) $y = \sec(x)$ d) $x = -y^2$ e) $x = -y^3$

---

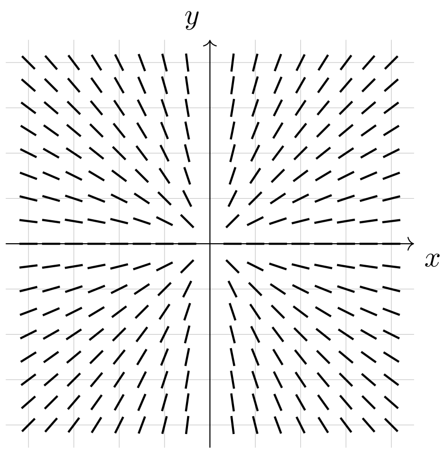
7. Which of the slope fields shown below corresponds to $\dfrac{dy}{dx} = -\dfrac{y}{x}$?



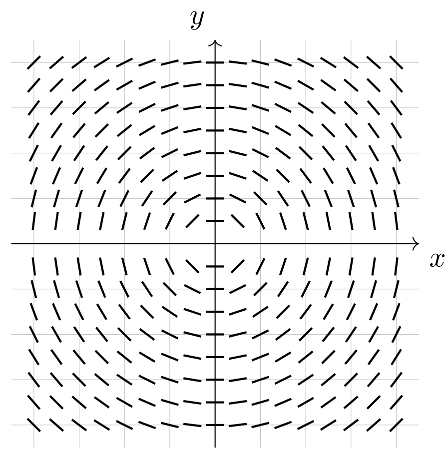
a)



b)

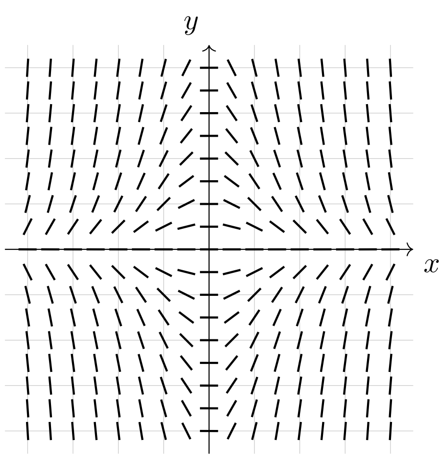


c)

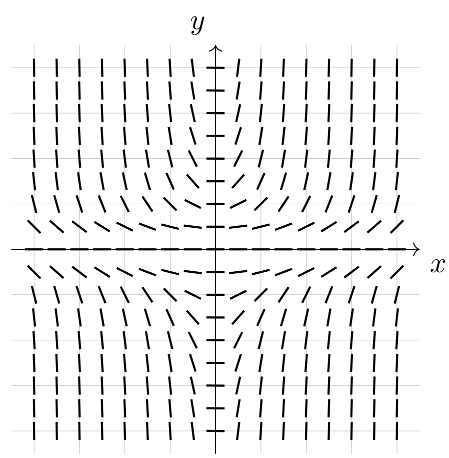


d)

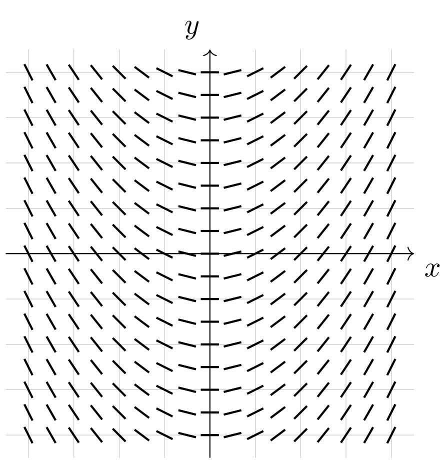
8. Which of the slope fields shown below corresponds to $\dfrac{dy}{dx} = xy$?



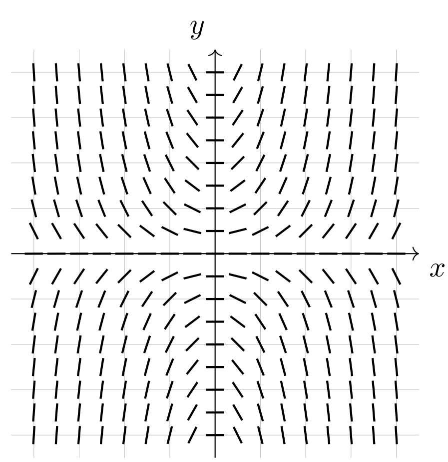
a)



b)

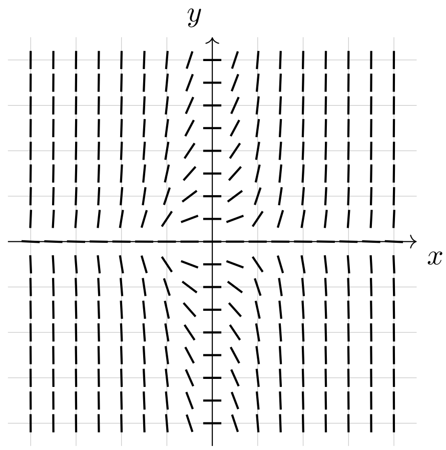


c)

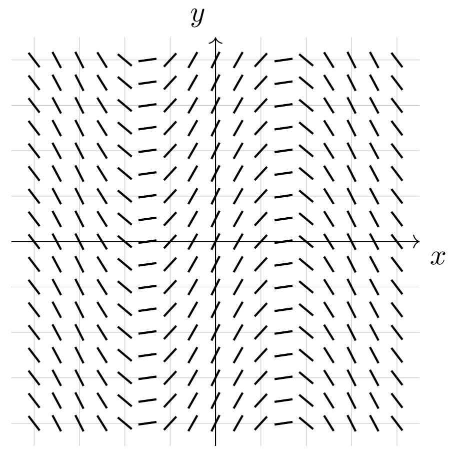


d)

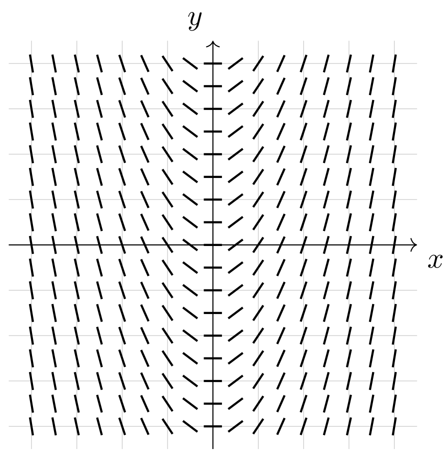
9. Which of the slope fields shown below corresponds to $|y| = e^{x^3}$?



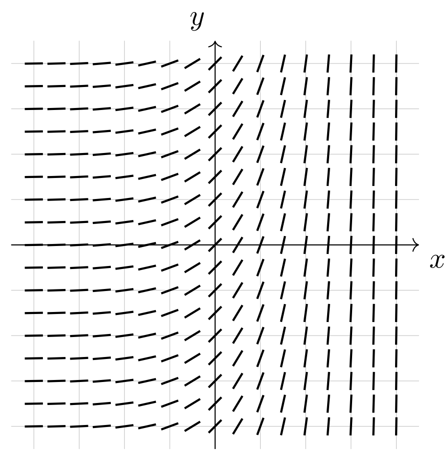
a)



b)



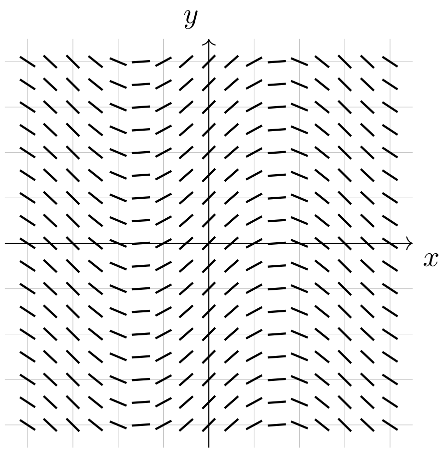
c)



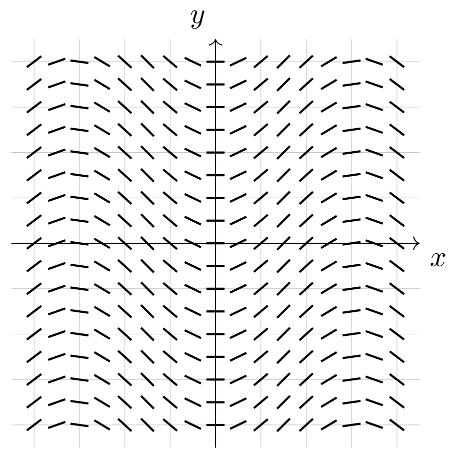
d)

---

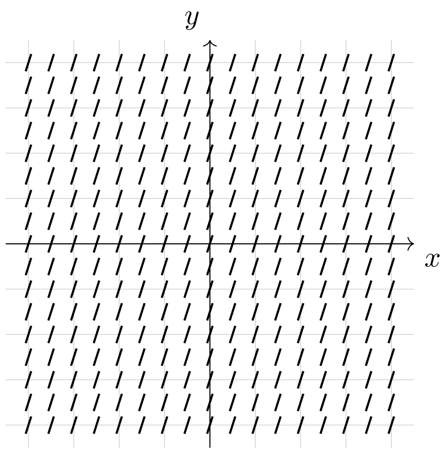
10. Which of the slope fields shown below corresponds to $y = \sec(x)$?



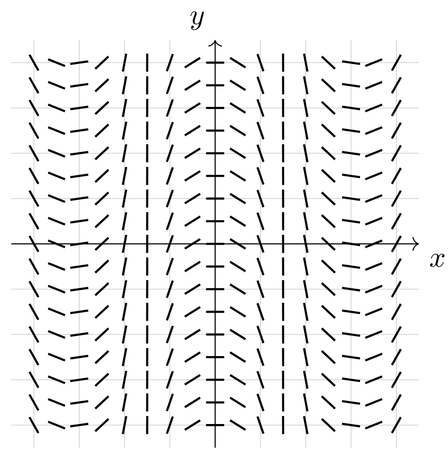
a)



b)



c)



d)

Full Name: AP Calculus BC

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Multiple Choice Section **20 Minutes; No Calculator**

Show All Work

1. Which of the following is **true**?

a) $\int \left(5^x + 2e^{-x}\right)\, dx = \frac{5^x}{\ln 5} + 2e^{-x} + C$

b) $\int \cot^3(x)\csc^2(x)\, dx = \frac{1}{4}\cot^4(x) + C$

c) $\int (x-1)\sqrt{x}\, dx = \frac{2}{3}x^{\frac{3}{2}} - \frac{1}{2}x^{\frac{1}{2}} + C$

d) $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\, dx = -2\cos\left(\sqrt{x}\right) + C$

---

2. $\int \frac{x}{x^2-4}\, dx =$

a) $-\frac{1}{4\left(x^2-4\right)^2} + C$

b) $-\frac{1}{2\left(x^2-4\right)} + C$

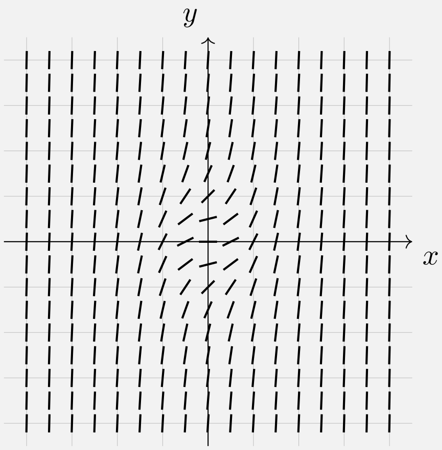
c) $\frac{1}{2}\ln\left|x^2-4\right| + C$

d) $2\ln\left|x^2-4\right| + C$

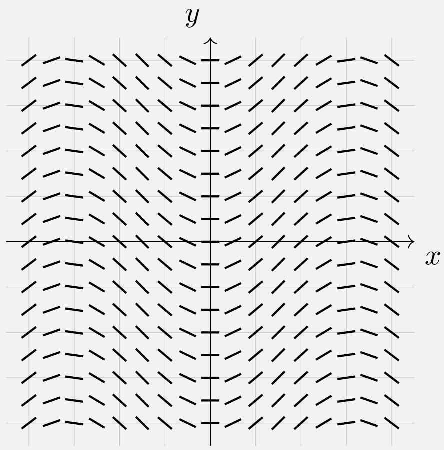
e) $\frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$

---

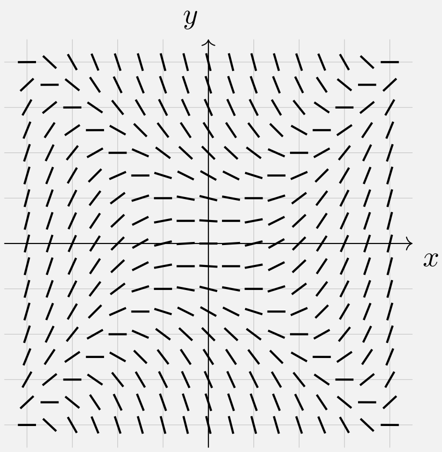
3. Which of the following slope fields presents $\dfrac{dy}{dx} = y^2 + 2x^2$?



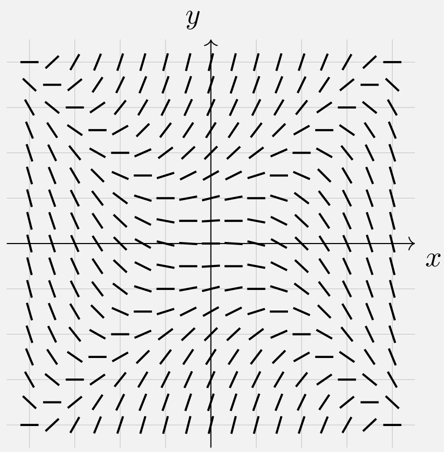
a)



b)

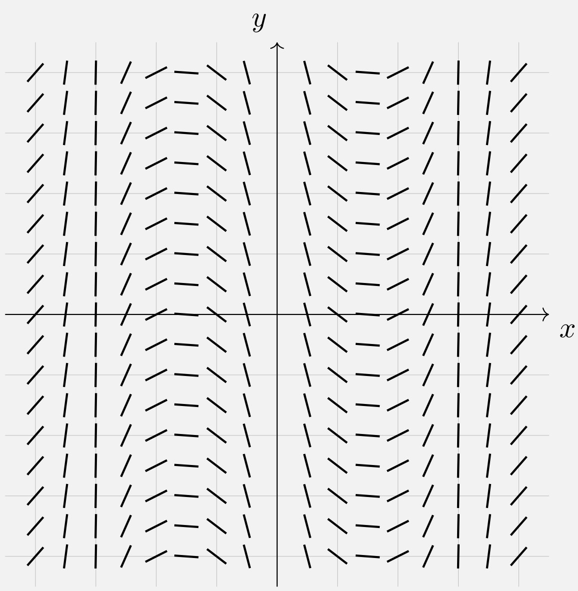


c)



d)

4. Which of the following equations corresponds to the solution to the slope field shown in the figure below?



a) $y = x^4 - 8x^2$ b) $y = 8x^2 - 4x^4$ c) $y = \csc(x)$ d) $y = -\sec(x)$

---

5. Which of the following is a solution to the differential equation $\dfrac{dy}{dx} = \dfrac{x-1}{y}$ with the initial condition $y(0) = -2$?

a) $y = -2e^{x^2-2x}$ b) $y = -2 + e^{x^2-2}$ c) $y = \sqrt{x^2 - 2x - 4}$

d) $y = -\sqrt{x^2 - 2x + 4}$ e) $y = -\sqrt{x^2 - 2x - 4}$

---

6. $\displaystyle\int \frac{4y^3 - 2y^2 - 5y}{\sqrt{y}}\, dy =$

a) $\left(y^4 - \dfrac{2}{3}y^3 - \dfrac{5}{2}y^2\right) 2y^{\frac{1}{2}}$ b) $4y^{\frac{5}{2}} - 2y^{\frac{3}{2}} - 5y^{\frac{1}{2}} + C$

c) $\dfrac{8}{7}y^{\frac{7}{2}} - \dfrac{4}{5}y^{\frac{5}{2}} - \dfrac{10}{3}y^{\frac{3}{2}} + C$ d) $10y^{\frac{3}{2}} - 3y^{\frac{1}{2}} - \dfrac{5}{2}y^{-\frac{1}{2}} + C$

---

7. Identify the first mistake (if any) in this process:

Problem:	$\dfrac{dy}{dx} = x - xy$
Step 1:	$\dfrac{1}{1-y}\,dy = x\,dx$
Step 2:	$-\ln\lvert 1-y\rvert = \dfrac{1}{2}x^2 + C$
Step 3:	$\lvert 1-y\rvert = e^{-\frac{1}{2}x^2+C}$
Step 4:	$y = 1 + Ke^{\frac{1}{2}x^2}$

a) Step 1 b) Step 2 c) Step 3 d) Step 4 e) No mistake

---

8. An object moves with velocity $v(t) = \sec^2(2t)$. It is known that the particle's position at time 0 is 2. What is the particle's position function?

a) $s(t) = \tan(2t) + 2$ b) $s(t) = \dfrac{1}{2}\tan(2t) + 2$ c) $s(t) = \sec^2(2t)\tan^2(2t) + 2$

d) $s(t) = \ln\lvert\sec(2t)\rvert + 2$ e) $s(t) = \dfrac{1}{2}\ln\lvert\sec(2t)\rvert + 2$

Free Response Section **35 Minutes; No Calculator**

Show All Work

1. Compute the following antiderivatives.

(a) $\int \left(2x^5 + 2^x - \frac{17}{\sqrt[5]{x^2}} - \frac{1}{5x^2}\right) dx$

(b) $\int e^{5x} \csc\left(e^{5x}\right) dx$

(c) $\displaystyle\int \left(\cos(3x) + \cos^2(5x) + \sin(7x)\sqrt{\cos(7x)} \right) dx$

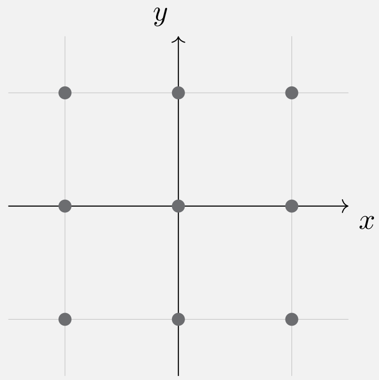
---

2. The acceleration of a particle is described by $a(t) = 98e^{-7t}$. Find the distance equation for $x(t)$ if $v(0) = 0$ and $x(0) = -2$.

---

3. Let's define a differential equation $\dfrac{dy}{dx} = \dfrac{y}{x^2+4}$.

(a) On the axis system provided, sketch the slope field for $\dfrac{dy}{dx}$ at all points plotted on the graph. You may assume that all gridlines have length 1.



(b) Find the particular solution $w = f(t)$ that passes through $y(0) = -2$.

Full Name: AP Calculus BC

Date: Chapter 2 Practice Test 2

Multiple Choice Section **20 Minutes; No Calculator**

1. Which of the following statements are **true**?

   I. $\int \left(x^3 + x\right) \sqrt[4]{x^4 + 2x^2 - 5}\, dx = \frac{1}{5}\left(x^4 - 2x^2 - 5\right)^{\frac{5}{4}} + C$

   II. $\int x^5 \sin\left(x^6\right)\, dx = -\frac{1}{6}\cos\left(x^6\right) + C$

   III. $\int \csc(x)\, dx = \ln|\csc(x) + \cot(x)| + C$

   a) I only   b) II only   c) III only   d) I and II only   e) II and III only

---

2. $\int \frac{x-2}{x-1}\, dx =$

   a) $-\ln|x-1| + C$ b) $x + \ln|x-1| + C$ c) $x - \ln|x-1| + C$

   d) $x - \sqrt{x-1} + C$ e) $x + \sqrt{x-1} + C$

---

3. If $\frac{dy}{dx} = \sin(x)\cos^3(x)$ and if $y = 1$ when $x = \pi$, what is the value of y when $x = 0$?

   a) -3 b) -2 c) 1 d) 2 e) 3

---

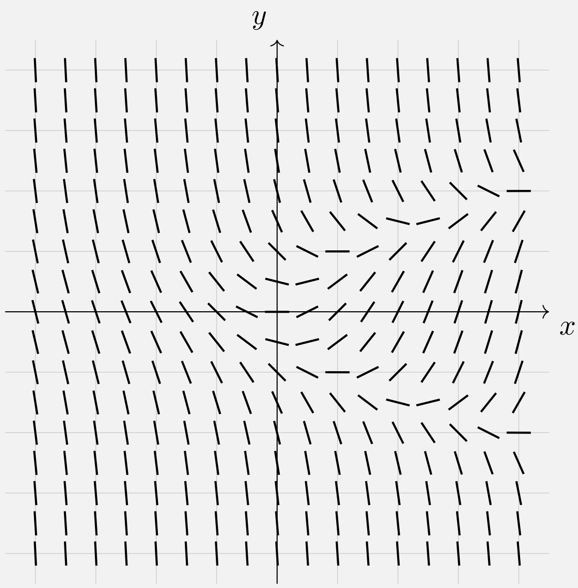
4. $\int x\sqrt{1-x^2}\,dx =$

a) $\dfrac{(1-x^2)^{\frac{3}{2}}}{3}+C$ b) $-\left(1-x^2\right)^{\frac{3}{2}}+C$ c) $\dfrac{x^2\,(1-x^2)^{\frac{3}{2}}}{3}+C$

d) $-\dfrac{x^2\,(1-x^2)^{\frac{3}{2}}}{3}+C$ e) $-\dfrac{(1-x^2)^{\frac{3}{2}}}{3}+C$

---

5. Which of the following differential equations corresponds to the slope field shown in the figure below?



a) $\dfrac{dy}{dx}=x-y^2$ b) $\dfrac{dy}{dx}=1-\dfrac{y}{x}$ c) $\dfrac{dy}{dx}=-y^3$ d) $\dfrac{dy}{dx}=x-\dfrac{1}{2}x^3$ e) $\dfrac{dy}{dx}=x+y$

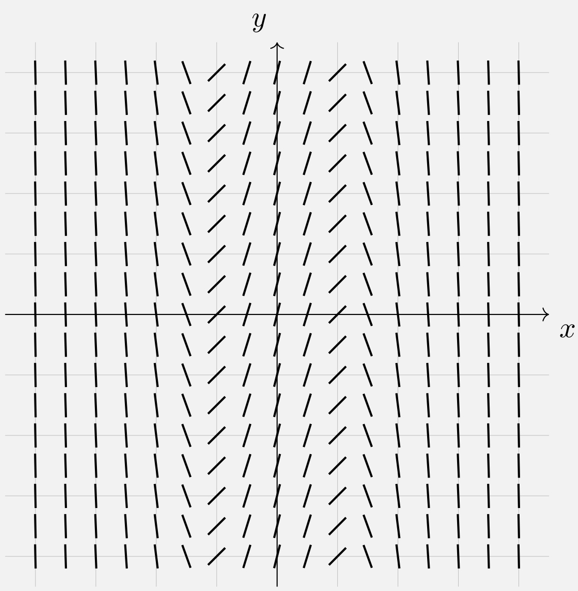
---

6. For $\int \sec^2(x)\tan^2(x)\,dx$, the correct u-substitution is

a) $u=\sec(x)$ b) $u=\tan(x)$ c) Either (a) or (b) d) Neither (a) nor (b)

---

7. Which of the following equations might be the solution to the slope field shown in the figure below?



a) $y = 4x - x^3$ b) $y = x^3 - 4x$ c) $y = 4x^4$ d) $y = x^3 - 15x^5$ e) $y = \sec(x)$

---

8. Identify the first mistake (if any) in this process:

Problem: $\dfrac{dy}{dx} = xy + x$

Step 1: $\dfrac{1}{y+1}\,dy = x\,dx$

Step 2: $\ln|y+1| = x^2 + C$

Step 3: $|y+1| = e^{x^2} + C$

Step 4: $y = e^{x^2} + C$

a) Step 1 b) Step 2 c) Step 3 d) Step 4 e) No mistake

Free Response Section **35 Minutes; No Calculator**

Show All Work

1. Compute the following antiderivatives.

(a) $\displaystyle\int \frac{t^3 - 4t - 3}{5t^{\frac{2}{3}}}\,dt$

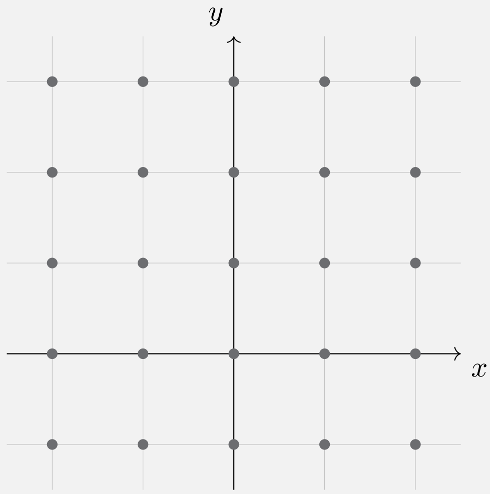
(b) $\displaystyle\int \frac{x^2}{(x^3 - 1)^{\frac{3}{2}}}\,dx$

(c) $\int x\sqrt{-3x^2+17}\,dx$

2. The acceleration of a particle is described by $a(t) = 48t^2 - 18t + 6$. Find the distance equation for $x(t)$ if $v(1) = 1$ and $x(1) = 3$.

3. Let's define a differential equation $\frac{dy}{dx} = \frac{y-2}{x+1}$.

(a) On the axis system provided, sketch the slope field for $\frac{dy}{dx}$ at all points plotted on the graph. You may assume that all gridlines have length 1.



(b) If the solution curve passes through the point (0, 0), sketch the solution curve on the same set of axes as your slope field.

(c) Find the equation for the solution curve of $\frac{dy}{dx} = \frac{y-2}{x+1}$ given that $y(0) = 5$.

Chapter 2 Answer Key

Chapter 3:

Integrals

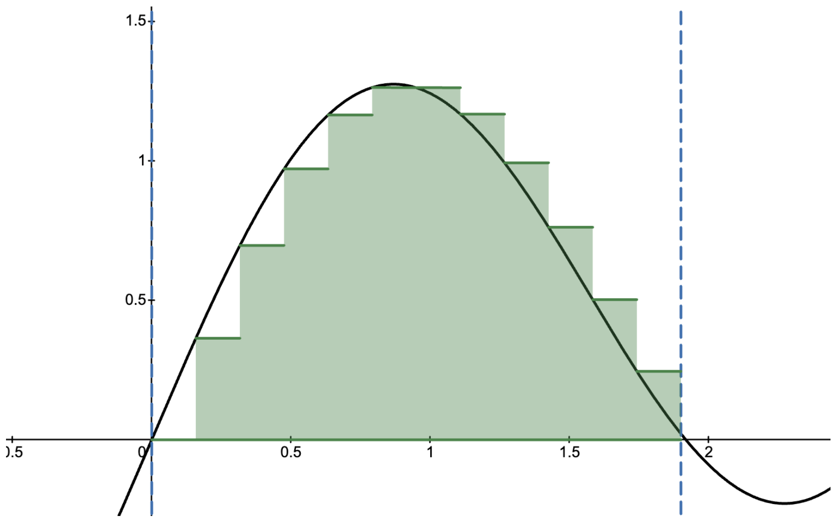
Chapter 3 Overview: Definite Integrals

In this chapter, we will study the Fundamental Theorem of Calculus, which establishes the link between the algebra and the geometry, with an emphasis on mechanics of how to find the definite integral. We will consider the differences implied between the context of the definite integral as an operation and as an area accumulator. We will learn some approximation techniques for definite integrals and see how they provide theoretical foundation for the integral. We will revisit graphical analysis in terms of the definite integral and view another typical AP context for it. Finally, we will consider what happens when trying to integrate at or near an asymptote.

As noted in the overview of the last chapter, antiderivatives are known as indefinite integrals because the answer is a function, not a definite number. But there is a time when the integral represents a number. That is when the integral is used in an analytic-geometrical context of area. Though it is not necessary to know the theory behind this in order to calculate the integral, the theory is a major subject of integral calculus, so we will explore it briefly in here.

The Limit Definition of the Definite Integral

We know, from geometry, how to find the exact area of various polygons, but we never considered figures where one side is not made of a line segment. Here we want to consider an area bounded by some curve $y = f(x)$ on the top, the x-axis on the bottom, some arbitrary $x = a$ on the left, and $x = b$ on the right.



As we can see above, the area approximated by rectangles whose height is the y-value of the equation and whose width we will call Δx. The more rectangles we make, the better the approximation. For a good animation of this concept, consider the following video:

Riemann sum approximation animation

The area of each rectangle would be $f(x) \cdot \Delta x$, and the total area of n rectangles would be

$$A = \sum_{i=1}^{n} f(x_i) \cdot \Delta x.$$

This equation is known as the **Riemann summation**. Although this equation looks complicated, it represents a rather simple idea. We are adding up the areas of many thin rectangles to approximate the total area under the curve $y = f(x)$ between two points. As we increase the number of rectangles n, each rectangle becomes narrower, and thus our approximation becomes more accurate.

But, how can we find the exact area? With the Riemann sum, we are only coming up with better and better approximations right now. If we could make an infinite number of rectangles (which would be infinitely thin), we could potentially find the exact area under this curve. Luckily, we just so happen to have the mathematical tools to do this: we can take the limit as n approaches infinity.

$$\lim_{n \to \infty} \sum_{i=1}^{n} f(x_i) \cdot \Delta x$$

This is where the definite integral comes in. The definite integral provides a precise way to calculate the exact area under a curve by taking the limit of the Riemann sum as the number of rectangles approaches infinity. In other words, instead of merely approximating the area with a finite number of rectangles, the definite integral captures what happens when the width of each rectangle becomes infinitesimally small. We write this limit in a compact form as:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i\right) \cdot \Delta x,$$

where a is the "lower bound" and b is the "upper bound." Mathematicians sometimes nuance this statement as

$$\boxed{\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(a + k\Delta x) \cdot \Delta x, \text{ where } \Delta x = \frac{b-a}{n}}$$

3.1: The Fundamental Theorem of Calculus

Up to this point, we've seen how integration can be used to find the exact area under a curve by taking the limit of Riemann sums. But what's truly remarkable is how integration connects so deeply with differentiation. This connection is captured in one of the most important results in all of calculus: the Fundamental Theorem of Calculus (FTC).

The Fundamental Theorem of Calculus

If $f(x)$ is a continuous function on $[a, b]$, then:

1. $\dfrac{d}{dx}\displaystyle\int_c^x f(t)\,dt = f(x)$ or $\dfrac{d}{dx}\displaystyle\int_c^u f(t)\,dt = f(u)\cdot\dfrac{du}{dx}$

2. If $F'(x) = f(x)$, then $\displaystyle\int_a^b f(x)\,dx = F(b) - F(a)$.

The first part of the Fundamental Theorem of Calculus simply says what we already know– that an integral is an anti-derivative. The second part of the Fundamental Theorem says that the answer to a definite integral is the difference between the anti-derivative at the upper bound and the anti-derivative at the lower bound.

The idea of the integral meaning the area may not make sense initially, mainly because we are used to geometry, where an area is always measured in square units. But, that is only because the length and width are always measured in the same kind of units, so multiplying length and width area always measured in the same kind of units, so multiplying length and width must yield square units. We are expanding our vision beyond that narrow view of things here. Consider a graph where the x-axis is time and the y-axis is velocity in feet per second. The area under the curve would be measured as seconds multiplied by feet per seconds, which is simply feet. So, the area under the curve is equal to the distance traveled in feet. In other words, the integral of velocity is distance.

OBJECTIVES

Evaluate Definite Integrals.
Find Average Value of a Continuous Function Over a Given Interval.
Differentiate Integral Expressions with the Variable in the Boundary.

Let us first consider part 2 of the Fundamental Theorem of Calculus, since it has a very practical application. This part of the Fundamental Theorem of Calculus gives us a clear method for evaluating definite integrals.

Ex 3.1.1: Evaluate $\displaystyle\int_2^8 (4x+3)\,dx$

Sol 3.1.1: First, let's start by treating this as a regular antiderivative

$$\int (4x+3)\,dx = 2x^2 + 3x$$

Note that our $+C$ will not be needed, as we will be taking a definite integral. Now, let's apply the Fundamental Theorem of Calculus.

$$\begin{aligned}
\int_2^8 f(x)\,dx &= F(8) - F(2) \\
&= \left(2x^2 + 3x\right)\Big|_2^8 \\
&= 2(8)^2 + 3(8) - \left(2(2)^2 + 3(2)\right) \\
&= \boxed{138}
\end{aligned}$$

The vertical bar that you see is called the evaluation bar, and it's used to indicate that we are evaluating the antiderivative at the upper and lower limits of integration.

Ex 3.1.2: Evaluate $\displaystyle\int_1^4 \frac{1}{\sqrt{x}}\,dx$

Sol 3.1.2:

$$\begin{aligned}
\int_1^4 \frac{1}{\sqrt{x}}\,dx &= 2\sqrt{x}\Big|_1^4 \\
&= 2\sqrt{(4)} - 2\sqrt{(1)} \\
&= \boxed{2}
\end{aligned}$$

Ex 3.1.3: Evaluate $\displaystyle\int_0^{\frac{\pi}{2}} \sin(x)\,dx$

Sol 3.1.3:

$$\int_0^{\frac{\pi}{2}} \sin(x)\,dx = -\cos(x)\Big|_0^{\frac{\pi}{2}}$$

$$= -\cos\left(\frac{\pi}{2}\right) + \cos(0)$$

$$= \boxed{1}$$

Ex 3.1.4: Evaluate $\displaystyle\int_1^2 \frac{4+u^2}{u^3}\,du$

Sol 3.1.4:

$$\int_1^2 \frac{4+u^2}{u^3}\,du = \int_1^2 \left(4u^{-3} + u^{-1}\right)\,du$$

$$= \left(-2u^{-2} + \ln|u|\right)\Big|_1^2$$

$$= -2(2)^{-2} + \ln 2 - (-2(1)^{-2} + \ln 1)$$

$$= \boxed{\frac{3}{2} + \ln 2}$$

Ex 3.1.5: Evaluate $\displaystyle\int_{-5}^5 \frac{1}{x^3}\,dx$

Sol 3.1.5: When initially looking at the problem, one may simply proceed with finding the definite integral.

$$\int_{-5}^5 \frac{1}{x^3}\,dx = -\frac{1}{x^2}\Big|_{-5}^5$$

$$= -\frac{1}{(-5)^2} + \frac{1}{(5)^2}$$

$$= 0$$

But, this is a trap! We have to be careful here, because the function $\dfrac{1}{x^3}$ is *not defined* over the the interval $[-5, 5]$, since $\dfrac{1}{0^3}$ is undefined. Therefore, the Fundamental Theorem of Calculus $\boxed{\text{does not apply}}$.

Just as we had many properties for the indefinite integral, we have many properties for the definite integral. There are three main properties that are utilized often on the AP exam.

Properties of Definite Integrals

1. $\displaystyle\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

2. $\displaystyle\int_a^a f(x)\,dx = 0$

3. $\displaystyle\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where $a < c < b$

Ex 3.1.6: If $\displaystyle\int_{-5}^{2} f(x)\,dx = -17$ and $\displaystyle\int_{5}^{2} f(x)\,dx = -4$, find $\displaystyle\int_{-5}^{5} f(x)\,dx$.

Sol 3.1.6:

$$\begin{aligned}\int_{-5}^{5} f(x)\,dx &= \int_{-5}^{2} f(x)\,dx + \int_{2}^{5} f(x)\,dx \\ &= \int_{-5}^{2} f(x)\,dx - \int_{5}^{2} f(x)\,dx \\ &= -17 - (-4) \\ &= \boxed{-13}\end{aligned}$$

Part I of the Fundamental Theorem of Calculus is very important for the **theory** of calculus, but is limited (hehe) in the context of this course to L'Hospital problems which will will explore in a later chapter. Here is how the formula may be applied.

Ex 3.1.7: Use the Fundamental Theorem of Calculus to find $f'(t)$ if $f(t) = \displaystyle\int_{2}^{3t^2} (4x + 3)\,dx$

Sol 3.1.7:

$$f'(t) = \frac{d}{dt}\int_{2}^{3t^2} (4x+3)\,dx$$

$$= \left(4\left(3t^2\right)+3\right)(6t)$$

$$= \boxed{72t^3+18t}$$

3.1 Free Response Homework

Use part II of the Fundamental Theorem of Calculus to evaluate the integral, or explain why the integral cannot be evaluated.

1. $\displaystyle\int_{-1}^{3} x^5\,dx$

2. $\displaystyle\int_{2}^{7} (5x-1)\,dx$

3. $\displaystyle\int_{-5}^{5} \frac{2}{x^3}\,dx$

4. $\displaystyle\int_{-3}^{-1} \frac{x^7-4x^3-3}{x}\,dx$

5. $\displaystyle\int_{1}^{2} \frac{3}{t^4}\,dt$

6. $\displaystyle\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \csc(y)\cot(y)\,dy$

7. $\displaystyle\int_{0}^{\frac{\pi}{4}} \sec^2(y)\,dy$

8. $\displaystyle\int_{1}^{9} \frac{3}{2z}\,dz$

9. $\displaystyle\int_{1}^{8} \frac{x^2-4}{\sqrt[3]{x}}\,dx$

10. $\displaystyle\int_{\pi}^{\frac{5\pi}{4}} \sin(y)\,dy$

11. $\displaystyle\int_{1}^{4} \frac{x^4-4x^2-5}{x^2}\,dx$

12. $\displaystyle\int_{3}^{5} \left(x^2+5x+6\right)\,dx$

13. $\displaystyle\int_{\pi}^{\frac{\pi}{4}} \cos(y)\,dy$

14. $\displaystyle\int_{1}^{4} \frac{x^3-2x^2-4x}{x^2}\,dx$

15. $\displaystyle\int_{1}^{2} \frac{x^2-4x+7}{x}\,dx$

16. $\displaystyle\int_{1}^{16} \frac{2x^2-1}{\sqrt[4]{x}}$

Use the following values for problems 17 - 27 to evaluate the given integrals.

$\displaystyle\int_{-2}^{5} f(x)\,dx = -2$	$\displaystyle\int_{1}^{5} f(x)\,dx = 3$
$\displaystyle\int_{-2}^{1} g(x)\,dx = 4$	$\displaystyle\int_{5}^{1} g(x)\,dx = 9$
$\displaystyle\int_{1}^{5} h(x)\,dx = 7$	$\displaystyle\int_{5}^{-2} h(x)\,dx = -6$

17. $\displaystyle\int_{-2}^{1} f(x)\,dx$

18. $\displaystyle\int_{-2}^{5} g(x)\,dx$

19. $\displaystyle\int_{-2}^{1} h(x)\,dx$

20. $\displaystyle\int_{1}^{5} (f(x)-g(x))\,dx$

21. $\int_{-2}^{5} (g(x) + h(x))\, dx$

22. $\int_{-2}^{1} (h(x) - f(x))\, dx$

23. $\int_{-2}^{5} (h(x) + f(x))\, dx$

24. $\int_{1}^{5} (2f(x) + 3h(x))\, dx$

25. $\int_{-2}^{1} (2f(x) - 3g(x))\, dx$

26. $\int_{-2}^{5} \left(\frac{1}{2}g(x) + 4h(x)\right)\, dx$

27. $\int_{1}^{5} \left(\frac{1}{3}h(x) + 2f(x)\right)\, dx$

28. $\int_{5}^{5} (f(x) + g(x) + h(x))\, dx$

Use part I of the Fundamental Theorem of Calculus to find the derivative of the function.

29. $g(y) = \int_{2}^{y} t^2 \sin(t)\, dt$

30. $g(x) = \int_{0}^{x} \sqrt{1+2t}\, dt$

31. $F(x) = \int_{x}^{2} \cos\left(t^2\right)\, dt$

32. $h(x) = \int_{2}^{\frac{1}{x}} \arctan(t)\, dt$

33. $y = \int_{3}^{\sqrt{x}} \frac{\cos(t)}{t}\, dt$

34. $f(x) = \int_{e}^{x^2} \ln\left(t^2+1\right)\, dt$

35. $f(x) = \int_{10}^{x^2} t \ln t\, dt$

36. $f(x) = \int_{e^x}^{5} \left(t^3 + t + 1\right)\, dt$

37. If $F(x) = \int_{1}^{t^2} \frac{\sqrt{1+u^4}}{u}\, du$, find $F'(t)$.

38. If $h(x) = \int_{\pi}^{\sqrt{x}} e^{5t}\, dt$, find $h'(x)$.

39. If $h(m) = \int_{5}^{\cos(m)} t^2 \cos^{-1}(t)\, dt$, find $h'(m)$.

40. If $h(y) = \int_{5}^{\ln y} \frac{e^t}{t^4}\, dt$, find $h'(y)$.

3.1 Multiple Choice Homework

1. If $\int_{-5}^{2} f(x)\, dx = -17$ and $\int_{5}^{2} f(x)\, dx = -4$, then $\int_{-5}^{5} f(x)\, dx =$

a) -21 b) -13 c) 0 d) 13 e) 21

---

2. Let f and g be continuous functions such that $\int_{0}^{6} f(x)\, dx = 9$, $\int_{3}^{6} f(x)\, dx = 5$, and $\int_{3}^{0} g(x)\, dx = -7$. What is the value of $\int_{0}^{3} \left(\frac{1}{2}f(x) - 3g(x)\right)\, dx$.

a) -23 b) -19 c) $-\frac{17}{2}$ d) 19 e) 23

---

3. Given that $\int_2^3 P(t)\,dt = 7$ and $\int_2^7 P(t)\,dt = -2$, what is $\int_7^3 P(t)\,dt$?

a) -9 b) -5 c) 5 d) 9 e) not enough information

---

4. Based on the information below, find $\int_1^{-2} (g(x) + f(x))\,dx$

$\int_{-2}^5 f(x)\,dx = -2$	$\int_1^5 f(x)\,dx = 3$
$\int_{-2}^1 g(x)\,dx = 4$	$\int_5^1 g(x)\,dx = 9$

a) -9 b) -1 c) 0 d) 1 e) 9

---

5. Based on the information below, find $\int_5^{-2} (g(x) - f(x))\,dx$

$\int_{-2}^5 f(x)\,dx = -2$	$\int_1^5 f(x)\,dx = 3$
$\int_{-2}^1 g(x)\,dx = 4$	$\int_5^1 g(x)\,dx = 9$

a) -3 b) 3 c) 6 d) -6 e) 14

---

6. Using the table values from questions 5 and 6, which of the following cannot be determined?

a) $\int_5^1 (g(x) + f(x))\,dx$ b) $\int_1^{-2} (g(x) - f(x))\,dx$

c) $\int_{-2}^5 3g(x)(-4(f(x)))\,dx$ d) $\int_1^5 (3g(x) + 4f(x))\,dx$

3.2: Definite Integrals and the Substitution Rule

Now, it's time to revisit u-substitution within the context of definite integrals. Although the process is largely similar, there are some nuances that we must consider.

OBJECTIVES

Evaluate Definite Integrals Using the Fundamental Theorem of Calculus.
Evaluate Definite Integrals Applying the Substitution Rule, When Appropriate.
Use Proper Notation When Evaluating These Integrals

Ex 3.2.1: Evaluate $\displaystyle\int_0^2 t^2\sqrt{t^3+1}\,dx$.

Sol 3.2.1: Now, this problem may look just like a regular u-substitution problem that we did in the previous chapter. However, when we switch our integration variable to u, we also need to make sure to switch our definite integration boundaries to match. Let's see what that means in the solution below.

$$\int_0^2 t^2\sqrt{t^3+1}\,dt = \frac{1}{3}\int_0^2 3t^2\sqrt{t^3+1}\,dt$$

$$\hookrightarrow u = t^3+1 \quad\Big|\quad \hookrightarrow du = 3t^2\,dt$$

Here is where we have to be careful. We cannot simply rewrite our definite integral in terms of u, since our upper and lower boundaries are in terms of t! Therefore, we need to also find our new boundaries for the u variables. Luckily, we can do this simply by plugging in our t values into our equation for u.

$$u(0) = (0)^3+1 = 1 \quad\Big|\quad u(2) = (2)^3+1 = 9$$

Now, we can continue integrating! (Note the boundary change in red.)

$$\begin{aligned}
\frac{1}{3}\int_{\color{red}0}^{\color{red}2} 3t^2\sqrt{t^3+1}\,dt &= \frac{1}{3}\int_{\color{red}1}^{\color{red}9} \sqrt{u}\,du \\
&= \frac{1}{3}\left(\frac{2}{3}u^{\frac{3}{2}}\right)\Bigg|_1^9 \\
&= \frac{2}{9}\left(9^{\frac{3}{2}} - 1^{\frac{3}{2}}\right) \\
&= \boxed{\frac{52}{9}}
\end{aligned}$$

Ex 3.2.2: Evaluate $\int_0^{\sqrt{\pi}} x\cos\left(x^2\right)\,dx$.

Sol 3.2.2:

$$\hookrightarrow u = x^2 \quad \Big| \quad \hookrightarrow du = 2x\,dx$$

$$u(0) = (0)^2 = 0 \quad \Big| \quad u\left(\sqrt{\pi}\right) = \left(\sqrt{\pi}\right)^2 = \pi$$

$$= \frac{1}{2}\int_0^{\pi} \cos(u)\,du$$

$$= \frac{1}{2}\sin(u)\Big|_0^{\pi}$$

$$= \boxed{0}$$

Ex 3.2.3: Evaluate $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2}\,dx$.

Sol 3.2.3:

$$\hookrightarrow u = \frac{1}{x} \quad \Big| \quad \hookrightarrow du = -\frac{1}{x^2}$$

$$u(1) = \frac{1}{(1)} = 1 \quad \Big| \quad u(2) = \frac{1}{(2)} = \frac{1}{2}$$

$$\int_1^2 \frac{e^{\frac{1}{x}}}{x^2}\,dx = -\int_1^{\frac{1}{2}} e^u\,du$$

$$= -e^u\Big|_1^{\frac{1}{2}}$$

$$= \boxed{-e^{\frac{1}{2}} + e}$$

Ex 3.2.4: Evaluate $\int_1^{\sqrt{13}} \frac{x}{x^2+3}\,dx$.

Sol 3.2.4:

$$\hookrightarrow u = x^2 + 3 \quad \Big| \quad \hookrightarrow du = 2x\,dx$$

$$u(1) = (1)^2 + 3 = 4 \quad \Big| \quad u\left(\sqrt{13}\right)^2 + 3 = 16$$

$$\begin{aligned}
\int_1^{\sqrt{13}} \frac{x}{x^2+3}\,dx &= \frac{1}{2}\int_1^{\sqrt{13}} \frac{2x}{x^2+3}\,dx \\
&= \frac{1}{2}\int_4^{16} \frac{1}{u}\,du \\
&= \frac{1}{2}\ln|u|\Big|_4^{16} \\
&= \boxed{\frac{1}{2}\ln(16) - \frac{1}{2}\ln 4}
\end{aligned}$$

Now, technically, $\frac{1}{2}\ln(16) - \frac{1}{2}\ln 4$ is the correct answer. However, when you are taking the multiple choice portion of the AP Calc BC exam, the expectation is that trivial logarithms should be simplified, so this answer may not appear as a choice. So, we must utilize our log rules:

$$\begin{aligned}
\frac{1}{2}\ln(16) - \frac{1}{2}\ln 4 &= \frac{1}{2}\ln\left(\frac{16}{4}\right) \\
&= \frac{1}{2}\ln 4 \\
&= \ln 4^{\frac{1}{2}} \\
&= \boxed{\ln 2}
\end{aligned}$$

The Average Value

One simple application of the definite integral is the Average Value Theorem. Recall that the average of a finite set of numbers is the total of the numbers divided by how many numbers we are averaging. Or, in technical terms:

$$\text{avg} = \frac{\sum_{i=1}^{n} x_i}{n}$$

But, what does it mean to take the average value of a continuous function? Let's say you drive from home to school—what was your average velocity? What was the average temperature today? What was your average height for the first 15 years of your life? All these questions can be answered with the following formula:

The Average Value Formula

The average value of a function f on a closed interval $[a, b]$ is defined by

$$f_{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

If we look at this formula in the context of the Fundamental Theorem of Calculus, it will start to make a little more sense.

$$\begin{aligned}\int_a^b f(x)\,dx &= F(b) - F(a) \\ \frac{1}{b-a}\int_a^b f(x)\,dx &= \frac{1}{b-a}(F(b) - F(a)) \\ &= \frac{F(b) - F(a)}{b-a}\end{aligned}$$

Notice that this is just the average slope for $F(x)$ on $x \in [a, b]$. Since the derivative $F'(x)$ gives the instantaneous rate of change of $F(x)$, the average slope of $F(x)$ is the same as average value of $F'(x)$. But, since the definition in the Fundamental Theorem of Calculus says that $F'(x) = f(x)$, this is actually just the average value of $f(x)$.

Ex 3.2.5: Find the average value of $f(x) = x^2 + 1$ on $[0, 5]$.

Sol 3.2.5: We first start with our formula:

$$f_{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

Then, we can substitute in the function and the interval.

$$f_{avg} = \frac{1}{5-0}\int_0^5 \left(x^2+1\right)\, dx$$

Using either $\boxed{\texttt{MATH}}$ $\boxed{\texttt{9}}$ on our TI-84 or integrating analytically gives us $f_{avg} = \boxed{\dfrac{28}{3}}$.

Ex 3.2.6: Find the average value of $h(\theta) = \sec(\theta)\tan(\theta)$ on $\left[0, \dfrac{\pi}{4}\right]$.

Sol 3.2.6:

$$\begin{aligned} h_{avg} &= \frac{1}{b-a}\int_a^b h(\theta)\, d\theta \\ &= \frac{1}{\frac{\pi}{4}-0}\int_0^{\frac{\pi}{4}} \sec(\theta)\tan(\theta) d\theta \\ &= \boxed{\frac{4}{\pi}\left(\sqrt{2}-1\right) \approx 0.524} \end{aligned}$$

3.2 Free Response Homework

1. $\displaystyle\int_0^1 x^2\left(1+2x^3\right)^5\,dx$

2. $\displaystyle\int_0^1 x^3\left(x^4+5\right)^3\,dx$

3. $\displaystyle\int_{-1}^1 x\sqrt{4-x^2}\,dx$

4. $\displaystyle\int_1^2 \frac{x^2}{\sqrt[3]{9-x^3}}\,dx$

5. $\displaystyle\int_0^3 \frac{10t+15}{\sqrt[4]{t^2+3t+1}}\,dt$

6. $\displaystyle\int_1^2 \frac{x+1}{\sqrt{x^2+2x+4}}\,dx$

7. $\displaystyle\int_1^3 \frac{5t}{t^2+1}\,dt$

8. $\displaystyle\int_{-1}^2 \frac{1}{2x+5}\,dx$

9. $\displaystyle\int_{\sqrt{3}}^2 ye^{y^2-3}\,dy$

10. $\displaystyle\int_0^1 \frac{v^2}{8-v^3}\,dv$

11. $\displaystyle\int_3^{e^2+2} \frac{1}{x-2}\,dx$

12. $\displaystyle\int_0^{\frac{\pi}{3}} \frac{\sin(\theta)}{\cos^2(\theta)}d\theta$

13. $\displaystyle\int_e^{e^4} \frac{1}{x\sqrt{\ln x}}\,dx$

14. $\displaystyle\int_0^{\pi} \sec^2\left(\frac{t}{4}\right)\,dt$

15. $\displaystyle\int_0^{\pi} \frac{\sin(x)}{2-\cos(x)}\,dx$

16. $\displaystyle\int_2^4 \frac{1}{x\ln x}\,dx$

17. $\displaystyle\int_0^{\ln 2} \frac{e^x}{1+e^{2x}}\,dx$

18. $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^5(x)\sin(x)\,dx$

19. $\displaystyle\int_0^{\frac{\pi}{8}} \sec^2(2x)\,dx$

20. $\displaystyle\int_{e^{\frac{\pi}{4}}}^{e^{\frac{\pi}{2}}} \frac{\csc^2(\ln y)}{y}\,dy$

21. $\displaystyle\int_0^{\pi} \frac{\cos(x)}{2+\sin(x)}\,dx$

22. $\displaystyle\int_0^{\pi} \frac{\sin(y)}{2+\cos(y)}\,dy$

23. $\displaystyle\int_0^{\sqrt{\frac{\pi}{4}}} m\sec\left(m^2\right)\tan\left(m^2\right)\,dm$

24. $\displaystyle\int_0^{\frac{\pi}{4}} \sec^2(x)\tan^3(x)\,dx$

25. $\displaystyle\int_{\frac{\pi}{2}}^{\pi} \cos^9(x)\sin(x)\,dx$

26. $\displaystyle\int_0^{\pi} \cos^6\left(\frac{x}{2}\right)\sin\left(\frac{x}{2}\right)\,dx$

27. $\displaystyle\int_{\pi}^{2\pi} \cos\left(\frac{1}{2}\theta\right)\,d\theta$

28. $\displaystyle\int_2^{e^3+1} \frac{(\ln(x-1))^4}{x-1}\,dx$

29. $\displaystyle\int_0^{e^2-1} \frac{1}{x+1}\,dx$

30. $\displaystyle\int_5^{e^3+4} \frac{1}{x-4}\,dx$

Find the average value of each of the following functions over the given interval.

31. $F(x) = (x-3)^2$ on $x \in [3,\ 7]$

32. $H(x) = \sqrt{x}$ on $x \in [0,\ 3]$

33. $F(x) = \sec^2(x)$ on $x \in \left[0,\ \frac{\pi}{4}\right]$

34. $F(x) = \frac{1}{x}$ on $x \in [1\,3]$

35. $f(t) = t^2 - \sqrt{t} + 5$ on $t \in [1,\ 4]$

36. $f(t) = t^2 - \sqrt{t} + 5$ on $t \in [4,\ 9]$

37. $f(x) = \cos(x)\sin^4(x)$ on $x \in [0,\ \pi]$

38. $g(x) = xe^{-x^2}$ on $x \in [1,\ 5]$

39. $G(x) = \dfrac{x}{(1+x^2)^3}$ on $x \in [0,\ 2]$

40. $h(x) = \dfrac{x}{(1+x^2)^2}$ on $x \in [0,\ 4]$

41. If a cookie taken out of a 450° F oven cools in a 60° F room, then according to Newton's Law of Cooling, the temperature of the cookie t minutes after it has been taken out of the oven is given by

$$T(t) = 60 + 390e^{-0.205t}.$$

What is the average value of the cookie's temperature during its first 10 minutes out of the oven?

42. We know as the seasons change so do the length of the days. Suppose the length of the day varies sinusoidally with time by the equation

$$L(t) = 10 - 3\cos\left(\frac{\pi t}{182}\right),$$

where t is the number of days after the winter solstice (December 22, 2007). What was the average day length from January 1, 2008 to March 31, 2008?

43. During one summer in the Sunset, the temperature is modeled by the function

$$T(t) = 50 + 15\sin\left(\frac{\pi}{12}t\right),$$

where T is measured in F° and t is measured in hours after 7 a.m. What is the average temperature in the Sunset during the six-hour chemistry class that runs from 9 a.m. to 3 p.m.?

3.2 Multiple Choice Homework

1. $\displaystyle\int_1^4 \frac{1}{(1+\sqrt{x})^2\sqrt{x}}\,dx$

a) $\frac{6}{5}$ b) $\frac{1}{3}$ c) $\frac{2}{3}$ d) $\frac{4}{9}$ e) $\frac{3}{2}$

2. If $\int_1^4 h(x)\,dx = 6$, then $\int_1^4 h(5-x)\,dx =$

a) -6 b) -1 c) 0 d) 3 e) 6

3. $\int_e^{e^2} \frac{1}{x \ln x}\,dx =$

a) $\ln(\ln 2)$ b) $\frac{2}{e^2}$ c) $\ln 2$ d) $\frac{1-2e}{2e^2}$ e) DNE

4. Determine the average value of $y = e^{6x}$ on $x \in [0,\, 4]$.

a) $\frac{e^{24}-1}{4}$ b) $\frac{e^{24}-1}{6}$ c) $\frac{e^{24}}{24}$ d) $\frac{e^{24}}{6}$ e) $\frac{e^{24}-1}{24}$

5. Determine the average value of $g(x) = (2x+3)^2$ on $x \in [-3,\, -1]$.

a) $\frac{7}{3}$ b) -4 c) 5 d) $\frac{14}{3}$ e) 3

6. Determine the average value of $g(x) = e^{7x}$ on $x \in [0,\, 4]$.

a) $\frac{1}{14}e^{14}$ b) $\frac{1}{7}\left(e^{14}-1\right)$ c) $\frac{1}{14}\left(e^{14}-1\right)$ d) $\frac{1}{2}\left(e^{14}-1\right)$ e) $\frac{1}{7}e^{14}$

7. If the function $y = x^3$ has an average value of 9 on $x \in [0,\, k]$, then $k =$.

a) 3 b) $\sqrt{3}$ c) $\sqrt[3]{18}$ d) $\sqrt[4]{36}$ e) $\sqrt[3]{36}$

8. Find the average rate of change of $y = x^2 + 5x + 14$ on $x \in [-1\,2]$

a) 3 b) 6 c) 9 d) $\frac{65}{6}$ e) 18

9. If the average of the function $f(x) = |x - a|$ on $[-1\,1]$ is $\frac{5}{4}$, what is/are the values of a?

a) ± 1 b) $\pm\frac{1}{2}$ c) $\pm\frac{1}{4}$ d) 0 e) None of these

---

10. What is the average rate of change of the function $f(x) = x^4 - 5x$ on the closed interval $[0,\ 3]$?

a) 8.5 b) 8.7 c) 22 d) 23 e) 66

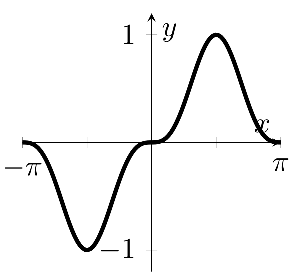
---

11. Determine the average value of $y = e^x \cos(x)$ on $x \in \left[0,\ \frac{\pi}{2}\right]$

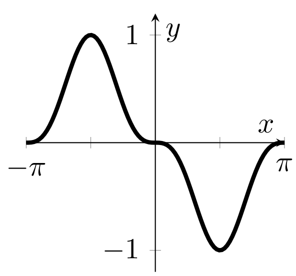
a) 0 b) 1.213 c) 1.905 d) 2.425 e) 3.810

---

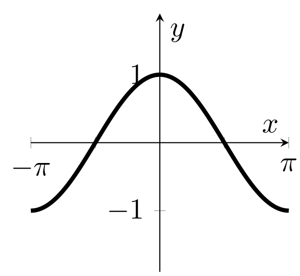
12. The graphs of five functions are shown below. Which function has a nonzero average value over the closed interval $x = [-\pi,\ \pi]$.



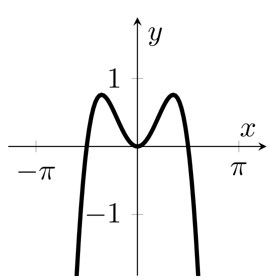
a)



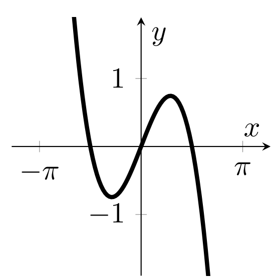
b)



c)



d)

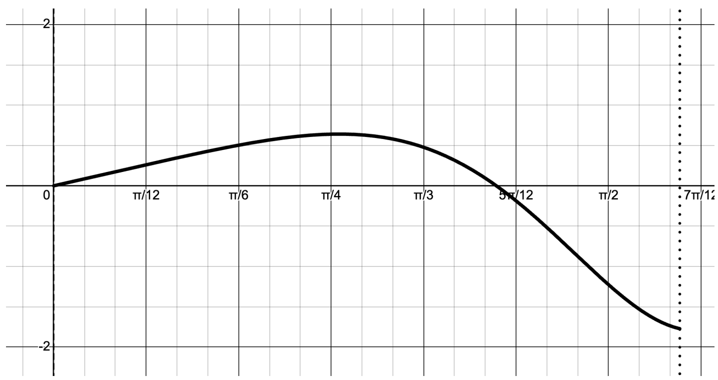


e)

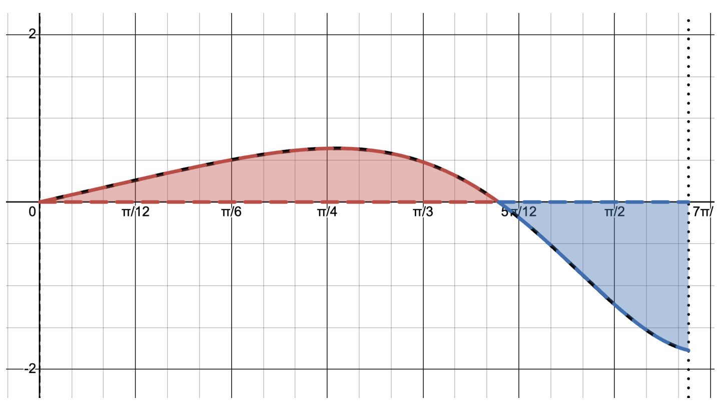
3.3: Context For Definite Integrals: Area, Displacement, and Net Change

Since we originally defined the definite integral in terms of "area under a curve," we need to consider what this idea of "area" really means in relation to the definite integral.

Recall **Ex 3.2.2**, where we had a function $y = x\cos\left(x^2\right)$ on $x \in [0\,\sqrt{\pi}]$. The graph looks like this:



In **Ex 3.2.2**, we found that $\displaystyle\int_0^{\sqrt{\pi}} x\cos\left(x^2\right)\,dx = 0$. But, there's clearly area under the curve, so how can the integral equal both the area and 0? Well, as it turns out, because the integral was created from rectangles with width dx and height $f(x)$, a negative $f(x)$ will result in a rectangle with "negative area." Take a look at the following graph:



It's clear that by our definition of the definite integral, the "area" in red would cancel out the "area" in blue. So, how would we find the actual, positive area? That is what we are going to talk about in this section.

OBJECTIVES

Relate Definite Integrals to Area Under a Curve.
Understand the Difference Between Displacement and Distance.
Understand Displacement and Distance in Other Contexts.

Ex 3.3.1: What is the area under $y = x\cos\left(x^2\right)$ on $x \in [0, \sqrt{x}]$?

Sol 3.3.1: First, let's make sure we're clear on terminology. In this context, "area under" means "area between the graph and x-axis." Now, to find the area under this graph, we have two methods.

The first method is to split the graph into two parts: one part above the x-axis and one part below the x-axis. We can integrate the part above the x-axis as usual, and for the part below the x-axis, we simply negate the result of the definite integral. The sum of these two parts should equal the area under the curve. We can begin by finding the point where the graph passes the x-axis.

$$x\cos\left(x^2\right) = 0$$

$$x = 1.253$$

Now, by the rule $\displaystyle\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where $a < c < b$, we can split our integral into two parts. Remember that we are negating the second term to compensate for the area being below the x-axis.

$$\int_0^{\sqrt{\pi}} x\cos\left(x^2\right)\,dx \to \int_0^{1.253} x\cos\left(x^2\right)\,dx + \left(-\int_{1.253}^{\sqrt{\pi}} x\cos\left(x^2\right)\,dx\right)$$

Finally, using $\boxed{\texttt{MATH}}$ $\boxed{9}$, we arrive at the answer $\boxed{1}$.

However, there is a much cleaner way to do this problem. As it turns out, if we simply take the integral of the absolute value of the integrand (the function that is being integrated), we arrive at the same answer. This is because the absolute value automatically accounts for both the parts of the graph that are above and below the x-axis, turning any negative "signed area" into positive area. Therefore, we can write

$$\int_0^{\sqrt{\pi}} \left|x\cos\left(x^2\right)\right|\,dx,$$

which turns out to also evaluate to $\boxed{1}$.

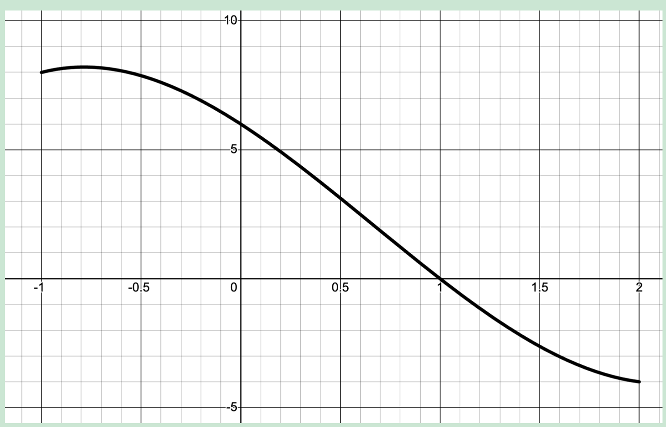
Now that we've seen an example, let's generalize some steps to finding the total area described by a definite integral.

Steps to Finding Total Area

1. Draw the function.
2. Find the zeroes between $x = a$ and $x = b$.
3. Set up separate integrals representing the area above and below the x-axis.
4. Change the sign on those integrals which represent the negative values (i.e., those where the curve is below the x-axis).
5. Solve the integral expression.

Ex 3.3.2: Find the area under $y = x^3 - 2x^2 - 5x + 6$ on $x \in [-1, 2]$.

Sol 3.3.2: Let's take a quick look at the graph:



It's clear that the graph crosses the x-axis at $x = 1$. Therefore, to get the total area, let's set up our two-term integration and solve.

$$\int_{-1}^{1} \left(x^3 - 2x^2 + 5x + 6\right)\, dx + \left(-\int_{1}^{2} \left(x^3 - 2x^2 + 5x + 6\right)\, dx\right)$$

$$= \left[\frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} + 6x\right]_{-1}^{1} - \left[\frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} + 6x\right]_{1}^{2}$$

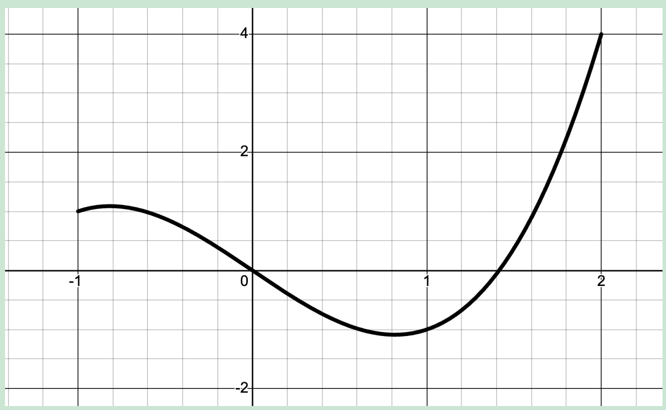
$$= \frac{32}{3} + \frac{29}{12}$$

$$= \boxed{\frac{157}{12}}$$

Now, two important things about this specific example. Instead of using the evaluation bar to denote evaluating the antiderivative, we instead can just use standard square brackets. These two notations can be used interchangeably. The second point is that the reason why we didn't use the quicker, absolute value method was because that method cannot be efficiently done by hand. On the AP exam, one can expect this problem to be on the no-calculator portion, and without a calculator, the absolute value method cannot be done.

Ex 3.3.3: Find the area under $y = x^3 - 2x$ on $x \in [-1, 2]$.

Sol 3.3.3: Once again, we begin with a sketch of the function:



Note that there are three regions here, so we will need to separate our integral into three terms. Our zeroes between $[-1, 2]$ are 0 and $\sqrt{2}$.

$$\int_{-1}^{0} \left(x^3 - 2x\right)\, dx + \left(-\int_{0}^{\sqrt{2}} \left(x^3 - 2x\right)\, dx\right) + \int_{\sqrt{2}}^{2} \left(x^3 - 2x\right)\, dx$$

$$= \left[\frac{x^4}{4} - x^2\right]_{-1}^{0} - \left[\frac{x^4}{4} - x^2\right]_{0}^{\sqrt{2}} + \left[\frac{x^4}{4} - x^2\right]_{\sqrt{2}}^{2}$$

$$= \frac{3}{4} - (-1) + 1$$

$$= \boxed{\frac{11}{4}}$$

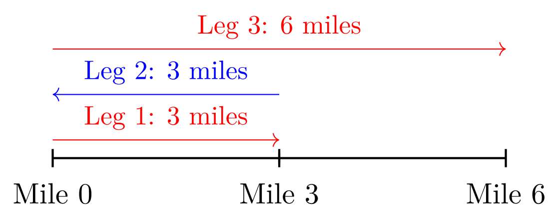
Distance and Displacement

Imagine you were leaving your house to go to school, and that school is 6 miles away. You leave your house and halfway to school you realize you have forgotten your AP Calc BC homework. You head back home, grab your assignment (that you just finished in the morning, of course), and then head to school.

There are two different questions that can be asked here. How far are you from where you started? And how far have you actually traveled? Although these questions seem extremely similar, their distinction is very important.

The first question is rather easy to answer. School is 6 miles from home, so we are 6 miles from where we started.

The second question is a little more difficult to answer. We first travel 3 miles to the the halfway point, but end up traveling another 3 miles back to home to grab the homework. Finally, you travel the 6 miles to school. We can see this in the visual below.



Therefore, the total mileage that you've covered is $3 + 3 + 6 = 12$ miles. These two questions are the root behind the concepts of **displacement** and **distance**.

Displacement $\rightarrow$ Definition: How far apart the starting position and ending position are. Note that this value can be positive or negative.

Distance $\rightarrow$ Definition: How far you travel in total. This value can only be positive.

Given v is the velocity function and x is the position function,

$$\text{Displacement} \rightarrow \int_a^b v\,dt \qquad\qquad \text{Total Distance} \rightarrow \int_a^b |v|\,dt$$

$$\text{Position at } t = a \rightarrow x(a) + \int_a^b v\,dt$$

Ex 3.3.4: A particle moves along a line so that its velocity at any time t is $v(t) = t^2 + t - 6$ (measured in meters per second).

(a) Find the displacement of the particle during the time period $1 \le t \le 4$.

(b) Find the distance traveled during the time period $1 \le t \le 4$.

Sol 3.3.4:
a)

$$\int_a^b v\,dt = \int_1^4 \left(t^2 + t - 6\right)\,dt$$

$$= \left[\frac{t^3}{3} + \frac{t^2}{2} - 6t\right]_1^4$$

$$= \boxed{-\frac{21}{2}\text{ meters}}$$

b) $v(t) = 0$ when $t = 2$.

$$\int_a^b |v|\,dt = -\int_1^2 \left(t^2 + t - 6\right)\,dt + \int_2^4 \left(t^2 + t - 6\right)\,dt$$

$$= -\left[\frac{t^3}{3} + \frac{t^2}{2} - 6t\right]_1^2 + \left[\frac{t^3}{3} + \frac{t^2}{2} - 6t\right]_2^4$$

$$= \boxed{\frac{89}{6}\text{ meters}}$$

Make sure to remember your units in your answer! The AP exam will dock points for unitless answers.

3.3 Free Response Homework

Find the area between the curve of the given equation and the x-axis on the given interval.

1a. $y = x^3$ on $x \in [0,\ 2]$

1b. $y = x^3$ on $xin[-1,\ 2]$

2a. $y = 4x - x^3$ on $x \in [0,\ 2]$

2b. $y = 4x - x^3$ on $x \in [-1,\ 2]$

3a. $y = \sin(x)$ on $x \in [0,\ \pi]$

3b. $y = \sin(x)$ on $x \in [-\pi,\ \pi]$

4a. $y = 2x^2 - x^3$ on $x \in [0,\ 2]$

4b. $2x^2 - x^3$ on $x \in [-1,\ 2]$

5. $y = x^3 - 2x^2 - 3x$ on $x = [-2,\ 2]$

6. $y = x^3 - 4x^2 + 4x$ on $x \in [-1, 2]$

7. $y = x^3 - 2x^2 - x + 2$ on $x \in [-3,\ 3]$

8. $y = \dfrac{\pi}{2}\cos(x)\sin(\pi + \pi\sin(x))$ on $x \in \left[-\dfrac{\pi}{2},\ \pi\right]$

9. $y = -\dfrac{x}{x^2+4}$ on $x \in [-2,\ 2]$

10. $y = \dfrac{4-x^2}{x^2+4}$ on $x \in [-3,\ 3]$

11. $y = \dfrac{\sin(\sqrt{x})}{\sqrt{x}}$ on $x \in [0.01,\ \pi^2]$

12. $y = x\sqrt{18 - 2x^2}$ on $x \in [-2,\ 1]$

13. $y = 3\sin(x)\sqrt{1-\cos(x)}$ on $x \in \left[-\dfrac{\pi}{2},\ \dfrac{\pi}{3}\right]$

14. $y = x^2 e^{x^3}$ on $x \in [0,\ 1.5]$

Answer the following questions regarding distance and displacement.

15. The velocity function (in meters per second) for a particle moving along a line is $v(t) = 3t - 5$.

(a) Find the displacement of the particle during the time period $0 \leq t \leq 3$.

(b) Find the distance traveled during the time period $0 \leq t \leq 3$.

16. The velocity function (in meters per second) for a particle moving along a line is $v(t) = t^2 - 2t - 8$.

(a) Find the displacement of the particle during the time period $1 \leq t \leq 6$.

(b) Find the distance traveled during the time period $1 \leq t \leq 6$.

For problems #17-20, show the setup to determine the area and solve the integral. Do not use the absolute value method.

17. Find the area under the curve $f(x) = e^{-x^2} - x$ on $x \in [-1,\ 2]$.

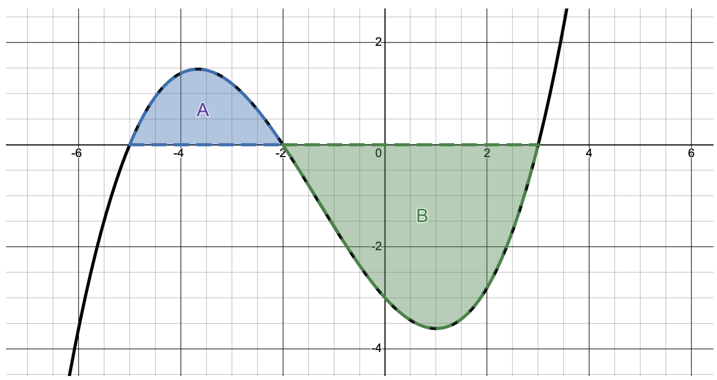
18. Find the area under the curve $f(x) = e^{-x^2} - 2x$ on $x \in [-1, 2]$.

19. Find the area under the curve $f(x) = \dfrac{x}{x^2+1} + \cos(x)$ on $x \in [0, \pi]$.

20. Find the area under the curve $g(x) = -1 - x\sin(x)$ on $x \in [0, 2\pi]$.

3.3 Multiple Choice Homework

1. The graph of $y = f(x)$ is shown below. A and B are positive numbers that represent the areas between the curve and the x-axis.

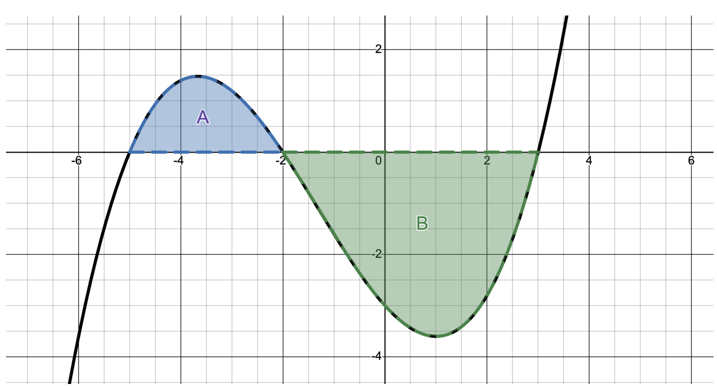


In terms of A and B, $\displaystyle\int_{-5}^{3} f(x)\,dx + \int_{-2}^{3} f(x)\,dx =$

a) A b) $A - B$ c) $2A - B$ d) $A + B$ e) $A - 2B$

---

2. The graph of $y = f(x)$ is shown below. A and B are positive numbers that represent the areas between the curve and the x-axis.

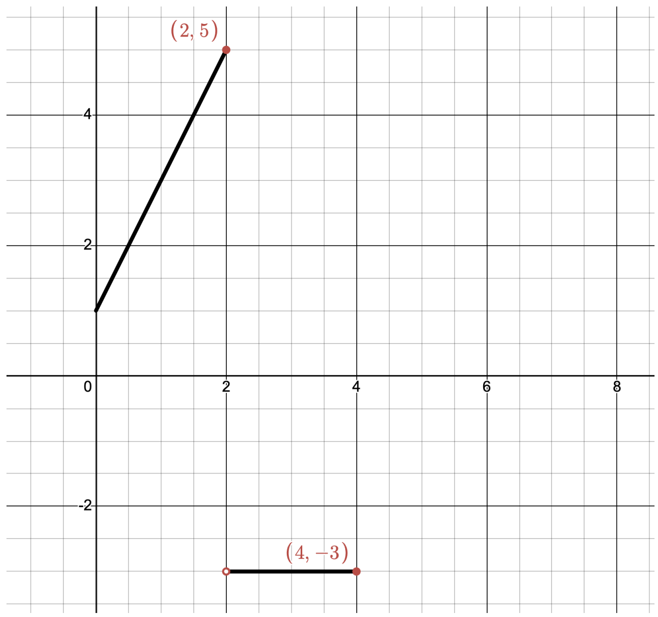


In terms of A and B, $2\int_{-5}^{3} f(x)\,dx + 3\int_{-2}^{3} f(x)\,dx =$

a) A b) $A - B$ c) $2A - B$ d) $A + B$ e) $A - 2B$

---

3. The graph of $f(x)$ on $0 \leq x \leq 4$ is shown.



a) -1 b) 0 c) 2 d) 6 e) 12

---

4. A particle moves along the x-axis so that at any time $t \geq 0$ its velocity is given by $v(t) = \ln(t+1) - 2t + 1$. What is the total **distance**, in meters, traveled by the particle during the time interval $0 \leq t \leq 3$?

a) -3.455 b) 0.704 c) 1.540 d) 2.667 e) 4.291

---

5. A particle travels along a straight line with a velocity of $v(t) = 3e^{-t^2}\sin(2t)$ meters per second. What is the total **distance**, in meters, traveled by the particle during the time interval $0 \leq t \leq 2$?

a) 0.835 b) 1.625 c) 1.661 d) 2.261 e) 5.350

---

6. A particle moves along the x-axis so that at any time $t \geq 0$ its velocity is given by $v(t) = \ln(t+1) - 2t + 1$. What is the total **displacement**, in meters, of the particle during the time interval $0 \leq t \leq 3$ seconds?

a) -3.455 b) 0.704 c) 1.540 d) 2.667 e) 4.291

---

7. A particle travels along a straight line with a velocity of $v(t) = 3e^{-t^2}\sin(2t)$ meters per second. What is the total **distance**, in meters, traveled by the particle during the time interval $0 \leq t \leq 2$?

a) 0.835 b) 1.625 c) 1.661 d) 2.261 e) 5.350

3.4: Accumulation of Rates

As we saw with the Riemann sums, in $\int f(x)\,dx$, $f(x_i) \cdot dx$ is the area of a rectangle (height times base). The $\int$ is the sum of the areas. The main concept in these "accumulation of rates" problem is that since a definite integral is a sum of values, the integral of a rate of change over some time t equals the total change over that time t.

OBJECTIVES

Analyze the Relationship Between Rates of Change and Integrals.

Beginning in 2002, CollegeBoard shifted emphasis on understanding of the accumulation aspect of the Fundamental Theorem to a new kind of rate problem. Previously, accumulation of rates problems were mostly in context of velocity and distance, though the 1996 Cola Consumption problem hinted at a new direction that these problems would take. The 2002 Amusement Park problem caught many students and teachers off guard, though. Almost every year since then, the test has included this kind of problem. Here is an example similar to the 2002 Amusement Park problem:

Ex 3.4.1:

The Amusement Park Problem (AP 2002)

The rate at which people enter a park is given by the function

$$E(t) = \frac{15600}{t^2 - 24t + 160},$$

and the rate at which they are leaving is given by

$$L(t) = \frac{9890}{t^2 - 38t + 370} - 76.$$

Both $E(t)$ and $L(t)$ are measured in people per hour where t is the number of hours past midnight. The functions are valid for when the park is open, $8 \le t \le 24$. At $t = 8$ there are no people in the park.

(a) How many people have entered the park at 4 pm ($t = 16$)? Round your answer to the nearest whole number.

(b) The price of admission is \$36 until 4 pm ($t = 16$). After that, the price drops to \$20. How much money is collected from admissions that day? Round your answer to the nearest whole number.

(c) Let $H(t) = \int_8^t E(x) - L(x)\,dx$ for $8 \leq t \leq 24$. The value of $H(16)$ to the nearest whole number is 5023. Find the value of $H'(16)$ and explain the meaning of $H(16)$ and $H'(16)$ in the context of the amusement park.

(d) At what time t, for $8 \leq t \leq 24$, does the model predict the number of people in the park is at at maximum?

Now, before we attempt this problem, there are many key phrases and concepts that we must understand. Take a look at the following table.

Key Phrases for Interpreting Accumulation of Rates Problems

Total change:	$\int_a^b R(t)\,dt$ or $\int_a^t (\text{incoming rate} - \text{outgoing rate})\,dx$
Total rate of change:	incoming rate − outgoing rate
Total amount:	initial value $+ \int_a^t (\text{incoming rate} - \text{outgoing rate})\,dx$
Instantaneous rate of change:	$\frac{dx}{dt}$ or $R(t)$
Average rate of change:	$\frac{f(b) - f(a)}{b - a}$
Average value of $f(x)$:	$\frac{1}{b-a}\int_a^b f(x)\,dx$
Amount increase/decrease:	rate of change positive/negative
Rate of change inc/dec:	$\frac{d}{dt}$[rate of change] positive/negative

Note that there is a difference between a function that is increasing or decreasing and its derivative that is increasing or decreasing.

One technique that is super useful in accumulation of rates problem is unit analysis. By analyzing the units of the problem, we are able to interpret what the question is asking, and provide an appropriate answer. For instance, if $R(t)$ is measured in miles per hour,

$$\int_a^b R(t)\,dt = \int_a^b \frac{\text{miles}}{\text{hour}}(\text{hours}) = \text{miles}$$

To that end, let's take a look at the above chart, but with the intended units of the answer.

Units for Accumulation of Rates Problems

Total change:	units of accumulated quantity (e.g., miles, people)
Total rate of change:	$\frac{\text{units of quantity}}{\text{units of time}}$ (e.g., miles/hour, people/day)
Total amount:	units of quantity (e.g., total miles, total gallons)
Instantaneous rate of change:	$\frac{\text{units of quantity}}{\text{units of time}}$
Average rate of change:	$\frac{\text{units of quantity}}{\text{units of time}}$
Average value of $f(x)$:	same units as $f(x)$
Amount increase/decrease:	units of quantity (increase or decrease)
Rate of change inc/dec:	$\frac{\text{units of rate}}{\text{units of time}}$ (e.g., acceleration: miles/hour^2)

Finally, let's go through some tips for answering these questions. There are four key parts to a good explanation:

1. The meaning of the equation.
2. The description of what is being measured.
3. The units.
4. The time frame involved.

Tips for Explaining Answers

1. **Echo the question in your answer.**
   Restate what the problem is asking so your explanation directly addresses it.

2. **Reason from the given information.**
   Base your explanation on the data, equations, or graphs provided—not on assumptions.

3. **Use and check units.**
   Always include proper units (e.g., miles, hours, people/day) and make sure they match the context.

4. **Specify the time frame.**
   When discussing change, clearly state *when* it occurs and include correct time units.

5. **Refer to functions, not numbers or graphs.**
   Describe how functions behave—whether they are positive/negative or increasing/decreasing—even if you are using a table or graph.

   - Instead of "the slope is positive," say "the function is increasing."
   - Instead of "the slope is increasing," say "the rate is increasing."
   - Avoid mentioning slope, second derivatives, or concavity unless the question specifically asks for them.

6. **Be concise and precise.**
   Express your idea in one clear sentence whenever possible. If your explanation turns into a paragraph, you are probably being unclear or unfocused.

With those instructions out the way, we can finally look at the solution for **Ex 3.4.1**.

Sol 3.4.1:
a) Since $E(t)$ is a rate in people per hour, the number of people who have entered the park will be an integral from $t = 8$ to $t = 16$. Therefore,

$$\text{Total entered} = \int_8^{16} E(t)\,dt$$

$$= 6126.105$$

$$\approx \boxed{6126 \text{ people}}$$

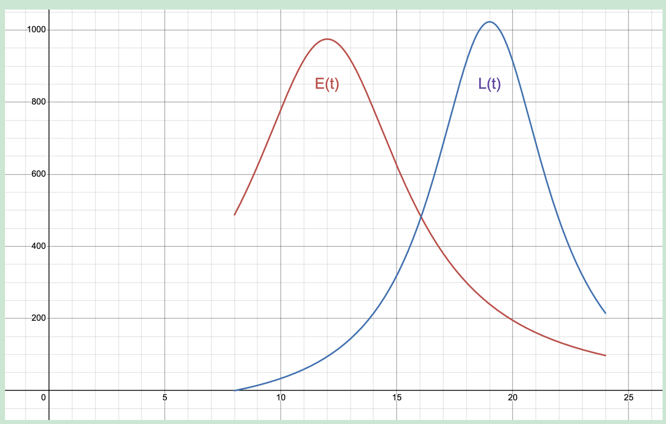
b) Since there are different entry fees for different times of day, we need to determine how many people paid each fee. So, we will have to do $\int_8^{16} E(t)\,dt$ and $\int_{16}^{24} E(t)\,dt$. Conveniently, we know that $\int_8^{16} E(t)\,dt = 6126$ people from the previous problem, so we can just find $\int_{16}^{24} E(t)\,dt$.

$$\int_{16}^{24} E(t)\,dt \approx 1808 \text{ people}$$

The total revenue that the park gets from admissions means multiplying the number of people by the admission charge. Therefore, we have

$$\begin{aligned}\text{Total revenue} &= \$36 \cdot 6126 + \$20 \cdot 1808 \\ &= \boxed{\$256696}\end{aligned}$$

c) While a graph is not necessary for solving this problem, sometimes it helps to visualize the situation. Below are the graphs of $E(x)$ and $L(x)$ on $8 \le t \le 24$:



Note how each function increases and then decreases. Since $H(t) = \int_8^t E(x) - L(x)\,dx$, we can use the Fundamental Theorem of Calculus to determine the derivative.

$$\begin{aligned}H'(t) &= \frac{d}{dt}\int_8^t E(x) - L(x)\,dx \\ &= E(t) - L(t) \\ &= E(16) - L(16) \\ &= 14\end{aligned}$$

Now let's interpret what these numbers mean. We know that $H(16) = 5023$ people. Since the derivative is a rate of change, we know that $H'(16) = 14$ means 14 people per hour. Because integrating a rate gives a total change, and $H(t)$ represents the integral of the difference between the entry and leaving rates, $H(t)$ tells us the *net number of people in the park* at 4 pm—that is, how many have entered minus how many have left since 8 am.

Meanwhile, $H'(t)$ represents the *instantaneous rate of change* of the park's population. This $H'(16) = 14$ means that **at 4 pm, the number of people in the park is increasing at a rate of 14 people per hour**.

In other words, more people are entering than leaving at that moment.

d) This question requires a little more advanced technique, so we will return to this problem in Chapter 4.

Ex 3.4.2:

The Cat Population Problem

The Peninsula Humane Society (PHS) is dedicated to the care and adoption of as many animals who they receive as possible. Since cats breed seasonally, the number of cats and kittens they receive into their facility in a given year varies roughly sinusoidally with time. The data available from 2019 show the rate $R(t)$, measured in healthy cats per month, varies with time, measured in months after New Years Day, according to the equation

$$R(t) = 120 - 88\cos\left(\frac{\pi}{6}(t-2)\right).$$

The rate $A(t)$ at which adoption occur, measured in cats per month, varies with time, measured in months after New Year's Day, according to the equation

$$A(t) = 125 - 85\cos\left(\frac{\pi}{6}(t-3)\right).$$

On New Years' Day ($t = 0$), there were 131 cats in the PHS nursery waiting to be adopted.

(a) $\displaystyle\int_0^{12} R(t)\,dt = 1440$ cats. Using the correct units, explain the meaning of this result in context of the problem.

(b) Assume $A'(10.3) = -28.008$. Using the correct units, explain the meaning of $A'(10.3)$ in context of the problem.

(c) $C(t) = 131 + \displaystyle\int_0^{12} R(t) - A(t)\,dt = 71$. Using the correct units, explain the meaning of this result in context of the problem.

(d) Using the correct units, explain the meaning of $\frac{1}{t}\int_0^t R(x)\,dx$ in the context of the problem.

(e) Suppose $R(7.4) = 185.4$ and $R'(7.4) = -30.8$. Using the correct units, explain the meaning of this information in context of the problem.

Sol 3.4.2:

a) $R(t)$ is the rate at which healthy cats per month are received at PHS. Therefore, $\int_0^{12} R(t)\,dt$ would be the total number of cats received over this time period. So, **1440 healthy cats were received at PHS between $t = 0$ and $t = 12$ (or, during the 12 months of 2019).**

b) $A(t)$ is the rate at which adoptions occur, measured in cats per month. Therefore, **the rate at which adoptions are occurring at $t = 10.3$ is decreasing by 28.008 cats per month, per month.**

c) $C(t) = 131 + \int_0^t R(x) - A(x)\,dx$ represents the total number of cats and kittens sheltered at PHS at any time t. So, we can say that **the number of cats and kittens sheltered at PHS at the end of the 12 months of 2019 is 71 cats and kittens.**

d) $\frac{1}{t}\int_0^t R(x)\,dx$ is the average number of cats and kittens being received at any time t. Therefore, $\frac{1}{t}\int_0^t R(x)\,dx$ **represents the average number of cats and kittens, in cats and kittens per month, that have been received at PHS during the interval $[0, t]$ of 2019.**

e) $R(7.4) > 0$ and $R'(7.4) < 0$. So, we can say that **at $t = 7.4$, the total number of cats and kittens that have been received at PHS is increasing at a decreasing rate of -30.8 cats and kittens per month, per month.**

Phew, that was a lot of words!! However, notice that each of the problems can be answered simply by looking back at the "Key Phrases" chart and matching the expression to the concept.

Ex 3.4.3:

The Synthetic Oil Problem

A certain industrial chemical reaction produces synthetic oil at a rate of

$$S(t) = \frac{15t}{1+3t}.$$

At the same time, the oil is removed from the reaction vessel by a skimmer that has a rate of

$$R(t) = 2 + 5\sin\left(\frac{4\pi}{25}t\right).$$

Both functions have units of gallons per hour, and the reaction runs from $t = 0$ to $t = 6$. At time $t = 0$, the reaction vessel contains 2500 gallons of oil.

(a) How much oil will the skimmer remove from the reaction vessel in this six hour period? Indicate units of measurement.

(b) Write an expression for $P(t)$, the total number of gallons of oil in the reaction vessel at time t.

(c) Find the rate at which the total amount of oil is changing at $t = 4$.

Sol 3.4.3:
a) To find the total amount removed over the time interval, we integrate the rate $R(t)$ over $0 \leq t \leq 6$.

$$\begin{aligned}\text{Amount removed} &= \int_0^6 \left(2 + 5\sin\left(\frac{4\pi}{25}t\right)\right)\, dt \\ &= \int_0^6 R(t)\, dt \\ &= \boxed{31.816 \text{ gallons}}\end{aligned}$$

b) Let $P(t)$ represent the total amount of oil in the reaction vessel at time t. Initially, $P(0) = 2500$ gallons. The net change of oil is production minus removal, which is $S(t) - R(t)$. Since the variable t is the upper boundary, we need to use a dummy variable x in the integrand. This gives us the expression

$$\boxed{P(t) = 2500 + \int_0^t (S(x) - R(x))\, dx}.$$

c) The rate of change of P is $P'(t) = S(t) - R(t)$. Therefore, at $t = 4$,

$$\begin{aligned}P'(4) &= S(4) - R(4) \\ &= \boxed{-1.909 \text{ gal/hr}}\end{aligned}$$

3.4 Free Response Homework

<u>1. The Puffin Problem</u>

The puffin population on the Skellig Islands off the coast of County Kerry, Ireland, can be modeled by a differentiable function P in terms of time t, where $P(t)$ is the number of puffins and t is measured in years, for $0 \le t \le 50$. There are 10,000 puffins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 500e^{0.05t} \text{ puffins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 110e^{0.09t} \text{ puffins per year.}$$

(a) Using the correct units, explain the meaning of $\displaystyle\int_0^{50} (B(t) - D(t))\,dt$.

(b) Using the correct units, explain the meaning of $\displaystyle\frac{1}{25}\int_0^{25} B(t)\,dt$.

(c) Using the correct units, explain the meaning of $D'(35)$,

(d) Using the correct units, explain the meaning of $\displaystyle\frac{B(35) - B(10)}{35 - 10}$.

(e) Suppose $D(35) = 2567$ and $D'(35) = 143.8$. Using the correct units, explain the meaning of these data in context of the number of puffin deaths.

<u>2. The Alcohol Metabolization Problem</u>

Metabolism is the body's process of converting ingested substances to other compounds. Alcohol is absorbed into the blood stream, then, through oxidation, it is detoxified and removed from the blood. Research shows that, after consuming three shots of alcohol in rapid succession, an average (180 lb) adult, fasting male has the alcohol metabolized at a rate modeled by $R(t) = 0.04\left(t^2 - 8t + 6\right)e^{-t}$, where $R(t)$ is the measured in percentage of alcohol in the blood stream per hour, t is measured in hours after consuming the third drink and is valid for $0 \le t \le 6$.

(a) $\displaystyle\int_0^{2.5} R(t)\,dt = 0.029$. Using the correct units, explain the meaning of this result.

(b) At $t = 2$ hours, $R(t)$ is negative and $R'(2) = 0.011$. Using the correct units, explain the meaning of this result in terms of the amount of alcohol in the blood stream.

(c) Using the correct units, explain the meaning of $\displaystyle\frac{2}{5}\int_0^{2.5} R(t)\,dt$.

(d) Using the correct units, explain the meaning of $\dfrac{R(5) - R(1)}{5 - 1}$.

3. The Novel-for-November Problem

A young novelist enters the Novel-for-November contest, where amateur writers attempt to write a full-length novel over the course of the month. After the months ends, she realizes that the rate at which she wrote varied over the month. She determines that a good model of the daily rate of her writing would be

$$W(t) = 7.5 + 7.5\cos\left(\frac{\pi}{15}(t-1)\right),$$

where t is measured in days and $t = 1$ is the morning of November 1st and $t = 31$ is the end of the day on November 30th. Furthermore, let

$$E(t) = 6 + 8\cos\left(\frac{\pi}{30}(t-10)\right)$$

model the rate at which her editor goes through the manuscript, beginning on November 10th ($t = 10$).

(a) How many pages does the writer complete during the month of November?

(b) Find the value of $W(17)$ and $W'(17)$. Using the correct units, explain the meaning of both.

(c) Find the number of pages which still need to be edited at the end of the day on November 30th.

(d) Set up, but do not solve, an integral equation that would determine how many more days would be needed to finish editing the manuscript.

4. The Tombstone Mine Problem

In 1881, the silver mines in Tombstone, Arizona, struck the local aquifer at 520 feet and began to flood. The owners of the Grand Central mine bought the Cornish engines from the Comstock mines to pump the water out. On a given day, the water was seeping into the mine at a constant rate of 100 gal/hr, and the pumps could drain the water at a rate described by the equation

$$D(t) = 414 + 375\sin\left(\frac{x^2}{72}\right) \text{ gal/hr.}$$

When the pumps start, there are 10,000 gallons of water in the mine.

(a) How many gallons of water were pumped out of the mine during the time interval $0 \leq t \leq 24$ hours?

(b) Is the level of water rising or falling at $t = 6$? Explain your reasoning.

(c) How many gallons of water are in the mine at $t = 14$ hours?

5. The Ellis Island Problem

In 1920, Dr. Quattrin's grandfather Andrea returned to America from Italy after fighting in World War I. He arrived in New York Harbor on the *SS Pannonia* and, despite having established residency in 1913, had to be processed through the Immigration Center at Ellis Island. There were 1123 non-citizen, third-class passengers on the *Pannonia* that had to go through processing. Immigrants entered the processing line at a rate modeled by the function

$$E(t) = 8843\left(\frac{t}{5}\right)^4\left(1 - \frac{t}{10}\right)^5.$$

where t, in $0 \le t \le 10$, is measured in hours after the ship began offloading immigrants. The new arrivals were processed out a rate of 250 people per hour. The *Pannonia* was the third ship in port, so there were already 2500 people in line when the *Pannonia* passenger got into line.

(a) How many passengers from the *Pannonia* had gotten in line for processing in the first 6.2 hours?

(b) Is the rate of change of people entering the processing line increasing or decreasing at $t = 6.2$?

(c) How many people were in line at $t = 6.2$?

6. The Trinary Star Problem

More than 30% of observed star systems have multiple stars, and 70% of those have more than two stars. When stars are closed together, they exchange mass in a process known as accretion. Consider a trinary system where S_1 is larger than S_2, and S_2 is larger than S_3. S_3 will lose mass to S_2, and S_2 will lose mass to S_1. While scientific readings are not available because of the time scale, let us suppose that S_2 loses mass to the larger S_1 at a rate of

$$L(t) = 1 + (0.01t)^2 + 0.23\sin\left(\frac{\pi}{25}t\right)$$

and gains mass from the smaller S_3 at a rate of

$$G(t) = 0.2 + 0.15\sqrt{t}$$

where $0 \le t \le 100$ years. $L(t)$ and $G(t)$ are measured in yottatons per year $\left(\frac{Y}{yr}\right)$. (A yottaton is 10^{26} tons, or 10^{-7} solar masses.)

(a) How much mass does S_2 lose to S_1 on $0 \leq t \leq 100$? State the units.

(b) At $t = 50$, is the mass S_2 is gaining from S_3 increasing at an increasing rate? Using the correct units, justify your answer.

(c) At what times on $0 \leq t \leq 100$ is S_2 losing as much mass to S_1 as it is gaining from S_3?

7. The Metformin Problem

A diabetic patient takes Metformin twice a day to control her blood sugar. The medication enters the bloodstream at a rate expressed by

$$M(t) = 8 - \frac{e^{0.47t}}{t+6},$$

where $M(t)$ is measured in centigrams per hour cg/hr and t is measured in hours for $0 \leq t \leq 12$. The liver cleans the medication out of the bloodstream at a rate of $L(t) = 7 - 0.46t\cos(t)$ cg/hr.

(a) How much Metformin enters the bloodstream during this 12-hour time period?

(b) After 9 hours, how much Metformin is still in her bloodstream?

(c) Find $L'(6)$ and explain the meaning of the answer, using the correct units.

(d) Set up, but do not solve, an integral equation that would determine the time when the dose of Metformin has been completely cleaned out of the bloodstream.

8. The San Francisco Intersection Problem

At an intersection in San Francisco, cars turn left at the rate

$$L(t) = 50\sqrt{t}\sin^2\left(\frac{t}{3}\right)$$

cars per hour for the time interval $0 \leq t \leq 18$.

(a) To the nearest whole number, find the total number of cars turning left on the time interval given above.

(b) Traffic engineers will consider turn restrictions if $L(t)$ equals or exceeds 125 cars per hour. Find the time interval where $L(t) \geq 125$, and find the average value of $L(t)$ for this time interval. Indicate units of measurement.

(c) San Francisco will install a traffic light if there is a two-hour time interval in which the product of the number of cars turning left and the number of cars traveling through the intersection exceeds 160,000. In every two-hour interval, 480 cars travel straight through the intersection. Does this intersection need a traffic light? Explain your reasoning.

9. The Post Office Letter Problem

Letters arrive at a post office at a rate of

$$P(t) = 8 + t\sin\left(\frac{t^3}{90}\right)$$

hundred letters per hour over the course of a workday. The day begins at 9 am ($t = 0$) and ends at 5 pm ($t = 8$). There are three hundred letters in the office at 9 am. Workers send letters out of the office at a constant rate of 5 hundred letters per hour.

(a) Find $P'(2)$. Using correct units, interpret the meaning of $P'(2)$ in the context of this problem.

(b) Find the total number of letters that arrive at the office between 9 am and noon ($t = 3$). Round to the nearest whole number of letters.

(c) Write an expression for $L(t)$, the total number of letters in the post office at time t.

10. The Flooded Basement Problem

The basement of a house is flooded, and water keeps pouring in at a rate of

$$w(t) = 95\sqrt{t}\sin^2\left(\frac{t}{6}\right)$$

gallons per hour. At the same time, water is being pumped out at a rate of

$$r(t) = 275\sin^2\left(\frac{t}{3}\right).$$

When the pump is started, at time $t = 0$, there is 1200 gallons of water in the basement. Water continues to pour in and be pumped out for the interval $0 \leq t \leq 18$.

(a) Is the amount of water increasing at $t = 15$? Why or why not?

(b) To the nearest whole number, how many gallons are in the basement at the time $t = 18$?

(c) For $t > 18$, the water stops pouring into the basement, but the pump continues to remove water until all of the water is pumped out of the basement. Let k be the time at which the tank becomes empty. Write, but do not solve, an equation involving an integral expression that can be used to find a value of k.

11. The Sewage Processing Problem

A tank at a sewage processing plant contains 125 gallons of raw sewage at time $t = 0$. During the time interval $0 \leq t \leq 12$ hours, sewage is pumped into the tank at the rate

$$E(t) = 2 + \frac{10}{1 + \ln(t+1)}.$$

During the same time interval, sewage is pumped out at a rate of

$$L(t) = 12 \sin\left(\frac{t^2}{47}\right).$$

(a) How many gallons of sewage are pumped into the tank during the time interval $0 \leq t \leq 12$ hours?

(b) Is the level of sewage rising or falling at $t = 6$? Explain your reasoning.

(c) How many gallons of sewage are in the tank at $t = 12$.

3.4 Multiple Choice Homework

1. For $t \geq 0$ hours, H is a differentiable function of t that gives the temperature, in degrees Celsius, at an Arctic weather station. Which of the following is the best interpretation of $H'(24)$?

   a) The change in temperature during the first day.

   b) The change in temperature during the 24th hour.

   c) The average rate at which the temperature changed during the 24th hour.

   d) The rate at which the temperature is changing during the first day.

   e) The rate at which the temperature is changing at the end of the 24th day.

---

2. For $t \geq 0$ hours, H is a differentiable function of t that gives the temperature, in degrees Celsius, at an Arctic weather station. Which of the following is the best interpretation of $\int_0^t H(x)\,dx$.

   a) The change in temperature during the first day.

   b) The change in temperature during the 24th hour.

   c) The average rate at which the temperature changed during the 24th hour.

d) The rate at which the temperature is changing during the first day.

e) The rate at which the temperature is changing at the end of the 24th day.

---

3. In the classic 2002 Amusement Park Problem, equations $E(t)$ and $L(t)$ were given, representing the rate at which people were entering and leaving the park respectively, for time $9 \leq t \leq 23$, the hours during which the park was open, with $t = 9$ corresponding to 9 am. Let us assume that $F(t) = E(t) - L(t)$. Which of the following is the best interpretation of $F(16)$?

a) The number of people in the park at 4 pm.

b) The number of people entering and leaving the park before 4 pm.

c) The average number of people in the park between 9 am and 4 pm.

d) The rate at which the number of people in the park is changing at 4pm.

e) The rate of change of how quickly the number of people in the park is changing at 4pm.

---

4. The cost, in dollars, to shred the confidential documents of a company is modeled by C, a differentiable function of the weights of documents in pounds. Of the following, which is the best interpretation of $C'(500) = 80$?

a) The cost to shred 500 pounds of documents is $80

b) The average cost to shred documents is $\dfrac{80}{500}$ dollar per pound.

c) Increasing the weight of documents by 500 pounds will increase the cost to shred the documents by approximately $80.

d) The cost to shred documents is increasing at a rate of $80 per pound when the weight of the documents is 500 pounds.

---

5. An ice field is melting at the rate $M(t) = 4 - \sin^3(t)$ acre-feet per day, where t is measured in days. How many acre-feet of this field will melt from the beginning of day 1 ($t = 0$) to the beginning of day 4 ($t = 3$)?

a) 6.846 b) 10.667 c) 10.951 d) 11.544 e) 11.999

6. Let $R(t)$ represent the rate in gal/hr at which water is leaking out of a tank, where t is measured in hours. Which of the following expressions represents the average rate of change of gallons of water per hour that leaks out in the first three hours?

a) $\displaystyle\int_0^3 R(t)\,dt$ b) $\displaystyle\frac{1}{3}\int_0^3 R(t)\,dt$ c) $\displaystyle\int_0^3 R'(t)\,dt$ d) $R(3) - R(0)$ e) $\displaystyle\frac{R(3) - R(0)}{3 - 0}$

---

7. The rate of natural gas sales for the year 1993 at a certain gas company is given by $P(t) = t^2 - 400t + 160000$, where $P(t)$ is measured in gallons/day and t is the number of days in 1993 from day 0 to 365. To the nearest gallon, what is the average rate of natural gas sales at this company for the 31 days of January 1993?

a) $4,777,730$ b) $4,617,930$ c) $154,120$ d) $148,965$ e) $148,561$

---

8. The rate at which ice is melting in a pond is given by $\dfrac{dV}{dt} = \sqrt{1+2^t}$, where V is the volume of the ice in cubic feet and t is the time in minutes. The amount of ice which has melted in the first five minutes is

a) 14.49 ft^3 b) 14.51 ft^3 c) 14.53 ft^3 d) 14.55 ft^3 e) 14.57 ft^3

---

9. The number of parts per million (ppm), $C(t)$, of chlorine in a pool changes at the rate of $C'(t) = 1 - 3e^{-0.2\sqrt{t}}$ ounces per day, where t is measured in days. There are 10 ppm of chlorine in the pool at time $t = 0$. How many ounces of chlorine are in the pool when $t = 9$?

a) -0.646 b) 9.354 c) -9.285 d) 9.285 e) 0.715

---

10. The amount of money in a bank account is increasing at the rate of $R(t) = 10000e^{0.06t}$ dollars per year, where t is measured in years. If $t = 0$ corresponds to the year 2005, then what is the approximate total amount of increase from 2005 to 2007?

a) $\$21,250$ b) $\$4,500$ c) $\$18,350$ d) $\$32,560$ e) $\$16,250$

---

11. The rate at which water is pumped into a tank is $r(t) = 20e^{0.02t}$, where t is in minutes and $r(t)$ in gallons per minute. Approximately how many gallons of water are pumped into the tank during the first five minutes?

a) 20 b) 22 c) 85 d) 105 e) 150

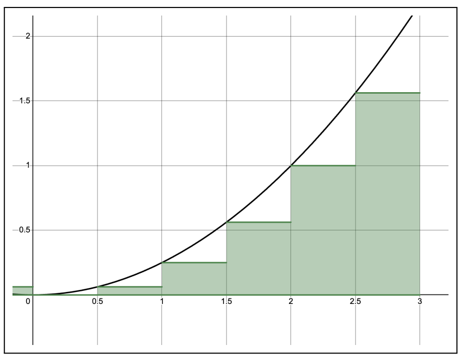
---

12. Oil is leaking from a tanker at the rate of $R(t) = 2000e^{-0.2t}$ gallons per hour, where t is measured in hours. How much oil, in gallons, leaks out of the tanker from $t = 0$ to $t = 10$?

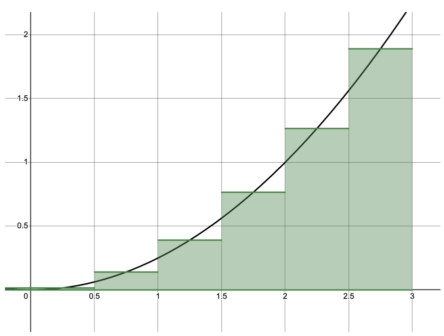
a) 54 b) 271 c) 865 d) $8,647$ e) $14,778$

3.5: Integral Approximations: Riemann Rectangles and Trapezoidal Sums

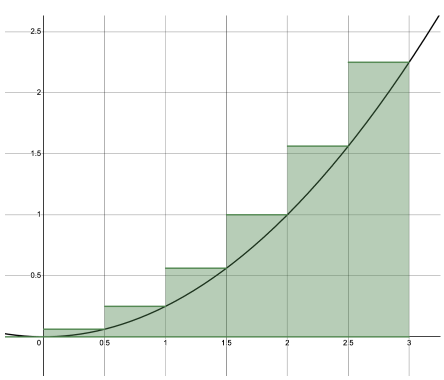
We have been focusing on anti-derivatives of functions were the equation is known. But let us suppose we need to evaluate an integral where either the function is unknown or cannot be anti-differentiated—such as $\displaystyle\int_{-2}^{3} e^{x^2}\,dx$. This is where the concepts that we touched on in the overview of the chapter come into play. If we have some exact y-values, we could approximate the area geometrically. This can be done by dividing the area in question into rectangles, and then finding the area of each rectangle. There are three way to draw the rectangles:



Left-Hand Rectangles



Midpoint Rectangles



Right-Hand Rectangles

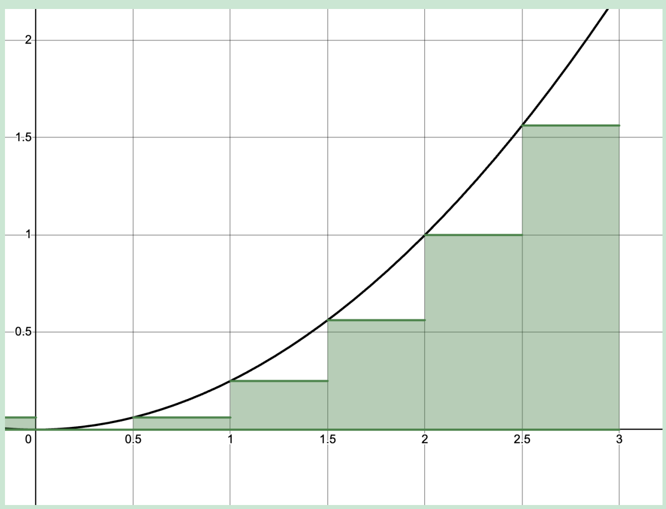
OBJECTIVES

Find Approximations of Integrals Using Different Rectangles.
Use Proper Notation When Dealing with Integral Approximation.

Ex 3.5.1: Use a left hand Riemann sum with four equal sub-intervals to approximate $\int_1^5 f(x)\,dx$ given the table of values below.

x	1	2	3	4	5
$f(x)$	1	4	9	16	25

Sol 3.5.1: Although this problem may seem a little difficult, it's actually quite simple. Let's first recall what a left hand Riemann summation should look like:



So, to find the areas of the rectangle, we multiply the width of the rectangle by the height. The width of the rectangle is the difference between adjacent x-values. Because we are asked to find a left hand Riemann sum, we are gonna start with the leftmost height. Therefore, we have

$$\int_1^5 x^2\,dx \approx (2-1)\cdot 1 + (3-2)\cdot 4 + (4-3)\cdot 9 + (5-4)\cdot 16 = \boxed{30}$$

Notice how we never use the $f(x)$ value of 25. This is because when we are using left-hand rectangles, we will never make a rectangle that

With this example, we can create some steps for approximation an integral with rectangles:

Steps to Approximating an Integral with Rectangles

1. Read the question and determine which type of rectangles to calculate

2. Write out the integral as a sum of lengths times widths.

3. Calculate the areas of each rectangle.

4. Add the areas together for your approximation.

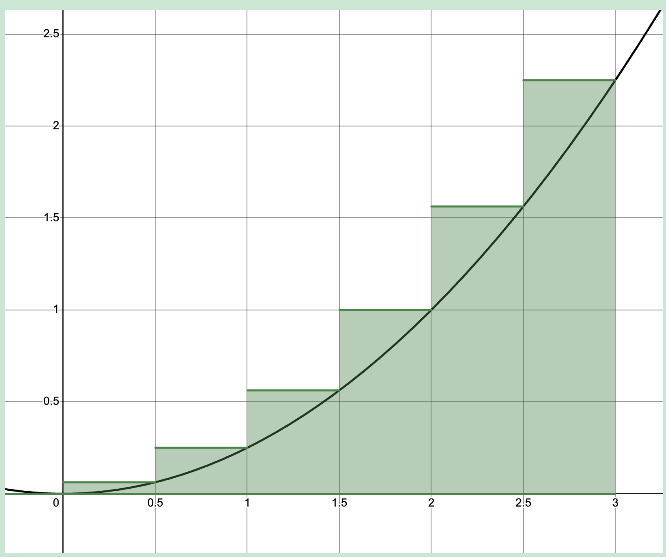
5. State answer using proper notation.

Although **Ex 3.5.1** had rectangles of equal width, this does not necessarily have to be true

Ex 3.5.2: Use a right hand Riemann sum with the sub-intervals defined below to approximate $\int_2^{14} f(x)\,dx$ given the table of values below.

x	2	5	10	14
$f(x)$	4	10	7	12

Sol 3.5.2: Let's remind ourselves what a right-hand rectangle approximation looks like:



Now, as this time we are asked to find a right hand Riemann sum, we start from the right. So, we have

$$\int_2^{14} f(x)\,dx \approx (14-10)\cdot 12 + (10-5)\cdot 7 + (5-2)\cdot 10 = \boxed{113}$$

The third kind of Riemann rectangle is where the height comes from the midpoints of the rectangles. Two conditions are needed for midpoint rectangles:

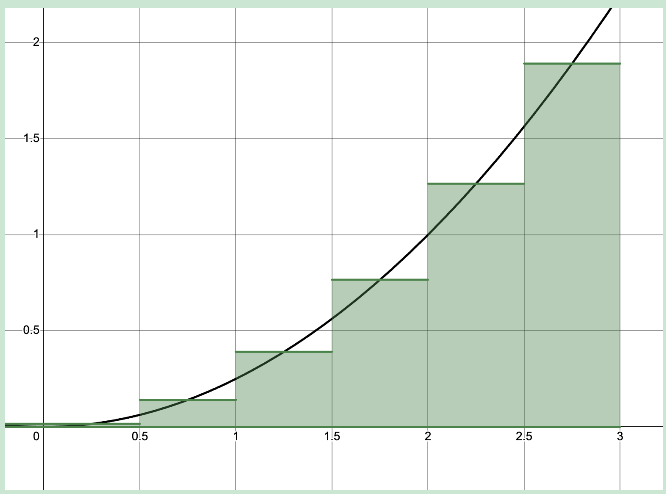
1. There needs to be an odd number of x values.
2. The even-numbered (i.e. 2nd, 4th) x-values must be midpoints of adjacent x-values.

Ex 3.5.3: The rate of consumption, in gallons per minute, recorded during an airplane flight is given by a twice differentiable and strictly increasing function $R(t)$. A table of selected values for $R(t)$ for the time interval $0 \leq t \leq 90$ is shown below.

t (minutes)	0	20	40	50	60	90	120
$R(t)$ (gallons per minute)	20	30	40	55	65	70	95

Use the Riemann sum with midpoint rectangles and the sub-intervals given by the table to approximate the value of $\displaystyle\int_0^{120} R(t)\,dt$.

Sol 3.5.3: Recall the structure of midpoint rectangles:



To find the Riemann sum with midpoint rectangles, we need to take the difference between the odd-numbered x-values as our width, and use the $R(t)$ values in between those x-values as our height. Therefore, we get

$$\int_0^{120} R(t) \approx (40-0)\cdot 30 + (60-40)\cdot 55 + (120-60)\cdot 70 = \boxed{6500}$$

Ex 3.5.4: Use the midpoint rule and the given data to approximate the value of

$$\int_0^{2.6} f(x)\,dx.$$

x	0	0.4	0.8	1.2	1.6	2.1	2.6
$f(x)$	3.5	2.3	3.2	4.3	4.7	5.9	4.1

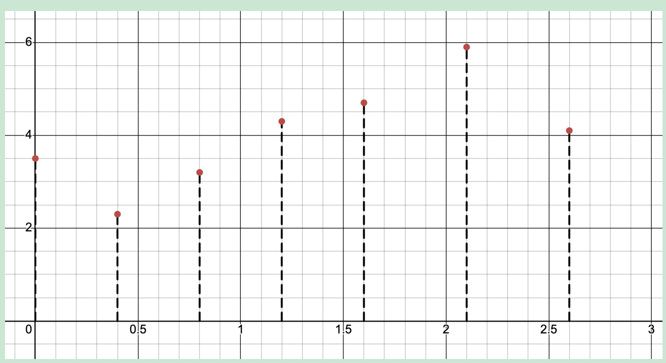
Sol 3.5.4: For this example, we simply follow the steps in **Ex 3.5.3**. This gives us

$$\int_0^{2.6} f(x)\,dx \approx (0.8-0)\cdot 2.3 + (1.6-0.8)\cdot 4.3 + (2.6-1.6)\cdot 5.9 = \boxed{11.18}$$

The left, right, and midpoint rectangle methods estimate the area under a curve by using rectangles with constant heights. The trapezoidal rule builds on this idea by connecting the tops of the rectangles with straight lines, creating shapes that better match the curve. It can be seen as the average of the left and right rectangle methods and serves as a simple step toward more accurate ways to approximate areas.

Ex 3.5.5: Using the table of data from **Ex 3.5.4** approximate $\int_0^{2.6} f(x)\,dx$ with 6 trapezoids.

Sol 3.5.5: Since this is a new method that we are approaching, let's try visualizing it. We can first start by drawing the heights.



Next, we are gonna connect the points to draw out our trapezoids.

Now, it becomes pretty clear how we can find the Riemann sum with this method. We use the difference between adjacent x-values as the height of the trapezoid, and adjacent pairs of $f(x)$-values as the two bases of the trapezoid. Therefore, let's use the area formula for a trapezoid,

$$A_{\text{trapezoid}} = \left(\frac{b_1 + b_2}{2}\right) \cdot h,$$

to calculate this Riemann sum:

$$\begin{aligned}\int_0^{2.6} f(x)\,dx &\approx \frac{3.5+2.3}{2}\cdot(0.4-0) + \frac{2.3+3.2}{2}\cdot(0.8-0.4) + \frac{3.2+4.3}{2}\cdot(1.2-0.8) \\ &+ \frac{4.3+4.7}{2}\cdot(1.6-1.2) + \frac{4.7+5.9}{2}\cdot(2.1-1.6) + \frac{5.9+4.1}{2}\cdot(2.6-2.1) \\ &= \boxed{8.15}\end{aligned}$$

However, one may notice that this process is quite cumbersome. We have to do multiple computations involving decimals just to get our answer, and sometimes this could lead to arithmetic errors. Luckily, there is a formula that could simplify this process a lot. However, note that this formula could only apply **if the heights of the trapezoid are all the same**. So, what exactly is this formula? Well, let's take a look.

Let's say we have points $x_0, x_1, x_2, \ldots, x_n$ and $f(x)$ values $f(x_0), f(x_1), f(x_2), \ldots, f(x_n)$. We know that we are integrating from x_0 to x_n, we can call those a and b, respectively. Since we know that the height of the trapezoids are the same, we know that the height of each trapezoid is the total sum of the heights divided by the number of trapezoids. Therefore, each trapezoid has a height of

$$\frac{b-a}{n}.$$

Now, let's start writing out the Riemann sum in its complete form. We do the method as we described in **Ex 3.5.5**, taking adjacent $x - values$ as the heights and adjacent pairs of $f(x)$-values as the bases. Therefore, we get

$$\int_a^b f(x)\,dx \approx \frac{f(x_0)+f(x_1)}{2}\left(\frac{b-a}{n}\right) + \frac{f(x_1)+f(x_2)}{2}\left(\frac{b-a}{n}\right) + \cdots + \frac{f(x_{n-1})+f(x_n)}{2}\left(\frac{b-a}{n}\right).$$

Factoring out $\dfrac{b-a}{n}$ and simplifying the fraction gives us

$$\int_a^b f(x)\,dx \approx \left(\frac{b-a}{n}\right)\left(\frac{f(x_0)+f(x_1)+f(x_1)+f(x_2)+\cdots f(x_{n-1})+f(x_n)}{2}\right).$$

We can simplify this even further to

$$\int_a^b f(x)\,dx \approx \left(\frac{b-a}{2n}\right)(f(x_0)+2f(x_1)+2(fx_2)+\cdots+2f(x_{n-1})+f(x_n)).$$

This gives us the Trapezoidal Rule for equal sub-intervals.

The Trapezoidal Rule for Riemann summations

$$\int_a^b f(x)\,dx \approx \left(\frac{b-a}{2n}\right)(f(x_0)+2f(x_1)+2(fx_2)+\cdots+2f(x_{n-1})+f(x_n)).$$

Let's see this formula in use with an example.

Ex 3.5.6: The following table gives values of a continuous function. Approximate the average value of the function using the Trapezoidal Rule.

x	10	20	30	40	50	60	70
$f(x)$	3.649	4.718	6.482	9.389	14.182	22.086	35.115

Sol 3.5.6: Note that this question is asking for the *average value* of the function. Therefore, we will have to use our average value formula.

$$\begin{aligned}
f_{avg} &= \frac{1}{b-a}\int_a^b f(x)\,dx \\
&= \frac{1}{70-10}\int_{10}^{70} f(x)\,dx \\
&\approx \frac{1}{70-10}\left(\frac{70-10}{2(6)}\right)(3.649+2\cdot 4.718+2\cdot 6.482+2\cdot 9.389 \\
&\qquad +2\cdot 14.182+2\cdot 22.086+35.115) \\
&= \boxed{12.707}
\end{aligned}$$

Over and Underestimations

Now that we've learned about the left, right, midpoint, and trapezoidal Riemann sum methods, it's time to explore how these approximations can overestimate or underestimate the true value of an integral. Depending on whether the function is increasing or decreasing, the rectangles or trapezoids we draw may cover too much or too little area under the curve. First, though, let's recall what we know about tangent line approximations:

→ Your approximation will be an **overestimate** if the curve is **concave down** (since your "tangent lines" will be above the curve).

→ Your approximation will be an **underestimate** if the curve is **concave up** (since your "tangent lines" will be below the curve).

Let's extend that concept to Riemann sums with the following statements.

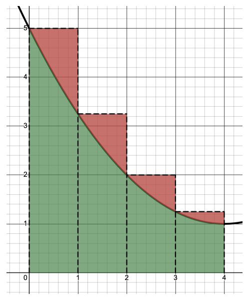
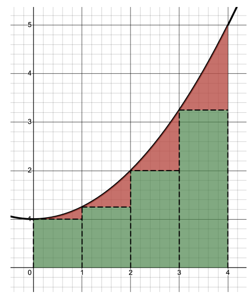
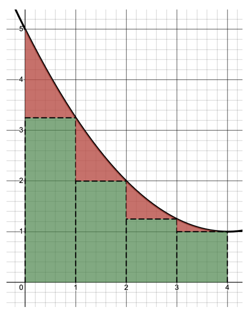
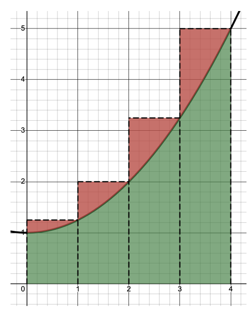
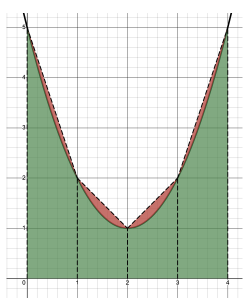
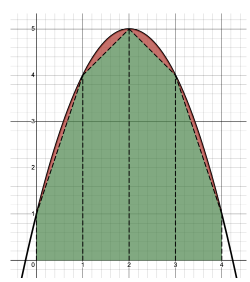
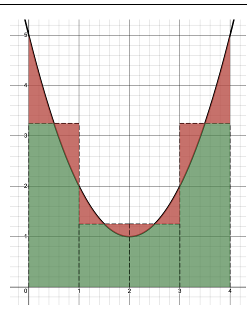
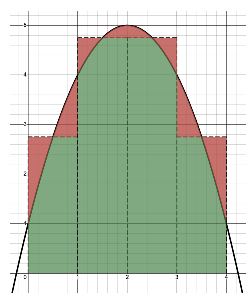
→ Left-hand rectangles are an **overestimate** if the curve is **decreasing** and an **underestimate** if the curve is **increasing**.

→ Right-hand rectangles are in **overestimate** if the curve is **increasing** and an **underestimate** if the curve is **decreasing**.

→ Midpoint Rectangles are an **overestimate** if the curve is **concave down** and an **underestimate** if the curve is **concave down**.

→ Trapezoids are an **overestimate** if the curve is **concave up** and an **underestimate** if the curve is **concave down**.

We can further illustrate this concept with the table in the following page:

	Decreasing	**Increasing**
Left Hand	 *Overestimate*	 *Underestimate*
Right Hand	 *Underestimate*	 *Overestimate*
	Concave Up	**Concave Down**
Trapezoid	 *Overestimate*	 *Underestimate*
Midpoint	 *Underestimate*	 *Overestimate*

3.5 Free Response Homework

1. The following table gives values of a continuous function:

x	0	1	2	3	4	5	6	7	8
$F(x)$	10	15	17	12	3	-5	8	-2	10

Estimate the average value of the function on $x \in [0, 8]$ using:

(a) right-hand Riemann rectangles.

(b) left-hand Riemann rectangles.

(c) midpoint Riemann rectangles.

2. The velocity of a car was read from its speedometer at 10-second intervals and recorded in the table. Use the midpoint rule to estimate the distance traveled by the car.

t (seconds)	0	10	20	30	40	50
$v(t)$ (mi/h)	0	38	52	58	55	51
t (seconds)	60	70	80	90	100	
$v(t)$ (mi/h)	56	53	50	47	45	

3. Below is a chart showing the rate of a rocket flying according to time in minutes:

t (minutes)	0	10	20	30	40	50	60
$v(t)$ (km/h)	30	28	32	18	52	48	28

Use this information to answer each of the questions below. Make sure you express your answer in correct units.

(a) Find an approximation for $\int_0^{60} v(t)\,dt$ using midpoint rectangles.

(b) Find an approximation for $\int_0^{30} v(t)\,dt$ using trapezoids.

(c) Find an approximation for $\int_{30}^{60} v(t)\,dt$ using left-hand rectangles.

(d) Find an approximation for $\int_0^{40} v(t)\,dt$ using right-hand rectangles.

4. Below is a chart showing the rate of water flowing through a pipeline according to time in minutes:

t (minutes)	0	8	16	24	32	40	48
$V(t)$ (m^3/min)	26	32	43	24	19	24	26

Use this information to answer each of the questions below. Make sure you express your answer in correct units.

(a) Find an approximation for $\int_0^{48} V(t)\,dt$ using midpoint rectangles.

(b) Find an approximation for $\int_0^{16} V(t)\,dt$ using right-hand rectangles.

5. Below is a chart of your speed driving to school in meters per second:

t (seconds)	0	30	90	120	220	300	360
$v(t)$ (m/sec)	0	21	43	38	30	24	0

Use this information to answer each of the questions below. Make sure you express your answer in correct units.

(a) Find an approximation for $\int_0^{360} v(t)\,dt$ using left-hand rectangles.

(b) Find an approximation for $\int_0^{220} v(t)\,dt$ using trapezoids.

6. Below is a chart showing the velocity of the Flash is he runs across the country:

t (seconds)	0	4	8	12	16	20	24
$v(t)$ (km/sec)	10	12	15	19	24	18	7

Use this information to answer each of the questions below. Make sure you express your answer in correct units.

(a) Find an approximation for $\int_0^{24} v(t)\,dt$ using midpoint rectangles.

(b) Find an approximation for $\int_0^{16} v(t)\,dt$ using trapezoids.

7. Below is a chart showing the rate of sewage flowing through a pipeline according to time in minutes:

t (minutes)	0	4	6	10	13	15	20
$V(t)$ (gallons/min)	83	68	82	40	38	30	68

Use this information to answer each of the questions below. Make sure you express your answer in correct units.

(a) Find an approximation for $\int_0^{20} V(t)\,dt$ using trapezoids.

(b) Find an approximation for $\int_0^{20} V(t)\,dt$ using left-hand rectangles.

8. Star Formation Rate (SFR) observations of red-shift allow scientists to track the total mass gained in a galaxy by the making of new stars. Below is a table of such data:

t	0	1	2	3	4	5	6	7	8
SFR	0.0029	0.0051	0.0055	0.0049	0.0042	0.0035	0.0029	0.0025	0.0021

SFR is measured in solar masses per cubic parsec per gigayear (millions of years), and t is measured in gigayears.

(a) Use midpoint rectangles to approximate the total mass of stars formed from $t = 0$ to $t = 8$.

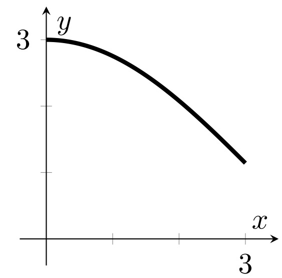
(b) Use right-hand rectangles to approximate the average solar masses per cubic parsec per gigayear.

9. Use (a) the trapezoidal rule and (b) the midpoint rule to approximate $\int_0^2 \sqrt[4]{1+x^2}\,dx$ with the specified value of $n = 8$ (8 intervals).

10. Use (a) the trapezoidal rule and (b) the midpoint rule to approximate $\int_1^2 \frac{\ln x}{1+x}\,dx$ with the specified value of $n = 10$ (10 intervals).

3.5 Multiple Choice Homework

1. The graph of the function f is shown below for $0 \leq x \leq 3$.



Of the following, which has the smallest value? Assume each approximation is done with 6 equal subintervals.

a) $\int_1^3 f(x)\,dx$

b) Left-hand Riemann approximation of $\int_1^3 f(x)\,dx$.

c) Right-hand Riemann approximation of $\int_1^3 f(x)\,dx$.

d) Midpoint Riemann approximation of $\int_1^3 f(x)\,dx$.

e) Trapezoidal approximation of $\int_1^3 f(x)\,dx$.

---

2. The table below gives the values for the rate at which water flowed into a lake, with readings taken at specific times.

Time (sec)	0	10	25	37	46	60
Rate (gal/sec)	500	400	350	280	200	180

A right-hand Riemann sum, with the five subintervals indicated by the data in the table, is used to estimate the total amount of water that flowed into the lake during the time period $0 \leq t \leq 60$. What is this estimate?

a) 1910 gal b) 14100 gal c) 16930 gal d) 18725 gal e) 20520 gal

---

3. A car is traveling on a straight road such that selected measures of the velocity have

values given on the table below.

t (sec)	10	20	40	70	80
$v(t)$ (m/sec)	90	88	100	90	85

Using four left-hand Riemann rectangles based on this table, the estimated distance traveled by the car between $t = 10$ and $t = 80$ seconds is

a) 6125 m b) 6380 m c) 6430 m d) 6495 m e) 6560 m

---

4. The function f, with some values represented in the table below, is continuous and differentiable on the closed interval [3, 12].

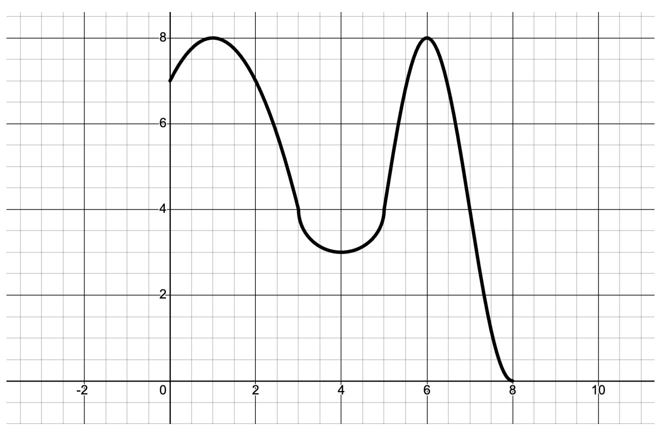
x	3	6	9	12
$f(x)$	12	8	7	5

What is the right Riemann approximation of $\int_3^{12} f(x)\,dx$?

a) 69 b) 90 c) 111 d) 126 e) 201

---

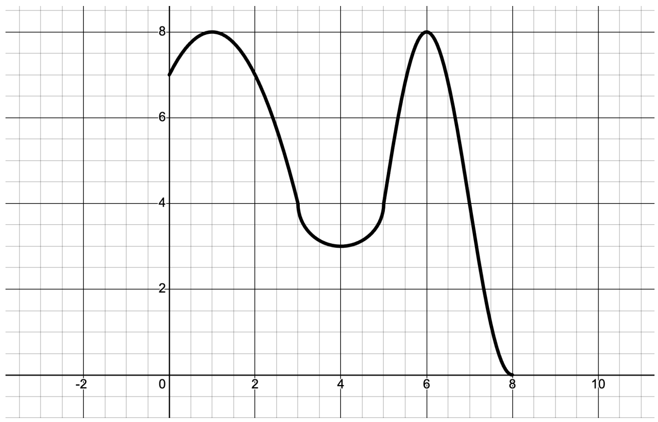
5. Consider the function f whose graph is shown below:



The approximate value of $\int_0^8 f(x)\,dx$, using eight right-hand rectangles with equal widths is

a) 18.5 b) 37 c) 40 d) 40.5 e) 44

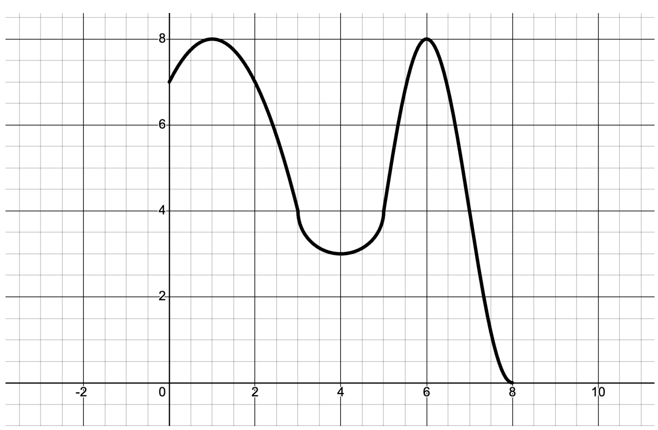
6. Consider the function f whose graph is shown below:



The approximate value of $\int_0^8 f(x)\,dx$, using eight left-hand rectangles with equal widths, is

a) 23 b) 37 c) 40 d) 40.5 e) 44

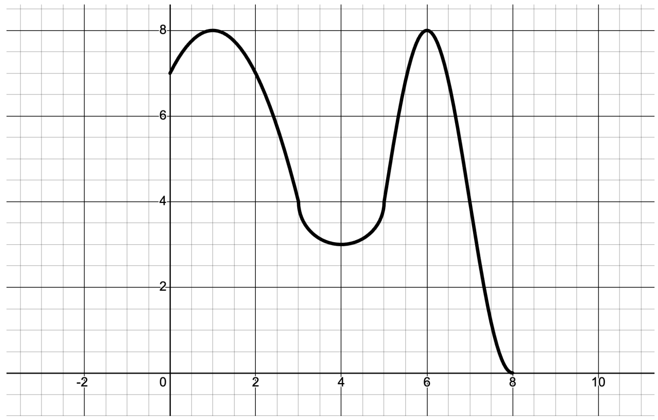
7. Consider the function f whose graph is shown below:



The approximate value of $\int_0^8 f(x)\,dx$, using eight trapezoids with equal widths, is

a) 37 b) 40 c) 40.5 d) 44 e) 48

8. Consider the function f whose graph is shown below:



The approximate value of $\int_0^8 f(x)\,dx$, using four midpoint rectangles with equal widths, is

a) 20 b) 37 c) 40 d) 40.5 e) 44

---

9. Let f be a differentiable function on the closed interval [2, 14] and which has values shown on the table below.

x	2	5	10	14
$f(x)$	12	28	34	30

Using the sub-intervals defined by the table, estimate $\int_2^{14} f(x)\,dx$ using a right-hand Riemann approximation.

a) 296 b) 312 c) 343 d) 374 e) 390

---

10. The following table lists the known values of a function $f(x)$.

x	1	2	3	4	5
$f(x)$	0	1.1	1.4	1.2	1.5

Use the trapezoidal rule to approximate $\int_1^5 f(x)\,dx$.

a) 4.1 b) 4.3 c) 4.5 d) 4.7 e) 4.9

---

11. A small plant is purchased from a nursery and the change in height of the plant is measured at the end of the each day for four days. The data, where $H(t)$ is measured in inches per day and t is measured in days, are listed below.

x	0	1	2	3	4
$f(x)$	0	1.3	1.5	2.1	2.6

Using the trapezoidal rule, which of the following represents an estimate of the average rate of growth of the plant over the four-day period?

a) $\frac{1}{4}(0 + 1.3 + 1.5 + 2.1 + 2.6)$

b) $\frac{1}{4}\left[\frac{1}{2}(0 + 1.3 + 1.5 + 2.1 + 2.6)\right]$

c) $\frac{1}{4}\left[\frac{1}{2}(0 + 2(1.3) + 2(1.5) + 2(2.1) + 2.6)\right]$

d) $\frac{1}{4}\left[\frac{1}{2}(0 + 2(1.3) + 2(1.5) + 2(2.1) + 2(2.6))\right]$

e) $\frac{1}{4}\left[\frac{1}{4}(0 + 2(1.3) + 2(1.5) + 2(2.1) + 2.6)\right]$

3.6: Intro to AP: Reasoning with Tabular Data

Reasoning with tabular data (aka Table Problems) has been one of the most commonly recurring topics on the AP exam. It is often the first free-response question on the exam.

There are two main kinds of table problems. The first kind appears in multiple choice questions that include tables of values meant to be plugged into the Chain, Product, or Quotient Rule. These tables often contain distractor values, so the challenge is identifying only the information you actually need. The second and more common type is a word problem involving an unknown function paired with specific data points. These problems closely connect to Accumulation of Rates and Rectilinear Motion topics, since they require interpreting how a function behaves based on limited numerical information. It is this second kind of problems which we will investigate here.

An analysis of the past ten years of AP exams reveals the most common subtopics on the test. They are, in order:

- Riemann and trapezoidal sums.
- Overestimate vs underestimate.
- Tangent approximations using secant lines.
- Graphic problems with the First Derivative Test.
- Interpretation of units.
- Mean value, intermediate value theorem.

For this section, we need to recall our key phrases and units for accumulation of rates problems, along with our tips for explaining answers.

OBJECTIVES

Analyze the Relationship Between Rates of Change and Integrals with Tabular Data.

Let us consider a problem similar to the ones in the Accumulation of Rates section. but with a table of data as part of the problem.

Ex 3.6.1: At 6am at the *Popular Potatoes* potato chip factory, there are already 5 tons of potatoes in the factory. More potatoes are delivered from 6am ($t = 6$) to noon ($t = 12$) at a rate modeled by

$$P(t) = 9 - \frac{9\sin(x-2)}{x-2}$$

tons of potatoes per hour. Workers arrive at 6am and begin to process the potatoes to turn them into potato chips. Their supervisor measures their rate of output every hour and records her findings in the chart below,

t	6	9	13	14	16
$C(t)$	7.9	6.5	3.9	3.1	1.3

where t represents the time after midnight (in hours) and $C(t)$ represents the rate of potatoes processed in tons per hour. The supervisor determines that the workers' rate of processing is a decreasing function throughout the day.

(a) How many tons of potatoes arrive at the *Popular Potatoes* factory between 6am and noon?

(b) Use a left Riemann sum with sub-intervals indicated by the table to approximate $\int_6^{16} C(t)\,dt$. Using correct units, explain the meaning of this value in the context of the problem.

(c) Is your approximation in part (b) an under or over-approximation? Explain.

(d) Approximate $C'(11.5)$. Explain the meaning of your answer

(e) The workers end their shift at 4pm. At that time, are there still potatoes in the factory left to process? Explain your reasoning.

Sol 3.6.1:
a) Since this question asks for a total amount, we know that we must find the integral of the rate of potatoes that are being delivered to the factory. Therefore, we have

$$\text{Total tons} = \int_6^{12} P(t)\,dt = \int_6^{12} 9 - \frac{9\sin(x-2)}{x-2} = \boxed{54.899 \text{ tons of potatoes}}$$

b) Using our setup for approximation with left-hand rectangles, we have

$$\int_6^{16} C(t)\,dt \approx (9-6)\cdot 7.9 + (13-9)\cdot 6.5 + (14-13)\cdot 3.9 + (16-14)\cdot 3.1 = 59.8$$

Therefore, **approximately 59.8 tons of potatoes were processed into chips between 6am and 4pm**.

Note the use of the word "approximately." Because the Riemann sum is an *approximation*, we must use "approximately" in our answer.

c) For this question, let's simply refer to our chart for over and underestimations. From the table, we know that function $C(t)$ is a decreasing function, and we also know that we are creating left-hand Riemann rectangles. Therefore, we know that our Riemann approximation must be an $\boxed{\text{overestimation}}$.

d) The first thing we notice about the problem is that $t = 11.5$ is not a value on our table. So, to approximate $C'(11.5)$, let's utilize the average rate of change formula, using the closest t values greater than and less than 11.5.

$$C'(11.5) \approx \frac{f(b) - f(a)}{b - a} = \frac{3.9 - 6.5}{13 - 9} = -0.65$$

Therefore, we can say that **at $t = 11.5$, the rate of potatoes processed is decreasing by approximately 0.65 tons per hour per hour**.

Once again, notice how we must use "approximately" in our answer!

e) To determine if there were still potatoes left to process, we need to take the total number of potatoes that have been processed and subtract that from the total number of potatoes there were to process.

From the problem, we know that we start with 5 tons of potatoes, and we found from part (a) that we accumulate an extra 54.899 tons of potatoes. We also know from part (b) that approximately 59.8 tons of potatoes were processed. Therefore, we have

$$\text{Potatoes left} = 5 + 54.899 - 59.8 = 0.099 \text{ tons}$$

Therefore, **there are still potatoes in the factory left to process**.

Some astute students may notice that our value of 59.8 was an estimate, so how do we know for certain that not all the potatoes were processed? Well, in part (c), we determined that this value was an overestimate. This means it represents the maximum possible processed amount. Since we are subtracting this maximum from $5 + 54.899$, even this largest possible subtraction still leaves a positive amount. Any more accurate value (which would be smaller) would leave even more potatoes remaining. Therefore, we can conclude with certainty that there are still potatoes left to process.

Ex 3.6.2: Below is a chart showing specific values of the rate of sewage flowing through a pipeline according to time in minutes.

t (minutes)	0	4	6	10	13	15	20
$C(t)$ (gallons/min)	83	68	83	48	38	30	38

Assume $V(t)$ is a continuous and differentiable function.

(a) Estimate $V'(7)$. Show the work that leads to your answer. Indicate the units.

(b) Use a trapezoidal sum with subintervals indicated by the table to approximate $\int_0^{20} V(t)\,dt$. Using correct units, explain the meaning of this value in the context of the problem.

(c) Find the value of $\int_0^{20} V'(t)\,dt$ and explain the meaning of this value in the context of the problem.

Sol 3.6.2:
a) Just like in part (d) in **Ex 3.6.1**, we will utilize the average rate of change formula with the closest t values great than and less than 7.

$$V'(7) \approx \frac{f(b) - f(a)}{b - a} = \frac{48 - 83}{10 - 6} = \boxed{-8.75 \text{ gal/min}^2}$$

b) Using our setup for the trapezoidal rule, we get

$$\begin{aligned}\int_0^{20} V(t)\,dt \approx{}& \frac{83 + 68}{2} \cdot (4 - 0) + \frac{68 + 83}{2} \cdot (6 - 4) + \frac{83 + 48}{2} \cdot (10 - 6) \\ &+ \frac{48 + 38}{2} \cdot (13 - 10) + \frac{38 + 30}{2} \cdot (15 - 13) + \frac{30 + 38}{2} \cdot (20 - 15) \\ &= 1082\end{aligned}$$

Therefore, we can say that **approximately 1082 gallons of sewage flowed through the pipeline between $t = 0$ minutes and $t = 20$ minutes**.

c) For this problem, since we see an integral and a derivative, we know that we must utilize the Fundamental Theorem of Calculus. Therefore, we have

$$\int_0^{20} V'(t)\,dt = V(20) - V(0) = 38 - 83 = -45 \text{ gal/min}$$

From this result, we can say that **the total change of the rate of flow between $t = 0$ and $t = 20$ minutes is -45 gallons per minutes**.

3.6 Free Response Homework

1. Below is a chart showing the rate of water flowing through a pipeline according to time in minutes. Use this information to answer each of the questions below.

t (minutes)	0	8	16	24	32	40	48
$V(t)$ (m^3/min)	26	32	43	24	19	24	26

(a) Estimate $V'(7)$. Show the work that leads to your answer. Indicate the units.

(b) Find $\displaystyle\int_8^{40} V'(t)\,dt$.

(c) Use a trapezoidal sum with sub-intervals indicated by the table to approximate $\displaystyle\int_0^{48} V(t)\,dt$. Using correct units, explain the meaning of this value in the context of the problem.

(d) Using correct units, explain the meaning of $\displaystyle\frac{1}{48}\int_0^{48} V(t)\,dt$ in the context of the problem.

2. A small plant is purchased from a nursery and the change in height of the plant is measured at the end of each day for four days. The data, where $H(t)$ is measured in millimeters per day and t is measured in days, are listed below.

t (days)	0	8	16	24	32	40	48
$H(t)$ (mm per day)	26	32	43	24	19	24	26

(a) Estimate $H'(3)$. Show the work that leads to your answer. Indicate the units.

(b) Explain how one would know that the plant's growth is not increasing at a decreasing rate.

(c) Use right-hand rectangles with sub-intervals indicated by the table to approximate $\displaystyle\int_0^{4} H(t)\,dt$. Using correct units, explain the meaning of this value in the context of the problem.

(d) Using correct units, explain the meaning of $\displaystyle\frac{1}{4}\int_0^{4} H(t)\,dt$ in the context of the problem.

3. The rate of consumption of fuel, in gallons per minute, recorded during an airplane flight is given by a twice differentiable and strictly increasing function $R(t)$. A table of selected values of $R(t)$ for the time interval $0 \le t \le 90$ is shown below.

t (minutes)	0	20	40	50	60	90
$H(t)$ (gallons/min)	20	30	40	55	65	70

(a) Estimate $R'(30)$. Show the work that leads to your answer. Indicate the units.

(b) Use right-hand Riemann rectangles to approximate $\int_0^{90} R(t)\,dt$ and indicate units of measure. Explain the meaning of $\int_0^{90} R(t)\,dt$ in terms of the fuel consumption.

(c) Use left-hand rectangles to find $\frac{1}{70}\int_{20}^{90} R(t)\,dt$. Using the correct units, explain the meaning of $\frac{1}{70}\int_{20}^{90} R(t)\,dt$ in terms of the fuel consumption.

4. A diabetic patient tests his blood glucose level every morning. After being put on insulin, the data below shows the glucose levels, $G(t)$, in milligrams per deciliter (mg/dL) over one week.

t (days)	1	2	3	4	5	6	7
$G(t)$ (mg/dL)	233	198	185	168	147	130	147

(a) Estimate $G'(3.7)$. Using the correct units, explain the meaning of the result.

(b) Use midpoint Riemann rectangles to approximate $\int_1^7 G(t)\,dt$. Using the correct units, explain the meaning of $\frac{1}{7}\int_1^7 G(t)\,dt$ in terms of the patient's glucose levels.

(c) Ignoring the last data point, $M(t) = 237.6e^{-0.082t}$ is a model of $G(t)$. Find $M'(3.7)$. Is $M(t)$ decreasing at an increasing rate? Show the work that leads to your conclusion.

5. Diabetic patients take a test called A1c every three months, which measures the three-month average percentage of glycated hemoglobin (that is, hemoglobin covered in glucose.) Let $A(t)$ represent the A1c score, measured as a percentage. The following data shows a patient's A1c score over the course of 21 months.

t (months)	0	3	6	9	12	15	18	21
$A(t)$ (%)	10.2	10.0	10.5	9.1	8.0	8.9	8.3	8.6

(a) Find $\int_0^{21} A'(t)\,dt$. Show the work that leads to your answer. Explain the meaning of $\int_0^{21} A'(t)\,dt$ in terms of A1c scores.

(b) Use a right-hand Riemann approximation to approximate $\int_0^{21} A(t)\,dt$. Using the correct units, explain the meaning of $\frac{1}{21}\int_0^{21} A(t)\,dt$ in terms of the patient's A1c score.

(c) A model of $A(t)$ is $B(t) = -0.305x + 10.571 + 0.1\sin(\pi x)$. Find $\frac{1}{21}\int_0^{21} B(t)\,dt$.

6. A family leases solar panels on their house. At the end of the year, they receieve a report, including the table below, which shows the monthly production $P(t)$ of electricity, in kilowatts per month (kW/month), from the panels.

t (months)	0	1	2	3	4	5	6
$P(t)$ (kW/month)	160.3	192.8	345.7	746.1	944.2	873.0	1128.6
t (months)	7	8	9	10	11	12	
$P(t)$ (kW/month)	928.3	851.3	751.3	535.5	216.4	150.7	

(a) Use a right-hand Riemann approximation to approximate $\int_0^{12} P(t)\,dt$. Indicate the units.

(b) A model of $P(t)$ is $k(t) = 660 - 489\cos\left(\frac{\pi}{6}t\right)$ is a model of $P(t)$. Find $\int_0^{12} k(t)\,dt$.

(c) Using the model $k(t) = 660 - 489\cos\left(\frac{\pi}{6}t\right)$, show that the production is decreasing at $t = 9$. Is the production decreasing at an increasing rate?

7. Dr. Quattrin analyzes his PG&E bill to track his consumption of both electricity ($C_e(t)$) and gas ($C_g(t)$) over the course of a year. The tables below are the result. $C_e(t)$ is measured in kilowatts (kW) and $C_g(t)$ is measured in therms (thm).

t (months)	0	1	2	3	4	5	6
$C_e(t)$ (kW)	390.7	660	667.1	538.4	420.5	412.1	347.8
$C_g(t)$ (thm)	87.6	84.6	109	116	79.8	53.9	42.9
t (months)	7	8	9	10	11	12	
$C_e(t)$ (kW)	287.5	303.1	322.4	342.5	390.3	384.2	
$C_g(t)$ (thm)	24.9	25.6	18	20.3	48.9	91.8	

(a) Approximate $C_e'(3.4)$ and $C_g'(3.4)$. Using correct units, explain the meaning of these estimations in terms of increasing and/or decreasing consumption of each commodity at $t = 3.4$

(b) Use right-hand Riemann rectangles to approximate $\int_0^{12} C_e(t)\,dt$. Indicate the units.

(c) Use midpoint Riemann rectangles to approximate $\int_0^{12} C_g(t)\,dt$. Indicate the units.

(d) Using the correct units, explain the meaning of $\frac{1}{12}\int_0^{12} C_g(t)\,dt$.

8. Dr. Quattrin decides to lease solar panels from Sunrun Solar. After a year, he reanalyzes his PG&E bill to track both his consumption of electricity ($C_e(t)$) and his production of electricity ($P_e(t)$) over the course of a year. The tables below show the consumption of electricity, measured in kilowatts (kWs).

t (months)	0	1	2	3	4	5	6
$C_e(t)$ (kW)	326.5	660.0	667.1	538.4	420.5	412.1	347.8
t (months)	7	8	9	10	11	12	
$C_e(t)$ (kW)	287.5	303.1	322.4	342.5	390.3	384.2	

The equation below is a model for the production in kW per month that PG&E buys back.

$$P_e(t) = 407 - 374.2\cos\left(\frac{\pi}{6}t\right)$$

(a) How much power does PG&E buy back from the Quattrins over the course of the year? Indicate the units.

(b) Using a trapezoidal approximation, approximate the amount of power the Quattrins consume over the course of the year. Based on this estimate and your answer for part (a), does Dr. Quattrin owe PG&E for electricity at the end of the year, or does PG&E owe Dr. Quattrin a refund?

(c) Electricity costs \$0.28 per kW. Write an expression for amount due on the PG&E bill at time t months.

9. Dr. Quattrin's paternal grandmother's family originated in the alpine town of Sauris, Italy, where the temperature in January changes at a rate of $W(t)$ degrees Celsius per hour. $W(t)$ is a twice-differentiable, increasing, and concave up function with selected values in the table below. At midnight ($t = 0$), the temperature in Sauris is $-8°$ C.

t (hours after midnight)	0	1	3	6	8
$W(t)$ (deg Celsius per hr)	-2.6	-3.1	-1.2	1.9	2.5

(a) At approximately what rate is the rate of change of the temperature changing at 2am ($t = 2$)? Indicate the units.

(b) Use a right Riemann sum with sub-intervals indicated by the table to approximate $\int_0^8 W(t)\,dt$. Using correct units, explain the meaning of this value in the context of the problem.

(c) Set up, but do not solve, an integral equation which would determine the temperature in Sauris at 1pm.

10. 1998 AP Calculus AB #3

11. 2001 AP Calculus AB #2

12. 2007 AP Calculus BC #5

Full Name: AP Calculus BC

Date: Chapter 3 Practice Test

Multiple Choice Section **20 Minutes; No Calculator**

Show All Work

1. $\int_1^8 \sqrt[3]{x^2} =$

a) $\frac{3}{5}$ b) $\frac{9}{2}$ c) 3 d) $\frac{93}{5}$ e) $\frac{96}{5}$

---

2. $\int_1^8 \frac{1}{(2+\sqrt[3]{x})\sqrt[3]{x^2}} =$

a) $\frac{7}{2}$ b) $\frac{21}{2}$ c) $\frac{1}{3}\ln\frac{3}{2}$ d) $\ln\frac{3}{2}$ e) $3\ln\frac{4}{3}$

---

3. Based on the information below, find $\int_1^{-2} (3g(x) + 2f(x))\,dx$

$\int_{-2}^5 f(x)\,dx = -2$	$\int_1^5 f(x)\,dx = 3$
$\int_{-2}^1 g(x)\,dx = 4$	$\int_5^1 g(x)\,dx = 9$

a) -8 b) -2 c) 0 d) 3 e) 5

---

4. For $t \geq 0$ hours, H is a differentiable function of t that gives the rate of change in temperature, in degrees Celsius per hour, at an Arctic weather station. In what units would $\dfrac{1}{t}\displaystyle\int_0^t H'(x)\,dx$ be measured?

a) hours
b) degrees Celsius
c) degrees Celsius per hour
d) degrees Celsius per hour per hour
e) hours per degrees Celsius

---

5. $\displaystyle\int_0^1 x^2 \tan\left(x^3\right)\,dx =$

a) $\dfrac{1}{3}\sec^2(1)$
b) $\dfrac{1}{3}\sec^2(1) - \dfrac{1}{3}$
c) $\dfrac{1}{3}\ln|\sec(1)| + \dfrac{1}{3}$
d) $\dfrac{1}{3}\ln|\sec(1)|$
e) $\dfrac{1}{3}\ln|\sec(3)|$

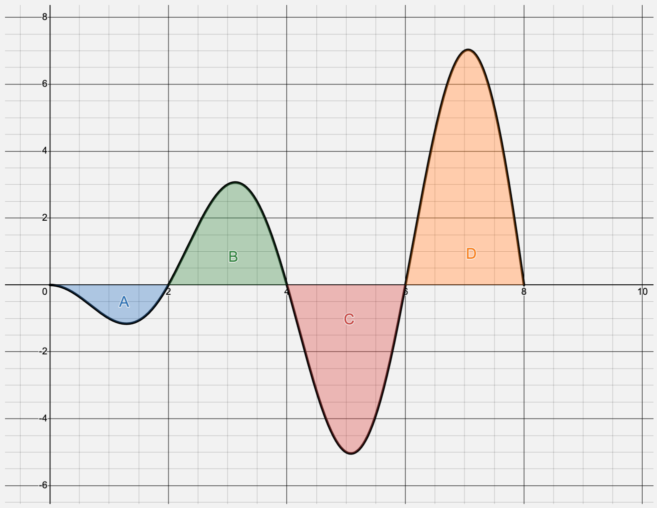
---

6. A ski resort uses a snow machine to control the snow level on a ski slope. Over a 24-hour period the volume of snow added to the slope per hour is modeled by the equation $S(t)$. The rate that the snow melts is modeled by $M(t)$. Both $M(t)$ and $S(t)$ are measured in yd^3/h and t is measured in hours for $0 \leq t \leq 24$. At time $t = 0$, the slope holds 50 yd^3 of snow. Which of the following expresses the average rate of change in the number of cubic yards of snow falling on the slope at t hours?

a) $\displaystyle\int_0^t S(x)\,dx$
b) $\dfrac{1}{t}\displaystyle\int_0^t S(x)\,dx$
c) $S(t)$
d) $S'(t)$
e) $\dfrac{\displaystyle\int_0^t S(x)\,dx - 50}{t - 0}$

---

7. In the graph below, A, B, C, and D are positive numbers that represent the areas between the curve $y = f(x)$ and the x-axis. If $A = 4$, $B = 8$, $C = 11$, and $D = 15$, what is the average value of $y = f(x)$ on $x \in [0, 8]$?

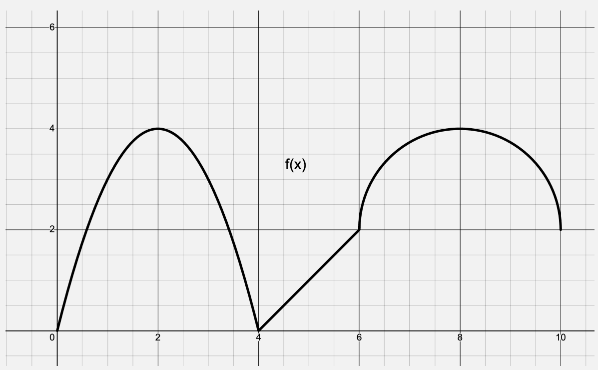


a) -1 b) 1 c) 8 d) 8 e) 48

---

8. Find the average rate of change of $y = \cos(3x)$ on $x \in \left[0, \dfrac{\pi}{6}\right]$.

a) $\dfrac{1}{3}$ b) 1 c) 3 d) $-\dfrac{6}{\pi}$ e) $\dfrac{2}{\pi}$

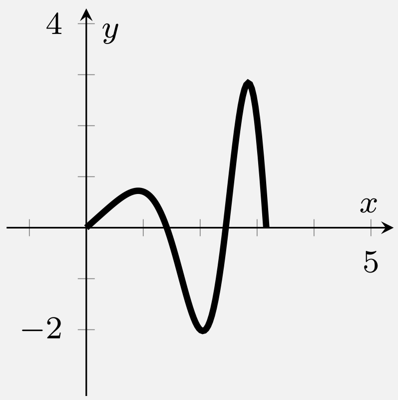


9. The approximate value of $\int_0^1 0f(x)\,dx$, using 5 right-hand rectangles with equal width, is

a) 20 b) 24 c) 28 d) 32 e) 36

Show All Work

1. Consider the function $g(x) = x\cos\left(\dfrac{\pi x^2}{4}\right)$ which has the graph below.



Note that the zeroes are at $x = 0,\ \sqrt{2},\ \sqrt{10}$.

(a) Show the u-sub setup for $\displaystyle\int_0^{\sqrt{10}} g(x)\,dx$ and solve it by calculator.

(b) Set up, but do not solve, an integral expression to determine the area between $y = g(x)$ and the x-axis on $x \in \left[0,\ \sqrt{10}\right]$.

2. <u>The Carbon Sequestration Problem</u>

One innovative approach to global warming is to capture carbon dioxide that is created as a byproduct of producing concrete and storing the CO_2 in the sandstone in depleted natural gas fields. At a particular site on a particular day, the CO_2 is injected into the sandstone at a rate of

$$I(t) = 121 \sin\left(\frac{\pi}{65}t^2\right).$$

The CO_2 stabilizes the ground and forces remaining natural gas upward where it can be extracted at a rate of

$$E(t) = 50 = 50 \cos\left(\frac{\pi}{8}t\right).$$

$I(t)$ and $E(t)$ are measured in metric tons per hour and t is measured in hours where $0 \le t \le 8$.

(a) How many metric tons of CO_2 is injected into the field over $0 \le t \le 8$?

(b) Find the value of $I(4)$ and $I'(4)$. Using the correct units, explain the meaning of both.

(c) Find the time if any, when the rate of injection of CO_2 is equal the rate of extraction of natural gas.

(d) Find the total change of gasses in the sandstone during this 8-hour day. Using the correct units, explain the results.

3. With 89.5 sacks in his 14-year career, Bryant Young is the 49ers all-time sack leader. The table below shows the rate $Y(g)$, measured in sacks per game, during his four most productive seasons, 1996 to 1999.

g (games completed)	28	44	56	68	84
$Y(g)$ (sacks per game)	0.5	0.72	0.33	0.79	0.69

(a) Using a right-hand Riemann approximation, determine the approximate number of sacks Bryant Young had during the four years. Round to the nearest whole number.

(b) Using the data on the table, estimate $Y'(75)$. Based on this estimate and the data on the table, was Young's sack total increasing at an increasing or decreasing rate? Using the correct units, explain your answer.

g (games completed)	28	44	56	68	84
$B(g)$ (sacks per game)	0.5	0.72	0.33	0.79	0.69

(c) The table above shows Nick Bosa's sack rates over his first four seasons. Use a right-hand Riemann approximation to approximate $\frac{1}{50}\int_0^{50} B(g)\,dg$. Using the correct units, explain the result in the context of the problem.

(d) Disregarding his injury-shortened 2020 season, a model for Bosa's sack rate per game can be given with the equation

$$S_B(g) = 0.15\sqrt{g}.$$

To the nearest whole number, how many more sacks than Bryant Young's does this model predict Nick Bosa will have between their 28th and 84th games?

Chapter 3 Answer Key

Chapter 4:

Advanced Derivative Applications

Chapter 4 Overview:

There are two main contexts for derivatives: graphing and motion. In this chapter, we will consider both these applications of the derivative. Much of this is a review of material covered in precalculus. Key topics include:

- Extremes
- The First and Second Derivative Tests
- Optimization
- Derivative-related theorems
- Intervals of increasing and decreasing
- Advanced graphing using the derivative
- Rectilinear motion

This chapter makes up a solid portion of the multiple choice, along with also being at least one full free response question (and often parts of others) on the AP exam.

4.1: Extrema and the Derivative Tests

One of the most valuable aspects of calculus is that it allows us to find extreme values of functions. The ability to find maximum or minimum values of functions has wide-ranging applications. Every industry has uses for finding extremes, from optimizing profit and loss, to maximizing output of a chemical reaction, to minimizing surface areas of packages. This one tool of calculus eventually revolutionized the way the entire world approached every aspect of industry. It allowed people to solve formerly unsolvable problems.

OBJECTIVES

Find Critical Values and Extreme Values for Functions
Use the 1st and 2nd Derivative Tests to Identify Maxima v.s. Minima

But, before we begin, let's review some definitions from last year, which will be useful in this chapter as well as subsequent chapters.

Critical Value → The x-coordinate of the extreme.

Maximum Value → The y-coordinate of the high point.

Minimum Value → The y-coordinate of the low point.

Relative Extremes → The highest or lowest point in a given section of the curve.

Absolute Extremes → The highest or lowest point of the whole curve.

Interval of Increasing → The interval of x-values for which the curve is rising from left to right.

Interval of Decreasing → The interval of x-values for which the curve is dropping from left to right.

Critical values of a function occur when:

1) $\dfrac{dy}{dx} = 0$

2) $\dfrac{dy}{dx}$ does not exist

3) the function reaches an endpoint of its domain

It's also helpful to remember that a **critical value** is referring specifically to the value of the x, while the **extreme value** refers to the value of the y.

As we already know, the first derivative is what allows us to algebraically find extremes, and the First Derivative Test allows us to interpret critical values as maxima or minima. Since the first derivative of a function tells us when that function is increasing or decreasing, we can figure out if a critical value is associated with a maximum or minimum depending on the sign change of the derivative. Let's quickly refresh our memory on the First Derivative Test.

The First Derivative Test

As the sign pattern of the first derivative is viewed left to right, the critical value represents a:

1) relative maximum if the sign changes from $+$ to $-$

2) relative minimum if the sign changes from $-$ to $+$

3) neither a relative max or min if the sign does not change

Ex 4.1.1: Apply the First Derivative Test to the function $y = 5x^4 - 10x^2$.

Sol 4.1.1:

$$\frac{dy}{dx} = 20x^3 - 20x = 0$$

$$20x\left(x^2 - 1\right) = 0$$

$$x = -1,\ 0,\ 1$$

Now, let's make our sign pattern.

y'	$-$	0	$+$	0	$-$	0	$+$
x		-1		0		1	

So, there appear to be critical values at x values of -1, 0, and 1. By utilizing the First Derivative Test, we can see that we have a $\boxed{\text{minimum at } -1}$, a $\boxed{\text{maximum at } 0}$, and another $\boxed{\text{minimum at } 1}$.

Ex 4.1.2: Find the maximum values for the function $g(x) = \sqrt{x^3 - 9x}$.

Sol 4.1.2: For this question, we have to be careful! Because we are dealing with a square root, we will have domain restrictions. We can find such restrictions by starting with a sign pattern for $g(x)$.

$g(x)$	$-$	0	$+$	0	$-$	0	$+$
x		-3		0		3	

Now, since we know that the argument of a square root has to be positive, we know that our domain is when $x^3 - 9x \geq 0$. Therefore, based on our sign pattern, our domain is $x \in [-3,\, 0] \cup [3, \infty)$.

With that out the way, we can now take the first derivative and apply the First Derivate Test.

$$g'(x) = \frac{3x^2 - 9}{2\sqrt{x^3 - 9x}}$$

$$g'(x) = 0 \text{ when } 3x^2 - 9 = 0 \therefore x = \pm\sqrt{3}$$

$$g'(x) = DNE \text{ when } x^3 - 9x = 0 \therefore x = \pm 3,\, 0$$

Since $x = \sqrt{3}$ is not in the domain of the function, our critical values are at $x = \pm 3,\, 0, -\sqrt{3}$. Therefore, our sign pattern looks like:

$g'(x)$	DNE	$+$	0	$-$	DNE				DNE	$+$
x	-3		$-\sqrt{3}$		0		$\sqrt{3}$		3	

From our sign pattern, we can conclude that at $x = -\sqrt{3}$, we have a maximum value, and by substituting this value into the function, we find the maximum value is at $\boxed{y = 3.224}$.

The First Derivative Test is not the only way to test whether critical values are associated with maximum or minimum values. If you recall, the second derivative can show us intervals of concavity. This is very useful because if we know that the curve is concave down at a critical value, it must be associated with a maximum. Similarly, if the curve is concave up, the critical value must be associated with a minimum value.

The Second Derivative Test

For a function f:

1) If $f'(c) = 0$ and $f''(c) > 0$, then f has a relative minimum at c

2) If $f'(c) = 0$ and $f''(c) < 0$, then f has a relative maximum at c

Now, we can revisit **Ex 1.5.7** part (c), which we previously could not do without the Second Derivative Test.

Ex 4.1.3: Consider the curve given by $x^2 + 4xy + y^2 = -12$.

(c) Find the value(s) of $\dfrac{d^2y}{dx^2}$ at the point(s) found in part (b). Does the curve have a local maximum, a local minimum, or neither at those points? Justify your answer.

(Ex 1.5.7)

Sol 4.1.3: What is really asked here is to apply the Second Derivative Test, because we cannot create a sign pattern for non-functions. The y is not isolated in the equation. Recall that the two points we found in part (b) were $(-4,\ 2)$ and $(4,\ -2)$.

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{dy}{dx} = -\frac{x+2y}{2x+y}\right]$$

$$= \frac{(2x+y)\left(1+2\dfrac{dy}{dx}\right) - (x+2y)\left(2+\dfrac{dy}{dx}\right)}{(2x+y)^2}$$

$$\left.\frac{d^2y}{dx^2}\right|_{(-4,2)} = \frac{(2(-4)+2)\,(1+2(0)) - (-4+2(2))\,(2+0)}{(2(-4)+2)^2}$$

$$= \frac{6}{(-6)^2} > 0$$

$$\left.\frac{d^2y}{dx^2}\right|_{(4,-2)} = \frac{(2(4)-2)\,(1+2(0)) - (4+2(-2))\,(2+0)}{(2(4)+2)^2}$$

$$= \frac{6}{(-6)^2} < 0$$

$\boxed{(-4, 2) \text{ will be a minimum}}$ because the second derivative is positive.

$\boxed{(4, -2) \text{ will be a maximum}}$ because the second derivative is negative.

Ex 4.1.4: Use the Second Derivative Test to determine if the critical values of $g(t) = 27t - t^3$ are at a maximum or minimum value.

Sol 4.1.4:

$$g'(t) = 27 - 3t^2 = 0$$

$$t = \pm 3$$

So, -3 and 3 are critical values. Now, let's take a look at the second derivative of $g(t)$ at those values.

$$g''(t) = -6t$$

$$g''(-3) = -6(-3) = 18$$

$$g''(3) = -6(3) = -18$$

Finally, we can apply the Second Derivative Test. Since $g'(-3) = 0$ and $g''(-3) > 0$, we can determine that $\boxed{t = -3 \text{ is a minimum}}$. Similarly, since $g'(3) = 0$ and $g''(3) < 0$, we can determine that $\boxed{t = 3 \text{ is a maximum}}$. Note that the numerical value of the second derivative is irrelevant — only the sign matters.

Ex 4.1.4: Determine if the critical values of $y = \left(3 - x^2\right) e^x$ are at a maximum or a minimum value.

Sol 4.1.4: First, let's find the critical values.

$$\frac{dy}{dx} = \left(3 - x^2\right) e^x + e^x(-2x) = 0$$

$$-e^x \left(x^2 + 2x - 3\right) = 0$$

$$-e^x(x+3)(x-1) = 0$$